disha
Publication Inc

SUPER 10 MOCK TESTS for NTA

JEE MAIN 2022

with Optional Questions

Corporate Office

DISHA PUBLICATION
45, 2nd Floor, Maharishi Dayanand Marg,
Corner Market, Malviya Nagar, New Delhi - 110017
Tel : 49842349 / 49842350

Typeset by Disha DTP Team

Table of Contents

Mock Test 1

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

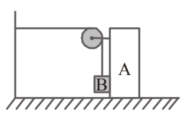
PHYSICS

Section - A

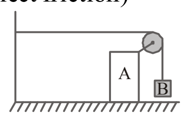
1. Two balls of same mass and carrying equal charge are hung by threads of length l from a fixed support. At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, x between the balls is proportional to :

   (1) l (2) l^2 (3) $l^{2/3}$ (4) $l^{1/3}$

2. A 20 kg block B is suspended from a cord attached to a 40 kg cart A. Find the ratio of the acceleration of block in cases (i) and (ii) shown in the figure immediately after the system is released from rest. (neglect friction)



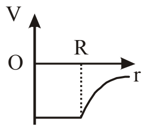
Case (i)

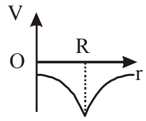


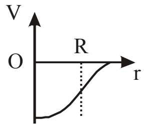
Case (ii)

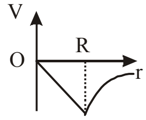
(1) $\frac{\sqrt{2}}{3}$ (2) $3\sqrt{2}$ (3) $\frac{3}{2}$ (4) $\frac{3}{2\sqrt{2}}$

3. The diagram showing the variation of gravitational potential of earth with distance from the centre of earth is

(1)


(2)


(3)


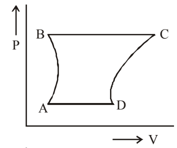
(4)


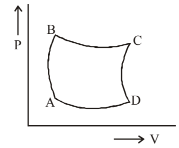
4. In "Al" and "Si", if temperature is changed from normal temperature to 70 K then
(1) The resistance of Al will increase and that of Si will decrease
(2) The resistance of Al will decrease and that of Si will increase
(3) Resistance of both decrease
(4) Resistance of both increase

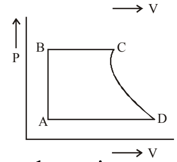
5. Two rods of length d_1 and d_2 and coefficients of thermal conductivities K_1 and K_2 are kept touching each other. Both have the same area of cross-section, the equivalent thermal conductivity is
(1) $K_1 + K_2$ (2) $K_1d_1 + K_2d_2$
(3) $\frac{d_1K_1 + d_2K_2}{d_1 + d_2}$ (4) $\frac{d_1 + d_2}{(d_1/K_1 + d_2/K_2)}$

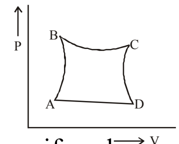
6. Charge q is uniformly spread on a thin ring of radius R. The ring rotates about its axis with a uniform frequency f Hz. The magnitude of magnetic induction at the centre of the ring is
(1) $\frac{\mu_0 qf}{2R}$ (2) $\frac{\mu_0 q}{2fR}$
(3) $\frac{\mu_0 q}{2\pi fR}$ (4) $\frac{\mu_0 qf}{2\pi R}$

7. A certain amount of gas is taken through a cyclic process (ABCDA) that has two isobars, one isochore and one isothermal. The cycle can be represented on a P-V indicator diagram as :

(1)


(2)


(3)


(4)


8. A goods train accelerating uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity u and the guard's room passes with velocity v. The middle wagon of the train passes the pole with a velocity.
(1) $\frac{u+v}{2}$ (2) $\frac{1}{2}\sqrt{u^2+v^2}$
(3) $\sqrt{uv}$ (4) $\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

9. A wheel is rotating at 900 r.p.m. about its axis. When power is cut off it comes to rest in 1 minute. The angular retardation in rad/s^2 is
(1) $\pi/2$ (2) $\pi/4$
(3) $\pi/6$ (4) $\pi/8$

10. Two springs of force constants 300 N/m (Spring A) and 400 N/m (Spring B) are joined together in series. The combination is compressed by 8.75 cm. The ratio of energy stored in A and B is $\frac{E_A}{E_B}$. Then $\frac{E_A}{E_B}$ is equal to :
(1) $\frac{4}{3}$ (2) $\frac{16}{9}$
(3) $\frac{3}{4}$ (4) $\frac{9}{16}$

11. A particle of mass m is acted upon by a force F given by the empirical law $F = \frac{R}{t^2} v(t)$. If this law is to be tested experimentally by observing the motion starting from rest, the best way is to plot :

(1) log v(t) against $\frac{1}{t}$ (2) v(t) against t^2

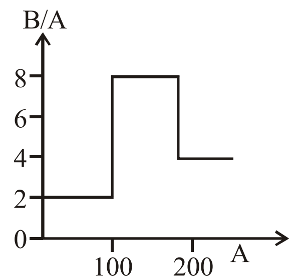
(3) log v(t) against $\frac{1}{t^2}$ (4) log v(t) against t

12. In case of a p-n junction diode at high value of reverse bias, the current rises sharply. The value of reverse bias is known as

(1) cut off voltage (2) zener voltage

(3) inverse voltage (4) critical voltage

13. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below.



(1) Fusion of two nuclei with mass numbers lying in the range of 1 < A < 50 will release energy

(2) Fusion of two nuclei with mass numbers lying in the range of 51 < A < 100 will release energy

(3) Fission of a nucleus lying in the mass range of 100 < A < 200 will release energy when broken into two equal fragments

(4) Fission of a nucleus lying in the mass range of 120 < A < 180 will release energy when broken into two equal fragments

14. If 10% of a radioactive material decays in 5 days, then the amount of the original material left after 20 days is approximately

(1) 60% (2) 66%

(3) 70% (4) 75%

15. When white light passes through a dispersive medium, it breaks up into various colours. Which of the following statements is true?

(1) Velocity of light for violet is greater than the velocity of light for red colour.

(2) Velocity of light for violet is less than the velocity of light for red.

(3) Velocity of light is the same for all colours

(4) Velocity of light for different colours has nothing to do with the phenomenon of dispersion

16. A plate of mass (M) is placed on a horizontal frictionless surface and a body of mass (m) is placed on this plate. The coefficient of dynamic friction between this body and the plate is μ. If a force 3μmg is applied to the body of mass (m) along the horizontal, the acceleration of the plate will be

(1) $\frac{\mu m}{M} g$ (2) $\frac{\mu mg}{M+m}$

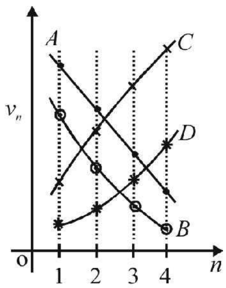
(3) $\frac{3\mu mg}{M}$ (4) $\frac{2\mu mg}{M+m}$

17. Lights of two different frequencies, whose photons have energies 1 eV and 2.5 eV respectively, successively illuminate a metal whose work function is 0.5 eV. The ratio of the maximum speeds of the emitted electrons will be

(1) 1 : 5 (2) 1 : 4

(3) 1 : 2 (4) 1 : 1

18. Which of the plots shown in the figure represents speed (v_n) of the electron in a hydrogen atom as a function of the principal quantum number (n)?



(1) B (2) D

(3) C (4) A

19. An engine has an efficiency of 1/6. When the temperature of sink is reduced by 62°C, its efficiency is doubled. Temperatures of source and sink are

(1) 99°C, 37°C (2) 124°C, 62°C

(3) 37°C, 99°C (4) 62°C, 124°C

20. A sinusoidal voltage of peak value 283 V and angular frequency 320/s is applied to a series LCR circuit. Given that R = 5 Ω, L = 25 mH and C = 1000 μF. The total impedance, and phase difference between the voltage across the source and the current will respectively be :

(1) 10 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

(2) 7 Ω and 45°

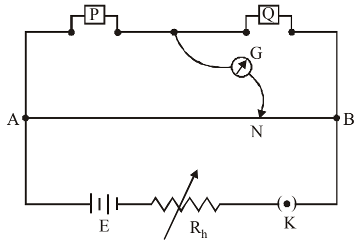
(3) 10 Ω and $\tan^{-1}\left(\frac{8}{3}\right)$

(4) 7 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

Section - B

21. A toy–car, blowing its horn, is moving with a steady speed of 5 m/s, away from a wall. An observer, towards whom the toy car is moving, is able to hear 5 beats per second. If the velocity of sound in air is 340 m/s, the frequency of the horn of the toy car is close to ________ Hz.

22. In a meter bridge experiment resistances are connected as shown in the figure. Initially resistance P = 4 Ω and the neutral point N is at 60 cm from A. Now an unknown resistance R is connected in series to P and the new position of the neutral point is at 80 cm from A. The value of unknown resistance R is ________ohm.

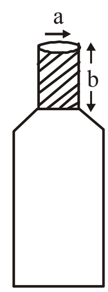


23. The circular head of a screw gauge is divided into 200 divisions and move 1 mm ahead in one revolution. If the same instrument has a zero error of –0.05 mm and the reading on the main scale in measuring diameter of a wire is 6 mm and that on circular scale is 45. The diameter of the wire is ________ mm.

24. The radius of curvature of a thin plano-convex lens is 20 cm (of curved surface) and the refractive index is 1.5. If the plane surface is silvered, then it behaves like a concave mirror of focal length ________ cm.

25. Three resistors of 4 Ω, 6 Ω and 12 Ω are connected in parallel and the combination is connected in series with a 1.5 V battery of 1 Ω internal resistance. The rate of Joule heating in the 4 Ω resistor is ________ watt.

26. A bottle has an opening of radius a and length b. A cork of length b and radius $(a + \Delta a)$ where ($\Delta a << a$) is compressed to fit into the opening completely (see figure). If the bulk modulus of cork is B and frictional coefficient between the bottle and cork is μ then the force needed to push the cork into the bottle is $(x\,\pi\mu Bb)\Delta A$.



Find the value of x.

27. A sinusoidal voltage of amplitude 25 volt and frequency 50Hz is applied to a half wave rectifier using P-n junction diode. No filter is used and the load resistor is 1000W. The forward resistance R_f of ideal diode is 10W. The percentage rectifier efficiency is _______.

28. An insect trapped in a circular groove of radius 12 cm moves along the groove steadily and completes 7 revolutions in 100 sec. What is the linear speed (in cm/s) of the motion?

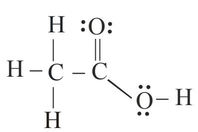
29. The magnetic field of earth at the equator is approximately 4×10^{-5} T. The radius of earth is 6.4×10^6 m. Then the dipole moment of the earth of the order of $10x$ Am^2. Find the value of x.

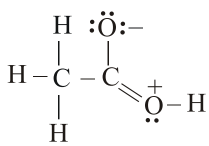
30. A particle starts S.H.M. from the mean position. Its amplitude is a and total energy E. At one instant its kinetic energy is 3E/4, its displacement at this instant is $y = \frac{a}{x}$. Find the value of x.

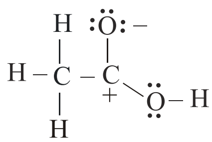
CHEMISTRY

Section - A

31. Which of the following resonance structure is lowest in energy?

A. 

B. 

C. 

(1) A
(2) B
(3) C
(4) All have same energy

32. Which of the following pairs have identical bond order ?
(1) N_2, O_2^{2+} (2) N_2, O_2^-
(3) N_2^-, O_2 (4) O^{2+}, N_2

33. In a compound AOH, electronegativity of 'A' is 2.1, the compound would be
(1) Acidic
(2) Neutral towards acid & base
(3) Basic
(4) Amphoteric

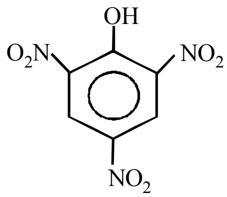
34. Which of the following orders is wrong?
(1) Electron affinity– N < O < F < Cl
(2) Ist ionisation potential – Be < B < N < O
(3) Basic property– MgO < CaO < FeO < Fe_2O_3
(4) Reactivity–Be < Li < K < Cs

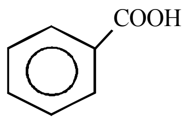
35. The following species will not exhibit disproportionation reaction
(1) ClO^- (2) ClO_2^-
(3) ClO_3^- (4) ClO_4^-

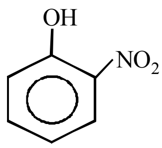
36. Given, $HF + H_2O \xrightarrow{K_a} H_3O^+ + F^-$;
$F^- + H_2O \xrightarrow{K_b} HF + OH^-$.
Which relation is correct ?
(1) $K_b = K_w$ (2) $K_b = \frac{1}{K_w}$
(3) $K_a \times K_b = K_w$ (4) $\frac{K_a}{K_b} = K_w$

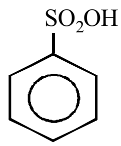
37. The oxidation states of sulphur in the anions SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order
(1) $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$
(2) $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$
(3) $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$
(4) $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

38. Among the electrolytes Na_2SO_4, $CaCl_2$, $Al_2(SO_4)_3$ and NH_4Cl, the most effective coagulating agent for Sb_2S_3 sol is
(1) Na_2SO_4 (2) $CaCl_2$
(3) $Al_2(SO_4)_3$ (4) NH_4Cl

39. Which of the following will not be soluble in sodium carbonate solution?
(1) 

(2) 

(3) 

(4) 

40. Although Al has a high oxidation potential it resists corrosion because of the formation of a tough, protective coat of
(1) $Al(NO_3)_2$ (2) AlN
(3) Al_2O_3 (4) $Al_2(CO_3)_2$

41. Which is used as medicine?
(1) PVC (2) Terylene
(3) Glyptal (4) Urotropine

42. In Lassaigne's test, the organic compound is fused with a piece of sodium metal in order to
(1) increase the ionisation of the compound
(2) decrease the melting point of the compound
(3) increase the reactivity of the compound
(4) convert the covalent compound into a mixture of ionic compounds

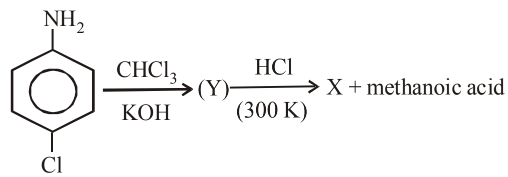
43. An aqueous solution of colourless metal sulphate M gives a white precipitate with NH_4OH. This was soluble in excess of NH_4OH. On passing H_2S through this solution a white ppt. is formed. The metal M in the salt is
(1) Ca (2) Ba
(3) Al (4) Zn

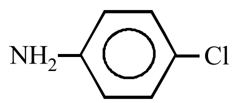
44. Which of the following oxidising reaction of $KMnO_4$ occurs in acidic medium?
(i) Fe^{2+} (green) is converted to Fe^{3+} (yellow).
(ii) Iodide is converted to iodate.
(iii) Thiosulphate oxidised to sulphate.
(iv) Nitrite is oxidised to nitrate.
(1) (i) and (iii) (2) (i) and (iv)
(3) (iv) only (4) (ii) and (iv)

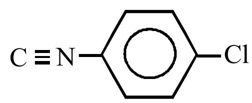
45. Which of the following compound cannot be used in preparation of iodoform?
(1) CH_3CHO (2) CH_3COCH_3
(3) HCHO (4) 2-propanol

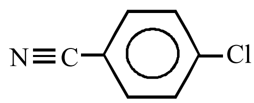
46. Anhydrous $AlCl_3$ cannot be obtained from which of the following reactions ?
(1) Heating $AlCl_3.6H_2O$
(2) By passing dry HCl over hot aluminium powder
(3) By passing dry Cl_2 over hot aluminium powder
(4) By passing dry Cl_2 over a hot mixture of alumina and coke

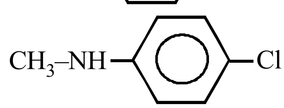
47. Identify X in the sequence given :



(1) 

(2) 

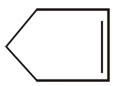
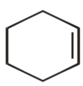
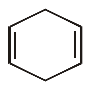
(3) 

(4) 

48. Select the rate law that corresponds to the data shown for the following reaction $A + B \longrightarrow C$

Expt. No.	[A]	[B]	Initial Rate
(i)	0.012	0.035	0.10
(ii)	0.024	0.070	0.80
(iii)	0.024	0.035	0.10
(iv)	0.012	0.070	0.80

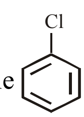
(1) Rate = $K[B]^3$ (2) Rate = $K[B]^4$
(3) Rate = $K[A][B]^3$ (4) Rate = $K[A]^2[B]^2$

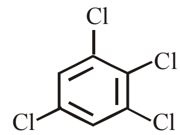
49. An alkene upon ozonolysis yield $CHO-CH_2-CH_2-CH_2-CHO$ only. The alkene is
(1) $CH_2=CH-CH_2-CH_2-CH_2-CH_2-CH_3$
(2) 
(3)  (4) 

50. An inorganic compound gives off O_2 when heated, turns an acidic solution of KI violet and reduces acidified $KMnO_4$. The compound is
(1) SO_3 (2) KNO_3
(3) H_2O_2 (4) All of these

Section - B

51. Ionization energy of gaseous Na atoms is 495.5 kJ mol^{-1}. Calculate the lowest possible frequency of light that ionizes a sodium atom in terms of $x \times 10^{15} s^{-1}$
($h = 6.626 \times 10^{-34}$ Js, $N_A = 6.022 \times 10^{23}$ mol^{-1})

52. The dipole moment of chlorobenzene  is 1.5 D. Find the dipole moment of



53. If 3.01×10^{20} molecules are removed from 98 mg of H_2SO_4, then calculate the number of moles of H_2SO_4 left in terms of $x \times 10^{-3}$.

54. The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, calculate the final volume of the gas.

55. In an amino acid, the carboxyl group ionises at $pK_{a_1} = 2.34$ and ammonium ion at $pK_{a_2} = 9.60$. Find the isoelectric point (pI) of the amino acid

56. AB, A_2 and B_2 are diatomic molecules. If the bond enthalpies of A_2, AB and B_2 are in the ratio 1:1 :0.5 and enthalpy of formation of AB from A_2 and B_2 is –100 kJ mol^{-1}. Calculate the bond energy of A_2

57. A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60 g mol^{-1}) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 g cm^{-3}, calculate molar mass of the substance

58. If the following half cells have the E° values as
$Fe^{+3} + e^- \longrightarrow Fe^{+2}$; E° = + 0.77V and
$Fe^{+2} + 2e^- \longrightarrow Fe$; E° = – 0.44V. Calculate the E° of the half cell $Fe^{+3} + 3e^- \longrightarrow Fe$

59. What is the oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2.

60. A solid AB crystallises as NaCl structure and the radius of the cation is 0.100nm. Calculate the maximum radius of the anion

MATHEMATICS

Section - A

61. If $2\sec 2\alpha = \tan\beta + \cot\beta$ then one of the values of $(\alpha+\beta) =$

(1) π (2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$ (4) None

62. The value of $\sum_{r=1}^{5} r\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} =$

(1) $5(n-3)$ (2) $5(n-2)$

(3) $5n$ (4) $5(2n-9)$

63. ${}^{14}C_7 + \sum_{i=1}^{3} {}^{17-i}C_6 =$

(1) ${}^{16}C_7$ (2) ${}^{17}C_7$

(3) ${}^{17}C_8$ (4) ${}^{16}C_8$

64. If $e^y(x+1)=1$, then, $\frac{d^2y}{dx^2}$ is

(1) $\frac{dy}{dx}$ (2) $\left(\frac{dy}{dx}\right)^2$

(3) $\left(\frac{dy}{dx}\right)^3$ (4) 1

65. Let ABC be a triangle with vertices at points A (2, 3, 5), B (–1, 3, 2) and C (λ, 5, μ) in three dimensional space. If the median through A is equally inclined with the axes, then (λ, μ) is equal to :

(1) (10, 7) (2) (7, 5)

(3) (7, 10) (4) (5, 7)

66. The angle between the two lines

$\frac{x+1}{2}=\frac{y+3}{2}=\frac{z-4}{-1}$ & $\frac{x-4}{1}=\frac{y+4}{2}=\frac{z+1}{2}$ is

(1) $\cos^{-1}\left(\frac{4}{9}\right)$ (2) $\cos^{-1}\left(\frac{3}{9}\right)$

(3) $\cos^{-1}\left(\frac{2}{9}\right)$ (4) $\cos^{-1}\left(\frac{1}{9}\right)$

67. The contrapositive of $(p\vee q)\Rightarrow r$ is

(1) $r\Rightarrow(p\vee q)$ (2) $\sim r\Rightarrow(p\vee q)$

(3) $\sim r\Rightarrow\sim p\wedge\sim q$ (4) $p\Rightarrow(q\vee r)$

68. If (2, 3, 5) are ends of the diameter of a sphere $x^2+y^2+z^2-6x-12y-2z+20=0$. Then coordinates of the other end are

(1) (4, 9, –3) (2) (4, 3, 5)

(3) (4, 3, –3) (4) (4, –3, 9)

69. $\int \frac{dx}{(x-\beta)\sqrt{(x-\alpha)(\beta-x)}}$ is

(1) $\frac{2}{\alpha-\beta}\sqrt{\frac{x-\alpha}{\beta-x}}+C$

(2) $\frac{2}{\alpha-\beta}\sqrt{(x-\alpha)(\beta-x)}+C$

(3) $\frac{\alpha-\beta}{2}(x-\alpha)\sqrt{\beta-x}$

(4) None of these.

70. Consider the following planes

$P: x+y-2z+7=0$

$Q: x+y+2z+2=0$

$R: 3x+3y-6z-11=0$

Space for Rough Work

(1) P and R are perpendicular
(2) Q and R are perpendicular
(3) P and Q are parallel
(4) P and R are parallel

71. If $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+.....+\infty=\frac{\pi^4}{90}$, then the value of $\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+......\infty$ is

(1) $\frac{\pi^4}{96}$ (2) $\frac{\pi^4}{45}$
(3) $\frac{89}{90}\pi^4$ (4) None of these

72. The domain of the function
$f(x)=\exp(\sqrt{5x-3-2x^2})$ is
(1) $[3/2,\infty)$ (2) $[1,3/2]$
(3) $(-\infty,1]$ (4) $(1,3/2)$

73. The value of the determinant
$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix}$ is zero, if $a\neq 1$
(1) $\sin x=0$ (2) $\cos x=0$
(3) $a=0$ (4) $\cos x=\frac{1+a^2}{2a}$

74. If $a=\cos\left(\frac{2\pi}{7}\right)+i\sin\left(\frac{2\pi}{7}\right)$, then the quadratic equation whose roots are $\alpha=a+a^2+a^4$ and $\beta=a^3+a^5+a^6$, is
(1) $x^2-x+2=0$ (2) $x^2+x-2=0$
(3) $x^2-x-2=0$ (4) $x^2+x+2=0$

75. If $AB=0$, then for the matrices
$A=\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ and
$B=\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$, $\theta-\phi$ is
(1) an odd multiple of $\frac{\pi}{2}$
(2) an odd multiple of π
(3) an even multiple of $\frac{\pi}{2}$
(4) 0

76. If $f(x)=xe^{x(1-x)}$, $x\in R$, then $f(x)$ is
(1) decreasing on $[-1/2, 1]$
(2) decreasing on R
(3) increasing on $[-1/2, 1]$
(4) increasing on R

77. The area bounded by the curves $x=y^2$ and $x=\frac{2}{1+y^2}$ is
(1) $\pi-\frac{2}{3}$ (2) $\pi+\frac{2}{3}$
(3) $-\pi-\frac{2}{3}$ (4) None of these

78. An inverted cone is 10 cm in diameter and 10 cm deep. Water is poured into it at the rate of $4cm^3$/min. When the depth of water level is 6 cm, then it is rising at the rate
(1) $\frac{9}{4\pi}cm^3/\min.$ (2) $\frac{1}{4\pi}cm^3/\min.$
(3) $\frac{1}{9\pi}cm^3/\min.$ (4) $\frac{4}{9\pi}cm^3/\min.$

79. The equation of tangent to $4x^2 - 9y^2 = 36$ which is perpendicular to straight line $5x + 2y - 10 = 0$ is

(1) $5(y-3) = 2\left(x - \frac{\sqrt{117}}{2}\right)$

(2) $2y - 5x + 10 - 2\sqrt{18} = 0$

(3) $2y - 5x - 10 - 2\sqrt{18} = 0$

(4) None of these

80. $\int_{\log\sqrt{\pi/2}}^{\log\sqrt{\pi}} e^{2x}\sec^2\left(\frac{1}{3}e^{2x}\right)dx$ is equal to :

(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$

(3) $\frac{3\sqrt{3}}{2}$ (4) $\frac{1}{2\sqrt{3}}$

Section - B

81. If a_1, a_2, a_3, ... are positive numbers in G.P. then the value of

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+1} & \log a_{n+2} & \log a_{n+3} \\ \log a_{n+2} & \log a_{n+3} & \log a_{n+4} \end{vmatrix}$$ is ______.

82. The probability that in the random arrangement of the letters of the word 'UNIVERSITY', the two I's does not come together is ______.

83. A point is selected at random from the interior of a circle. The probability that the point is close to the centre, than the boundary of the circle, is ________.

84. Three persons A, B, C throw a die in succession. The one getting 'six' wins. If A starts then the probability of B winning is __________.

85. If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ coincide with the foci of the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$, then value of b^2 is ________.

86. If $f(x) = |x - 2|$ and $g(x) = f(f(x))$ then for $x > 10$, $g'(x)$ is.

87. If ab^2c^3, $a^2b^3c^4$, $a^3b^4c^5$ are in A.P. ($a, b, c > 0$), then the minimum value of $a + b + c$ is.

88. The sum of the coefficients in the expansion of $\left(x^2 - \frac{1}{3}\right)^{199} \times \left(x^3 + \frac{1}{2}\right)^{200}$ is ________.

89. If the median and the range of four numbers $\{x, y, 2x + y, x - y\}$, where $0 < y < x < 2y$, are 10 and 28 respectively, then the mean of the numbers is _________.

90. If $\int \frac{dx}{\cos^3 x\sqrt{2\sin 2x}} = (\tan x)^A + C(\tan x)^B + k$, where k is a constant of integration, then value of $A + B + C$ is _________.

Mock Test 2

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

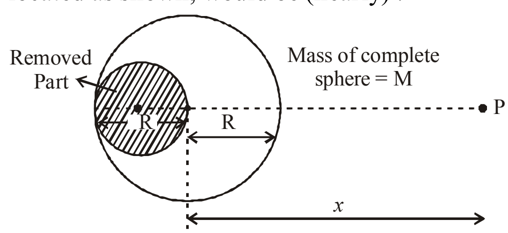
Section - A

1. N divisions on the main scale of a vernier callipers coincide with N + 1 divisions on the vernier scale. If each division on the main scale is of 'a' units, the least count of the instrument is

(1) $\frac{a}{N+1}$ (2) $\frac{a}{N-1}$

(3) $\frac{2a}{N-1}$ (4) $\frac{2a}{N+1}$

2. The gravitational field, due to the 'left over part' of a uniform sphere (from which a part as shown, has been 'removed out'), at a very far off point, P, located as shown, would be (nearly) :

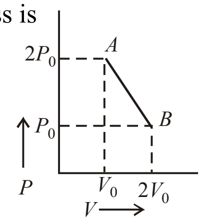


(1) $\frac{5}{6}\frac{GM}{x^2}$ (2) $\frac{8}{9}\frac{GM}{x^2}$

(3) $\frac{7}{8}\frac{GM}{x^2}$ (4) $\frac{6}{7}\frac{GM}{x^2}$

Space for Rough Work

3. n moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. Maximum temperature of the gas during the process is



(1) $\frac{9P_0V_0}{nR}$ (2) $\frac{3P_0V_0}{2nR}$

(3) $\frac{9P_0V_0}{2nR}$ (4) $\frac{9P_0V_0}{4nR}$

4. Which of these nuclear reactions is possible –

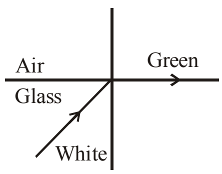
(1) ${}^{23}_{11}Na + {}^{1}_{1}H \longrightarrow {}^{20}_{10}Ne + {}^{4}_{2}He$

(2) ${}^{10}_{5}B + {}^{4}_{2}He \longrightarrow {}^{13}_{7}N + {}^{1}_{1}H$

(3) ${}^{10}_{5}B + {}^{1}_{0}n \longrightarrow {}^{11}_{5}B + \beta^- + \overline{\nu}$

(4) ${}^{14}_{7}N + {}^{1}_{1}H \longrightarrow {}^{12}_{6}C + \beta^- + \overline{\nu}$

5. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains



(1) yellow, orange, red

(2) violet, indigo, blue

(3) All colours

(4) All colours except green

6. What is the conductivity of a semiconductor sample having electron concentration of 5×10^{18} m^{-3}, hole concentration of 5×10^{19} m^{-3}, electron mobility of 2.0 m^2 V^{-1} s^{-1} and hole mobility of 0.01 m^2 V^{-1} s^{-1} ?

(Take charge of electron as 1.6×10^{-19} C)

(1) 1.68 $(\Omega - m)^{-1}$ (2) 1.83 $(\Omega - m)^{-1}$

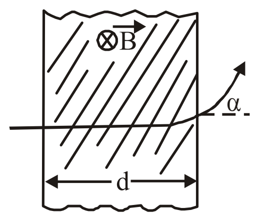
(3) 0.59 $(\Omega - m)^{-1}$ (4) 1.20 $(\Omega - m)^{-1}$

7. A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is E_1 = 2J and at a displacement y from mean position, its potential energy is E_2 = 8J. The potential energy E at a displacement (x + y) from mean position is

(1) 10J (2) 14J

(3) 18J (4) 4J

8. A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width 'd'. If α be the angle of deviation of proton from initial direction of motion (see figure), the value of $sin\ \alpha$ will be :



(1) $qV\sqrt{\frac{Bd}{2m}}$

(2) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$

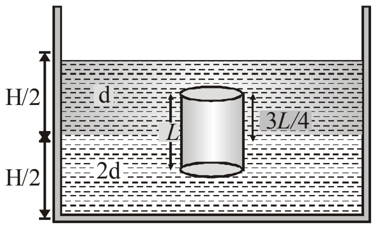
(3) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$

(4) $Bd\sqrt{\frac{q}{2mV}}$

9. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(1) $4F$ (2) $6F$ (3) $9F$ (4) F

10. A homogeneous solid cylinder of length L (L < H/2). Cross-sectional area A/5 is immersed such that it floats with its axis vertical at the liquid-liquid interface with length L/4 in the denser liquid as shown in the fig. The lower density liquid is open to atmosphere having pressure P_0. Then density of solid is given by



(1) $\frac{5}{4}d$

(2) $\frac{4}{5}d$

(3) d

(4) $\frac{d}{5}$

11. In a Young's double slit experiment with light of wavelength λ, fringe pattern on the screen has fringe width β. When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance

(1) $\frac{\beta(\mu-1)}{\lambda}\left(\frac{t_1}{t_2}\right)$ (2) $\frac{\mu\beta}{\lambda}\frac{t_1}{t_2}$

(3) $\frac{\beta(\mu-1)}{\lambda}(t_1-t_2)$ (4) $(\mu-1)\frac{\lambda}{\beta}(t_1+t_2)$

12. A dipole consisting of two charges +q and –q joined by a massless rod of length ℓ, is seen oscillating with a small amplitude in a uniform electric field of magnitude E. The period of oscillation

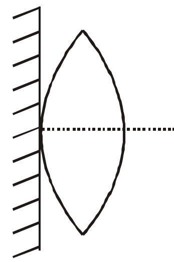
(1) is proportional to E

(2) is proportional to 1/E

(3) is $\pi\sqrt{\frac{m\ell}{3qE}}$

(4) is proportional to $\frac{1}{\sqrt{E}}$ but $\neq \pi\sqrt{\frac{m\ell}{3qE}}$

13. A thin convex lens of focal length 'f' is put on a plane mirror as shown in the figure. When an object is kept at a distance 'a' from the lens - mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of 'a' is :



(1) 3f

(2) $\frac{3}{2}$f

(3) f

(4) 2f

14. A sinusoidal wave is propagating in negative x-direction in a string stretched along x-axis. A particle of string at x = 2m is found at its mean position and it is moving in positive y-direction at t = 1 sec. The amplitude of the wave, the wavelength and the angular frequency of the wave are 0.1 meter, $\pi/2$ meter and 2π rad/sec respectively.

The equation of the wave is

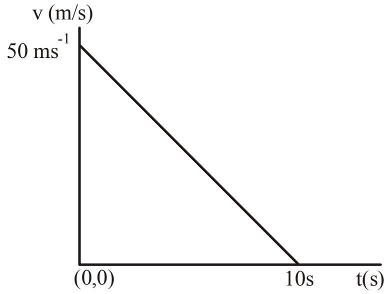
(1) $y = 0.1 \sin(4\pi(t-1) + 8(x-2))$

(2) $y = 0.1 \sin((t-1) - (x-2))$

(3) $y = 0.1 \sin(2\pi(t-1) + 4(x-2))$

(4) None of these

15. Velocity–time graph for a body of mass 10 kg is shown in figure. Work–done on the body in first two seconds of the motion is :



(1) – 9300 J (2) 12000 J
(3) –4500 J (4) –12000 J

16. A charged particle of mass m and having a charge Q is placed in an electric field E which varies with time as $E = E_0 \sin \omega t$. What is the amplitude of the S.H.M. executed by the particle?

(1) $\frac{QE_0}{m\omega^2}$ (2) $\frac{1}{2}\frac{QE_0}{m\omega^2}$

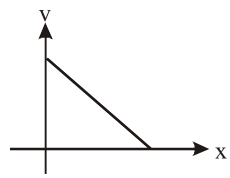
(3) $\frac{2QE_0}{m\omega^2}$ (4) None of these

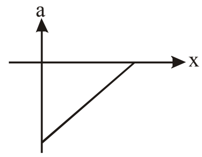
17. In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $3\lambda/4$, the speed of the fastest emitted electron will become –

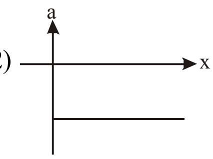
(1) $v\sqrt{\frac{3}{4}}$ (2) $v\sqrt{\frac{4}{3}}$

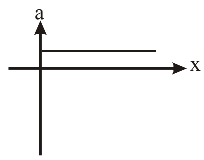
(3) less than $v\sqrt{\frac{4}{3}}$ (4) greater than $v\sqrt{\frac{4}{3}}$

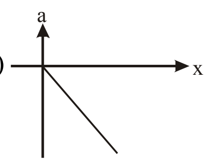
18. A particle moves along x-axis with initial position x = 0. Its velocity varies with x-coordinate as shown in graph. The acceleration 'a' of this particle varies with x as –



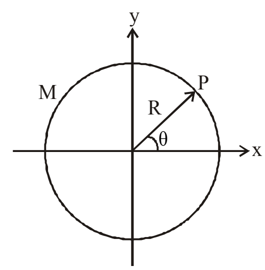
(1) 

(2) 

(3) 

(4) 

19. A ring of mass M and radius R lies in x-y plane with its centre at origin as shown. The mass distribution of ring is non-uniform such that at any point P on the ring, the mass per unit length is given by $\lambda = \lambda_0 \cos^2\theta$ (where λ_0 is a positive constant). Then the moment of inertia of the ring about z-axis is –



(1) MR^2

(2) $\frac{1}{2}MR^2$

(3) $\frac{1}{2}\frac{M}{\lambda_0}R$

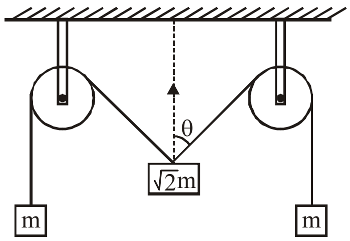
(4) $\frac{1}{\pi}\frac{M}{\lambda_0}R$

20. The electric potential V is given as a function of distance x by $V = (5x^2 + 10x - 9)$ volt. Value of electric field at x = 1m is

(1) –20 V/m (2) 6 V/m
(3) 11 V/m (4) 20 V/m

Section - B

21. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be _____ degree.



22. A current of 4A produces a deflection of 30° in the galvanometer. The figure of merit is _____ A/rad.

23. A uniform thin rod AB of length L has linear mass density $\mu(x) = a + \frac{bx}{L}$, where x is measured from A. If the CM of the rod lies at a distance of $\left(\frac{7}{12}\right)L$ from A, then a and b are related as b = xa. Find the value of x.

24. A transmitting antenna at the top of a tower has height 32 m and height of the receiving antenna is 50 m. What is the maximum distance (in km) between them for satisfactory communication in line of sight (LOS) mode?

25. A ball of mass 160 g is thrown up at an angle of 60° to the horizontal at a speed of 10 ms^{-1}. The angular momentum of the ball at the highest point of the trajectory with respect to the point from which the ball is thrown is _____ kg m^2/s. (g = 10 ms^{-2})

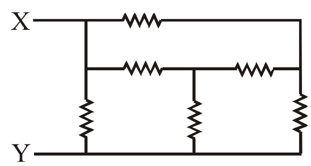
26. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to 2R. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is $\sqrt{\frac{x}{2}}$. Find the value of x.

27. A mass of 50g of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from 30°C to 25°C. A mass of 100g of another liquid in an identical vessel with identical surroundings takes the same time to cool from 30° C to 25° C. The specific heat of the liquid is _____ kcal/kg.
(The water equivalent of the vessel is 30g.)

28. The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength is _____ V/m.

29. An ideal monatomic gas with pressure P, volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i. If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a. The ratio $\frac{P_a}{P_i}$ is _____.

30. Six resistors of 10Ω each are connected as shown. The equivalent resistance between points X and Y is _____.



CHEMISTRY

Section - A

31. Consider the following reactions

(i) $Cd^{2+}(aq) + 2e^{-} \longrightarrow Cd(s)$, $E° = -0.40\ V$

(ii) $Ag^{+}(aq) + e^{-} \longrightarrow Ag(s)$, $E° = 0.80\ V$

For the galvanic cell involving the above reactions. Which of the following is not correct?

(1) Molar concentration of the cation in the cathodic compartment changes faster than that of the cation in the anodic compartment.

(2) E_{cell} increase when Cd^{2+} solution is diluted.

(3) Twice as many electrons pass through the cadmium electrode as through silver electrode.

(4) E_{cell} decreases when Ag^{+} solution is diluted.

32. A container contains certain gas of mass 'm' of high pressure. Some of the gas has been allowed to escape from the container and after some time the pressure of the gas becomes half and its absolute temperature 2/3 rd. The amount of the gas escaped is

(1) 2/3 m (2) 1/2 m
(3) 1/4 m (4) 1/6 m

33. The hybrid state of S in SO_3 is similar to that of

(1) C in C_2H_2 (2) C in C_2H_4
(3) C in CH_4 (4) C in CO_2

34. Choose the correct options –

(1) Tranquillizers are called psychotherapeutic drugs.

(2) Vitamin C, Vitamin E and β-carotene are antioxidants.

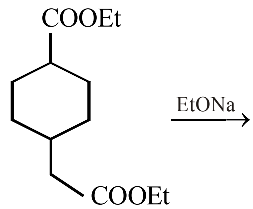
(3) Sodium or potassium salt of a long chain fatty acid is called soap.

(4) All of these.

35. Electrode potential of the half cell Pt (s) | Hg (l) | Hg_2Cl_2(s) | Cl^- (aq) can be increased by :

(1) Increasing $[Cl^-]$
(2) Decreasing $[Cl^-]$
(3) Increasing Hg_2Cl_2 (s)
(4) Decreasing Hg (l)

36. What is the name of the following reaction :



(1) Knoevenagel reaction
(2) Perkin reaction
(3) Reformatsky reaction
(4) Dieckmann's condensation.

37. Choose the correct options –

(1) The change of atmospheric temperature with altitude is called the lapse rate.

(2) The gases responsible for greenhouse effects are CO_2, water vapour, CH_4 and ozone.

(3) The order of contribution to the acid rain is $H_2SO_4 < HNO_3 < HCl$

(4) Both (1) and (2)

Space for Rough Work

38. Optically active isomer (A) of (C_5H_9Cl) on treatment with one mole of H_2 gives an optically inactive compound (B) compound (A) will be:

(1) $CH_3-\underset{\underset{CH_2Cl}{|}}{CH}-CH=CH_2$

(2) $Cl-\underset{\underset{CH_3}{|}}{CH}-CH=CH-CH_3$

(3) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_2-CH=CH_2$

(4) $CH_3-CH_2-\underset{\underset{Cl}{|}}{CH}-CH=CH_2$

39. Which statement is/are correct about ICl_3 molecule ?

(1) All I – Cl bonds are equivalent.
(2) Molecule is polar and non-planar.
(3) All adjacent bond angles are equal.
(4) All hybrid orbitals of central atom having equal *s*-character.

40. Isopropyl alcohol is obtained by reacting which of the following alkenes with concentrated H_2SO_4 followed by boiling with H_2O?

(1) Ethylene (2) Propylene
(3) 2-Methylpropene (4) Isoprene

41. In electro-refining of metal the impure metal is used to make the anode and a strip of pure metal as the cathode, during the electrolysis of an aqueous solution of a complex metal salt. This method cannot be used for refining of

(1) silver (2) copper
(3) aluminium (4) sodium

42. Which of the following compounds does not give nitrogen gas on heating?

(1) NH_4NO_2 (2) $(NH_4)_2SO_4$
(3) NH_4ClO_4 (4) $(NH_4)_2Cr_2O_7$

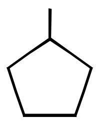
43. Flocculation value of $BaCl_2$ is much less than that of KCl for sol A and flocculation value of Na_2SO_4 is much less than that of NaBr for sol B. The correct statement among the following is :

(1) Both the sols A and B are negatively charged.
(2) Sol A is positively charged and Sol B is negatively charged.
(3) Both the sols A and B are positively charged.
(4) Sol A is negatively charged and sol B is positively charged.

44. $HC\equiv CH \xrightarrow[H_2SO_4]{HgSO_4} (A) \xrightarrow[5\,°C]{dil.\ OH^-} (B)$

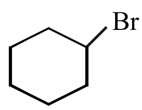
Give the IUPAC name of "B" is –

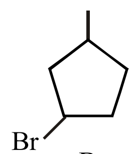
(1) 2-butanal
(2) 3-hydroxy butanal
(3) 3-formyl-2-propanol
(4) 4-oxo-2-propanol

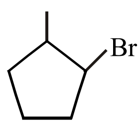
45.  $\xrightarrow{Br_2/h\nu}$ Major (X) $\xrightarrow[KOH/\Delta]{Alcoholic}$

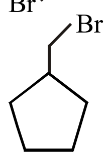
Major (Y) $\xrightarrow[Peroxide]{H-Br}$ Major (Z)

Major final product (Z) is –

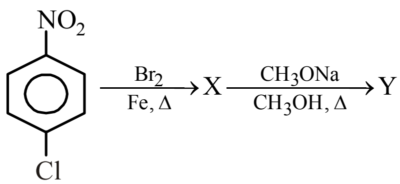
(1) 

(2) 

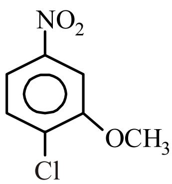
(3) 

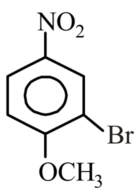
(4) 

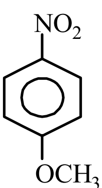
46. Choose the incorrect option –
(1) When a dilute solution of an acid is added to a dilute solution of a base, neutralization reaction takes place.
(2) In acid-base titrations, at the end point, the amount of acid becomes chemically equivalent to the amount of base present.
(3) In the case of a strong acid and a strong base titration, at the end point the solution becomes acidic.
(4) In acid-base titrations the end point is determined by the hydrogen ion concentration of the solution.

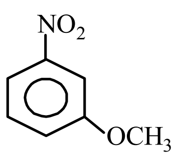
47. 

Product (Y) of this reaction is –

(1) 

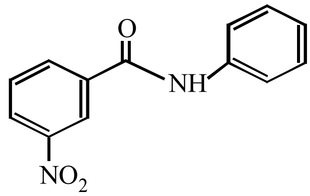
(2) 

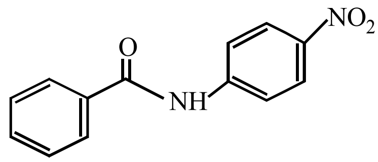
(3) 

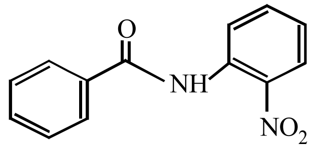
(4) 

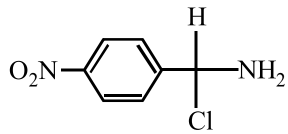
48. $A \xrightarrow[\text{(ii) } H^{\oplus}]{\text{(i) NaOI}} B \xrightarrow{SOCl_2} C \xrightarrow[NaOH]{C_6H_5NH_2}$

$D \xrightarrow[\text{Conc. } H_2SO_4 \text{ (1 equivalent)}]{\text{Conc. } HNO_3} E$ (Major) E can be –

(1) 

(2) 

(3) 

(4) 

49. $\underset{(C_5H_{10}O)}{(X)} \xrightarrow{H_3O^+} Y + Z$ (Y and Z both give the Iodoform test). The compound X is –
(1) $CH_3-CH=CH-O-CH_2-CH_3$
(2) $CH_3-\underset{CH_3}{\overset{H}{C}}-O-CH_2-CH_3$
(3) $CH_3-\underset{CH_2}{\overset{}{\underset{\|}{C}}}-O-CH_2-CH_3$
(4) Both (1) and (3)

50. If the elevation in boiling point of a solution of non-volatile, non-electrolytic and non-associating solute in a solvent (K_b = x K kg mol^{-1}) is y K, then the depression in freezing point of solution of same concentration would be (K_f of the solvent = z K kg mol^{-1})

(1) $\frac{2xz}{y}$K (2) $\frac{yz}{x}$K

(3) $\frac{xz}{y}$K (4) $\frac{yz}{2x}$K

Section - B

51. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3. Find the length of the side of the unit cell is _____.

52. When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form is _____.

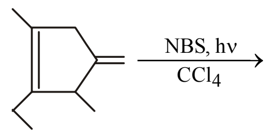
CHO — CH_2 — CHOH — CHOH — CHOH — CH_2OH

53. A 25.0 mm × 40.0 mm piece of gold foil is 0.25 mm thick. The density of gold is 19.32 g/cm^3. How many gold atoms are in the sheet in terms of $x \times 10^{22}$? (Atomic weight : Au = 197.0)

54. In a chemical reaction A is converted into B. The rates of reaction, starting with initial concentrations of A as 2×10^{-3} M and 1×10^{-3} M, are equal to 2.40×10^{-4} Ms^{-1} and 0.60×10^{-4} Ms^{-1} respectively. Find the order of reaction with respect to reactant A.

55. Rate constant k = 2.303 min^{-1} for a particular reaction. The initial concentration of the reactant is 1 mol/litre then calculatle the rate of reaction after 1 minute.

56. How many free radicals can be produced during following reaction (ignoring resonating structure) ?



57. An electron has a speed of 30,000 cm sec^{-1} accurate upto 0.001%. What is the uncertainty (in cm) in locating it's position?

58. The standard enthalpy of formation of NH_3 is –46.0 kJ/mol. If the enthalpy of formation of H_2 from its atoms is – 436 kJ/mol and that of N_2 is –712 kJ/mol, calculate the average bond enthalpy of N - H bond in NH_3.

59. The rate coefficient (k) for a particular reaction is 1.3×10^{-4} M^{-1} s^{-1} at 100 °C and 1.3×10^{-3} M^{-1} s^{-1} at 150 °C. What is the energy of activation (E_A) (in kJ/mol) for this reaction? (R = molar gas constant = 8.314 JK^{-1} mol^{-1})

60. The e.m.f. of the cell Zn | Zn^{2+} (0.01M) | | Fe^{2+} (0.001M) | Fe at 298 K is 0.2905 then the value of equilibrium constant for the cell reaction 10^x. Find x.

Space for Rough Work

MATHEMATICS

Section - A

61. Given $f(x) = \begin{cases} \sqrt{10-x^2} & \text{if } -3 < x < 3 \\ 2-e^{x-3} & \text{if } x \geq 3 \end{cases}$

The graph of $f(x)$ is –
(1) continuous and differentiable at $x = 3$
(2) continuous but not differentiable at $x = 3$
(3) differentiable but not continuous at $x = 3$
(4) neither differentiable nor continuous at $x = 3$

62. The greatest and the least absolute value of $z+1$, where $|z+4| \leq 3$ are respectively
(1) 6 and 0 (2) 10 and 6
(3) 4 and 3 (4) None of these

63. The equation
$(5x-1)^2 + (5y-2)^2 = (\lambda^2 - 4\lambda + 4)(3x+4y-1)^2$
represents an ellipse if $\lambda \in$
(1) $(0, 1]$ (2) $(-1, 2)$
(3) $(2, 3)$ (4) $(-1, 0)$

64. If $(-4, 5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of second diagonal is
(1) $x+3y=21$ (2) $2x-3y=7$
(3) $x+7y=31$ (4) $2x+3y=21$

65. Which of the following is always true?
(1) $(\sim p \Rightarrow q) = \sim q \Rightarrow \sim p$
(2) $(\sim p \vee q) \equiv \vee p \vee \sim q$
(3) $\sim (p \Rightarrow q) \equiv p \wedge \sim q$
(4) $\sim (p \vee q) \equiv \sim p \wedge \sim q$

66. Find $\int e^{\sin x}\left(\dfrac{x\cos^3 x - \sin x}{\cos^2 x}\right) dx$
(1) $x\,e^{\sin x} - e^{\sin x}\sec x + C$
(2) $x\,e^{\cos x} - e^{\sin x}\sec x + C$
(3) $x^2 e^{\sin x} + e^{\sin x}\sec x + C$
(4) $2x\,e^{\sin x} - e^{\sin x}\tan x + C$

67. The function $f : [2, \infty) \rightarrow (0, \infty)$ defined by $f(x) = x^2 - 4x + a$, then the set of values of 'a' for which f(x) becomes onto is
(1) $(4, \infty)$ (2) $[4, \infty)$
(3) $\{4\}$ (4) ϕ

68. If α and β are the real roots of the equation $x^2 - (k-2)x + (k^2+3k+5) = 0$ $(k \in R)$.
Find the maximum and minimum values of $(\alpha^2 + \beta^2)$.
(1) 18, 50/9 (2) 18, 25/9
(3) 27, 50/9 (4) None of these

69. The sum of the coefficient of all the terms in the expansion of $(2x - y + z)^{20}$ in which y do not appear at all while x appears in even powers and z appears in odd powers is –
(1) 0 (2) $\dfrac{2^{20}-1}{2}$
(3) 2^{19} (4) $\dfrac{3^{20}-1}{2}$

70. All the five digit numbers in which each successive digit exceeds is predecessor are arranged in the increasing order. The $(105)^{th}$ number does not contain the digit

(1) 1 (2) 2
(3) 6 (4) All of these

71. If $\hat{a}, \hat{b}, \hat{c}$ are non-coplanar unit vectors such that $\hat{a} \times (\hat{b} \times \hat{c}) = \frac{1}{\sqrt{2}}(\hat{b} + \hat{c})$ then the angle between the vectors $\hat{a}, \hat{b}$ is

(1) $3\pi/4$ (2) $\pi/4$
(3) $\pi/8$ (4) $\pi/2$

72. The greatest and the least values of $|z_1 + z_2|$ if $z_1 = 24 + 7i$ and $|z_2| = 6$ respectively are

(1) 31, 19 (2) 19, 31
(3) –19, –31 (4) –31, –19

73. Let P = (–1, 0), Q = (0, 0) and R = $(3, 3\sqrt{3})$ be three points. The equation of the bisector of the angle PQR is

(1) $\frac{\sqrt{3}}{2}x + y = 0$ (2) $x + \sqrt{3y} = 0$
(3) $\sqrt{3}x + y = 0$ (4) $x + \frac{\sqrt{3}}{2}y = 0$.

74. If $\int \sqrt{2}\sqrt{1+\sin x}\,dx = -4\cos(ax+b) + C$ then the value of (a, b) is :

(1) $\frac{1}{2}, \frac{\pi}{4}$ (2) $1, \frac{\pi}{2}$
(3) 1, 1 (4) None of these

75. If the tangent at any point on the curve $x^4 + y^4 = a^4$ cuts off intercepts p and q on the co-ordinate axes then the value of $p^{-4/3} + q^{-4/3}$ is

(1) $a^{-4/3}$ (2) $a^{-1/2}$
(3) $a^{1/2}$ (4) None

76. The area bounded by the curve $y^2(2a - x) = x^3$ and the line x = 2a is

(1) $3\pi a^2$ sq. unit (2) $\frac{3\pi a^2}{2}$ sq. unit
(3) $\frac{3\pi a^2}{4}$ sq. unit (4) $\frac{6\pi a^2}{5}$ sq. unit

77. The expression satisfying the differential equation $(x^2 - 1)\frac{dy}{dx} + 2xy = 1$ is

(1) $x^2y - xy^2 = c$
(2) $(y^2 - 1)x = y + c$
(3) $(x^2 - 1)y = x + c$
(4) None of these

78. Let $f: R \to R$ and $f_n(x) = f(f_{n-1}(x))\ \forall\ n \geq 2, n \in N$, the roots of equation $f_3(x)\,f_2(x)\,f(x) - 25f_2(x)\,f(x) + 175\,f(x) = 375$, which also satisfy equation $f(x) = x$ will be

(1) 5 (2) 15
(3) 10 (4) Both (1) and (2)

79. The value of

$$\int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\ dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\ dt \text{ is}$$

(1) π (2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$ (4) 1

80. If a is real and $\sqrt{2}ax + \sin By + \cos Bz = 0$, $x + \cos By + \sin Bz = 0$, $-x + \sin By - \cos Bz = 0$, then the set of all values of a for which the system of linear equations has a non-trivial solution, is–

(1) $[1, 2]$ (2) $[-1, 1]$

(3) $[1, \infty]$ (4) $[2^{-1/2}, 2^{1/2}]$

Section - B

81. If two vertical poles 20 m and 80 m high stand apart on a horizontal plane, then the height (in m) of the point of intersection of the lines joining the top of each pole to the foot of other is _____.

82. If $x + iy = \dfrac{3}{\cos\theta + i\sin\theta + 2}$ then the value of $4x - x^2 - y^2$ is _____.

83. Box contains 2 one rupee, 2 five rupee, 2 ten rupee and 2 twenty rupee coin. Two coins are drawn at random simultaneously. The probability that their sum is Rs. 20 or more, is _____.

84. The value of the definite integral, $\displaystyle\int_{\theta_1}^{\theta_2} \frac{d\theta}{1+\tan\theta} = \frac{501\pi}{K}$ where $\theta_2 = \dfrac{1003\pi}{2008}$ and $\theta_1 = \dfrac{\pi}{2008}$. The value of K is _____.

85. The expansion of $(1 + x)^n$ has 3 consecutive terms with coefficients in the ratio 1 : 2 : 3 and can be written in the form ${}^nC_k : {}^nC_{k+1} : {}^nC_{k+2}$. The sum of all possible values of (n + k) is _____.

86. The mean and standard deviation of 6 observations are 8 and 4 respectively. If each observation is multiplied by 3, the new standard deviation of the resulting observations is _____.

87. The value of $\displaystyle\lim_{x\to 0} \frac{\tan x\sqrt{\tan x} - \sin x\sqrt{\sin x}}{x^3.\sqrt{x}}$ is _____.

88. Three people each flip two fair coins. The probability that exactly two of the people flipped one head and one tail, is _____.

89. If $P(S)$ denotes the set of all subsets of a given set S, then the number of one-to-one functions from the set $S = \{1, 2, 3\}$ to the set $P(S)$ is _____.

90. A triangle ABC satisfies the relation $2\sec 4C + \sin^2 2A + \sqrt{\sin B} = 0$ and a point P is taken on the longest side of the triangle such that it divides the side in the ratio 1 : 3. Let Q and R be the circumcentre and orthocentre of ΔABC. If PQ : QR : RP = 1 : α : β, then the value of $\alpha^2 + \beta^2$ is _____.

Mock Test 3

Time : 3 hrs. **Max. Marks : 300**

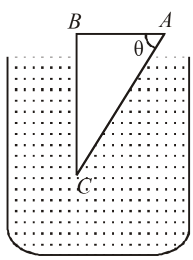
INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

1. A glass prism of refractive index 1.5 is immersed in water (refractive index $\frac{4}{3}$) as shown in figure. A light beam incident normally on the face *AB* is totally reflected to reach the face *BC*, if



(1) $\sin\theta \geq \frac{5}{9}$

(2) $\sin\theta \geq \frac{2}{3}$

(3) $\sin\theta \geq \frac{8}{9}$

(4) $\sin\theta \geq \frac{1}{3}$

2. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of 2R?

(1) $\frac{5GmM}{6R}$

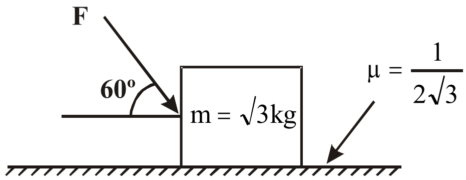
(2) $\frac{2GmM}{3R}$

(3) $\frac{GmM}{2R}$

(4) $\frac{3GmM}{2R}$

Space for Rough Work

3. What is the maximum value of the force F such that the block shown in the arrangement, does not move?



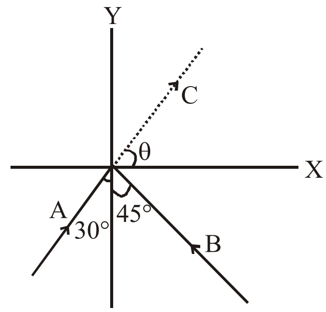
(1) 20 N (2) 10 N (3) 12 N (4) 15 N

4. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground. Then pickup the incorrect alternative
   (1) The magnitude of angular momentum of the particle about O is increasing
   (2) The magnitude of torque of the gravitational force on the particle about O is decreasing
   (3) The moment of inertia of the particle about O is decreasing
   (4) The magnitude of angular velocity of the particle about O is increasing

5. Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is :

   (1) $T \propto \frac{1}{R}$ (2) $T \propto \frac{1}{R^3}$
   (3) $T \propto e^{-R}$ (4) $T \propto e^{-3R}$

6. Two particles A and B of equal mass M are moving with the same speed v as shown in the figure. They collide completely inelastically and move as a single particle C. The angle θ that the path of C makes with the X-axis is given by:



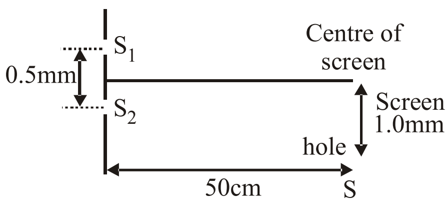
   (1) $\tan\theta = \frac{\sqrt{3}+\sqrt{2}}{1-\sqrt{2}}$
   (2) $\tan\theta = \frac{\sqrt{3}-\sqrt{2}}{1-\sqrt{2}}$
   (3) $\tan\theta = \frac{1-\sqrt{2}}{\sqrt{2}(1+\sqrt{3})}$
   (4) $\tan\theta = \frac{1-\sqrt{3}}{1+\sqrt{2}}$

7. The fundamental frequency of a sonometer wire of length ℓ is n_0. A bridge is now introduced at a distance of $\Delta\ell$ ($<< \ell$) from the centre of the wire. The lengths of wire on the two sides of the bridge are now vibrated in their fundamental modes. Then, the beat frequency nearly is –

(1) $n_0\Delta\ell/\ell$ (2) $8n_0\Delta\ell/\ell$

(3) $2n_0\Delta\ell/\ell$ (4) $n_0\Delta\ell/2\ell$

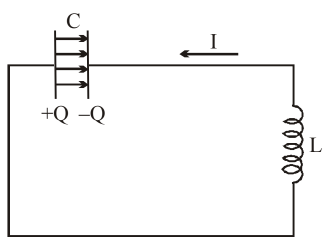
8. In Young's double slit experiment shown in figure S_1 and S_2 are coherent sources and S is the screen having a hole at a point 1.0mm away from the central line. White light (400 to 700nm) is sent through the slits. Which wavelength passing through the hole has strong intensity?



(1) 400 nm (2) 700 nm

(3) 500 nm (4) 667 nm

9. In the LC circuit, the current in the direction shown and the charges on the capacitor plates have the signs shown. At this time



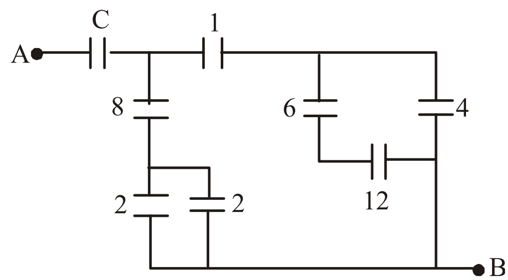
(1) I is increasing and Q is increasing

(2) I is increasing and Q is decreasing

(3) I is decreasing and Q is increasing

(4) I is decreasing and Q is decreasing

10. Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be 1 μF is:



(1) $\frac{32}{23}\mu F$ (2) $\frac{31}{23}\mu F$

(3) $\frac{33}{23}\mu F$ (4) $\frac{34}{23}\mu F$

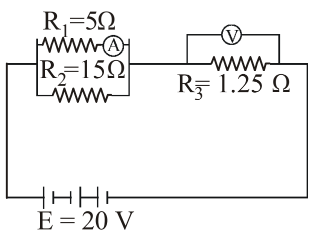
11. The position of a projectile launched from the origin at $t = 0$ is given by $\vec{r} = \left(40\hat{i} + 50\hat{j}\right)$m at $t = 2$s. If the projectile was launched at an angle θ from the horizontal, then θ is (take $g = 10\ ms^{-2}$)

(1) $\tan^{-1}\frac{2}{3}$ (2) $\tan^{-1}\frac{3}{2}$

(3) $\tan^{-1}\frac{7}{4}$ (4) $\tan^{-1}\frac{4}{5}$

12. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc
(1) continuously decreases
(2) continuously increases
(3) first increases and then decreases
(4) remains unchanged

13. An ideal ammeter (zero resistance) and an ideal voltmeter (infinite resistance) are connected as shown. The ammeter and voltmeter reading for $R_1 = 5\,\Omega$, $R_2 = 15\,\Omega$, $R_3 = 1.25\,\Omega$ and $E = 20$ V are given as

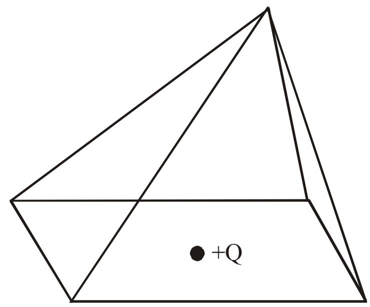


(1) 6.25 A, 3.75 V (2) 3.00 A, 5 V
(3) 3.75 A, 3.75 V (4) 6.25 A, 6.25 V

14. A monoatomic ideal gas is filled in a nonconducting container. The gas can be compressed by a movable nonconducting piston. The gas is compressed slowly to 12.5% of its initial volume.

Find final temperature of the gas if it is T_0 initially–
(1) $4T_0$ (2) $3T_0$
(3) $2/3\ T_0$ (4) T_0

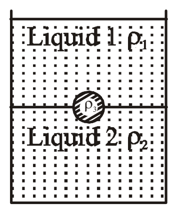
15. A point charge +Q is positioned at the center of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is –



(1) $\frac{Q}{16\varepsilon_0}$ (2) $\frac{Q}{4\varepsilon_0}$

(3) $\frac{Q}{8\varepsilon_0}$ (4) None of these

16. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and, ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_1 and ρ_3?



(1) $\rho_3 < \rho_1 < \rho_2$
(2) $\rho_1 > \rho_3 > \rho_2$
(3) $\rho_1 < \rho_2 < \rho_3$
(4) $\rho_1 < \rho_3 < \rho_2$

17. A point particle of mass 0.1 kg is executing SHM of amplitude of 0.1m. When the particle passes through the mean position, its kinetic energy is 18×10^{-3} J. The equation of motion of this particle when the initial phase of oscillation is 45° can be given by –

(1) $y = 0.1 \cos\left(6t + \frac{\pi}{4}\right)$

(2) $y = 0.1 \sin\left(6t + \frac{\pi}{4}\right)$

(3) $y = 0.4 \sin\left(t + \frac{\pi}{4}\right)$

(4) $y = 0.2 \sin\left(\frac{\pi}{2} + 2t\right)$

18. What happens when the applied load increases and upto breaking stress in the experiment to determine the Young's modulus of elasticity ?

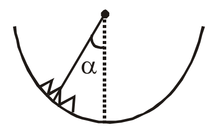
(1) The area of wire goes on decreasing and wire extends and breaks.

(2) The area of wire goes on increasing and wire breaks.

(3) The wire extends and area remains constant.

(4) The area remains same and wire length is also same.

19. An insect crawls up a hemispherical surface very slowly. The coefficient of friction between the insect and the surface is 1/3. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α so that the insect does not slip is given by



(1) $\cot \alpha = 3$ (2) $\sec \alpha = 3$

(3) $\operatorname{cosec} \alpha = 3$ (4) $\cos \alpha = 3$

20. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2s. The magnet is cut along its length into three equal parts and these parts are then placed on each other with their like poles together. The time period of this combination will be

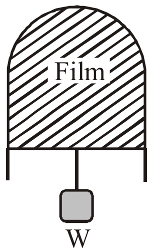
(1) $2\sqrt{3}$ s (2) $\frac{2}{3}$ s

(3) 2 s (4) $\frac{2}{\sqrt{3}}$ s

Section - B

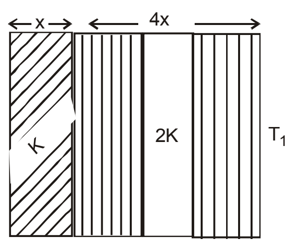
21. A uniform wire of length l and radius r has a resistance of 100 Ω. It is recast into a wire of radius $\frac{r}{2}$. The resistance of new wire will be ________ Ω.

22. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of 1.5×10^{-2} N (see figure). The length of the slider is 30 cm and its weight negligible. The surface tension of the liquid film is _________Nm^{-1}.



23. A piece of wood from a recently cut tree shows 20 decays per minute. A wooden piece of same size placed in a museum (obtained from a tree cut many years back) shows 2 decays per minute. If half life of C^{14} is 5730 years, then age of the wooden piece placed in the museum is approximately ________ years.

24. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively, are T_2 and $T_1 (T_2 > T_1)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right) f$. Find the value of f.



25. If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities?

26. A car, starting from rest, accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is 15 S, then $S = \frac{ft^2}{x}$. Find the value of x.

27. Consider an optical communication system operating at a wavelength of 800 nm. Suppose, only 1% of the optical source frequency is the available channel bandwidth for optical communication. How many channels can be accommodated for transmitting audio signals requiring a bandwidth of 8 kHz ?

28. The current voltage relation of a diode is given by $I = \left(e^{1000V/T} - 1\right)$ mA, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in ________ mA?

29. Currents of a 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2 m.

The force acting on the conductor B, which is situated at 10 cm distance from A will be $x \times 10^{-5}$ N. Find the value of x.

30. Hot water cools from 60°C to 50°C in the first 10 minutes and to 42°C in the next 10 minutes. The temperature of the surroundings is ______ °C.

CHEMISTRY

Section - A

31. Let ν_1 be the frequency of the series limit of the Lyman series, ν_2 be the frequency of the first line of the Lyman series, and ν_3 be the frequency of the series limit of the Balmer series, then –

(1) $\nu_3 = \frac{1}{2}(\nu_1 - \nu_3)$ (2) $\nu_2 - \nu_1 = \nu_3$

(3) $\nu_1 - \nu_2 = \nu_3$ (4) $\nu_1 + \nu_2 = \nu_3$

32. Match List I with List II and select the correct answer :

List I (Ions)	List II (Shapes)
A. ICl_2^-	1. Linear
B. BrF_2^+	2. Pyramidal
C. ClF_4^-	3. Tetrahedral
D. $AlCl_4^-$	4. Square planar
	5. Angular

	A	B	C	D
(1)	1	2	4	5
(2)	4	5	2	3
(3)	1	2	4	3
(4)	5	1	3	4

33. Which one of the following is the correct statement?

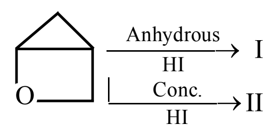
(1) Boric acid is a protonic acid.

(2) Beryllium exhibits coordination number of six.

(3) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase.

(4) $B_2H_6.2NH_3$ is known as 'inorganic benzene'.

34. Choose the correct option for



(1) I and II are identical

(2) I and II are different

(3) Mechanism of formation of I and II are not known

(4) None of these

35. Reducing the pressure from 1.0 atm to 0.5 atm would change the number of molecules in one mole of ammonia to
(1) 25% of its initial value
(2) 50% of its initial value
(3) 75% of its initial value
(4) None of the above

36. Structure of some important polymers are given. Which one represents Buna-S?
(1) $\left(CH_2-\overset{\overset{CH_3}{|}}{C}=CH-CH_2\right)_n$
(2) $\left(CH_2-CH=CH-CH_2-\underset{\underset{C_6H_5}{|}}{CH}-CH_2\right)_n$
(3) $\left(CH_2-CH=CH-CH_2-\underset{\underset{CN}{|}}{CH}-CH_2\right)_n$
(4) $\left(CH_2-\overset{\overset{Cl}{|}}{C}=CH-CH_2\right)_n$

37. In electrolysis of NaCl when Pt electrode is taken, then H_2 is liberated at cathode while with Hg cathode it forms sodium amalgam. This is because
(1) Hg is more inert than Pt
(2) more voltage is required to reduce H^+ at Hg than at Pt
(3) Na is dissolved in Hg while it does not dissolve in Pt
(4) conc. of H^+ ions is larger when Pt electrode is taken

38. Which of the following option is having maximum number of unpaired electrons –
(1) A tetrahedral d^6 ion
(2) $[Co(NH_3)_6]^{3+}$
(3) A square planar d^7 ion
(4) A co-ordination compound with magnetic moment of 5.92 B.M.

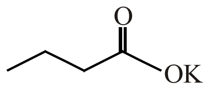
39. Which of the following carbide does not release any hydrocarbon on reaction with water?
(1) SiC (2) Be_2C
(3) CaC_2 (4) Mg_2C_3

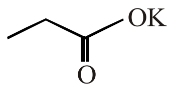
40. An ether (A), $C_5H_{12}O$, when heated with excess of hot concentrated HI produced two alkyl halides which when treated with NaOH yielded compounds (B) and (C). Oxidation of (B) and (C) gave a propanone and an ethanoic acid respectively. The IUPAC name of the ether (A) is:
(1) 2-ethoxypropane
(2) ethoxypropane
(3) methoxybutane
(4) 2-methoxybutane

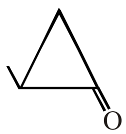
41. Aqueous solution of (M) + $(NH_4)_2S \rightarrow$ yellow ppt (B) $\xrightarrow{(NH_4)_2S_2}$ insoluble.
The cation present in (M) is –
(1) CdS (2) SnS_2
(3) Cd^{2+} (4) Sn^{2+}

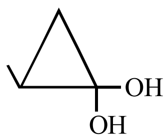
42. Both geometrical and optical isomerisms are shown by
(1) $[Co(en)_2Cl_2]^+$ (2) $[Co(NH_3)_5Cl]^{2+}$
(3) $[Co(NH_3)_4Cl_2]^+$ (4) $[Cr(ox)_3]^{3-}$

43. $Me-CH=CH_2+CHCl_3 \xrightarrow[\text{excess, }\Delta]{\text{aq. KOH}} A$ (Major products) is –

(1) 

(2) 

(3) 

(4) 

44. The decomposition of N_2O_5 in carbon tetrachloride was followed by measuring the volume of O_2 gas evolved :

$$2N_2O_5(CCl_4) \rightarrow 2N_2O_4(CCl_4)+O_2(g).$$

The maximum volume of O_2 gas obtained was 100 cm^3. In 500 minutes, 90 cm^3 of O_2 were evolved. The first order rate constant (in min^{-1}) for the disappearance of N_2O_5 is :

(1) $\frac{2.303}{500}$ (2) $\frac{2.303}{500}\log\frac{100}{90}$

(3) $\frac{2.303}{500}\log\frac{90}{100}$ (4) $\frac{100}{10\times 500}$

45. The basic character of the transition metal monoxides follows the order

(Atomic Nos., Ti = 22, V = 23, Cr = 24, Fe = 26)

(1) $TiO > VO > CrO > FeO$

(2) $VO > CrO > TiO > FeO$

(3) $CrO > VO > FeO > TiO$

(4) $TiO > FeO > VO > CrO$

46. $NH_4ClO_4 + HNO_3\text{ (dilute)} \longrightarrow X + HClO_4$

$X \xrightarrow{\text{heat}} Y\text{ (gas)}$ Gas (Y) is –

(1) O_2 (2) N_2

(3) NO_2 (4) N_2O

47. Which of the following is/are formed when ozone reacts with the unburnt hydrocarbons in polluted air ?

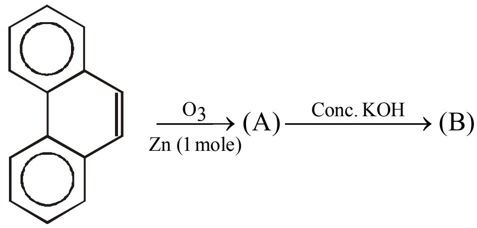
(i) Formaldehyde (ii) Acrolein

(iii) Peroxyacetyl nitrate (iv) Formic acid

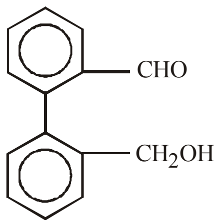
(1) (i) and (iv) (2) (ii) only

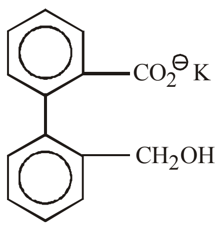
(3) (iii) only (4) (i), (ii) and (iii)

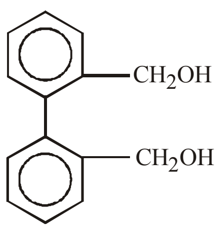
Space for Rough Work

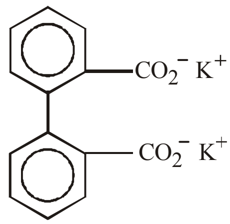
48. 

End product (B) of above reaction is –

(1) 

(2) 

(3) 

(4) 

49. Precautions to be taken in the study of reaction rate for the reaction between potassium iodate (KIO_3) and sodium sulphite (Na_2SO_3) using starch solution as indicator at different concentrations and temperature –

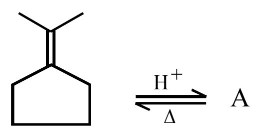
(1) The concentration of sodium thiosulphate solution should always be less than the concentration of the potassium iodide solution.

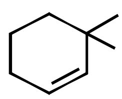
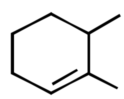
(2) Freshly prepared starch solution should be used.

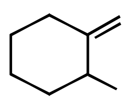
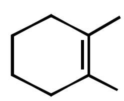
(3) Experiments should be performed with the fresh solutions of H_2O_2 and KI.

(4) All of these

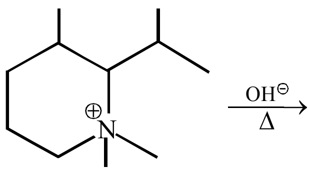
50. Predict the product (A) of the following reaction



(1) 
(2) 

(3) 
(4) 

Section - B

51. 

Find the number of α-H in alkene which is major product in this reaction?

52. At 675K, $H_2(g)$ and $CO_2(g)$ react to form CO(g) and H_2O (g), K_p for the reaction is 0.16. If a mixture of 0.25 mole of $H_2(g)$ and 0.25 mol of CO_2 is heated at 675K, calculate the mole % of CO(g) in equilibrium mixture

53. In the spinel structure, oxides ions are cubical-closest packed whereas 1/8th of tetrahedral voids are occupied by A^{2+} cation and 1/2 of octahedral voids are occupied by B^{3+} cations. The general formula of the compound having spinel structure is AB_nO_{2n}. Find the value of n.

54. The value of P° for benzene is 640 mm of Hg. The vapour pressure of solution containing 2.5g substance in 39g benzene is 600mm of Hg. Find the molecular mass of X.

55. Find the pH of a 2 litre solution which is 0.1 M each with respect to CH_3COOH and $(CH_3OO)_2Ba$. ($K_a = 1.8 \times 10^{-5}$)

56. H_3PO_2 is the molecular formula of an acid of phosphorus. What is its basicity?

57. A gas present in a cylinder fitted with a frictionless piston expands against a constant pressure of 1 atm from a volume of 2 litre to a volume of 6 litre. In doing so, it absorbs 800 J heat from surroundings. Determine increase in internal energy of process.

58. Calculate the number of metamers represented by molecular formula $C_4H_{10}O$.

59. A current of 2.0 A passed for 5 hours through a molten metal salt deposits 22.2 g of metal (At wt. = 177). Find the oxidation state of the metal in the metal salt.

60. The instantaneous rate of disappearance of MnO_4^- ion in the following reaction is $4.56 \times 10^{-3} Ms^{-1}$

$$2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$$

The rate of appearance I_2 is $x \times 10^{-2}$ Ms^{-1}

MATHEMATICS

Section - A

61. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, then which of the following equation can represent L_1 ?

(1) $x + 7y = 0$ (2) $x - y = 0$

(3) $x - 7y = 0$ (4) Both (1) and (2)

62. Let α_1, α_2 and β_1, β_2 be the roots of $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ respectively. If the system of equations $\alpha_1 y + \alpha_2 z = 0$ and $\beta_1 y + \beta_2 z = 0$ has a non-trivial solution, then

(1) $\dfrac{b^2}{q^2} = \dfrac{ac}{pr}$ (2) $\dfrac{c^2}{r^2} = \dfrac{ab}{pq}$

(3) $\dfrac{a^2}{p^2} = \dfrac{bc}{qr}$ (4) None of these

63. If $f''(x) < 0, \forall x \in (a, b)$, then $f'(x) = 0$ occurs

(1) exactly once in (a, b)

(2) atmost once in (a, b)

(3) atleast once in (a, b)

(4) None of these

64. If the function $f : [0, 16] \to R$ is differentiable. If $0 < \alpha < 1$ and $1 < \beta < 2$, then $\int_0^{16} f(t)\, dt$ is equal to–

(1) $4[\alpha^3 f(\alpha^4) - \beta^3 f(\beta^4)]$

(2) $4[\alpha^3 f(\alpha^4) + \beta^3 f(\beta^4)]$

(3) $4[\alpha^4 f(\alpha^3) + \beta^4 f(\beta^3)]$

(4) $4[\alpha^2 f(\alpha^2) + \beta^2 f(\beta^2)]$

65. Three distinct points $P(3u^2, 2u^3)$, $Q(3v^2, 2v^3)$ and $R(3w^2, 2w^3)$ are collinear then –

(1) $uv + vw + wu = 0$ (2) $uv + vw + wu = 3$

(3) $uv + vw + wu = 2$ (4) $uv + ww + wu = 1$

66. Let 'a' denote the roots of equation

$$\cos(\cos^{-1} x) + \sin^{-1} \sin\left(\frac{1+x^2}{2}\right) = 2\sec^{-1}(\sec x)$$

then possible values of $[\,|10a|\,]$ where $[\,.\,]$ denotes the greatest integer function will be

(1) 1 (2) 5

(3) 10 (4) Both (1) and (3)

67. The image of the pair of lines represented by: $ax^2 + 2hxy + by^2 = 0$ by the line mirror $y = 0$ is

(1) $ax^2 - 2hxy - by^2 = 0$

(2) $bx^2 - 2hxy + ay^2 = 0$

(3) $bx^2 + 2hxy + ay^2 = 0$

(4) $ax^2 - 2hxy + by^2 = 0$

68. If $x^2 - 2x\cos\theta + 1 = 0$, then the value of $x^{2n} - 2x^n \cos n\theta + 1$, $n \in N$ is equal to –

(1) $\cos 2n\theta$

(2) $\sin 2n\theta$

(3) 0

(4) some real number greater than 0

69. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$

and $I_4 = \int_1^2 2^{x^3} dx$ then

(1) $I_2 > I_1$ (2) $I_1 > I_2$

(3) $I_3 = I_4$ (4) $I_3 > I_4$

70. The value of the definite integral

$$\int_0^{3\pi/4} [(1+x)\sin x + (1-x)\cos x]\, dx \text{ is –}$$

(1) $2(\sqrt{2}+1)$ (2) $\sqrt{2}+1$

(3) $2\sqrt{2}$ (4) 0

71. The locus of the centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y - 2 = 0$ orthogonally is –
(1) $9x + 10y - 7 = 0$ (2) $x - y + 2 = 0$
(3) $9x - 10y + 11 = 0$ (4) $9x + 10y + 7 = 0$

72. The sum of the first n terms of the series
$$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...$$
is $\frac{n(n+1)^2}{2}$ when n is even. When n is odd the sum is
(1) $\left[\frac{n(n+1)}{2}\right]^2$ (2) $\frac{n^2(n+1)}{2}$
(3) $\frac{n(n+1)^2}{4}$ (4) $\frac{3n(n+1)}{2}$

73. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is –
(1) $x^2 + 2y^2 - ax = 0$ (2) $2x^2 + y^2 - 2ax = 0$
(3) $2x^2 + 2y^2 - ay = 0$ (4) $2x^2 + y^2 - 2ay = 0$

74. If $z + 1/z = 2\cos\theta$, then the value of $|(z^{2n} - 1)/(z^{2n} + 1)|$
(1) $|\tan n\theta|$ (2) $\tan n\theta$
(3) $|\cot n\theta|$ (4) $\cot n\theta$

75. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 = 2$
(1) cuts at right angle (2) touch each other
(3) cut at an angle $\frac{\pi}{3}$ (4) cut at an angle $\frac{\pi}{4}$

76. If the pair of lines $ax^2 + 2(a+b)xy + by^2 = 0$ lie along diameters of a circle and divide the circle into four sectors such that the area of one of the sectors is thrice the area of another sector then
(1) $3a^2 - 10ab + 3b^2 = 0$
(2) $3a^2 - 2ab + 3b^2 = 0$
(3) $3a^2 + 10ab + 3b^2 = 0$
(4) $3a^2 + 2ab + 3b^2 = 0$

77. Which of the following is a contradiction?
(1) $(p \wedge q) \wedge \sim (p \vee q)$ (2) $p \vee (-p \wedge q)$
(3) $(p \Rightarrow q) \Rightarrow p$ (4) None of these

78. If $1, \omega, \omega^2, ...\omega^{n-1}$ are the n roots of unity, then $(1-\omega)(1-\omega^2)........(1-\omega^{n-1})$ equals
(1) 0 (2) 2
(3) n (4) n^2

79. Set of values of m for which two points P and Q lie on the line $y = mx + 8$ so that $\angle APB = \angle AQB = \frac{\pi}{2}$ where $A \equiv (-4, 0)$, $B \equiv (4, 0)$ is –
(1) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty) - \{-2, 2\}$
(2) $[-\sqrt{3}, -\sqrt{3}] - \{-2, 2\}$
(3) $(-\infty, -1) \cup (1, \infty) - \{-2, 2\}$
(4) $\{-\sqrt{3}, \sqrt{3}\}$

Space for Rough Work

80. The trace $T_r(A)$ of a 3×3 matrix $A = (a_{ij})$ is defined by the relation $T_r(A) = a_{11} + a_{22} + a_{33}$ (i.e., $T_r(A)$ is sum of the main diagonal elements). Which of the following statements cannot hold ?
(1) $T_r(kA) = kT_r(A)$ (k is a scalar)
(2) $T_r(A + B) = T_r(A) + T_r(B)$
(3) $T_r(I_3) = 3$
(4) $T_r(A^2) = T_r(A)^2$

Section - B

81. The area above the x–axis enclosed by the curves $x^2 - y^2 = 0$ and $x^2 + y - 2 = 0$ is_____.

82. If number of permutations 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time are such that the digit. 1 appearing somewhere to the left of 2, 3 appearing to the left of 4 and 5 somewhere to the left of 6, is equal to k.7!, then value of k is _____.
(e.g., 815723946 would be one such permutation)

83. Vertices of a parallelogram taken in order are A (2, –1, 4), B (1, 0, –1), C (1, 2, 3) and D. If distance of the point P (8, 2, –12) from the plane of the parallelogram is $\frac{k\sqrt{6}}{9}$, then value of k is__.

84. Given
$\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{C} = \hat{i} + 2\hat{j} + \hat{k}$.
The value of $|\vec{A} \times [\vec{A} \times (\vec{A} \times \vec{B})].\vec{C}|$ is _____.

85. If area of triangle formed by common tangents to the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 + 2x = 0$ is $\lambda\sqrt{3}$, then value of λ is ____.

86. A box contains 6 red, 5 blue and 4 white marbles. Four marbles are chosen at random without replacement. The probability that there is atleast one marble of each colour among the four chosen, is ______.

87. Period of $\frac{\sin\theta + \sin 2\theta}{\cos\theta + \cos 2\theta}$ is $\frac{K\pi}{3}$,then k is ____.

88. If the tangent at the point $\left(4\cos\theta, \frac{16}{\sqrt{11}}\sin\theta\right)$ to the ellipse $16x^2 + 11y^2 = 256$ is also a tangent to the circle $x^2 + y^2 - 2x = 15$, then the value of θ is $\pm\frac{\pi}{K}$. The value of k is _____.

89. One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that it gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that he has the disease. The probability that he actually has the disease, is _____.

90. Let $a_n = \int_0^{\pi/2} (1 - \sin t)^n \sin 2t \, dt$ then

$\lim_{n\to\infty} \sum_1^n \frac{a_n}{n}$ is ______.

Space for Rough Work

Mock Test 4

Time : 3 hrs. **Max. Marks : 300**

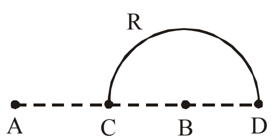
INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

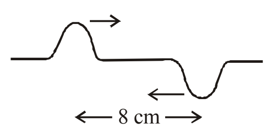
1. Charges +q and –q are placed at points A and B respectively which are a distance 2L apart, C is the midpoint between A and B. The work done in moving a charge +Q along the semicircle CRD is



(1) $\frac{qQ}{2\pi\varepsilon_0 L}$ (2) $\frac{qQ}{6\pi\varepsilon_0 L}$

(3) $-\frac{qQ}{6\pi\varepsilon_0 L}$ (4) $\frac{qQ}{4\pi\varepsilon_0 L}$

2. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in the figure. The speed of each pulse is 2 cm/s. After 2 seconds, the total energy of the pulses will be



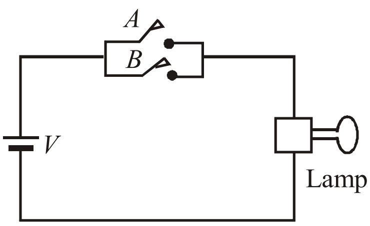
(1) Zero
(2) Purely kinetic
(3) Purely potential
(4) Partly kinetic and partly potential

3. A metallic wire of density d is lying horizontal on the surface of water. The maximum length of wire so that it may not sink will be

(1) $\sqrt{\frac{2Tg}{\pi d}}$ (2) $\sqrt{\frac{2\pi T}{dg}}$

(3) $\sqrt{\frac{2T}{\pi dg}}$ (4) any length

4. Which logic gate with inputs A and B performs the same operation as that performed by the following circuit?



(1) NAND gate (2) OR gate
(3) NOR gate (4) AND gate

5. The specific heat capacity of a metal at low temperature (T) is given as $C_p(\text{kJK}^{-1}\text{kg}^{-1}) = 32\left(\frac{T}{400}\right)^3$. A 100 g vessel of this metal is to be cooled from 20 K to 4 K by a special refrigerator operating at room temperature (27°C). The amount of work required to cool in vessel is

(1) equal to 0.002 kJ
(2) greater than 0.148 kJ
(3) between 0.148 kJ and 0.028 kJ
(4) less than 0.028 kJ

6. A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to:
(g = *gravitational acceleration*)

(1) $\left[1-\left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$ (2) $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

(3) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{A}{Mg}$ (4) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{Mg}{A}$

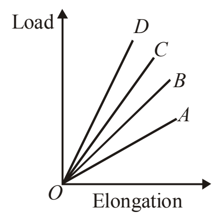
7. A bucket full of hot water is kept in a room and it cools from 75°C to 70°C in T_1 minutes, from 70°C to 65°C in T_2 minutes and from 65°C to 60°C in T_3 minutes. Then

(1) $T_1 = T_2 = T_3$ (2) $T_1 < T_2 < T_3$
(3) $T_1 > T_2 > T_3$ (4) $T_1 < T_3 < T_2$

8. The length of an elastic string is x when the tension is 5N. Its length is y when the tension is 7N. What will be its length, when the tension is 9N?

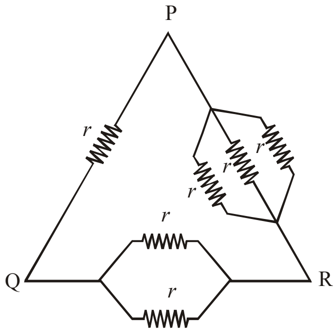
(1) $2y + x$ (2) $2y - x$
(3) $7x - 5y$ (4) $7x + 5y$

9. The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line



(1) *OA* (2) *OC*
(3) *OD* (4) *OB*

10. Six equal resistances are connected between points P, Q and R as shown in figure. Then net resistance will be maximum between :



(1) P and R (2) P and Q
(3) Q and R (4) Any two points

11. Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference 4 V. The de-Broglie wavelengths are λ_e and λ_p for electrons and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by : (given m_e is mass of electron and m_p is mass of proton).

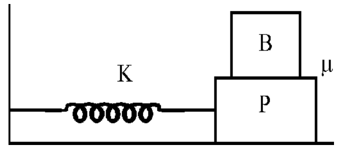
(1) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$ (2) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_e}{m_p}}$

(3) $\frac{\lambda_e}{\lambda_p} = \frac{1}{2}\sqrt{\frac{m_e}{m_p}}$ (4) $\frac{\lambda_e}{\lambda_p} = 2\sqrt{\frac{m_p}{m_e}}$

12. A straight section PO of a circuit lies along the x-axis from x = –a/2 to x = +a/2, and carries a steady current 'I'. The magnitude of magnetic field due to the section PO at a point to y = + a will be

(1) proportional to a (2) proportional to a^2
(3) proportional to 1/a (4) equal to zero

13. A flat plate P of mass 'M' executes SHM in a horizontal plane by sliding over a frictionless surface with a frequency V. A block 'B' of mass 'm' rests on the plate as shown in figure. Coefficient of friction between the surface of B and P is μ. What is the maximum amplitude of oscillation that the plate block system can have if the block B is not to slip on the plate :



(1) $\frac{\mu g}{4\pi^2 V^2}$ (2) $\frac{\mu g}{4\pi^2 V}$

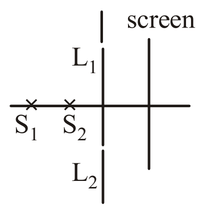
(3) $\frac{\mu}{4\pi^2 V^2 g}$ (4) $\frac{\mu g}{2\pi^2 V^2}$

14. A glass slab has the left half of refractive index n_1, and the right half of $n_2 = 3n_1$. The effective refractive index of the whole slab is

(1) $\frac{n_1}{2}$ (2) 2n

(3) $\frac{3n_1}{2}$ (4) $\frac{2n_1}{3}$

15. In the arrangement shown L_1, L_2 are slits and S_1 and S_2 are two independent sources. On the screen, interference fringes



(1) will not be there
(2) will not be there if the intensity of light reaching the screen from S_1 and S_2 are equal.
(3) will be there under all circumstances
(4) we will have only the central fringe

16. For good demodulation of AM signal of carrier frequency f, the value of RC should be

(1) $RC = \frac{1}{f}$ (2) $RC < \frac{1}{f}$

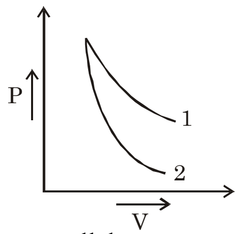
(3) $RC \geq \frac{1}{f}$ (4) $RC >> \frac{1}{f}$

17. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2. If the half life of the specimen is T, the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is proportional to

(1) $(R_1T_1 - R_2T_2)$ (2) $(R_1 - R_2)$

(3) $(R_1 - R_2)/T$ (4) $(R_1 - R_2) \times T$

18. P–V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to



(1) He and Ar

(2) He and O_2

(3) O_2 and N_2

(4) O_2 and He

19. A magnetic needle lying parallel to a magnetic field requiers W units of work to turn it through 60^0. The torque needed to maintain the needle in this position will be

(1) $\sqrt{3}W$ (2) W

(3) $\frac{\sqrt{3}}{2}W$ (4) 2W

20. Two small equal point charges of magnitude q are suspended from a common point on the ceiling by insulating mass less strings of equal lengths. They come to equilibrium with each string making angle θ from the vertical. If the mass of each charge is m, then the electrostatic potential at the centre of line joining them will be $\left(\frac{1}{4\pi \in_0} = k\right)$.

(1) $2\sqrt{k\,mg\tan\theta}$ (2) $4\sqrt{k\,mg\tan\theta}$

(3) $4\sqrt{k\,mg/\tan\theta}$ (4) $\sqrt{k\,mg/\tan\theta}$

Section - B

21. A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2. The car will catch up the bus after a time of _______ s.

22. A transformer is used to light a 140 W, 24 V bulb from a 240 V a.c. mains. The current in the main cable is 0.7 A. The efficiency of the transformer is _____%.

23. A rod of length L is placed on x – axis between x = 0 and x = L. The linear density i.e., mass per unit length denoted by ρ, of this rod, varies as, ρ = a + bx. The dimensions of b is $ML^{-2}T^x$. Find the value of x.

24. The surface charge density of a thin charged disc of radius R is σ. The value of the electric field at the centre of the disc is $\frac{\sigma}{2\in_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc is reduces by _____%.

25. A body is thrown vertically upwards from the surface of earth in such a way that it reaches upto a height equal to $10R_e$. The velocity imparted to the body will be _______ km/s

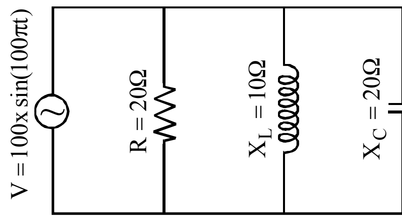
26. What is the ratio of the circumference of the first Bohr orbit for the electron in the hydrogen atom to the de Brogile wavelength of electrons having the same velocity as the electron in the first Bohr orbit of the hydrogen atom?

27. A projectile can have the same range 'R' for two angles of projection. If 'T_1' and 'T_2' to be time of flights in the two cases, then the product of the two time of flights is $\frac{xR}{g}$. Find the value of x.

28. The electric field associated with an e.m. wave in vacuum is given by $\vec{E} = \hat{i}\ 40\cos(kz - 6 \times 10^8 t)$, where E, z and t are in volt/m, meter and seconds respectively. The value of wave vector k is _______ m^{-1}.

29. Two points of a rod move with velocities 3v and v perpendicular to the rod and in the same direction, separated by a distance r. Then the angular velocity of the rod $\frac{xv}{r}$. Find the value of x.

30. In the given circuit, the current drawn from the source is _____A.

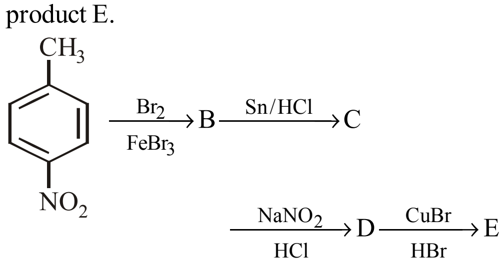


CHEMISTRY

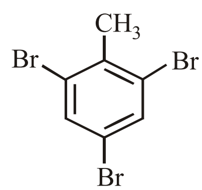
Section - A

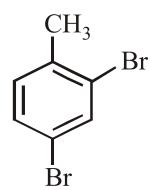
31. Which of the following structures does not contain any chiral C atom but represent the chirality in the structure.
(1) 2 – Ethyl – 3 – hexene
(2) 2, 3-Pentadiene
(3) 1,3 – Butadiene
(4) Pent – 3 – en – 1 – yne

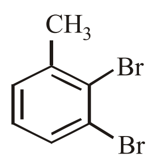
32. In a set of reactions p-nitrotoluene yielded a product E.

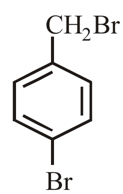


The product E would be:

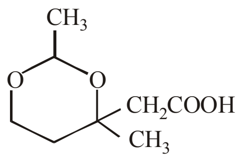
(1) 

(2) 

(3) 

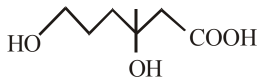
(4) 

33. Compound A is formed by the interaction of

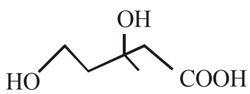


[A]

(1) CH_3COOH and 

(2) CH_3CHO and 

(3) CH_3COCH_2COOH and

(4) CH_3CHO and 

34. N_2 and O_2 are converted to mono cations N_2^+ and O_2^+ respectively, which of the following is wrong?

(1) In N_2^+, the N – N bond weakens

(2) In O_2^+, the O – O bond order increases

(3) In O_2^+, paramagnetism decreases

(4) N_2^+ becomes diamagnetic

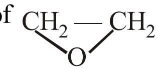
35. The reaction in which hydrogen peroxide acts as a reducing agent is

(1) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

(3) $2FeSO_4 + H_2SO_4 + H_2O_2 \rightarrow Fe_2(SO_4)_3 + 2H_2O$

(4) $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

36. Reaction of  with RMgX leads to formation of

(1) RCHOHR (2) $RCHOHCH_3$

(3) RCH_2CH_2OH (4) 

37. If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:

(1) $\frac{1}{2}\frac{e^2}{r}$ (2) $-\frac{e^2}{r}$

(3) $\frac{me^2}{r}$ (4) $-\frac{1}{2}\frac{e^2}{r}$

38. When chlorine water is added to an aqueous solution of sodium iodide in the presence of chloroform, a violet colouration is obtained. On adding more of chlorine water and vigorous shaking, the violet colour disappears. This shows the conversion of ...... into ......

(1) I_2, HIO_3 (2) I_2, HI

(3) HI, HiO_3 (4) I_2, HOI

39. When tert-butyl chloride is made to react with sodium ethoxide, the major product is

(1) dimethyl ether

(2) di-tert-butyl ether

(3) tert-butylmethyl ether

(4) isobutylene

40. Which statement is true regarding following reactions

$$trans-2-\text{Butene}\xrightarrow{HCO_3H}A$$

$$cis-2-\text{Butene}\xrightarrow{HCO_3H}B$$

(1) Compounds A and B are formed by *syn* addition and they are racemic mixture and meso respectively
(2) Compounds A and B are formed by *anti* addition and are racemic mixture and meso respectively
(3) Compounds A and B are formed by *anti* addition and are meso and racemic mixture respectively
(4) Compounds A and B are formed by *syn* addition and are meso and racemic mixture respectively

41. An organic compound is treated with $NaNO_2$ and dil. HCl at 0°C. The resulting solution is added to an alkaline solution of β-naphthol where by a brilliant red dye is produced. It shows the presence of
(1) $-NO_2$ group
(2) aromatic $-NH_2$ group
(3) $-CONH_2$ group
(4) aliphatic $-NH_2$ group

42. Point out the incorrect statment among the following :
(1) The oxidation state of oxygen is +2 in OF_2.
(2) Acidic character follows the order $H_2O < H_2S < H_2Se < H_2Te$.
(3) The tendency to form multiple bonds increases in moving down the group from sulphur to tellurium (towards C and N).
(4) Sulphur has a strong tendency to catenate while oxygen shows this tendency to a limited extent.

43. Removal of Fe, Cu, W from Sn metal after smelting is by .... because ....
(1) poling; of more affinity towards oxygen for impurities
(2) selective oxidation; of more affinity towards oxygen for impurities
(3) electrolytic refining; impurities undissolved in elec-trolyte
(4) liquation; Sn having low melting point compared to impurities.

44. Among KO_2, AlO_2^-, BaO_2 and NO_2^+, unpaired electron is present in
(1) NO_2^+ and BaO_2
(2) KO_2 and AlO_2^-
(3) KO_2 only
(4) BaO_2 only

45. If a 0.1 M solution of glucose (Mol. wt 180) and 0.1 molar solution of urea (Mol. wt. 60) are placed on two sided semipermeable membrane to equal heights, then it will be correct to say that
(1) there will be no net movement across the membrane.
(2) glucose will flow across the membrane into urea solution.
(3) urea will flow across the membrane into glucose solution.
(4) water will flow from urea solution to glucose solution.

46. Melamine plastic crockery is a codensation polymer of
(1) HCHO and melamine
(2) HCHO and ethylene
(3) melamine and ethylene
(4) None of these

47. When pink complex, $[Co(H_2O)_6]^{2+}$ is dehydrated the colour changes to blue. The correct explanation for the change is :
(1) The octahedral complex becomes square planar.
(2) A tetrahedral complex is formed.
(3) Distorted octahedral structure is obtained.
(4) Dehydration results in the formation of polymeric species.

48. In the form of dichromate, Cr (VI) is a strong oxidising agent in acidic medium but Mo (VI) in MoO_3 and W (VI) in WO_3 are not because ____________ .
(i) Cr (VI) is more stable than Mo(VI) and W (VI).
(ii) Mo (VI) and W(VI) are more stable than Cr(VI).
(iii) Higher oxidation states of heavier members of group-6 of transition series are more stable.
(iv) Lower oxidation states of heavier members of group-6 of transition series are more stable.
(1) (i) and (ii) (2) (ii) and (iii)
(3) (i) and (iv) (4) (ii) and (iv)

49. The correct statement among the following is :
(1) The alkali metals when strongly heated in oxygen form superoxides.
(2) Caesium is used in photoelectric cells.
(3) $NaHCO_3$ is more soluble in water than $KHCO_3$.
(4) The size of hydrated ions of alkali metals increases from top to bottom.

50. If the following half cells have E° values as

$$A^{3+} + e^- \longrightarrow A^{2+},\ E° = y_2 V$$

$$A^{2+} + 2e^- \longrightarrow A,\ E° = -y_1 V$$

The E° of the half cell

$$A^{3+} + 3e^- \longrightarrow A \text{ will be}$$

(1) $\frac{2y_1 - y_2}{3}$ (2) $\frac{y_2 - 2y_1}{3}$
(3) $2y_1 - 3y_2$ (4) $y_2 - 2y_1$

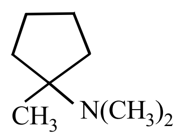
Section - B

51. The radii of Na^+ and Cl^- ions are 100 pm and 200 pm respectively. Calculate the edge length of NaCl unit cell.

52. When 1.8 g of steam at the normal boiling point of water is converted into water at the same temperature, calculate Δs for the above change in J/K. (ΔH_{vap} for water = 40.8 kJ mol^{-1})

53. Given
$n = 5, m_\ell = +1$
Find the maximum number of electron(s) in an atom that can have the quantum numbers as given.

54. What amount of sugar ($C_{12}H_{22}O_{11}$) (in g) is required to prepare 2 L of its 0.1 M aqueous solution?

55.  $\xrightarrow[\text{(ii) heat}]{\text{(i) } H_2O_2}$ Product

Find the number of unsaturated carbon in the product.

56. A tetrapeptide has —COOH group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, find the number of possible sequences (primary structures) with —NH_2 group attached to a chiral center.

57. 4 g of H_2 effused through a pinhole in 10 s at some constant temperature and pressure. Calculate the amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure.

58. The vapour pressure of benzene at a certain temperature is 640 mm of Hg. A non volatile and non electrolyte solid weighing 2.175 g is added to 39.08 g of benzene. If the vapour pressure of the solution is 600 mm of Hg, calculate the molecular weight of solid substance.

59. When CO_2 dissolves in water, the following equilibrium is established

$CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$; for which the equilibrium constant is 3.8×10^{-6} and pH = 6.0. What would be the ratio of concentration of bicarbonate ion to carbon dioxide?

60. For a first order reaction, calculate the ratio between the time taken to complete 3/4th of the reaction and time taken to complete half of the reaction.

MATHEMATICS

Section - A

61. The roots of the given equation $(p-q)x^2 + (q-r)x + (r-p) = 0$ are :

(1) $\frac{p-q}{r-p}, 1$ (2) $\frac{q-r}{p-q}, 1$

(3) $\frac{r-p}{p-q}, 1$ (4) None of these

62. If a, b, c, d and p are distinct non zero real numbers such that $(a^2+b^2+c^2)p^2 - 2(ab+bc+cd)p + (b^2+c^2+d^2) \le 0$ then a,b,c,d are in

(1) A.P. (2) G.P.

(3) H.P. (4) satisfy $ab = cd$

63. Which of the following is correct?

(1) If $a^2 + 4b^2 = 12ab$, then

$\log(a+2b) = \frac{1}{2}(\log a + \log b)$

(2) If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$,

then $x^a . y^b . z^c = abc$

(3) $\frac{1}{\log_{xy} xyz} + \frac{1}{\log_{yz} xyz} + \frac{1}{\log_{zx} xyz} = 2$

(4) All are correct

64. If $0 < \alpha, \beta, \gamma < \pi/2$ such that $\alpha + \beta + \gamma = \frac{\pi}{2}$ and cot α, cot β, cot γ are in arithmetic progression, then the value of $\cot\alpha \cot\gamma$ is

(1) 1 (2) 3

(3) $\cot^2\beta$ (4) $\cot\alpha + \cot\gamma$

65. The equation of the chord of the hyperbola $25x^2 - 16y^2 = 400$ that is bisected at point (5, 3) is:

(1) $135x - 48y = 481$ (2) $125x - 48y = 481$

(3) $125x - 4y = 48$ (4) None of these

66. The number of solutions of the equation

$$\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}\,x + 4$$

(1) forms an empty set (2) is only one
(3) is only two (4) is more than 2

67. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the roots of $z^5 - 1 = 0$, then the value of $\frac{\omega-\alpha_1}{\omega^2-\alpha_1}\cdot\frac{\omega-\alpha_2}{\omega^2-\alpha_2}\cdot\frac{\omega-\alpha_3}{\omega^2-\alpha_3}\cdot\frac{\omega-\alpha_4}{\omega^2-\alpha_4}$ is (where ω is imaginary cube root of unity)

(1) 1 (2) ω
(3) ω^2 (4) None of these

68. Which of the following is correct?

(1) If A and B are square matrices of order 3 such that | A | = –1, | B | = 3, then the determinant of 3 AB is equal to 27.
(2) If A is an invertible matrix, then det (A^{-1}) is equal to det (A)
(3) If A and B are matrices of the same order, then $(A+B)^2 = A^2 + 2AB + B^2$ is possible if AB = I
(4) None of these

69. Let $f(x) = |x - 1|$. Then

(1) $f(x^2) = (f(x))^2$
(2) $f(x + y) = f(x) + f(y)$
(3) $f(|x|) = |f(x)|$
(4) None of these

70. If $\sin^{-1}\frac{2a}{1+a^2} + \sin^{-1}\frac{2b}{1+b^2} = 2\tan^{-1}x$, then x is equal to

(1) $\frac{a-b}{1+ab}$ (2) $\frac{b}{1+ab}$
(3) $\frac{b}{1-ab}$ (4) $\frac{a+b}{1-ab}$

71. Let $f(x) = \frac{2}{x+1}$, $g(x) = \cos x$ and $h(x) = \sqrt{x+3}$ then the range of the composite function fogoh, is

(1) R^+ (2) $R - \{0\}$
(3) $[1, \infty)$ (4) $R^+ - \{1\}$

72. The set of points where $f(x) = (x-1)^2 (x + |x-1|)$ is thrice differentiable, is

(1) R (2) $R - \{0\}$
(3) $R - \{1\}$ (4) $R - \{0,1\}$

73. Let $f(x) = 1/(x-1)$ and $g(x) = 1/(x^2 + x - 2)$. Then the set of points where (gof)(x) is discontinuous, is

(1) {1} (2) {–2,1}
(3) {1/2, 1, 2} (4) {1/2 , 1}

74. $\sum_{r=0}^{m} {}^{n+r}C_n$ is equal to :

(1) ${}^{n+m+1}C_{n+1}$ (2) ${}^{n+m+2}C_n$
(3) ${}^{n+m+3}C_{n-1}$ (4) None of these

75. The value of $\int_{1/n}^{(an-1)/n} \frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}dx$ is

(1) $\frac{a}{2}$ (2) $\frac{1}{2n}(na+2)$

(3) $\frac{na-2}{2n}$ (4) None of these.

76. The value of

$\int_{-\pi/4}^{\pi/4} (x|x|+\sin^3 x + x\tan^2 x + 1)\, dx$ is

(1) 0 (2) 1

(3) $\pi/4$ (4) $\pi/2$

77. Let $(1-x-2x^2)^6 = 1 + a_1 x + a_2 x^2 + \ldots + a_{12} x^{12}$.

Then $\frac{a_2}{2^2}+\frac{a_4}{2^4}+\frac{a_6}{2^6}+......+\frac{a_{12}}{2^{12}}$ is equal to

(1) –1 (2) –1/2

(3) 0 (4) 1/2

78. The equation of a common tangent to $y^2 = 4x$ and the curve $x^2 + 4y^2 = 8$ can be

(1) $x-2y+2=0$ (2) $x+2y+4=0$

(3) $x-2y=4$ (4) $x+2y=4$

79. The function $f(x) = (x-3)^2$ satisfies all the conditions of mean value theorem in [3, 4]. A point on $y=(x-3)^2$, where the tangent is parallel to the chord joining (3, 0) and (4,1) is

(1) $\left(\frac{7}{2},\frac{1}{2}\right)$ (2) $\left(\frac{7}{2},\frac{1}{4}\right)$

(3) (1, 4) (4) (4, 1)

80. If $x+y-z+xyz=0$, then $\frac{2x}{1-x^2}+\frac{2y}{1-y^2}-\frac{2z}{1-z^2}$ is equal to

(1) $\frac{xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(2) $\frac{-xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(3) $\frac{8xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(4) $\frac{-8xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

Section - B

81. If the number of 5-element subsets of the set $A = \{a_1, a_2, ..., a_{20}\}$ of 20 distinct elements is k times the number of 5-element subsets containing a_4, then k is _____.

82. The value of $\cos 36^0 \cos 42^0 \cos 78^0$ is _____.

$\left[\text{Given : } \sin 18° = \frac{\sqrt{5}-1}{4} \text{ and } \cos 36° = \frac{\sqrt{5}+1}{4}\right]$

83. If $x = 1/5$, the value of $|\cos(\cos^{-1}x + 2\sin^{-1}x)|$ is _____.

84. Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$, and $B(1,7)$ and $D(4,-2)$ are points on the circle then, if tangents be drawn at B and D, which meet at C, then area of quadrilateral $ABCD$ is _____.

85. If the system of linear equations :

$$x_1 + 2x_2 + 3x_3 = 6$$
$$x_1 + 3x_2 + 5x_3 = 9$$
$$2x_1 + 5x_2 + ax_3 = b$$

is consistent and has infinite number of solutions, then value of a + b is _____.

86. If $A^k = 0$ (A is nilpotent with index k), $(I - A)^p = I + A + A^2 + \ldots + A^{k-1}$, then value of $|p|$ is _____.

87. Let $f(x) = \dfrac{x - \{x+1\}}{x - \{x+2\}}$; where {x} is the fractional part of x, then value of $\lim\limits_{x \to 1/3} f(x)$ is _____.

88. Let $y(x)$ be a solution of $\dfrac{(2+\sin x)}{(1+y)}\dfrac{dy}{dx} = \cos x$.

If $y(0) = 2$, then value of $y\left(\dfrac{\pi}{2}\right)$ is _____.

89. $\displaystyle\int \frac{1}{f(x)}\,dx = \log(f(x))^2 + C.$ If $f(0) = 5$ then value of constant term of f(x) is _____.

90. In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0,$ then value of

$\sin^2 A + \sin^2 B + \sin^2 C$ is _____.

Mock Test 5

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

1. A lift is moving down with acceleration a. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively
   (1) g, g (2) $g - a, g - a$
   (3) $g - a, g$ (4) a, g
2. Relative permittivity and permeability of a material are ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material?
   (1) $\varepsilon_r = 0.5, \mu_r = 1.5$ (2) $\varepsilon_r = 1.5, \mu_r = 0.5$
   (3) $\varepsilon_r = 0.5, \mu_r = 0.5$ (4) $\varepsilon_r = 1.5, \mu_r = 1.5$
3. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0\left(\frac{5}{4} - \frac{r}{R}\right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance $r(r < R)$ from the origin is given by

Space for Rough Work

(1) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$ (2) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$

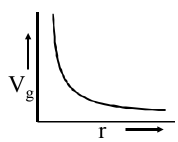
(3) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$ (4) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

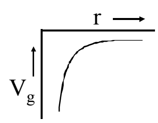
4. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M. The piston and the cylinder have equal cross sectional area A. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency

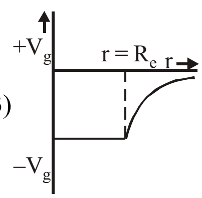
(1) $\frac{1}{2\pi}\frac{A\gamma P_0}{V_0 M}$ (2) $\frac{1}{2\pi}\frac{V_0 M P_0}{A^2\gamma}$

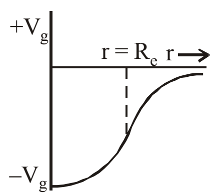
(3) $\frac{1}{2\pi}\sqrt{\frac{A^2\gamma P_0}{MV_0}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{MV_0}{A\gamma P_0}}$

5. The correct graph between the gravitational potential (V_g) due to a hollow sphere and distance from its centre will be

(1) 

(2) 

(3) 

(4) 

6. In a cubical vessel are enclosed n molecules of a gas each having a mass m and an average speed v. If ℓ is the length of each edge of the cube, the pressure exerted by the gas will be

(1) $\frac{nmv^2}{\ell^3}$ (2) $\frac{nm^2v}{2\ell^3}$

(3) $\frac{mnv^2}{3\ell^3}$ (4) $\frac{nmv}{2\ell}$

7. A sphere, a cube and a thin circular plate all made of the same material and having the same mass, are initially heated to a temperature of 200°C. Which of these objects will cool slowest when left in air at room temperature?

(1) the sphere

(2) the cube

(3) the circular plate

(4) all will cool at same rate

8. An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

(1) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

(2) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(3) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(4) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

9. A particle of mass m_1 collides head-on with another stationary particle of mass m_2 ($m_2 > m_1$). The collision is perfectly inelastic. The fraction of kinetic energy which is converted into heat in this collision is

(1) $m_2/(m_1+m_2)$ (2) $m_1/(m_1+m_2)$

(3) $m_1/(m_1-m_2)$ (4) $m_2/(m_1-m_2)$

10. The instantaneous values of current and voltage in an A.C. circuit are $I = 4\sin\omega t$ and $E = 100\cos\left(\omega t+\frac{\pi}{3}\right)$ respectively. The phase difference between voltage and current is

(1) $\frac{\pi}{3}$ (2) $\frac{2\pi}{3}$

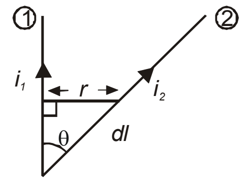
(3) $\frac{5\pi}{6}$ (4) $\frac{7\pi}{6}$

11. Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge?

(1) Wb/m^2 (2) Henry (H)

(3) H/m^2 (4) Weber (Wb)

12. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1?



(1) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \tan\theta$

(2) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \sin\theta$

(3) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \cos\theta$

(4) $\frac{\mu_0}{4\pi r} i_1 i_2\, dl \sin\theta$

13. In Young's double slit experiment, one of the slit is wider than other, so that amplitude of the light from one slit is double of that other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by :

(1) $\frac{I_m}{9}(4+5\cos\phi)$

(2) $\frac{I_m}{3}\left(1+2\cos^2\frac{\phi}{2}\right)$

(3) $\frac{I_m}{5}\left(1+4\cos^2\frac{\phi}{2}\right)$

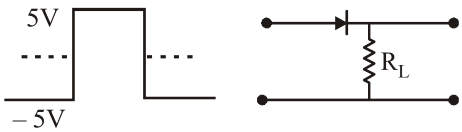
(4) $\frac{I_m}{9}\left(1+8\cos^2\frac{\phi}{2}\right)$

14. A pendulum consists of a wooden bob of mass m and length 'ℓ'. A bullet of mass m_1 is fired towards the pendulum with a speed v_1. The bullet emerges out of the bob with a speed $v_1/3$, and the bob just completes motion along a vertical circle. Then v_1 is

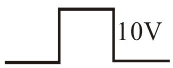
(1) $\left(\frac{m}{m_1}\right)\sqrt{5g\ell}$ (2) $\frac{3}{2}\left(\frac{m}{m_1}\right)\sqrt{5g\ell}$

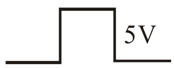
(3) $\frac{2}{3}\left(\frac{m_1}{m}\right)\sqrt{5g\ell}$ (4) $\left(\frac{m_1}{m}\right)\sqrt{g\ell}$

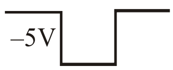
15. In a *p-n* junction diode, a square input signal of 10 V is applied as shown in figure.



The output signal across R_L will be

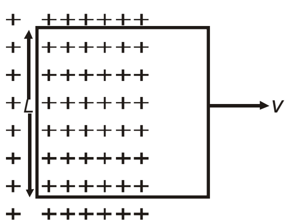
(1) 

(2) 

(3) 

(4) 

16. A conducting square loop of side *L* and resistance *R* moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction *B* constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced emf is



(1) zero (2) *RvB*
(3) *vBL*/*R* (4) *vBL*

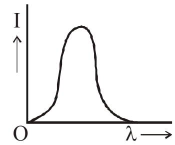
17. Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are

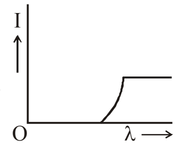
(1) 1490, 1510 (2) 1510, 1490
(3) $\frac{1}{1490}, \frac{1}{1510}$ (4) $\frac{1}{1510}, \frac{1}{1490}$

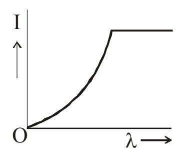
18. A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The ray emerging out of the prism makes an angle of 30° with the incident ray. The emergent ray is

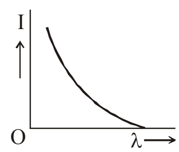
(1) Normal to the face through which it emerges
(2) Inclined at 30° to the face through which it emerges
(3) Inclined at 60° to the face through which it emerges
(4) Inclined at 90° to the normal at face through which it emerges

19. The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows :

(1) 

(2) 

(3) 

(4) 

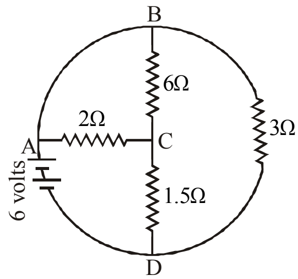
20. A proton, a deutron and an α particle accelerated through the same potential difference enter a region of uniform magnetic field, moving at right angles to B. What is the ratio of their K.E.?

(1) 1 : 2 : 2 (2) 2 : 2 : 1
(3) 1 : 2 : 1 (4) 1 : 1 : 2

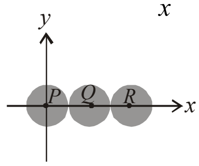
Space for Rough Work

Section - B

21. In the circuit shown, the total current supplied by the battery is _________ A.



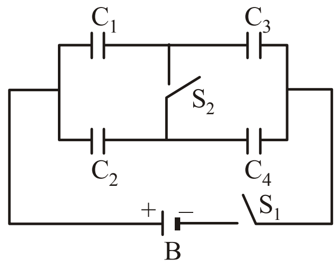
22. Three identical spheres, each of mass 1 kg are kept as shown in figure, touching each other, with their centres on a straight line. If their centres are marked P, Q, R respectively, the distance of centre of mass of the system from P is $\frac{PQ+PR}{x}$. Find the value of x.



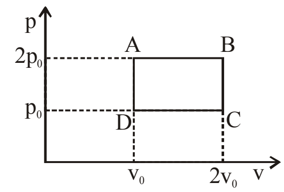
23. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it had decayed and time t_1 when $\frac{1}{3}$ of it had decayed is _____ min.

24. A wooden block of volume 1000 cc is suspended from a spring balance. It weighs 12 N in air. It is then held suspended in water with half of it inside water. What would be the reading in spring balance (in N) now?

25. In the figure battery B supplies 12 V. Take $C_1 = 1\ \mu F$, $C_2 = 2\ \mu F$, $C_3 = 3\ \mu F$, $C_4 = 4\ \mu F$. Charge on capacitor C_1 when only S_1 is closed is _____ μC.



26.



The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is $x\ p_0v_0$. Find the value of x.

27. A hydrogen-like atom has one electron revolving around a stationary nucleus. The energy required to excite the electron from the second orbit to the third orbit is 47.2 eV. The atomic number of the atom is _________.

28. A boy playing on the roof of a 10 m high building throws a ball with a speed of 10m/s at an angle of 30º with the horizontal. How far (in m) from the throwing point will the ball be at the height of 10 m from the ground ?

$[g = 10\text{m/s}^2,\ \sin 30^o = \frac{1}{2},\ \cos 30^o = \frac{\sqrt{3}}{2}]$

Space for Rough Work

29. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is x $(I_1 + I_2)$. Find the value of x.

30. A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's modulii of copper and steel are respectively 1.0×10^{11} Nm^{-2} and 2.0×10^{11} Nm^{-2}, the total extension of the composite wire is _______ mm.

CHEMISTRY

Section - A

31. In which of the following arrangements, the sequence is not strictly according to the property written against it?
   (1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power
   (2) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength
   (3) HF < HCl < HBr < HI: increasing acid strength
   (4) B < C < O < N: increasing first ionisation enthalpy.

32. Aluminothermy used for on the spot welding of large iron structure is based on the fact that-
   (1) As compared to iron, aluminium has greater affinity for oxygen.
   (2) As compared to aluminium, iron has greater affinity for oxygen.
   (3) Reaction between aluminium and oxygen is endothermic.
   (4) Reaction between iron and oxygen is endothermic.

33. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+, Ca^{2+}, Mg^{2+}, Be^{2+}?
   (1) $Ca^{2+} < Mg^{2+} < Be^{+} < K^{+}$
   (2) $Mg^{2+} < Be^{2+} < K^{+} < Ca^{2+}$
   (3) $Be^{2+} < K^{+} < Ca^{2+} < Mg^{2+}$
   (4) $K^{+} < Ca^{2+} < Mg^{2+} < Be^{2+}$.

34. A 1.0 M solution with respect to each of the metal halides AX_3, BX_2, CX_3 and DX_2 is electrolysed using platinum electrodes. If
   $E^\circ_{A^{3+}/A} = 1.50$ V, $E^\circ_{B^{2+}/B} = 0.3$ V,
   $E^\circ_{C^{3+}/C} = -0.74$ V, $E^\circ_{D^{2+}/D} = -2.37$ V.
   The correct sequence in which the various metals are deposited at the cathode is :
   (1) A, B, C, D (2) A, B, C
   (3) D, C, B, A (4) C, B, A

35. An ideal gas undergoes isothermal expansion at constant pressure. During the process :
   (1) enthalpy increases but entropy decreases.
   (2) enthalpy remains constant but entropy increases.
   (3) enthalpy decreases but entropy increases.
   (4) Both enthalpy and entropy remain constant.

36. Among the reactions given below for B_2H_6, the one which does not take place is

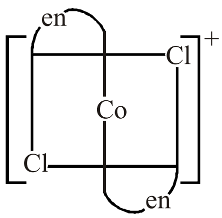
(1) $B_2H_6 + HCl \longrightarrow B_2H_5Cl + H_2$

(2) $2B_2H_6 + 6NH_3 \xrightarrow{\Delta} B_3N_3H_6$ (borazine)

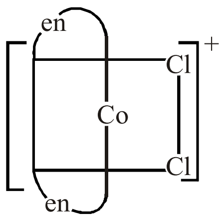
(3) $B_2H_6 + 2N(CH_3)_3 \longrightarrow 2(CH_3)_3NBH_3$

(4) $B_2H_6 + 6C_2H_4 \xrightarrow{H_3O^+} 3C_2H_5OH + 2B(OH)_3$

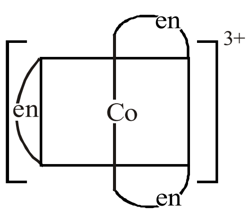
37. Which of the following ions are optically active?



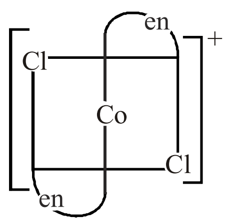
I



II



III



IV

(1) I only (2) II only

(3) II and III (4) IV only

38. The pair of compounds having metals in highest oxidation state is:

(1) MnO_2 and CrO_2Cl_2

(2) $[NiCl_4]^{2-}$ and $[CoCl_4]^{2-}$

(3) $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_2]^{2-}$

(4) $[FeCl_4]^-$ and Co_2O_3

39. A metal gives two chlorides A and B. A gives black precipitate with NH_4OH and B gives white. With KI, B gives a red precipitate soluble in excess of KI. A and B are respectively :

(1) $HgCl_2$ and Hg_2Cl_2 (2) Hg_2Cl_2 and $HgCl_2$

(3) $HgCl_2$ and $ZnCl_2$ (4) $ZnCl_2$ and $HgCl_{24}$

40. In which reaction, there is change in oxidation number of N

(1) $2NO_2 \rightarrow N_2O_4$

(2) $NH_4OH \rightarrow NH_4^+ + OH^-$

(3) $N_2O_5 + H_2O \rightarrow 2HNO_3$

(4) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

41. Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic media. The final products obtained from it in the three conditions are, respectively

(1) MnO_4^{2-}, Mn^{3+} and Mn^{2+}

(2) MnO_2, MnO_2 and Mn^{2+}

(3) MnO_2, MnO_2^+ and Mn^{3+}

(4) MnO, MnO_2 and Mn^{2+}

42. An organic compound 'X' on ozonolysis followed by reduction with Zn/H_2O gives 2 moles of $H-\overset{\overset{O}{\|}}{C}-H$ and $H-\overset{\overset{O}{\|}}{C}-CH_2-\overset{\overset{O}{\|}}{C}-H$. 'X' is

(1) $CH_2=CH-CH_2-CH=CH_2$

(2) $CH_2=CH-CH_2-CH_2-CH=CH_2$

(3) $H-C\equiv C-C\equiv C-H$

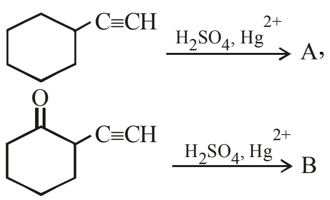
(4) $CH_2=CH-CH=CH_2$

43. Which one of the following compounds is resistant to nucleophilic attack by hydroxyl ions?
(1) Methyl acetate (2) Acetonitrile
(3) Diethyl ether (4) Acetamide

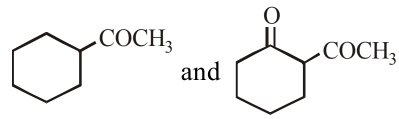
44. To detect iodine in presence of bromine, the sodium extract is treated with $NaNO_2$ + glacial acetic acid + CCl_4. Iodine is detected by the appearance of
(1) yellow colour of CCl_4 layer.
(2) purple colour of CCl_4.
(3) brown colour in the organic layer of CCl_4.
(4) deep blue colour in CCl_4.

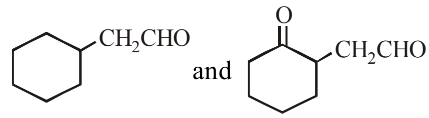
45. An organic compound A ($C_4H_{10}O$) has two enantiomeric forms and on dehydration it gives B(major product) and C (minor product). B and C are treated with HBr/ Peroxide and the compounds so produced were subjected to alkaline hydrolysis then-
(1) B will give an isomer of A
(2) C will give an isomer of A
(3) Neither of them will give isomer of A
(4) Both B and C will give isomer of A

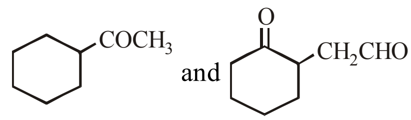
46. In which of the following polymers, empirical formula resembles with monomer?
(1) Bakelite (2) Teflon
(3) Nylon-6, 6 (4) Dacron

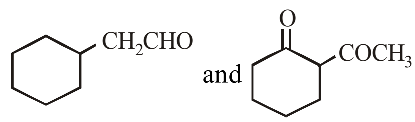
47. 

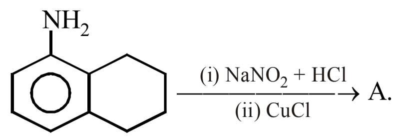
The respective compounds A and B are

(1) 

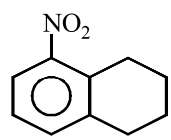
(2) 

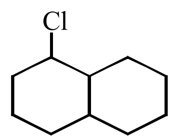
(3) 

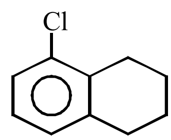
(4) 

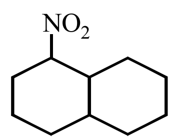
48. 

Compound 'A' is

(1) 

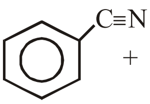
(2) 

(3) 

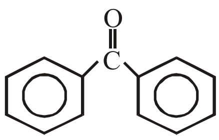
(4) 

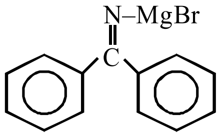
49. Allyl phenyl ether can be prepared by heating:
(1) $C_6H_5Br + CH_2 = CH - CH_2 - ONa$
(2) $CH_2 = CH - CH_2 - Br + C_6H_5ONa$
(3) $C_6H_5 - CH = CH - Br + CH_3 - ONa$
(4) $CH_2 = CH - Br + C_6H_5 - CH_2 - ONa$

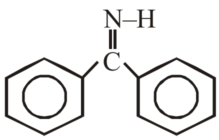
Space for Rough Work

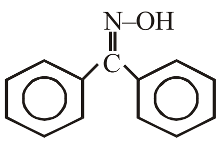
50.  + C_6H_5MgBr $\xrightarrow{\text{Ether}} A \xrightarrow{H_3O^+} B$

The final product (B) in above reaction is

(1) 

(2) 

(3) 

(4) 

Section - B

51. If radiation corresponding to second line of "Balmer series" of Li^{2+} ion, knocked out electron from first excited state of H-atom, then find the kinetic energy of ejected electron.

52. What is the sum of number of sigma (σ) and pi (π) bonds present in sulphuric acid molecule ?

53. Copper crystallises in *fcc* with a unit length of 361pm. What is the radius of copper atom ?

54. To find the standard potential of M^{3+}/M electrode, the following cell is constituted : Pt/ $M/M^{3+}(0.001 \text{ mol L}^{-1})/Ag^+(0.01 \text{ mol L}^{-1})/Ag$

The emf of the cell is found to be 0.421 volt at 298 K calculate the standard potential of half reaction

$M^{3+} + 3e^- \longrightarrow M$ at 298 K

(Given $E^\circ_{Ag^+/Ag}$ at 298 K = 0.80 Volt)

55. The number of π electrons present in 6.4 g of calcium carbide is $x \times N_A$ where (N_A = Avagadro's number). Find the value of x.

56. Two elements A & B form compounds having molecular formulae AB_2 and AB_4. When dissolved in 20.0 g of benzene 1.00 g of AB_2 lowers f.p. by 2.3 °C whereas 1.00 g of AB_4 lowers f.p. by 1.3 °C. The molal depression constant for benzene in 1000 g is 5.1. Calculate the sum of atomic masses of A and B.

57. An element (atomic mass =100 g/mol) having *bcc* structure has unit cell edge 400 pm. Calculate the density (in g/cm^3) of the element.

58. A substance "A" decomposes in solution following the first order kinetics. Flask I contains 1 L of 1 M solution of A and flask II contains 100 mL of 0.6 M solution. After 8 hr, the concentration, of A in flask I becomes 0.25 M. What will be the time for concentration of A in flask II to become 0.3 M?

59. Concentration of NH_4Cl and NH_4OH in a buffer solution is in the ratio of 1 : 1, K_b for NH_4OH is 10^{-10}. Calculate the pH of the buffer.

60. At 5×10^5 bar pressure and density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value of $\Delta U - \Delta H$ for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T'.

MATHEMATICS

Section - A

61. Which of the following statements is false?
(1) The length of sub-tangent to the curve $x^2y^2 = 16a^4$ at the point (–2a, 2a) is 2a.
(2) $x + y = 3$ is a normal to the curve $x^2 = 4y$
(3) Curves $y = -4x^2$ and $y = e^{-x/2}$ are orthogonal.
(4) If $a \in (-1, 0)$, then tangent at each point of the curve $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$ makes an acute angle with the positive direction of x-axis.

62. If $\int \frac{(x^2-1)dx}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}$ $= \log|\tan^{-1} f(x)| + C$, then
(1) $f(x) = x^2+1$ (2) $f(x) = \frac{x^2+1}{2x}$
(3) $f(x) = \frac{x^2+1}{x}$ (4) $f(x) = \frac{1}{2}(x^2+1)$

63. If $P(\theta)$ and $Q\left(\frac{\pi}{2}+\theta\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then locus of the mid- point of PQ is
(1) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$ (2) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$
(3) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (4) None of these

64. Area included between $y = \frac{x^2}{4a}$ and $y = \frac{8a^3}{x^2+4a^2}$ is
(1) $\frac{a^2}{3}(6\pi-4)$ (2) $\frac{a^2}{3}(4\pi+3)$
(3) $\frac{a^2}{3}(8\pi+3)$ (4) None of these

65. Given that $\vec{a}$ is $\perp$ to $\vec{b}$ and p is a non zero scalar if $p\vec{r}+(\vec{r}.\vec{b})\vec{a}=\vec{c}$ then $\vec{r}$ equals
(1) $\vec{c}/p - \left[(\vec{b}.\vec{c})\vec{a}\right]/p^2$
(2) $\vec{a}/p - \left[(\vec{c}.\vec{a})\vec{b}\right]/p^2$
(3) $\vec{b}/p - \left[(\vec{a}.\vec{b})\vec{c}\right]/p^2$
(4) None of these

66. The domain of the function $f(x) = \sqrt{x-\sqrt{1-x^2}}$ is
(1) $\left[-1,-\frac{1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},1\right]$
(2) [–1, 1]
(3) $\left(-\infty,-\frac{1}{2}\right]\cup\left[\frac{1}{\sqrt{2}},+\infty\right)$
(4) $\left[\frac{1}{\sqrt{2}},1\right]$

67. Which of the following is an empty set?
(1) The set of prime numbers which are even
(2) The solution set of the equation
$$\frac{2(2x+3)}{x+1}-\frac{2}{x+1}+3=0;\ x\in \mathbf{R}$$
(3) $(A\times B)\cap(B\times A)$, where A and B are disjoint.
(4) The set of real which satisfy $x^2+ix+i-1=0$

68. $f(x) = x^2[x]$
(1) increases in (0,1) (2) decreases in (0 , 1)
(3) increases in (–1,0) (4) None of these

69. If $y = e^x + \sin x$, then d^2x/dy^2 is equal to
(1) $e^x - \sin x$
(2) $-(e^x + \cos x)^{-2}$
(3) $-(e^x - \sin x)\,(e^x + \cos x)^{-2}$
(4) $(\sin x - e^x)\,(\cos x + e^x)^{-3}$

70. The sum to n terms of the series
$\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \ldots\ldots\ldots\ldots$ is
(1) $n - 1 - 2^{-n}$ (2) 1
(3) $n - 1 + 2^{-n}$ (4) $1 + 2^{-n}$

71. If $f : R \to R$ be a function defined by $f(x) = 4x^3 - 7$. Then
(1) f is one-one -into (2) f is many-one - into
(3) f is many-one onto (4) f is bijective

72. Let $f : \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right] \to [0, 4]$ be a function defined as
$f(x) = \sqrt{3}\sin x - \cos x + 2$. Then $f^{-1}(x)$ is given by
(1) $\sin^{-1}\left(\frac{x-2}{2}\right) - \frac{\pi}{6}$
(2) $\sin^{-1}\left(\frac{x-2}{2}\right) + \frac{\pi}{6}$
(3) $\frac{2\pi}{3} + \cos^{-1}\left(\frac{x-2}{2}\right)$
(4) None of these

73. The roots α and β of the quadratic equation $px^2 + qx + r = 0$ are real and of opposite signs. The roots of $\alpha(x-\beta)^2 + \beta(x-\alpha)^2 = 0$ are
(1) positive (2) negative
(3) of opposite signs (4) non-real

74. The equation $k\sin\theta + \cos 2\theta = 2k - 7$ possesses a solution if :
(1) $2 \le k \le 6$ (2) $k > 2$
(3) $k > 6$ (4) $k < 2$

75. If $y = \sin^{-1}\frac{2x}{1+x^2}$, then which of the following is not correct?
(1) $\frac{dy}{dx} = \frac{2}{1+x^2}$ for $|x| < 1$
(2) $\frac{dy}{dx} = \frac{-2}{1+x^2}$ for $|x| > 1$
(3) $\frac{dy}{dx} = 2$ for $x = -1$
(4) $\frac{dy}{dx}$ does not exist at $|x| = 1$

76. If the slope of the tangent at (x, y) to a curve passing through $\left(1, \frac{\pi}{4}\right)$ is given by $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, then the equation of the curve is
(1) $y = \tan^{-1}\log(e/x)$
(2) $y = e^{1+\cot(y/x)}$
(3) $y = x\tan^{-1}\log(e/x)$
(4) $y = e^{1+\tan(y/x)}$

77. If $y = \tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$, then $\frac{dy}{dx}$ at $x = 0$ is :
(1) $-\frac{3}{5}\log 2$ (2) $\frac{2}{5}\log 2$
(3) $-\frac{3}{2}\log 2$ (4) None of these

78. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and the vectors $\vec{A} = (1, a, a^2)$; $\vec{B} = (1, b, b^2)$; $\vec{C} = (1, c, c^2)$ are non-coplanar then the product abc =

(1) 0 (2) 1
(3) –1 (4) None

79. A and B are two independent witnesses (i.e. there is no collusion between them) in a case. The probability that A will speak the truth is x and the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true is

(1) $\frac{x-y}{x+y}$ (2) $\frac{xy}{1+x+y+xy}$

(3) $\frac{x-y}{1-x-y+2xy}$ (4) $\frac{xy}{1-x-y+2xy}$

80. Let x_1 and y_1 be real numbers. If z_1 and z_2 are complex numbers such that $|z_1| = |z_2| = 4$, then $|x_1z_1 - y_1z_2|^2 + |y_1z_1 + x_1z_2|^2 =$

(1) $32(x_1^2 + y_1^2)$ (2) $16(x_1^2 + y_1^2)$

(3) $4(x_1^2 + y_1^2)$ (4) $8\left(x_1^2 + y_1^2\right)$

Section - B

81. Let f be a composite function of x defined by $f(u) = \frac{1}{u^2 + u - 2}, u(x) = \frac{1}{x-1}$.
Then the number of points x where f is discontinuous is ________.

82. If the function $f(x) = Pe^{2x} + Qe^{x} + Rx$ satisfies the conditions $f(0) = -1, f'(\log 2) = 31$ and $\int_0^{\log 4} (f(x) - Rx)dx = \frac{39}{2}$, then valve of P + Q + R is ______.

83. The sum of 10 items is 12 and sum of their squares is 18, then standard deviation will be ________.

84. If the system of equations :
$p^3x + (p+1)^3 y = (p+2)^3$
$px + (p+1) y = p+2$
$x + y = 1$
is consistent then valve of |P| is ______.

85. If $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$, then $\lim\limits_{\alpha \to 0} \frac{f(1-\alpha) - f(1)}{\alpha^3 + 3\alpha}$ is ________.

86. A survey of 500 television viewers produced the following information, 285 watch football, 195 watch hockey, 115 watch basket-ball, 45 watch football and basket ball, 70 watch football and hockey, 50 watch hockey and basket ball, 50 do not watch any of the three games. The number of viewers, who watch exactly one of the three games are ________.

87. The value of $|x|$ for which $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1} x)$ is _______.

88. The value of $\lim\limits_{x \to 0} \frac{1-\cos^3 x}{x \sin x \cos x}$ is _______.

89. If $f(x) = \frac{1}{1-x}$, the number of points of discontinuity of $f\{f[f(x)]\}$ is ________.

90. The number of roots of equation $\cos x + \cos 2x + \cos 3x = 0$ is $(0 \le x \le 2\pi)$ ______.

Mock Test 6

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

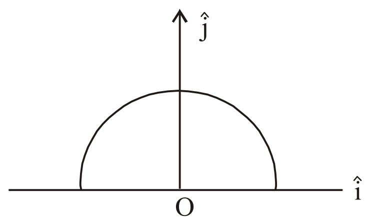
Section - A

1. A 1 kg block attached to a spring vibrates with a frequency of 1 Hz on a frictionless horizontal table. Two springs identical to the original spring are attached in parallel to an 8 kg block placed on the same table. The frequency of vibration of the 8 kg block is :

(1) $\frac{1}{4}$Hz (2) $\frac{1}{2\sqrt{2}}$Hz

(3) $\frac{1}{2}$Hz (4) $\sqrt{2}$Hz

2. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is



(1) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$ (2) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

(3) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$ (4) $\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

3. If a bar magnet of magnetic moment 80 units be cut into two halves of equal lengths, the magnetic moment of each half will be

(1) 80 units (2) 40 units

(3) 160 units (4) 20 units

4. Hydrogen $\left({}_1H^1\right)$, Deuterium $\left({}_1H^2\right)$, singly ionised Helium $\left({}_2He^4\right)^+$ and doubly ionised lithium $\left({}_3Li^6\right)^{++}$ all have one electron around the nucleus. Consider an electron transition from n = 2 to n = 1. If the wavelengths of emitted radiation are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 respectively then approximately which one of the following is correct?

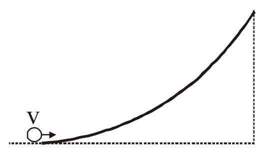
(1) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(2) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(3) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

(4) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

5. A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is



(1) Ring

(2) Solid sphere

(3) Hollow sphere

(4) Disc

6. The amplitude of velocity of a particle acted on by a force F cos ωt along the x direction, is given by $x = \frac{1}{\sqrt{[a\omega^2 - b\omega + c]}}$ where a, b, c, are constants and $b^2 < 4$ ac. For what value of ω does the resonance occur ?

(1) $\omega = \frac{b}{2a}$ (2) $\omega = \frac{b}{a}$

(3) $\omega = c$ (4) $\omega = 0$

7. Three bars each of area of cross - section A and length L are connected in series. The thermal conductivities of their materials are K, 2K, 1.5K. If the temperatures of the external ends of the first and last bar are 200 & 18ºC, then the temperatures of both the junctions are

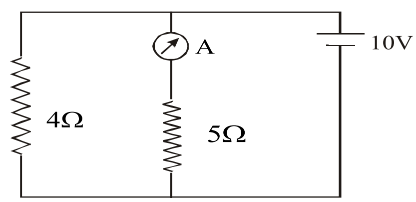
(1) $T_1 = 116$ºC, $T_2 = 74$ºC

(2) $T_1 = 120$ºC, $T_2 = 84$ºC

(3) $T_1 = 132$ºC, $T_2 = 98$ºC

(4) $T_1 = 164$ºC, $T_2 = 62$ºC

8. In the following diagram the reading of the ammeter is (when the internal resistance of the battery is zero)

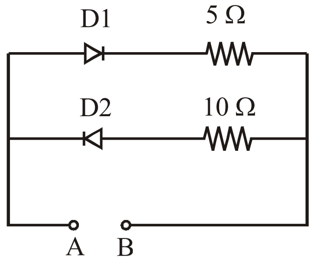


(1) $\frac{40}{29}$ A (2) $\frac{10}{9}$ A

(3) $\frac{5}{3}$ A (4) 2 A

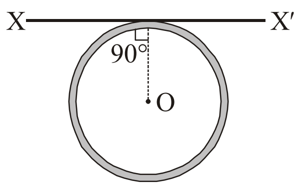
Space for Rough Work

9. A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be:



(1) 0.4 A and 0.2 A (2) 0.2 A and 0.4 A
(3) 0.1 A and 0.2 A (4) 0.2 A and 0.1 A

10. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is



(1) $\frac{\rho L^3}{8\pi^2}$

(2) $\frac{\rho L^3}{16\pi^2}$

(3) $\frac{5\rho L^3}{16\pi^2}$

(4) $\frac{3\rho L^3}{8\pi^2}$

11. In a uniform magnetic field of induction B a wire in the form of a semicircle of radius r rotates about the diameter of the circle with an angular frequency ω. The axis of rotation is perpendicular to the field. If the total resistance of the circuit is R, the mean power generated per period of rotation is

(1) $\frac{(B\pi r\omega)^2}{2R}$ (2) $\frac{(B\pi r^2\omega)^2}{8R}$

(3) $\frac{B\pi r^2\omega}{2R}$ (4) $\frac{(B\pi r\omega^2)^2}{8R}$

12. A steel wire is suspended vertically from a rigid support. When loaded with a weight in air, it extends by l_a and when the weight is immersed completely in water, the extension is reduced to l_w. Then the relatively density of material of the weight is

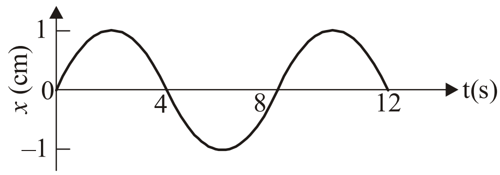
(1) $\frac{l_a}{l_w}$ (2) $\frac{l_a}{l_a - l_w}$

(3) $\frac{l_w}{l_a - l_w}$ (4) $\frac{l_w}{l_a}$

13. An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8 \times 10^{-12}$ C^2/N-m^2, then average energy density of electric field will be:

(1) 35.2×10^{-10} J/m^3
(2) 35.2×10^{-11} J/m^3
(3) 35.2×10^{-12} J/m^3
(4) 35.2×10^{-13} J/m^3

14. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4/3$ s is



(1) $\frac{\sqrt{3}}{32}\pi^2$ cm/s^2 (2) $\frac{-\pi^2}{32}$ cm/s^2

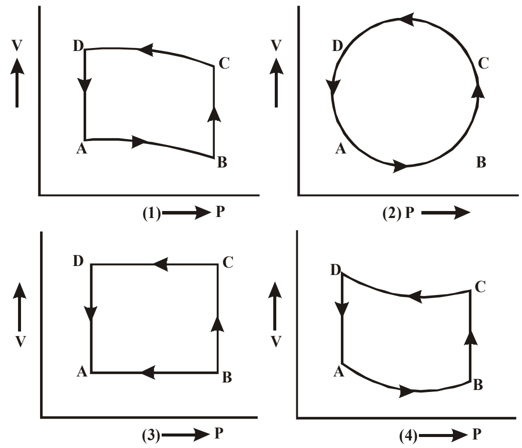
(3) $\frac{\pi^2}{32}$ cm/s^2 (4) $-\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

15. In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that D >> d >> λ. If the fringe width is β, the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is:

(1) $\frac{\beta}{6}$ (2) $\frac{\beta}{3}$

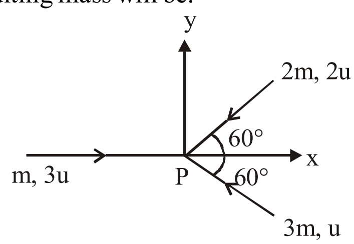
(3) $\frac{\beta}{4}$ (4) $\frac{\beta}{2}$

16. In diagrams (1 to 4), variation of volume with changing pressure is shown. A gas is taken along the path ABCD. The change in internal energy of the gas will be



(1) Positive in all the cases (1) to (4)
(2) Positive in cases (1), (2), (3) but zero in case(4)
(3) Negative in cases (1), (2), (3) but zero in case (4)
(4) Zero in all the cases

17. Three masses m, 2m and 3m are moving in x-y plane with speed 3u, 2u and u respectively as shown in figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be:



(1) $\frac{u}{12}\left(\hat{i}+\sqrt{3}\hat{j}\right)$ (2) $\frac{u}{12}\left(\hat{i}-\sqrt{3}\hat{j}\right)$

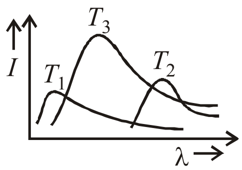
(3) $\frac{u}{12}\left(-\hat{i}+\sqrt{3}\hat{j}\right)$ (4) $\frac{u}{12}\left(-\hat{i}-\sqrt{3}\hat{j}\right)$

18. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is

(1) $\frac{gR^2}{R+x}$ (2) $\frac{gR}{R-x}$

(3) gx (4) $\left(\frac{gR^2}{R+x}\right)^{1/2}$

19. The plots of intensity versus wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperature are such that



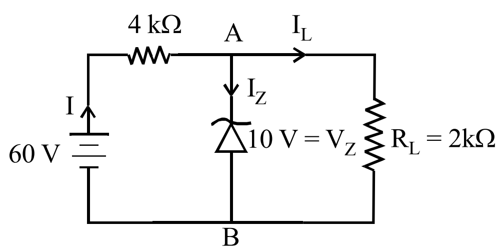
(1) $T_1 > T_2 > T_3$

(2) $T_1 > T_3 > T_2$

(3) $T_2 > T_3 > T_1$

(4) $T_3 > T_2 > T_1$

20. A Zener diode is connected to a battery and a load as shown below:



The currents, I, I_Z and I_L are respectively.

(1) 15 mA, 5 mA, 10 mA

(2) 15 mA, 7.5 mA, 7.5 mA

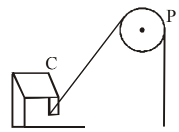
(3) 12.5 mA, 5 mA, 7.5 mA

(4) 12.5 mA, 7.5 mA, 5 mA

Section - B

21. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μT. What will be its value at the centre of loop (in μT)?

22. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N. With what value of maximum safe acceleration (in ms^{-2}) can a man of 60 kg climb on the rope?



23. The velocity of projection of a body is increased by 2%, other factors remaining unchanged, what will be the percentage change in the maximum height attained?

24. The insulation property of air breaks down at $E = 3 \times 10^6$ volts/ metre. The maximum charge that can be given to a sphere of diameter 5 m is approximately______ coulomb.

25. A compound microscope has an objective and eye-piece as thin lenses of focal lengths 1 cm and 5 cm respectively. The distance between the objective and the eye-piece is 20 cm. The distance at which the object must be placed in front of the objective if the final image is located at 25 cm from the eye-piece, is numerically ______ cm.

26. When light of 3000 Å is incident on sodium chloride, the stopping potential is 1.85 volt and when light of 4000 Å is incident, then the stopping potential becomes 0.82 volt. The threshold wavelength for sodium is ______ Å. (h = 6.6 × 10^{-34} J-sec.)

27. In four complete revolution of the cap, the distance travelled on the pitch scale is 2 mm. If there are 50 divisions on the circular scale, then the least count of the screw gauge is _____ mm.

28. An electric charge $10^{-3}\ \mu$ C is placed at the origin (0, 0) of $X-Y$ co-ordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and (2, 0) respectively. The potential difference between the points A and B will be ______ volts.

29. The Thallium-201 half-life is 74 hours. If the sample has an activity of 80 millicuries initially, what will be the activity (in mci) after 9.25 days ?

30. An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km, it blows a whistle whose echo is heard by the driver after 5 seconds. If the speed of sound in air is 330 m/s, then the speed of the engine (in m/s) is _____ .

CHEMISTRY

Section - A

31. Four species are listed below:

i. HCO_3^- ii. H_3O^+

iii. HSO_4^- iv. HSO_3F

Which one of the following is the correct sequence of their acid strength?

(1) iv < ii < iii < i (2) ii < iii < i < iv

(3) i < iii < ii < iv (4) iii < i < iv < ii

32. Which one of the following pairs is mismatched

(1) Fossil fuel burning - release of CO_2

(2) Nuclear power - radioactive wastes

(3) Solar energy - Greenhouse effect

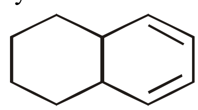
(4) Biomass burning - release of CO_2

33. Which of the following undergoes hydrolysis by S_N1 mechanism :

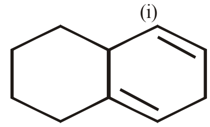
(1) $CH_3CH_2CH_2CH_2Cl$ (2) $CH_3CH_2CH_2Cl$

(3) CH_3-O-CH_2Cl (4) CH_3Cl

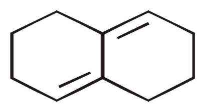
34. Which of the following hydrocarbon can react with maleic anhydride ?



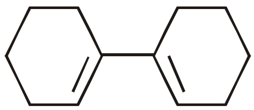
(i)



(ii)



(iii)



(iv)

(1) only (i) (2) (i) and (iv)
(3) (i), (iii) and (iv) (4) all the four

35. The process of the extraction of Au and Ag is based on their solubility in
(1) NH_3 or NH_4OH (2) KCN or NaCN
(3) HCl or HNO_3 (4) H_2SO_4

36. Which of the following compounds having highest and lowest melting point respectively ?
(1) CsF (2) LiF
(3) HCl (4) HF
Correct answer is
(1) 2, 3 (2) 1, 4
(3) 4, 3 (4) 2, 1

37. Silver bromide when dissolve in hypo solution gives complex ..... in which oxidation state of silver is ....
(1) $Na_3[Ag(S_2O_3)_2]$, (I)
(2) $Na_3[Ag(S_2O_3)_3]$, (III)
(3) $Na_3[Ag(S_2O_3)_2]$, (II)
(4) $Na_3[Ag(S_2O_3)_4]$, (I)

38. An organic compound (A) reacts with sodium metal and forms (B). On heating with conc. H_2SO_4, (A) gives diethyl ether, (A) and (B) are
(1) C_2H_5OH and C_2H_5ONa
(2) C_3H_7OH and CH_3ONa
(3) CH_3OH and CH_3ONa
(4) C_4H_9OH and C_4H_9ONa

39. Which one of the following esters cannot undergo Claisen self-condensation?
(1) $CH_3-CH_2-CH_2-CH_2-COOC_2H_5$
(2) $C_6H_5COOC_2H_5$
(3) $C_6H_5CH_2COOC_2H_5$
(4) $C_6H_{11}CH_2COOC_2H_5$

40. In recovery of silver from photographic film, you have decided to dissolve the silver ion with dilute nitric acid. Addition of dilute HCl to precipitate AgCl seems to result in unacceptable losses. You might improve recovery by addition of_______ in the latter step.
(1) $NaNO_3$ (2) NaCl
(3) Ag_2SO_4 (4) sodium acetate

41. One mole of fluorine is reacted with two moles of hot and concentrated KOH. The products formed are KF, H_2O and O_2. The molar ratio of KF, H_2O and O_2 respectively is
(1) 1 : 1 : 2 (2) 2 : 1 : 0.5
(3) 1 : 2 : 1 (4) 2 : 1 : 2

42. In the reaction,

$$CH_2=CH_2 \xrightarrow{HOCl} A \xrightarrow{B} \begin{array}{l} CH_2OH \\ | \\ CH_2OH \end{array}$$

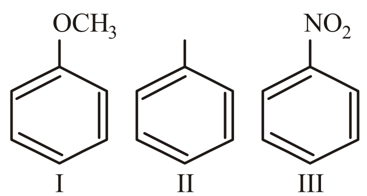
The molecule 'A' and the reagent 'B' are
(1) H_2C-CH_2 (bridged by O) and hot water
(2) CH_3CH_2Cl and NaOH
(3) CH_3CH_2OH and H_2SO_4
(4) $\begin{array}{cc} CH_2 - & CH_2 \\ | & | \\ Cl & OH \end{array}$ and $NaHCO_3$

Space for Rough Work

43. Alkali metals dissolve in liquid NH_3 then which of the following observations is not true?
(1) H_2 gas is liberated.
(2) Solution turns into blue due to solvated electrons.
(3) It becomes diamagnetic.
(4) Solution becomes conducting.

44. At very high pressures, the compressibility factor of one mole of a gas is given by :
(1) $1+\frac{Pb}{RT}$ (2) $\frac{Pb}{RT}$
(3) $1-\frac{Pb}{RT}$ (4) $1-\frac{b}{(VRT)}$

45. Among the following compounds (I-III) the correct order of reactivity with electrophile is



(1) II > III > I (2) III < I < II
(3) I > II > III (4) I = II > III

46. One desires to prepare a positively charged sol of silver iodide. This can be achieved by
(1) adding a little $AgNO_3$ solution to KI solution in slight excess.
(2) adding a little KI solution to $AgNO_3$ solution in slight excess.
(3) mixing equal volumes of equimolar solutions of $AgNO_3$ and KI.
(4) none of these.

47. Which of the following is an artificial edible colour?
(1) Melamine (2) Saffron
(3) Carotene (4) Tetrazine

48. Lassaigne's test for nitrogen is positive for which compound?
(1) NH_2OH (2) NH_2NH_2
(3) H_2NCONH_2 (4) All the three

49. It is always advisable not to keep egg yolk or mustard in silver cutlery because
(1) silver reacts with water of egg yolk to form AgOH.
(2) silver reacts with sulphur of egg yolk forming black Ag_2S.
(3) silver reacts with egg yolk forming Ag_2SO_4 which is a poisonous substance.
(4) silver attracts UV light of the atmosphere, thereby spoiling the food.

50. Which of the following complexes will give white precipitate with $BaCl_2$ (aq)?
(1) $[Co(NH_3)_4SO_4]NO_2$
(2) $[Cr(NH_3)_4SO_4]Cl$
(3) $[Cr(NH_3)_5Cl]SO_4$
(4) Both (2) & (3)

Space for Rough Work

Section - B

51. Calculate the molality of 1 litre solution of 93% H_2SO_4 (weight/volume). The density of the solution is 1.84 g/mL.
52. Find out the number of stereoisomers obtained by bromination of *trans*-2-butene.
53. The enthalpy of combustion of $H_2(g)$, to give $H_2O(g)$ is –249 kJ mol^{-1} and bond enthalpies of H – H and O = O are 433 kJ mol^{-1} and 492 kJ mol^{-1} respectively. The bond enthalpy of O – H is
54. A certain metal when irradiated to light ($v = 3.2\times 10^{16}$ Hz) emits photoelectrons with twice kinetic energy as did photoelectrons when the same metal is irradiated by light ($v = 2.0 \times 10^{16}$ Hz) The v_0 (threshold frequency) of metal is $x \times 10^{15}$. Find the value of x.
55. Malachite has the formula $Cu_2CO_3(OH)_2$. What percentage by mass of malachite is copper?
56. The freezing point of a solution, prepared from 1.25 g of a non-electrolyte and 20 g of water, is 271.9 K. If molar depression constant is 1.86 K mole^{-1}, find the molar mass of the solute
57. A solution of Ni $(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?
58. In O_2^-, O_2 and O_2^{2-} molecular species. Find the total number of antibonding electrons.
59. x × 10^{21} unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00 g. Find the value of x.
[Atomic masses: Na = 23, Cl = 35.5]
60. The electrode potential of Mg^{2+}/Mg electrode, in which conc. of Mg^{2+} is 0.01M, is
$(E^{o}_{Mg^{2+}/Mg} = -2.36V)$

MATHEMATICS

Section - A

61. If $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$, then $\frac{d^2y}{dx^2}$ is equal to :

(1) y (2) $\sqrt{1+y^2}$

(3) $\frac{x}{\sqrt{1+y^2}}$ (4) y^2

62. The number of real roots of the equation
$1 + a_1 x + a_2 x^2 + \ldots + a_n x^n = 0$
where $|x| < \frac{1}{3}$ and $|a_n| < 2$, is

(1) n if n is even
(2) 0 for any natural number n
(3) 1 if n is odd
(4) None of these.

Space for Rough Work

63. If $e^{-\frac{\pi}{2}} < \theta < \frac{\pi}{2}$, then

(1) $\cos \log \theta > \log \cos \theta$

(2) $\cos \log \theta < \log \cos \theta$

(3) $\cos \log \theta = \log \cos \theta$

(4) $\cos \log \theta = \frac{2}{3} \log \cos \theta$

64. The value of $\sum_{r=0}^{s} \sum_{\substack{s=1 \\ r \le s}}^{n} {}^nC_s \, {}^sC_r$ is

(1) $3^n - 1$ (2) $3^n + 1$

(3) 3^n (4) $3(3^n - 1)$

65. Number of ways to distribute 10 distinct balls to 3 persons. One getting 2, 2nd getting 3 and 3rd getting 5 balls is

(1) $\frac{10!}{2!.3!.5!}$ (2) $\frac{10!}{2!.(3!)^2.5!}$

(3) $\frac{10!}{2!.5!}$ (4) $10!$

66. The general solution of the differential equation, $\sin 2x \left(\frac{dy}{dx} - \sqrt{\tan x}\right) - y = 0$, is :

(1) $y\sqrt{\tan x} = x + c$

(2) $y\sqrt{\cot x} = \tan x + c$

(3) $y\sqrt{\tan x} = \cot x + c$

(4) $y\sqrt{\cot x} = x + c$

67. In any triangle ABC, if $\cos A = \frac{\sin B}{2 \sin C}$, then

(1) $a = b = c$ (2) $c = a$

(3) $a = b$ (4) $b = c$

68. If ω is a cube root of unity, then a root of the following equation $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$ is

(1) $x = 1$ (2) $x = \omega$

(3) $x = \omega^2$ (4) $x = 0$.

69. The lines $\ell x + my + n = 0$, $mx + ny + \ell = 0$ and $nx + \ell y + m = 0$ are concurrent if

(1) $\ell - m - n = 0$

(2) $\ell + m - n = 0$

(3) $\ell - m + n = 0$

(4) $\ell^2 + m^2 + n^2 = \ell m + mn + n\ell$

70. If (x, y) are the co-ordinates of a point in the plane, then $\begin{vmatrix} 3 & 4 & 2 \\ 5 & 8 & 2 \\ x & y & 2 \end{vmatrix} = 0$ represent

(1) a. st. line || to y-axis (2) a st. line || to x-axis

(3) a st. line (4) a circle

71. The equation of the circle whose radius is 5 and which touches the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at the point (5,5) is

(1) $x^2 + y^2 + 18x + 16y + 120 = 0$

(2) $x^2 + y^2 - 18x - 16y + 120 = 0$

(3) $x^2 + y^2 - 18x + 16y + 120 = 0$

(4) $x^2 + y^2 + 18x - 16y + 120 = 0$

72. The normal at $\left(2, \frac{3}{2}\right)$ to the ellipse, $\frac{x^2}{16} + \frac{y^2}{3} = 1$ touches a parabola, whose equation is

(1) $y^2 = -104x$ (2) $y^2 = 14x$
(3) $y^2 = 26x$ (4) $y^2 = -14x$

73. If p : I study and q : I fail

Then negation of 'I study or I fail' is

(1) I do not study and I do not fail
(2) I do not study or I do not fail
(3) Either I study and I do not rail or I study and I do not fail
(4) I study and I do not fail

74. The general solution of the trigonometric equation sin x – cos x = 1 is given by

(1) $x = 2n\pi, n \in I$
(2) $x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}, n \in I$
(3) $x = 2n\pi + \frac{\pi}{2}, n \in I$
(4) $x = n\frac{\pi}{2}, n \in I$

75. If $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{C} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are linearly independent vectors and $|\vec{C}| = \sqrt{3}$, then

(1) $\alpha = 1, \beta = -1$ (2) $\alpha = 1, \beta = \pm 1$
(3) $\alpha = -1, \beta = \pm 1$ (4) $\alpha = \pm 1, \beta = 1$

76. p : Every quadratic equation has one real root and q : Every quadratic equation has two real roots, then truth value of p and q are

(1) p is true and q is false
(2) p is false and q is true
(3) p and q both true
(4) p and q both false

77. Let A, B and C be $n \times n$ matrices. Which one of the following is a correct statement?

(1) If AB = AC, then B = C
(2) If $A^3 + 2A^2 + 3A + 5I = 0$; then A is invertible.
(3) If $A^2 = 0$, then A = 0
(4) None of these

78. If for a real number y, [y] is the greatest integer less than or equal to y, then the value of the integral $\int_{\pi/2}^{3\pi/2} [2\sin x]dx$ is

(1) $-\pi$ (2) π
(3) $\pi/2$ (4) $-\pi/2$

79. If $f(x) = ax^2 + b, b \neq 0, x \leq 1;$
$= bx^2 + ax + c, x > 1,$
then f(x) is continuous and differentiable at x = 1 if

(1) c = 0, a = 2b (2) a = b, c arbitrary
(3) a = b, c = 0 (4) a = b, c $\neq$ 0

Space for Rough Work

80. The values of the parameter a such that the roots α, β of the equation $2x^2 + 6x + a = 0$ satisfy the inequality $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < 2$ are
(1) $a > 0$ (2) $a < 9/2$
(3) $a < 0$ or $a > 9/2$ (4) None of these

Section - B

81. It has been found that A and B play a game 12 times, A wins 6 times, B wins 4 times and they draw twice. A and B take part in a series of 3 games. The probability that they will win alternately is ______ .

82. If period of the function $\left|\sin^3 \frac{x}{2}\right| + \left|\cos^5 \frac{x}{5}\right| = k\pi$, then value of x is ______ .

83. If the least difference between the roots, in the first quadrant $\left(0 \le x \le \frac{\pi}{2}\right)$, of the equation $4\cos x(2 - 3\sin^2 x) + (\cos 2x + 1) = 0$ is $\frac{\pi}{n}$, then value of n is ______ .

84. If [x] denotes the greatest integer $\le$ x and
$$\lim_{x\to 3} \frac{1}{n^3}([1^2 x] + [2^2 x] + [3^2 x] + ... + [n^2 x]) = \frac{\pi}{n},$$
then value of n is ______ .

85. For natural numbers m, n if $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \ldots$ and $a_1 = a_2 = 10$, then value of $m + n$ is ______ .

86. The value of expression
$$3\left[\sin^4\left(\frac{3\pi}{2} - \alpha\right) + \sin^4(3\pi + \alpha)\right]$$
$$-2\left[\sin^6\left(\frac{\pi}{2} + \alpha\right) + \sin^6(5\pi - \alpha)\right]$$ is ______ .

87. The value of $[\vec{A} - \vec{B}, \vec{B} - \vec{C}, \vec{C} - \vec{A}]$ where $|\vec{A}| = 1$, $|\vec{B}| = 2$ and $|\vec{C}| = 3$ is ______ .

88. If $f(x) = \frac{2}{x-3}$, $g(x) = \frac{x-3}{x+4}$
$h(x) = -\frac{2(2x+1)}{x^2 + x - 12}$ and
$\lim_{x \to 3} [f(x) + g(x) + h(x)] = k$ then value of $|k|$ is ______ .

89. If P and Q be two given points on the curve $y = x + \frac{1}{x}$ such that $\overrightarrow{OP}.\hat{i} = 1$ and $\overrightarrow{OQ}.\hat{i} = -1$ where $\hat{i}$ is a unit vector along the x–axis and the length of vector $2\overrightarrow{OP} + 3\overrightarrow{OQ} = n\sqrt{5}$, then value of n is ______ .

90. The area bounded by parabola $y^2 = x$, st. line $y = 4$ and y-axis is ______ .

Mock Test 7

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

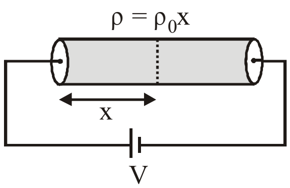
1. Using mass(M), length(L), time(T) and electric current (A) as fundamental quantities the dimensions of permittivity will be
   (1) $MLT^{-1}A^{-1}$ (2) $MLT^{-2}A^{-2}$
   (3) $M^{-1}L^{-3}T^{4}A^{2}$ (4) $M^{2}L^{-2}T^{-2}A^{2}$

2. $y = 2\text{ (cm) }\sin\left[\frac{\pi t}{2} + \phi\right]$
   What is the maximum acceleration of the particle executing the SHM?
   (1) $\frac{\pi}{2}$ cm/s^2 (2) $\frac{\pi^2}{2}$ cm/s^2
   (3) $\frac{\pi^2}{4}$ cm/s^2 (4) $\frac{\pi}{4}$ cm/s^2

3. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b respectively. When a current I passes through the coil, the magnetic field at the center is
   (1) $\frac{\mu_o NI}{b}$ (2) $\frac{2\mu_o NI}{a}$
   (3) $\frac{\mu_0 NI}{2(b-a)}\ln\frac{b}{a}$ (4) $\frac{\mu_0 IN}{2(b-a)}\ln\frac{a}{b}$

Space for Rough Work

4. A cylindrical solid of length L and radius a having varying resistivity given by $\rho = \rho_0 x$, where ρ_0 is a positive constant and x is measured from left end of solid. The cell shown in the figure having emf V and negligible internal resistance. The electric field as a function of x is best described by



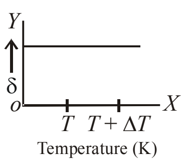
(1) $\frac{2V}{L^2}x$ (2) $\frac{2V}{\rho_0 L^2}x$

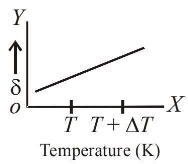
(3) $\frac{V}{L^2}x$ (4) None of these

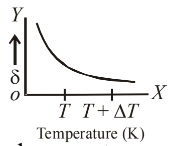
5. A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass m, if the string does not slip on the pulley, is:

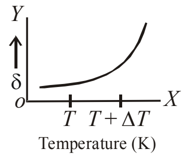
(1) g (2) $\frac{2}{3}g$ (3) $\frac{g}{3}$ (4) $\frac{3}{2}g$

6. An ideal gas is initially at temperature T and volume V. Its volume is increased by ΔV due to an increase in temperature ΔT, pressure remaining constant. The quantity $\delta = \frac{\Delta V}{V \Delta T}$ varies with temperature as

(1) 

(2) 

(3) 

(4) 

7. A Hydrogen atom and a Li^{++} ion are both in the second excited state. If ℓ_H and ℓ_{Li} are their respective electronic angular momenta, and E_H and E_{Li} their respective energies, then

(1) $\ell_H > \ell_{Li}$ and $|E_H| > |E_{Li}|$

(2) $\ell_H = \ell_{Li}$ and $|E_H| < |E_{Li}|$

(3) $\ell_H = \ell_{Li}$ and $|E_H| > |E_{Li}|$

(4) $\ell_H < \ell_{Li}$ and $|E_H| < |E_{Li}|$

8. The temperature of reservoir of Carnot's engine operating with an efficency of 70% is 1000 kelvin. The temperature of its sink is

(1) 300 K (2) 400 K

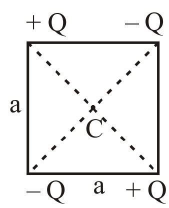
(3) 500 K (4) 700 K

9. A radar has a power of 1kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500 m. The maximum distance upto which it can detect object located on the surface of the earth (Radius of earth = 6.4×10^6m) is :

(1) 80 km (2) 16 km

(3) 40 km (4) 64 km

10. What is the electric potential at the centre of the square?

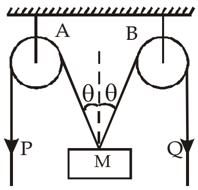


(1) zero (2) kq/a $\sqrt{2}$

(3) kq/ a^2 (4) None of these

Space for Rough Work

11. In the arrangement shown in the Fig, the ends *P* and *Q* of an unstretchable string move downwards with uniform speed *U*. Pulleys *A* and *B* are fixed. Mass *M* moves upwards with a speed



(1) $2U\cos\theta$

(2) $U/\cos\theta$

(3) $2U/\cos\theta$

(4) $U\cos\theta$

12. In Young's double slit experiment, the slits are illuminated by white light. The distance between two slits is d and screen is D distance apart from the slits. Some wavelengths are missing on the screen in front of one of the slits. These wavelengths are

(1) $\lambda = \frac{d^2}{D}$ (2) $\lambda = \frac{2d^2}{D}$

(3) $\lambda = \frac{d^2}{2D}$ (4) $\lambda = \frac{2d^2}{3D}$

13. A rectangular block of mass m and area of cross-section A floats in a liquid of density ρ. If it is given a small vertical displacement from equilibrium it undergoes oscillation with a time period T. Then

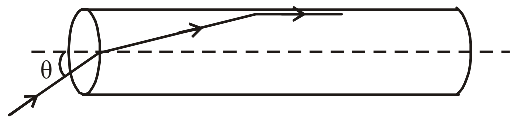
(1) $T \propto \frac{1}{\sqrt{A}}$ (2) $T \propto \frac{1}{\rho}$

(3) $T \propto \frac{1}{\sqrt{m}}$ (4) $T \propto \sqrt{\rho}$

14. There are two wires of the same length. The diameter of second wire is twice that of the first. On applying the same load to both the wires, the extension produced in them will be in ratio of

(1) 1 : 4 (2) 1 : 2 (3) 2 : 1 (4) 4 : 1

15. A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.

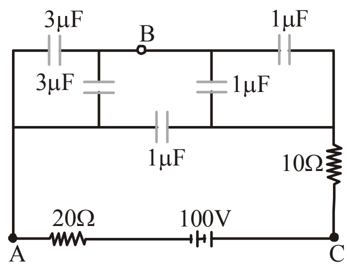


The incident angle θ for which the light ray grazes along the wall of the rod is

(1) $\sin^{-1}\left(\sqrt{3}/2\right)$ (2) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(1/2\right)$

16. In the figure below, what is the potential difference between the point A and B and between B and C respectively in steady state



(1) $V_{AB} = V_{BC} = 100\,V$

(2) $V_{AB} = 75, V_{BC} = 25\,V$

(3) $V_{AB} = 25V, V_{BC} = 75V$

(4) $V_{AB} = V_{BC} = 50V$

17. A block is placed on a frictionless horizontal table. The mass of the block is m and springs are attached on either side with force constants K_1 and K_2. If the block is displaced a little and left to oscillate, then the angular frequency of oscillation will be

(1) $\left(\frac{K_1+K_2}{m}\right)^{\frac{1}{2}}$ (2) $\left[\frac{K_1K_2}{m(K_1+K_2)}\right]^{\frac{1}{2}}$

(3) $\left[\frac{K_1K_2}{(K_1-K_2)m}\right]^{\frac{1}{2}}$ (4) $\left[\frac{K_1^2+K_2^2}{(K_1+K_2)m}\right]^{\frac{1}{2}}$

18. A bullet looses $\left(\frac{1}{n}\right)^{th}$ of its velocity passing through one plank. The number of such planks that are required to stop the bullet can be:

(1) $\frac{n^2}{2n-1}$ (2) $\frac{2n^2}{n-1}$
(3) infinite (4) n

19. An ideal coil of 10H is connected in series with a resistance of 5Ω and a battery of 5V. 2second after the connection is made, the current flowing in ampere in the circuit is
(1) $(1-e^{-1})$ (2) $(1-e)$
(3) e (4) e^{-1}

20. A spacecraft of mass 'M', moving with velocity 'v' suddenly breaks into two pieces. After the explosion mass 'm' becomes stationary. What is the velocity of the other part of the spacecraft?

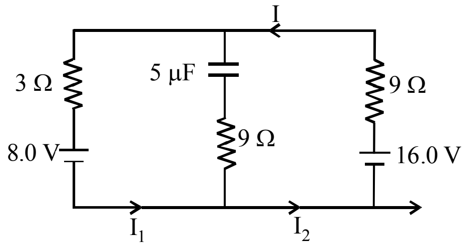
(1) $\frac{Mv}{M-m}$ (2) v

(3) $\frac{mv}{M}$ (4) $\frac{M-m}{m}v$

Section - B

21. In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is 220V and 50 Hz respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30°. On taking out the inductor from the circuit the current leads the voltage by 30°. The power dissipated in the LCR circuit is ______ watt.
22. In an isothermal expansion of 10g of gas from volume V to 2V the work done by the gas is 575J. What is the root mean square speed (in m/s) of the molecules of the gas at that temperature?
23. An automobile travelling with a speed of 60 km/h, can brake to stop within a distance of 20m. If the car is going twice as fast i.e., 120 km/h, the stopping distance (in metre) will be
24. At the centre of a circular current carrying coil of radius 5 cm magnetic field due to earth is 0.5×10^{-5} W/m². What should be the current (in A) flowing through the coil so that it annuls the earth's magnetic field?
25. An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m³. The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth (in metre) is the bubble below the surface of the liquid? ($g = 9.8\ ms^{-2}$)
26. The escape velocity for a body of mass 1 kg from the earth surface is 11.2 kms^{-1}. The escape velocity for a body of mass 100 kg would be _______ kms^{-1}.
27. The circuit shown here has two batteries of 8.0 V and 16.0 V and three resistors 3 Ω, 9 Ω and 9 Ω and a capacitor of 5.0 μF.

Space for Rough Work



How much is the current (I) (in A) in the circuit in steady state?

28. A beam of light of intensity 12 watt/cm^2 is incident on a totally reflecting plane mirror of area 1.5 cm^2, then the force (in newton) acting on the mirror will be

29. A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 kVm^{-1}. The density of liquid is 1.26×10^3 $kg\,m^{-3}$. The radius of the drop is ______ metre. (neglect buoyancy).

30. Two waves of wavelengths 99 cm and 100 cm both travelling with velocity 396 m/s are made to interfere. The number of beats produced per second is

CHEMISTRY

Section - A

31. Ethylene dichloride and ethylidine chloride are isomeric compounds. The false statement about these isomers is that they
    (1) react with alcoholic potash and give the same product.
    (2) are position isomers.
    (3) contain the same percentage of chlorine.
    (4) are both hydrolysed to the same product.

32. By what ratio the average velocity of the molecule in gas changes when the temperature is raised from 50 to 200 ºC ?
    (1) $\frac{1.21}{1}$ (2) $\frac{1.46}{1}$
    (3) $\frac{2}{1}$ (4) $\frac{4}{1}$

33. Which of the following would not give 2-phenylbutane as the major product in a Friedel-Crafts alkylation reaction ?
    (1) 1-butene + HF
    (2) 2-butanol + H_2SO_4
    (3) Butanoyl chloride + $AlCl_3$ then Zn, HCl
    (4) Butyl chloride + $AlCl_3$

34. Moissan boron is
    (1) amorphous boron of ultra purity
    (2) crystalline boron of ultra purity
    (3) amorphous boron of low purity
    (4) crystalline boron of low purity

35. Which reaction characteristics are changed by the addition of a catalyst to a reaction at constant temperature ?
    (i) Activation energy
    (ii) Equilibrium constant
    (iii) Reaction enthalpy
    (1) (i) only (2) (iii) only
    (3) (i) and (ii) only (4) all of these

36. A metal M reacts with N_2 to give a compound 'A' (M_3N). 'A' on heating at high temperature gives back 'M' and 'A' on reacting with H_2O gives a gas 'B'. 'B' turns $CuSO_4$ solution blue on passing through it. M and B can be :
(1) Al & NH_3 (2) Li & NH_3
(3) Na & NH_3 (4) Mg & NH_3

37. Asthma patient use a mixture of .... for respiration.
(1) O_2 and CO_2 (2) O_2 and He
(3) O_2 and NH_3 (4) O_2 and CO

38. When ethanal reacts with CH_3MgBr and C_2H_5OH/dry HCl, the product formed are :
(1) ethyl alcohol and 2-propanol
(2) ethane and hemi acetal
(3) 2-propanol and acetal
(4) propane and methyl acetate

39. Amongst the following salts of iron, which is most unstable in aqueous solutions?
(1) $K_3[Fe(CN)_6]$ (2) $Fe_2(SO_4).9H_2O$
(3) $FeSO_4.7H_2O$ (4) FeI_3

40. Given the molecular formula of the hexa-coordinated complexes (i) $CoCl_3.6NH_3$, (ii) $CoCl_3.5NH_3$, (iii) $CoCl_3.4NH_3$
If the number of co-ordinated NH_3 molecules in i, ii and iii respectively are 6, 5, 4, the primary valencies in (i), (ii) and (iii) are :
(1) 6, 5, 4 (2) 3, 2, 1
(3) 0, 1, 2 (4) 3, 3, 3

41. Polyethylene is
(1) Random copolymer
(2) Homopolymer
(3) Alternate copolymer
(4) Crosslinked copolymer

42. The reason for "drug induced poisoning" is :
(1) Binding reversibly at the active site of the enzyme
(2) Bringing conformational change in the binding site of enzyme
(3) Binding irreversibly to the active site of the enzyme
(4) Binding at the allosteric sites of the enzyme

43. With a change in hybridization of the carbon bearing the charge, the stability of a carbanion increases in the order :
(1) $sp < sp^2 < sp^3$ (2) $sp < sp^3 < sp^2$
(3) $sp^3 < sp^2 < sp$ (4) $sp^2 < sp < sp^3$

44. Which of the following compounds are not arranged in order of decreasing reactivity towards electrophilic substitution?
(1) Methoxybenzene > Toluene > Bromobenzene
(2) Phenol > N-Propylbenzene > Benzoic acid
(3) Chlorotoluene > Para Nitro toluene > 2-chloro-4-Nitrotoluene
(4) Benzoic acid > Phenol > N-Propylbenzene

45. Which of the given sets of temperature and pressure will cause a gas to exhibit the greatest deviation from ideal gas behaviour ?
(1) 100 ºC & 4 atm (2) 100 ºC & 2 atm
(3) –100 ºC & 4 atm (4) 0 ºC & 2 atm

46. Which is not true for beryllium ?
(1) Beryllium is amphoteric.
(2) It forms unusual carbide, Be_2C.
(3) $Be(OH)_2$ is basic.
(4) Beryllium halides are electron deficient.

47. Which of the following pairs show reverse properties on moving along a period from left to right and from up to down in a group :
(1) Nuclear charge and electron affinity
(2) Ionization energy and electron affinity
(3) Atomic radius and electron affinity
(4) None of these

Space for Rough Work

48. An organic amino compound reacts with aqueous nitrous acid at low temperature to produce an oily nitrosoamine. The compound is:
(1) CH_3NH_2 (2) $CH_3CH_2NH_2$
(3) $CH_3CH_2NH.CH_2CH_3$ (4) $(CH_3CH_2)_3N$

49. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3. The anion could be
(1) CO_3^{2-} (2) Cl^-
(3) SO_4^{2-} (4) S^{2-}

50. Energy of activation is lowest for which reaction?
(1) $RCH_2\overset{+}{O}H_2 \rightarrow R\overset{+}{C}H_2$
(2) $R_2CH\overset{+}{O}H_2 \rightarrow R_2\overset{+}{C}H$
(3) $R_3C\overset{+}{O}H_2 \rightarrow R_3C^+$
(4) All have same E_{act}

Section - B

51. The pK_a of HCOOH is 3.8 and pK_b of NH_3 is 4.8, find the pH of aqueous solution of 1M $HCOONH_4$.

52. The number of electrons passing per second through a cross-section of copper wire carrying 10^{-6} amperes of current is found to be $x \times 10^2$. Find the value of x.

53. Number H-atoms present in 0.046 g of ethanol is $x \times 10^{21}$. Find the value of x.

54. Calculate the radius (in pm) of the largest sphere which fits properly at the centre of the edge of a body centred cubic unit cell. (Given edge length is 100 pm)

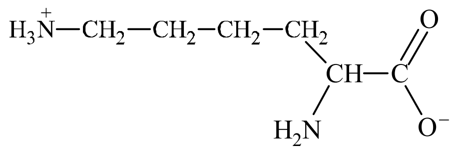
55. What is the molarity of H_2SO_4 solution if 25mL is exactly neutralized with 32.63 mL of 0.164 M, NaOH ?

56. 1.500g of hydroxide of a metal gave 1.000g of its oxide on heating. What is the equivalent mass of the metal ?

57. The enthalpy change on freezing of 1 mole of water at 5 °C to ice at –5 °C is :
[Given $\Delta_{fus}H = 6$ kJ mol^{-1} at 0 °C,
$C_p(H_2O, l) = 75.3$ J mol^{-1} K^{-1},
$C_p(H_2O, s) = 36.8$ J mol^{-1} K^{-1}]

58. The ground state energy of hydrogen atom is –13.6 eV. Calculate the energy of second excited state of He^+ ion in eV.

59. The total number of basic groups in the following form of lysine is



60. What quantity (in mL) of a 45% acid solution of a mono-protic strong acid must be mixed with a 20% solution of the same acid to produce 800 mL of a 29.875% acid solution?

MATHEMATICS

Section - A

61. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in
(1) $\left(0, \frac{\pi}{2}\right)$ (2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(3) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (4) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

62. A tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ meets x-axis at P and y-axis at Q. Lines PR and QR are drawn such that OPRQ is a rectangle (where O is the origin). Then R lies on :
(1) $\frac{4}{x^2} + \frac{2}{y^2} = 1$ (2) $\frac{2}{x^2} - \frac{4}{y^2} = 1$
(3) $\frac{2}{x^2} + \frac{4}{y^2} = 1$ (4) $\frac{4}{x^2} - \frac{2}{y^2} = 1$

63. With the usual notation $\int_1^2 \left([x^2] - [x]^2\right) dx$ is equal to
(1) $4 + \sqrt{2} - \sqrt{3}$ (2) $4 - \sqrt{2} + \sqrt{3}$
(3) $4 - \sqrt{2} - \sqrt{3}$ (4) None of these

64. The standard deviation of a variate x is σ. The standard deviation of the variable $\frac{ax+b}{c}$; a, b, c are constants, is
(1) $\left(\frac{a}{c}\right)\sigma$ (2) $\left|\frac{a}{c}\right|\sigma$
(3) $\left(\frac{a^2}{c^2}\right)\sigma$ (4) None

65. Assume R and S are (non-empty) relations in a set A. Which of the following relation given below is false
(1) If R and S are transitive, then $R \cup S$ is transitive.
(2) If R and S are transitive, then $R \cap S$ is transitive.
(3) If R and S are symmetric, then $R \cup S$ is symmetric.
(4) If R and S are reflexive, Then $R \cap S$ is reflexive.

66. Given that f and g are continuous functions on [0 a] satisfying $f(a-x) = f(x)$ and $g(x)+g(a-x) = 2$. Then $\int_0^a f(x)g(x)dx$ is equal to :
(1) $\int_0^a f(x)dx$ (2) $\int_0^a g(x)dx$
(3) 0 (4) None of these

67. If $c_1 = y = \frac{1}{1+x^2}$ and $c_2 = y = \frac{x^2}{2}$ be two curves lying in XY-plane, then
(1) area bounded by curve $y = \frac{1}{1+x^2}$ and $y = 0$ is $\frac{\pi}{2}$
(2) area bounded by c_1 and c_2 is $\frac{\pi}{2} - 1$
(3) area bounded by c_1 and c_2 is $1 - \frac{\pi}{2}$
(4) area bounded by curve $y = \frac{1}{1+x^2}$ and x-axis is $\frac{\pi}{2}$

Space for Rough Work

68. Let f and g be functions from the interval $[0, \infty)$ to the interval $[0, \infty)$, f being an increasing and g being a decreasing function. If $f\{g(0)\} = 0$ then
(1) $f\{g(x)\} \geq f\{g(0)\}$
(2) $g\{f(x)\} \leq g\{f(0)\}$
(3) $f\{g(2)\} = 7$
(4) None of these

69. $\frac{dy}{dx} + y = 2e^{2x}$ then y is
(1) $ce^{-x} + \frac{2}{3}e^{2x}$
(2) $(1+x)e^{-x} + \frac{2}{3}e^{2x} + c$
(3) $ce^{-x} + \frac{2}{3}e^{2x} + c$
(4) $e^{-x} + \frac{2}{3}e^{2x} + c$

70. If $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$ then x is
(1) 0, 2a (2) a, 2a
(3) 0, 3a (4) None of these

71. The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length 3a is
(1) $x^2 + y^2 = 9a^2$ (2) $x^2 + y^2 = 16a^2$
(3) $x^2 + y^2 = 4a^2$ (4) $x^2 + y^2 = a^2$

72. If a positive integer n is divisible by 9, then the sum of the digits of n is divisible by 9. So which statement is it contrapositive.
(1) (sum of digits of n is divisible by 9) $\Rightarrow$ (n is divisible by 9)
(2) (sum of digits of n is not divisible by 9) $\Rightarrow$ (n is not divisible by 9)
(3) (sum of digits of n is divisible by 9) $\Rightarrow$ (n is divisible by 9)
(4) None of these

73. Fifteen coupons are numbered 1, 2 ..... 15, respectively. Seven coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9, is
(1) $\left(\frac{9}{16}\right)^6$ (2) $\left(\frac{8}{15}\right)^7$
(3) $\left(\frac{3}{5}\right)^7$ (4) None of these

74. If $a \leq 0$ then roots of $x^2 - 2a|x-a| - 3a^2 = 0$ is
(1) $(-1+\sqrt{6})a$ (2) $(\sqrt{6}-1)a$
(3) a (4) None of these

75. If $n(A) = 1000$, $n(B) = 500$ and if $n(A \cap B) \geq 1$ and $n(A \cup B) = p$, then
(1) $500 \leq p \leq 1000$ (2) $1001 \leq p \leq 1498$
(3) $1000 \leq p \leq 1498$ (4) $1000 \leq p \leq 1499$

76. If $y = \log_2\{\log_2(x)\}$, then $\frac{dy}{dx}$ is
(1) $\frac{\log_2 e}{x \ln x}$ (2) $\frac{2.3026}{x \ln x \ln 2}$
(3) $\frac{1}{\ln(2x)^x}$ (4) None of these

77. $f(x) = \sin|x|.f(x)$ is not differentiable at
(1) x = 0 only (2) all x
(3) multiples of π (4) multiples of $\frac{\pi}{2}$

Space for Rough Work

78. The angle between the pair of tangents drawn to the ellipse $3x^2 + 2y^2 = 5$ from the point (1,2) is

(1) $\tan^{-1}\left(\frac{12}{5}\right)$ (2) $\tan^{-1}\left(6\sqrt{5}\right)$

(3) $\tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$ (4) $\tan^{-1}\left(12\sqrt{5}\right)$

79. The least integral value α of x such that $\frac{x-5}{x^2+5x-14} > 1$, satisfies :

(1) $\alpha^2 + 3\alpha - 4 = 0$
(2) $\alpha^2 - 5\alpha + 4 = 0$
(3) $\alpha^2 - 7\alpha + 6 = 0$
(4) $\alpha^2 + 5\alpha - 6 = 0$

80. The x satisfying $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$ are

(1) 1, 0 (2) 1, –1

(3) $0, \frac{1}{2}$ (4) None of these

Section - B

81. The degree of differential equation satisfying the relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is ______ .

82. If the second term in the expansion $\left(\sqrt[13]{a} + \frac{a}{\sqrt{a^{-1}}}\right)^n$ is $14a^{5/2}$, then value of $\frac{{}^nC_3}{{}^nC_2}$ is ______ .

83. Given two independent events, if the probability that exactly one of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is ______ .

84. α, β be the roots of $x^2 - 3x + a = 0$ and γ, δ be the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing G.P. then value of a + b is _______ .

85. The probability of A = Probability of B = Probability of C = $\frac{1}{4}$

$P(A) \cap P(B) \cap P(C) = 0$, $P(B \cap C) = 0$ and $P(A \cap C) = \frac{1}{8}$, $P(A \cap B) = 0$

the probability that atleast one of the events A, B, C exists is _______ .

86. If $\sum_{r=0}^{n} \frac{r+2}{r+1} {}^nC_r = \frac{2^8 - 1}{6}$, then n is _______ .

87. $f(x) = \frac{\sin 3x}{\sin x}$, when $x \neq 0$
$= k$, when $x = 0$
for the function to be continuous, k should be _______ .

88. If $\int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{1+\sin 2x}} dx = k\pi$ then value of k is _______ .

89. The S.D. of the following data is nearly

x_i	140	145	150	155	160	165	170	175
f_i	4	6	15	30	36	24	8	2

is _______ .

90. A line passes through (2,2) and is perpendicular to the line 3x + y = 3, its y intercept is _______ .

Mock Test 8

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

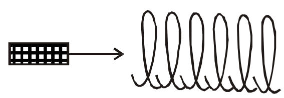
Section - A

1. A given ideal gas with $\gamma = \dfrac{C_p}{C_v} = 1.5$ at a temperature T. If the gas is compressed adiabatically to one-fourth of its initial volume, the final temperature will be

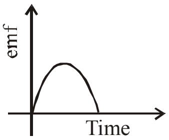
   (1) $2\sqrt{2}T$ (2) $4\,T$

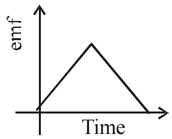
   (3) $2\,T$ (4) $8\,T$

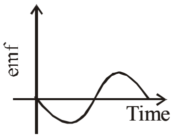
2. A small bar magnet is being slowly inserted with constant velocity inside a solenoid as shown in figure. Which graph best represents the relationship between emf induced with time

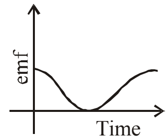


Space for Rough Work

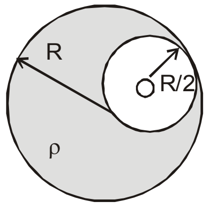
(1) 

(2) 

(3) 

(4) 

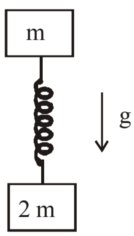
3. There is a spherical cavity of radius R/2 in uniformly charged spherical region having charge density ρ and radius R as shown in figure. A small charged particle having q_0 charge is released at point O. It will collide with the wall of cavity with kinetic energy



(1) $\frac{\rho R^2 q_0}{3 \epsilon_0}$ (2) $\frac{\rho R^2 q_0}{6\pi \epsilon_0}$

(3) $\frac{\rho R^2 q_0}{4\pi \epsilon_0}$ (4) $\frac{\rho R^2 q_0}{12 \epsilon_0}$

4. Two bodies of mass m and 2m connected by an unextended spring of spring constant 'k' are allowed to fall simultaneously in a uniform gravitational field g. The extension 'x' in the spring when the bodies A and B are falling



(1) varies with time as $x = A \sin(\omega t + \phi)$

(2) is constant and has value $\sqrt{\frac{2mg}{k}}$

(3) is zero

(4) increases linearly with time

5. A radioactive nucleus of mass M emits a photon of frequency ν and the nucleus recoils. The recoil energy will be

(1) $Mc^2 - h\nu$ (2) $h^2\nu^2 / 2Mc^2$

(3) zero (4) $h\nu$

6. In Young's double slit experiment, if intensity of maxima on screen is I (assume equal intensity of the source) and when the sources become incoherent, average intensity on the screen is I_2, then the ratio $\frac{I_1}{I_2}$ is

(1) 2 (2) $\frac{1}{2}$ (3) 1 (4) 0

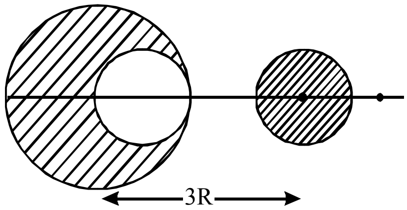
7. Orbits of a particle moving in a circle are such that the perimeter of the orbit equals an integer number of de-Broglie wavelengths of the particle. For a charged particle moving in a plane perpendicular to a magnetic field, the radius of the n^{th} orbital will therefore be proportional to :

(1) n^2 (2) n

(3) $n^{1/2}$ (4) $n^{1/4}$

8. From a sphere of mass M and radius R, a smaller sphere of radius $\frac{R}{2}$ is carved out such that the cavity made in the original sphere is between its centre and the periphery (See figure). For the configuration in the figure where the distance between the centre of the original

sphere and the removed sphere is 3R, the gravitational force between the two sphere is:



(1) $\frac{41\ GM^2}{3600\ R^2}$ (2) $\frac{41\ GM^2}{450\ R^2}$

(3) $\frac{59\ GM^2}{450\ R^2}$ (4) $\frac{GM^2}{225\ R^2}$

9. Two rods of the same length and diameter having thermal conductivities K_1 and K_2 are joined in parallel. The equivalent thermal conductivity of the combination is

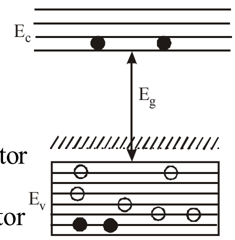
(1) $\frac{K_1 K_2}{K_1 + K_2}$ (2) $K_1 + K_2$

(3) $\frac{K_1 + K_2}{2}$ (4) $\sqrt{K_1 K_2}$

10. A photoelectric surface is illuminated successively by monochromatic light of wavelengths λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is :

(1) $\frac{hc}{2\lambda}$ (2) $\frac{hc}{\lambda}$ (3) $\frac{hc}{3\lambda}$ (4) $\frac{3hc}{\lambda}$

11. In the energy band diagram of a material shown below, the open circles and filled circles denote holes and electrons respectively. The material is



(1) an insulator

(2) a metal

(3) an n-type semiconductor

(4) a p-type semiconductor

12. The figure of merit of a galvanometer is defined as
(1) the voltage required to produce unit deflection in the galvanometer
(2) the current required to produce unit deflection in the galvanometer
(3) the power required to produce unit deflection in the galvanometer
(4) None of these

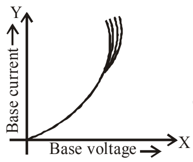
13. The activity of a radioactive element decreases to one-third of the original activity I_0 in a period of nine years. After a further lapse of nine years its activity will be

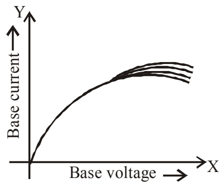
(1) I_0 (2) $\frac{2}{3}I_0$

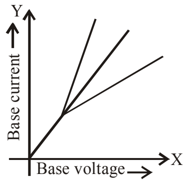
(3) $\frac{I_0}{9}$ (4) $\frac{I_0}{6}$

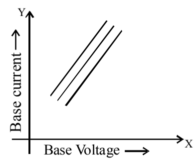
14. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is
(1) $v_0 + g/2 + f$ (2) $v_0 + 2g + 3f$
(3) $v_0 + g/2 + f/3$ (4) $v_0 + g + f$

15. Which of the following graph represents the input characteristics of a common emmiter transistor correctly ?

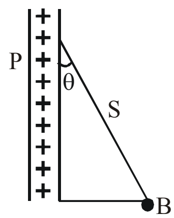
(1) 

(2) 

(3) 

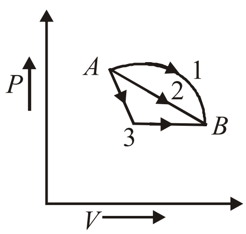
(4) 

16. A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to



(1) $\cot\theta$ (2) $\cos\theta$
(3) $\tan\theta$ (4) $\sin\theta$

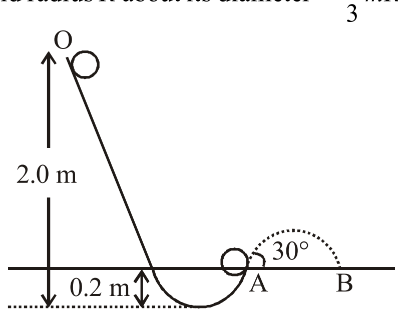
17. An ideal gas goes from state A to state B via three different processes as indicated in the P-V diagram :



If Q_1, Q_2, Q_3 indicate the heat absorbed by the gas along the three processes and ΔU_1, ΔU_2, ΔU_3 indicate the change in internal energy along the three processes respectively, then

(1) $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
(2) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
(3) $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$
(4) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

18. A tennis ball (treated as hollow spherical shell) starting from O rolls down a hill. At point A the ball becomes air borne leaving at an angle of 30° with the horizontal. The ball strikes the ground at B. What is the value of the distance AB ? (Moment of inertia of a spherical shell of mass m and radius R about its diameter $= \frac{2}{3}mR^2$)



(1) 1.87m (2) 2.08m
(3) 1.57m (4) 1.77m

19. In an organ pipe, three successive resonance frequencies are observed at 425, 595 and 765Hz. The length of organ pipe is (Velocity of sound in air = 340 m/s).

(1) 2.0m (2) 1.5m
(3) 1.0m (4) 2.5m

20. A ball is thrown from a point with a speed 'v_0' at an elevation angle of θ. From the same point and at the same instant, a person starts running with a constant speed $\frac{'v_0'}{2}$ to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection θ ?
(1) No (2) Yes, 30°
(3) Yes, 60° (4) Yes, 45°

Section - B

21. An object is moving towards a concave mirror of focal length 24 cm. When it is at a distance of 60 cm from the mirror its speed is 9 cm/s. The speed of its image at the instant is ______ cm/s toward away from the mirror.

22. An aluminium foil of relative emittance of 0.1 is placed in between two concentric spheres at temperature 300 K and 200 K respectively. What is the temperature in kelvin of this foil when steady state is reached? Assume the spheres to be black bodies. ($\sigma = 5.672 \times 10^{-8}$)

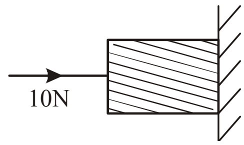
23. The space between the parallel plates of a capacitor is tightly filled with three dielectric slabs A,B, C of thickness 5mm, 3mm and 2mm with dielectric constants 2, 3 and 5 respectively. If a potential difference of 351 volts is applied to plates then electric field intensity in slab A is ______ v/m.

24. Given that radius of earth is 6.4×10^6 m and a transmitting antenna at the top of a tower has a height 40 m and the height of the receiving antenna is 50 m. What is the maximum distance (in metre) between them for satisfactory communication in los mode?

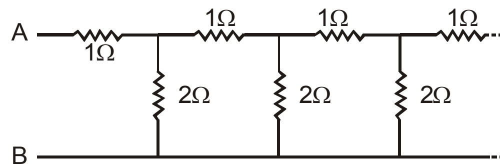
25. A structural steel rod has a radius of 10 mm and length of 1.0 m. A 100 kN force stretches it along its length. Young's modulus of structural steel is 2×10^{11} Nm^{-2}. The percentage strain is about

26. In the X-rays tube before striking the target we accelerate the electrons through a potential difference of V volt. For kV we will have X-rays of largest wavelength?

27. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is ______ N.



28. The effective resistance between points A and B of infinite network is .... ohm.



29. While measuring the length of the rod by vernier callipers the reading on main scale is 6.4 cm and the eight division on vernier is in line with marking on main scale division. If the least count of callipers is 0.01 and zero error –0.04 cm, the length of the rod is ______ cm.
(1) 5.25 cm (2) 6.52 cm
(3) 5.52 cm (4) 6.44 cm

30. A moving particle of mass m, makes a head on elastic collision with another particle of mass $2m$, which is initially at rest. The percentage loss in energy of the colliding particle on collision, is close to ______ .

CHEMISTRY

Section - A

31. Match the items in Column I with its main use listed in Column II:

Column I	Column II
(A) Silica gel	(i) Transistor
(B) Silicon	(ii) Ion–exchanger
(C) Silicone	(iii) Drying agent
(D) Silicate	(iv) Sealant

(1) (A)–(iii), (B)–(i), (C)–(iv), (D)–(ii)
(2) (A)–(iv), (B)–(i), (C)–(ii), (D)–(iii)
(3) (A)–(ii), (B)–(i), (C)–(iv), (D)–(iii)
(4) (A)–(ii), (B)–(iv), (C)–(i), (D)–(iii)

32. Which of the following has highest boiling point?
(1) H_2Se (2) H_2Te (3) H_2S (4) H_2O

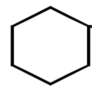
33. In which of the following will the oxidation state change ?
(1) Reaction of Cu^{2+} with NaOH
(2) Reaction of Cu^{2+} with iron metal
(3) Reaction of Cu^{2+} with KI
(4) both (2) and (3)

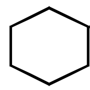
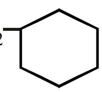
34. Reduction of esters with sodium and alcohol is referred to as
(1) Bouveault-Blanc reduction
(2) Mendius reaction
(3) Clemmensen's reduction
(4) MPV reduction

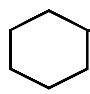
35. *p*-Chloroaniline and anilinium hydrochloride can be distinguished by
(1) Sandmeyer reaction (2) $NaHCO_3$
(3) $AgNO_3$ (4) Carbylamine test

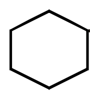
36. Chemical reduction is not suitable for
(1) conversion of bauxite to aluminium
(2) conversion of cuprite into copper
(3) conversion of haematite to iron
(4) conversion of zinc oxide to zinc

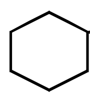
37. Which is the only group which is deactivating yet ortho-para directing when attached to benzene ring?
(1) NO_2 (2) CH_3
(3) Cl (4) None of these

38.  $-C \equiv C - H + CH_3 - CH_2MgBr$ gives

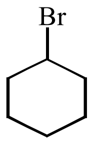
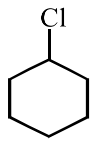
(1)  $-C \equiv CMgBr +$ $CH_3 - CH_2-$ 

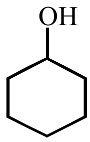
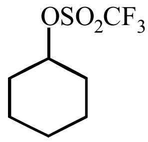
(2)  $-C \equiv CMgBr + C_2H_6$

(3)  $-MgBr + CH_3 - CH_2 - C \equiv CH$

(4)  $-C_2H_5 + HC \equiv CMgBr$

39. Which of the following will be most reactive towards nucleophillic substitution.

(1)  (2) 

(3)  (4) 

40. Ozone reacts with dry iodine to give

(1) IO_2 (2) I_2O_3 (3) I_2O_4 (4) I_4O_9

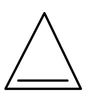
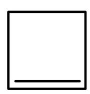
41. Dipole moment is shown by
(1) 1, 4-dichlorobenzene
(2) *cis*-1,2-dichloroethene
(3) *trans*-1,2-dichloroethene
(4) *trans*-2,3-dichloro-2 butene

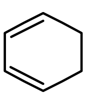
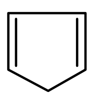
42. In what manner will increase of pressure effect the following equilibrium?

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

(1) Shift in forward direction
(2) Shift in reverse direction
(3) Increase in yield of hydrogen
(4) No effect

43. Which of the following compounds is most reactive to an aqueous solution of sodium carbonate?

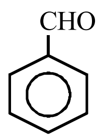
(1)  (2) 

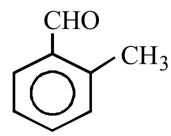
(3)  (4) 

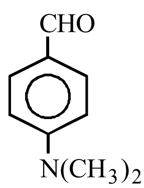
44. AgBr dissolves in $Na_2S_2O_3$ solution and forms (A). On boiling an aquous solution of (A), a ppt. (B) is obtained. The colour of B is
(1) Red (2) White
(3) Black (4) Colourless

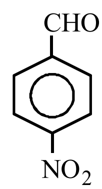
45. Chloride of an element gives neutral solution in water. In the periodic table the element belongs to
(1) third group
(2) fifth group
(3) first group
(4) first transition series

46. Arrange the following in order of decreasing order of reactivity towards Perkin reaction


I


II


III


IV

(1) IV > III > I > II (2) II > III > IV > I
(3) III > II > I > IV (4) IV > III > I > II

47. In the preparation of *p*-nitro acetanilide from aniline titration is not done by nitrating mixture (a mixture of conc. H_2SO_4 and conc. HNO_3) because
(1) on nitration it gives *o*-nitro acetanilide
(2) It gives a mixture of *o* and *p*-nitro aniline
(3) $-NH_2$ group gets oxidised
(4) it forms a mixture of *o* and *p*-nitro acetanilide.

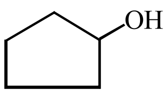
Space for Rough Work

48. Which of the following sequences represents the correct order of increasing stability of +3 oxidation state?

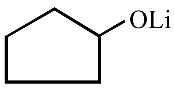
(1) Al < In < Ga (2) As < Sb < Bi
(3) Bi < Sb < As (4) none of these

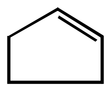
49. An alloy of copper, silver and gold is found to have copper constituting the *ccp* lattice. If silver atoms occupy the edge centres and gold is present at body centre the alloy has a formula

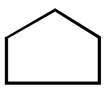
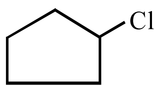
(1) Cu_4Ag_2Au (2) Cu_4Ag_4Au
(3) Cu_4Ag_3Au (4) CuAgAu

50. 

$\xrightarrow[\text{LiAlH}_4]{\text{TsCl}}$ A, Then A is

(1) 

(2) 

(3) 
(4) 

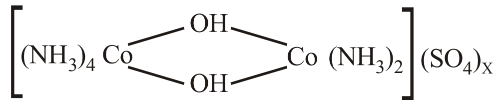
Section - B

51. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. Find the molar mass of gaseous molecule

52. How many stereoisomers are possible for 1, 3-dichloro- cyclopentane?

53. What is the work function (in eV) of the metal if the light of wavelength 4000Å generates photoelectrons of velocity 6×10^5 ms^{-1} from it ?
(Mass of electron=9×10^{-31} kg;
Velocity of light = 3×10^8 ms^{-1};
Planck's constant=6.626×10^{-34} Js;
Charge of electron =1.6×10^{-19} JeV^{-1})

54. What is the value of x in the following complex?



Consider the oxidation state of Cobalt in the complex same as that of iron in $K_3[Fe(CN)_6]$?

55. The hydrogen electrode is dipped in a solution of pH = 3 at 25 ºC. Find the potential of the cell would be

56. 50 mL of 0.2 M ammonia solution is treated with 25 mL of 0.2 M HCl. If pK_b of ammonia solution is 4.75. Find the pH of the mixture will be :

57. When a solution containing *w* g of urea in 1kg of water is cooled to – 0.372ºC, 200g of ice is separated. If K_f for water is 1.86 *K* kg mol^{-1}, calculate *w*.

58. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. Calculate the activation energy of the reaction (Assume activation energy and pre-exponential factor are independent of temperature; ln 2 = 0.693; R = 8.314 J mol^{-1} K^{-1})

59. An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. Calculate the work done in kJ.

60. Calculate the amount of arsenic pentasulphide that can be obtained when 35.5 g arsenic acid is treated with excess H_2S in the presence of conc. HCl (assuming 100% conversion)

MATHEMATICS

Section - A

61. If $f(x)=\sqrt{1-\sqrt{1-x^2}}$, then at $x=0$,
(1) $f(x)$ is differentiable as well as continuous
(2) $f(x)$ is differentiable but not continuous
(3) $f(x)$ is continuous but not differentiable
(4) $f(x)$ is neither continuous nor differentiable

62. In a $\Delta ABC, \dfrac{\sin A}{\sin C}=\dfrac{\sin(A-B)}{\sin(B-C)}$, then a^2, b^2, c^2 are such that
(1) they are in G.P.
(2) they are in H.P.
(3) they are in A.P.
(4) $b^2 = a^2 + c^2$

63. Coefficient of x^{25} in the expansion of expression
$$\sum_{r=0}^{50} {}^{50}C_r (2x-3)^r (2-x)^{n-r}$$ is
(1) ${}^{50}C_{25}$ (2) $-{}^{50}C_{24}$
(3) $-{}^{50}C_{25}$ (4) ${}^{50}C_{30}$

64. The equation $x^3+5x^2+px+q=0$ and $x^3+7x^2+px+r=0$ have two roots in common. If their third roots are x_1 and x_2, then $(x_1, x_2)=$
(1) $(-5,-7)$ (2) $(5,7)$
(3) $(5,-7)$ (4) $(-5,7)$

65. The number of ways of selecting 4 cards out of 52 cards of an ordinary pack, so that exactly 3 of them are of same denomination is
(1) 52×48 (2) ${}^{13}C_3\times{}^{4}C_3\times 48$
(3) ${}^{52}C_3\times 48$ (4) ${}^{52}C_3\times{}^{48}C_3$

66. The domain of the function
$$f(x)=\exp(\sqrt{5x-3-2x^2})$$ is
(1) $[3/2,\infty)$ (2) $[1,3/2]$
(3) $(-\infty,1]$ (4) $(1,3/2)$

67. Solution of differential equation
$$\frac{dy}{dx}=\sin x+2x,$$ is
(1) $y=x^2-\cos x+c$ (2) $y=\cos x+x^2+c$
(3) $y=\cos x+2$ (4) $y=\cos x+2+c$

68. If $z=x+iy$ and $w=\dfrac{(1-iz)}{(z-i)}$, then $|w|=1$, show that in complex plane
(1) z is situated on imaginary axis
(2) z is situated on real axis
(3) z is situated on unit circle
(4) None of these

Space for Rough Work

69. If z, ωz and $\overline{\omega}$z are the vertices of a triangle, then the area of the triangle will be (where ω is cube root of unity) :

(1) $\frac{3|z|^2}{2}$ (2) $\frac{3\sqrt{3}|z|^2}{2}$

(3) $\frac{\sqrt{3}|z|^2}{2}$ (4) None of these

70. From a point (h, 0) common tangents are drawn to circles $x^2+y^2=1$ and $(x-2)^2+y^2=4$, value of h is

(1) 2 (2) –2

(3) $-\frac{2}{3}$ (4) $\frac{2}{3}$

71. The altitude of a cone is 20cm and its semi-vertical angle is 30°. If the semi-vertical angle is increasing at the rate of 2° per second, then the radius of the base is increasing at the rate of–

(1) 30 cm/sec (2) $\frac{160}{3}$ cm/sec

(3) 10 cm/sec (4) 160 cm/sec.

72. If $\int \frac{1}{1+\sin x}dx = \tan\left(\frac{x}{2}+a\right)+b$ then

(1) $a=-\frac{\pi}{4}$, $b\in\mathbf{R}$ (2) $a=\frac{\pi}{4}$, $b\in\mathbf{R}$

(3) $a=\frac{5\pi}{4}$, $b\in\mathbf{R}$ (4) None of these

73. If $f(x) = \begin{cases} \frac{1-[x]}{1+x}, & x\neq -1 \\ 1, & x=-1 \end{cases}$, then the value of $f(|2k|)$ will be (where [] shows the greatest integer function]

(1) Continuous at x = –1

(2) Continuous at x = 0

(3) Discontinuous at $x=\frac{1}{2}$

(4) All of these

74. A variable circle passes through the fixed point $A(p, q)$ and touches x-axis. The locus of the other end of the diameter through A is

(1) $(y-q)^2=4px$ (2) $(x-q)^2=4py$

(3) $(y-p)^2=4qx$ (4) $(x-p)^2=4qy$

75. Let $\vec{A}=2\hat{i}+\hat{k}$, $\vec{B}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{C}=4\hat{i}-3\hat{j}+7\hat{k}$.

The vector $\vec{R}$ which satisfies the equations $\vec{R}\times\vec{B}=\vec{C}\times\vec{B}$ and $\vec{R}.\vec{A}=0$ is given by

(1) $-2\hat{i}+\hat{k}$

(2) $-\hat{i}-8\hat{j}+2\hat{k}$

(3) $\frac{1}{\sqrt{6}}(\hat{i}-\hat{j}+2\hat{k})$

(4) None of these

Space for Rough Work

76. $\sum_{k=33}^{65}\left(\sin\frac{2k\pi}{8}-i\cos\frac{2k\pi}{8}\right)=$

(1) $1+i$ (2) $1-i$

(3) $1+\frac{i}{\sqrt{2}}$ (4) $\frac{1-i}{\sqrt{2}}$

77. The complete solution set for $\sin^{-1}(x)=3\sin^{-1}(a)$ is

(1) $0\le a\le\frac{1}{2}$ (2) $-\frac{1}{2}\le a\le 0$

(3) $-\frac{1}{2}\le a\le\frac{1}{2}$ (4) None of these

78. The equation of the plane through the line of intersection of planes

$ax+by+cz+d=0,\ a'x+b'y+c'z+d'=0$ and parallel to the line $y=0, z=0$ is

(1) $(ab'-a'b)x+(bc'-b'c)y+(ad'-a'd)=0$

(2) $(ab'-a'b)x+(bc'-b'c)y+(ad'-a'd)z=0$

(3) $(ab'-a'b)y+(ac'-a'c)z+(ad'-a'd)=0$

(4) None of these

79. If $\tan^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right)=e^a$ where a is constant, then $\frac{dy}{dx}$

(1) $\frac{x}{y}$ (2) $\frac{y}{x}$

(3) $\frac{x^2}{y^2}$ (4) $\frac{y^2}{x^2}$

80. The coordinates of the foot of perpendicular from the point (1, 0, 0) to the line

$\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ are

(1) $(2,-3,8)$ (2) $(1,-1,-10)$

(3) $(5,-8,-4)$ (4) $(3,-4,-2)$

Section - B

81. The smallest value of 'p' for which $y^2+xy+px^2-x-2y+p=0$ represent 2 straight lines is ____ .

82. If the length of a focal chord of parabola $y^2=4x$ is $\frac{25}{4}$ and has positive slope, then the slope will be ______ .

83. The roots of the equation

$x^5-40x^4+px^3+qx^2+rx+s=0$ are in geometric progression and the sum of their reciprocal is 10. Then value of |S| is ______ .

84. Let $|\vec{a}|=2$, $|\vec{b}|=5$. The values of |k| for which the vectors $\vec{a}+k\vec{b}$ and $\vec{a}-k\vec{b}$ are perpendicular is ______ .

85. From a pack of 52 cards two cards are drawn at random. The probability of both cards being spades is ______ .

86. If $f(A) = 3$, $f'(A) = -2$, $g(A) = -1$, $g'(A) = 4$, then value of $\lim_{x \to a} \dfrac{g(x)f(a) - g(a)f(x)}{x - a}$ is ______ .

87. If $\Delta(x) = \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} = A + Bx + Cx^2$ + ... then value of B is ______ .

88. The sides of a triangle ABC lie on the lines $3x + 4y = 0$, $4x + 3y = 0$ and $x = 3$. Let (h, k) be the centre of the circle inscribed in Δ ABC. The value of $(h + k)$ is ______ .

89. If $A = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 8 & 10 \\ -6 & -12 & -15 \end{bmatrix}$. Then rank of A is ______ .

90. If a coin is tossed n times, the probability that head appears odd no. of times is ______ .

Space for Rough Work

Mock Test 9

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

1. Two sources of equal emf are connected to an external resistance R. The internal resistances of the two sources are R_1 and R_2 ($R_2 > R_1$). If the potential difference across the source having internal resistance R_2 is zero then

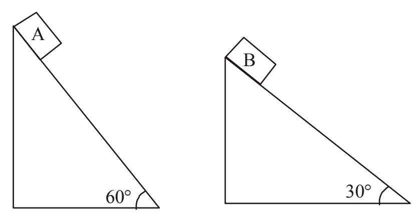
   (1) $R = R_1R_2/(R_1 + R_2)$

   (2) $R = R_1R_2/(R_2 - R_1)$

   (3) $R = R_2 \times (R_1 + R_2)/(R_2 - R_1)$

   (4) $R = R_2 - R_1$

2. Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B ?



   (1) 4.9 ms^{-2} (in horizontal direction)
   (2) 9.8 ms^{-2} (in vertical direction)
   (3) Zero
   (4) 4.9 ms^{-2} (in vertical direction)

3. A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap -water solution. Which of the following shows the relative nature of the liquid columns in the two tubes.

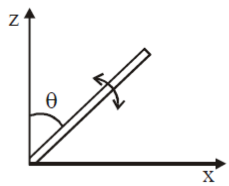
(1) 

(2) 

(3) 

(4) 

4. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is



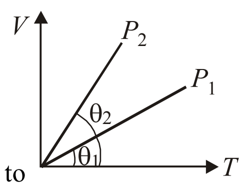
(1) $\frac{3g}{2\ell}\cos\theta$ (2) $\frac{2g}{3\ell}\cos\theta$

(3) $\frac{3g}{2\ell}\sin\theta$ (4) $\frac{2g}{2\ell}\sin\theta$

5. The rectangular surface of area 8 cm × 4 cm of a black body at temperature 127°C emits energy E per second. If the length and breadth are reduced to half of the initial value and the temperature is raised to 327°C, the rate of emission of energy becomes

(1) $\frac{3}{8}E$ (2) $\frac{81}{16}E$ (3) $\frac{9}{16}E$ (4) $\frac{81}{64}E$

6. The figure shows the volume V versus temperature T graphs for a certain mass of a perfect gas at two constant pressures of P_1 and P_2. What inference can you draw from the graphs?



(1) $P_1 > P_2$
(2) $P_1 < P_2$
(3) $P_1 = P_2$
(4) No interference can be drawn due to insufficient information.

7. What should be the maximum acceptance angle at the air core interface of an optical fibre if n_1 and n_2 are the refractive indices of the core and the cladding, respectively

(1) $\sin^{-1}(n_2/n_1)$ (2) $\sin^{-1}\sqrt{n_1^2 - n_2^2}$

(3) $\left[\tan^{-1}\frac{n_2}{n_1}\right]$ (4) $\left[\tan^{-1}\frac{n_1}{n_2}\right]$

Space for Rough Work

8. When Uranium is bombarded with neutrons, it undergoes fission. The fission reaction can be written as :

$_{92}U^{235} + _0n^1 \rightarrow _{56}Ba^{141} + _{36}Kr^{92} + 3x + Q(\text{energy})$

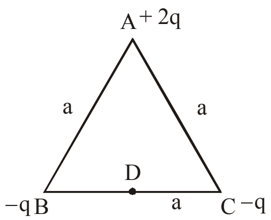
where three particles named x are produced and energy Q is released. What is the name of the particle x ?

(1) electron (2) α-particle
(3) neutron (4) neutrino

9. Given that K = energy, V = velocity, T = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension?

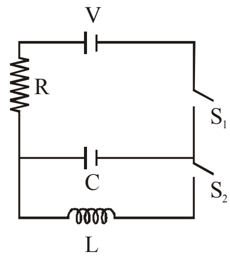
(1) $[KV^{-2}T^{-2}]$ (2) $[K^2V^2T^{-2}]$
(3) $[K^2V^{-2}T^{-2}]$ (4) $[KV^2T^2]$

10. Three charges of (+ 2q), (– q) and (– q) are placed at the corners A, B and C of the equilateral triangle of side a as shown in the given figure. Then the dipole moment of this combination is



(1) qa (2) zero
(3) qa $\sqrt{3}$ (4) $\frac{2}{\sqrt{3}}$ qa

11. In an LCR circuit as shown below both switches S_1 and S_2 are open initially. Now switch S_1 is closed, S_2 kept open. (q is charge on the capacitor and τ = RC is capacitive time constant). Which of the following statements is correct ?



(1) Work done by the battery is half of the energy dissipated in the resistor
(2) At t = τ, q = CV/2
(3) At t = 2τ, q = CV $(1 - e^{-2})$
(4) At t = $\frac{\tau}{2}$, q = CV $(1 - e^{-1})$

12. Torques τ_1 and τ_2 are required for a magnetic needle to remain perpendicular to the magnetic fields at two different places. The magnetic fields at the those places are B_1 and B_2 respectively; then ratio $\frac{B_1}{B_2}$ is

(1) $\frac{\tau_2}{\tau_1}$ (2) $\frac{\tau_1}{\tau_2}$
(3) $\frac{\tau_1 + \tau_2}{\tau_1 - \tau_2}$ (4) $\frac{\tau_1 - \tau_2}{\tau_1 + \tau_2}$

13. The transverse displacement $y(x, t)$ of a wave is given by $y(x,t) = e^{-\left(ax^2 + bt^2 + 2\sqrt{ab}\,xt\right)}$

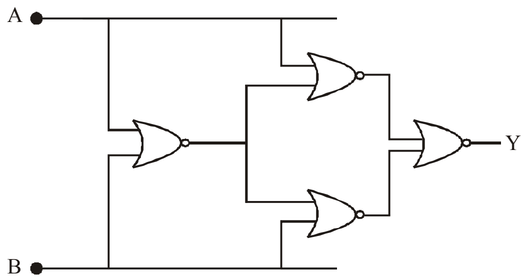
This represents a:

(1) wave moving in $-x$ direction with speed $\sqrt{\frac{b}{a}}$
(2) standing wave of frequency $\sqrt{b}$

(3) standing wave of frequency $\frac{1}{\sqrt{b}}$

(4) wave moving in $+x$ direction speed $\sqrt{\frac{a}{b}}$

14. A system of four gates is set up as shown. The 'truth table' corresponding to this system is :



(1)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(4)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

15. A particle of mass m is under a potential $v(x) = (1/2)k_1x^2$ for $x > 0$ and $v(x) = k_1x^2$ for $x < 0$. When disturbed a little from the position $x = 0$, it will

(1) not execute SHM

(2) execute SHM with $T = 2\pi\sqrt{\frac{m}{3k_1}}$

(3) execute SHM with $T^2 = 2\pi^2 m/(k_1^2)$

(4) execute SHM with $T = \frac{\pi\sqrt{m}}{\sqrt{k_1\left(1+\frac{1}{\sqrt{2}}\right)}}$

16. Two identical short bar magnets, each having magnetic moment M, are placed a distance of 2d apart with axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is

(1) $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$ (2) $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$

(3) $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$ (4) $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

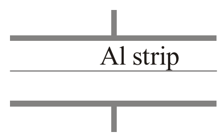
17. Inside a cylinder closed at both ends is a movable piston. On one side of the piston is a mass m of a gas, and on the other side a mass 2m of the same gas. What fraction of volume of the cylinder will be occupied by the larger mass of the gas when the piston is in equilibrium? The temperature is the same throughout.

(1) 1/4 (2) 1/2 (3) 2/3 (4) 1/3

18. A hydrogen atom emits green light when it changes from n = 4 energy level to the n = 2 level. Which colour of light would the atom emit when it changes from n = 5 level to the n = 2 level?

(1) red (2) yellow (3) green (4) violet

19. As shown in the figure, a very thin sheet of aluminium is placed in between the plates of the condenser. Then the capacity



(1) Will increase
(2) Will decrease
(3) Remains unchanged
(4) May increase or decrease

20. A coil having n turns and resistance $R\Omega$ is connected with a galvanometer of resistance $4R\Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is

(1) $-\dfrac{(W_2 - W_1)}{Rnt}$ (2) $-\dfrac{n(W_2 - W_1)}{5\ Rt}$

(3) $-\dfrac{(W_2 - W_1)}{5\ Rnt}$ (4) $-\dfrac{n(W_2 - W_1)}{Rt}$

Section - B

21. Radioactive material 'A' has decay constant '8 λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After $\dfrac{1}{x\lambda}$ time, the ratio of number of nuclei of material 'A' to that 'B' is $\dfrac{1}{e}$. Find the alue of x.

22. A ball projected from ground at an angle of 45° just clears a wall in front. If point of projection is 4 m from the foot of wall and ball strikes the ground at a distance of 6 m on the other side of the wall, the height of the wall is ______metre.

23. A particle of charge 16×10^{-16} C moving with velocity 10 ms^{-1} along x-axis enters a region where magnetic field of induction $\vec{B}$ is along the y-axis and an electric field of magnitude 10^4 Vm^{-1} is along the negative z-axis. If the charged particle continues moving along x-axis, the magnitude of $\vec{B}$ is ________ Wb m^{-2}

24. A satellite is revolving round the earth with velocity v. The minimum percentage increase in its velocity necessary for the escape of satellite will be

25. The work function of a metallic surface is 5.01 eV, photoelectrons are emitted when light of wavelength 2000Å falls on it. The minimum potential difference (in volts) required to stop the fastest photoelectrons is (h = 4.14×10^{-15} eV–s)

26. A refrigerator takes heat from water at 0°C inside it and rejects it to the room at a temperature of 27°C. The latent heat of ice is 336×10^3 J kg^{-1}. If 5 kg of water at 0°C is converted into ice at 0°C by the refrigerator, then the energy consumed by the refrigerator is close to ________ joule.

27. The potential energy of a 1 kg particle free to move along the x-axis is given by $V(x) = \left(\dfrac{x^4}{4} - \dfrac{x^2}{2}\right) J$.
The total mechanical energy of the particle is 2 J. Then, the maximum speed (in m/s) is

28. The radiation corresponding to $3 \to 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by these electrons is 10.0 mm, the work function of the metal is close to _____ eV.

29. A stuntman plans to run across a roof top and horizontally jumps on to another roof 4.9 m below the first one and at a distance of 6.2 m away. What is the minimum velocity (in m/s) he must have before the jump ?

30. A thin sheet of glass (μ = 1.5) of thickness 6 microns introduced in the path of one of interfering beams in a double slit experiment shifts the central fringe to a position previously occupied by fifth bright fringe. Then the wave length of the light used is _______ Å.

CHEMISTRY

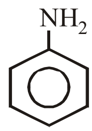
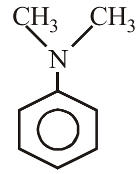
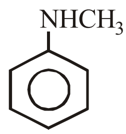
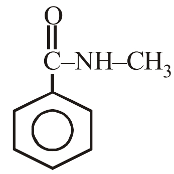
Section - A

31. A hydrocarbon with five carbon atoms in the molecule, decolourizes alkaline $KMnO_4$, but does not give a precipitate with ammonical Cu_2Cl_2 solution. The hydrocarbon is possibly
(1) 1-pentyne (2) 1, 3-pentadiyne
(3) 2-pentyne (4) 1, 4-pentadiyne

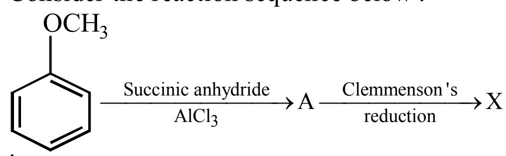
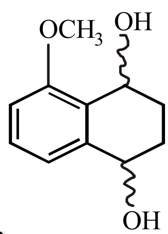
32. The threshold frequency of a metal is $1 \times 10^{15}\ s^{-1}$. The ratio of maximum kinetic energies of the photoelectrons when the metal is irradiated with radiations of frequencies $1.5 \times 10^{15}\ s^{-1}$ and $2 \times 10^{15}\ s^{-1}$ respectively would be
(1) 3 : 4 (2) 1 : 2 (3) 2 : 1 (4) 4 : 3

33. A solution containing a group–IV cation gives a precipitate on passing H_2S. A solution of this precipitate in dil.HCl produces a white precipitate with NaOH solution and bluish–white precipitate with basic potassium ferrocyanide. The cation is :
(1) Co^{2+} (2) Ni^{2+} (3) Mn^{2+} (4) Zn^{2+}

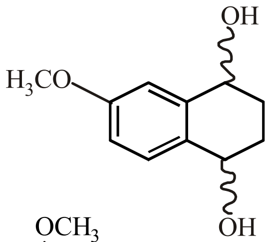
34. Identify the feasible reaction among the following:
(1) $K_2CO_3 \xrightarrow{\Delta} K_2O + CO_2$
(2) $Na_2CO_3 \xrightarrow{\Delta} Na_2O + CO_2$
(3) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$
(4) $Rb_2CO_3 \xrightarrow{\Delta} Rb_2O + CO_2$

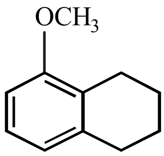
35. Which of the following gives N-nitrosoamine on reaction with nitrous acid?
(1) 
(2) 
(3) 
(4) 

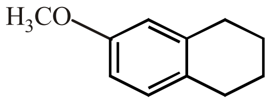
36. Consider the reaction
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ in a closed container at equilibrium. What would be the effect of addition of $CaCO_3$ on the equilibrium concentration of CO_2 ?
(1) Increases
(2) Decreases
(3) Data insufficient
(4) Remains unaffected

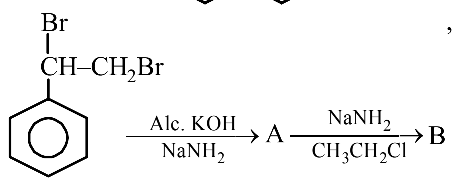
37. Consider the reaction sequence below :

is
(1) 

Space for Rough Work

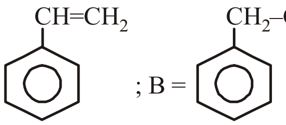
(2) 

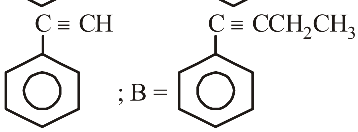
(3) 

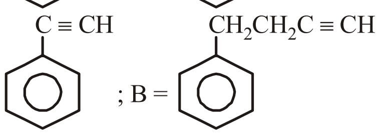
(4) 

38. 

A and B are

(1) A = 

(2) A = 

(3) A = 

(4) None of these

39. Which one of the following on treatment with 50% aqueous sodium hydroxide yields the corresponding alcohol and acid?

(1) C_6H_5CHO (2) $CH_3CH_2CH_2CHO$

(3) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$ (4) $C_6H_5CH_2CHO$

40. The complexes of Nickel (II) can be

(1) Square planar, tetrahedral and octahedral

(2) Square planar and octahedral

(3) Tetrahedral and octahedral

(4) Square planar only

41. Which of the following represents the correct order of stability?

(1) $H-C\equiv C^{-}:>H_2C=CH^{-}:>H_3C-CH_2^{-}:$

(2) $H_3C-CH_2^{-}:>H_2C=CH^{-}:>H-C\equiv C^{-}:$

(3) $H_3C-CH_2^{-}:>H_2C=CH^{-}:=H-C\equiv C^{-}:$

(4) None of these

42. Which of the metal is extracted by Hall-Heroult process?

(1) Al (2) Cu (3) Ni (4) Zn

43. $\Delta G=\Delta H-T\Delta S$ and $\left[\dfrac{d(\Delta G)}{dT}\right]_p=-\Delta S$

The enthalpy of cell reaction, ΔH, is then given by

(1) $T\left[\dfrac{dE_{\text{cell}}}{dT}\right]_p-E_{\text{cell}}$

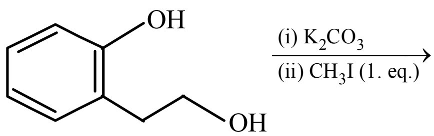
(2) $nF\left[T\left(\dfrac{dE_{\text{cell}}}{dT}\right)_p-E_{\text{cell}}\right]$

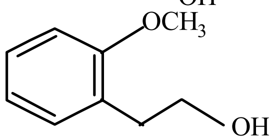
(3) $nF\left[E_{\text{cell}}-T\left(\dfrac{dE_{\text{cell}}}{dT}\right)_p\right]$

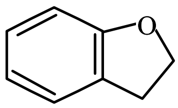
(4) $nFT\left(\dfrac{dE_{\text{cell}}}{dT}\right)_p$

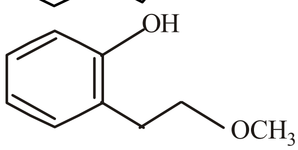
44. Given $E^{\circ}_{Au^{3+}/Au} = 1.52V$ and $E^{\circ}_{Au^{3+}/Au^{+}} = 1.36V$.
Point out the correct statement of the following
(1) Au^{3+} disproportionates into Au^{4+} and Au^{2+} in aqueous solution
(2) Au^{3+} disproportionates into Au^{4+} and Au^{+} in aqueous solution
(3) Au^{+} disproportionates into Au^{3+} and Au in aqueous solution
(4) Au^{+} disproportionates into Au^{2+} and Au in aqueous solution

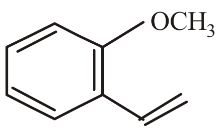
45. The major product of the following reaction is :



(1) 

(2) 

(3) 

(4) 

46. The hybridization states of the central atoms in the complexes $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$ and $[Co(NO_2)_6]^{3-}$ are
(1) d^2sp^3, sp^3 and d^4s^2 respectively
(2) d^2sp^3, sp^3d and sp^3d^2 respectively
(3) d^2sp^3, sp^3d^2 and dsp^2 respectively
(4) all d^2sp^3

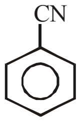
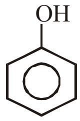
47. Among the acids given below
CH_3CH_2COOH (X); $CH_2 = CHCOOH$ (Y) and $CH \equiv C\text{-}COOH$ (Z)
The correct order of increasing acid strength is
(1) $X < Y < Z$ (2) $X < Z < Y$
(3) $Y < X < Z$ (4) $Z < Y < X$

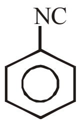
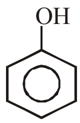
48. Which series of reactions correctly represents chemical reactions related to iron and its compound?
(1) $Fe \xrightarrow{\text{dil. } H_2SO_4} FeSO_4 \xrightarrow{H_2SO_4,\, O_2} Fe_2(SO_4)_3 \xrightarrow{\text{heat}} Fe$
(2) $Fe \xrightarrow{O_2,\,\text{heat}} FeO \xrightarrow{\text{dil. } H_2SO_4} FeSO_4 \xrightarrow{\text{heat}} Fe$
(3) $Fe \xrightarrow{Cl_2,\,\text{heat}} FeCl_3 \xrightarrow{\text{heat, air}} FeCl_2 \xrightarrow{Zn} Fe$
(4) $Fe \xrightarrow{O_2,\,\text{heat}} Fe_3O_4 \xrightarrow{CO,\, 600°C} FeO \xrightarrow{CO,\, 700°C} Fe$

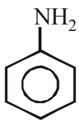
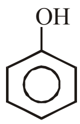
49. The form of BN which is as hard as diamond is
(1) hexagonal form
(2) Cubic form with ZnS structure
(3) both of these
(4) none of these

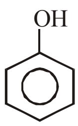
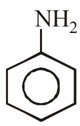
50. A mixture of two aromatic compounds A and B is separated by dissolving in chloroform followed by extraction with aq. KOH solution. The alkaline aqueous layer gives a mixture of two isomeric

compounds on treatment with carbon tetrachloride. The organic layer containing compound A gives an unpleasant odour on treatment with alcoholic solution of KOH. Compounds A and B respectively are

(1)  and 

(2)  and 

(3)  and 

(4)  and 

Section - B

51. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

$$\frac{1}{2}Cl_2(g) \xrightarrow{\frac{1}{2}\Delta_{diss}H^\circ} Cl(g) \xrightarrow{\Delta_{eg}H^\circ} Cl^-(g) \xrightarrow{\Delta_{hyd}H^\circ} Cl^-(aq)$$

(using the data,

$\Delta_{diss} H^\circ_{Cl_2} = 240 \text{ kJ mol}^{-1}$,

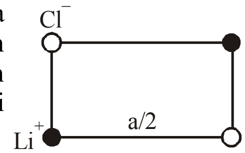
$\Delta_{eg} H^\circ_{Cl} = -349 \text{ kJ mol}^{-1}$,

$\Delta_{hyd} H^\circ_{Cl^-} = -381 \text{ kJ mol}^{-1}$), ΔH will be _____.

52. Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is 6.104×10^{-30} Cm. The percentage ionic character in HF will be ________. (electron charge = 1.60×10^{-19} C)

53. The vapour pressure (at the standard boiling point of water) of an aqueous solution containing 28% by mass of a non-volatile normal solute (molecular mass = 28) will be ________.

54. Lithium chloride has a cubic structure as shown below. If the edge length is 400 pm., then the radii of Cl^- ions is



55. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. Find the molar mass of gaseous molecule.

56. Normality of a mixed solution of sulphuric acid and hydrochloric acid is 0.6 N. 20 mL of this solution gives 0.4305 g of AgCl on reacting with $AgNO_3$ solution. The strength of H_2SO_4 in g/ L in the mixed solution is ________ .

57. What is the value of *n* in the molecular formula $Be_nAl_2Si_6O_{18}$?

58. The half life of a first order reaction is 10 min. If initial amount is 0.08 mol L^{-1} and concentration at some instant is 0.01 mol L^{-1}. What is the time elapsed ?

59. For soaps critical micelle concentration (CMC) is 10^{-x} (min.) to 10^{-y} (max.) mol/L. What is the value of x?

60. A weak base BOH is titrated with a strong acid HA. When 10 ml of HA is added, the pH is found to be 9.00 and when 25 ml is added, pH is 8.00. Find the volume of the acid required to reach the equivalence point

Space for Rough Work

MATHEMATICS

Section - A

61. If $P=\{x\in\mathbf{R}:f(x)=0\}$ and $Q=\{x\in\mathbf{R}:g(x)=0\}$ then $P\cup Q$ is
(1) $\{x\in\mathbf{R}:f(x)+g(x)=0\}$
(2) $\{x\in\mathbf{R}:f(x)\,g(x)=0\}$
(3) $\left\{x\in\mathbf{R}:(f(x))^2+(g(x))^2=0\right\}$
(4) None of these

62. If $\operatorname{cosec}\theta=\dfrac{p+q}{p-q}$ $(p\neq q\neq 0)$, then $\left|\cot\left(\dfrac{\pi}{4}+\dfrac{\theta}{2}\right)\right|$ is equal to:
(1) $\sqrt{\dfrac{p}{q}}$ (2) $\sqrt{\dfrac{q}{p}}$ (3) $\sqrt{pq}$ (4) pq

63. The expression satisfying the differential equation $(x^2-1)\dfrac{dy}{dx}+2xy=1$ is
(1) $x^2y-xy^2=c$
(2) $(y^2-1)x=y+c$
(3) $(x^2-1)\,y=x+c$
(4) None of these

64. $\int e^{(\log x+ax^2)}\cos(bx^2+c)\,dx$ is equal to
(1) $\dfrac{1}{\sqrt{a^2+b^2}}e^{ax^2}\cos(bx^2+c+\tan^{-1}\dfrac{b}{a})+A$
(2) $\dfrac{1}{2\sqrt{a^2+b^2}}e^{ax^2}\cos(bx^2-c-\tan^{-1}\dfrac{b}{a})+A$
(3) $\dfrac{1}{\sqrt{a^2+b^2}}e^{ax^2}\cos(bx^2+c-\tan^{-1}\dfrac{b}{a})+A$
(4) $\dfrac{1}{2\sqrt{a^2+b^2}}e^{ax^2}\cos(bx^2+c-\tan^{-1}\dfrac{b}{a})+A$

65. If $\int\sqrt{1+\sec x}\,dx=2(fog)(x)+C$, then
(1) $f(x)=\sec x-1$ (2) $f(x)=2\tan^{-1}x$
(3) $g(x)=\sqrt{\sec x-1}$ (4) None of these

66. If $y^2=P(x)$ is a polynomial of degree 3, then $2\dfrac{d}{dx}\left[y^3\dfrac{d^2y}{dx^2}\right]$ is equal to
(1) $P(x)+P'(x)$ (2) $P(x)+P'''(x)$
(3) $P(x)\,P'''(x)$ (4) None of these

67. Which of the following is an even function?
(1) $f(x)=\log\left(\dfrac{1-x}{1+x}\right)$
(2) $f(x)=\log\,(x+\sqrt{1+x^2})$
(3) $f(x)=\dfrac{x}{e^x-1}+\dfrac{x}{2}+1$
(4) $f(x)=e^{2x}+\sin x$

68. A plane meets the coordinate axes in points A, B, C and the centroid of the triangle ABC is (α,β,γ). The equation of the plane is
(1) $\dfrac{x}{\alpha}+\dfrac{y}{\beta}+\dfrac{z}{\gamma}=3$ (2) $\alpha x+\beta y+\gamma z=3\alpha\beta\gamma$
(3) $\dfrac{x}{\alpha}+\dfrac{y}{\beta}+\dfrac{z}{\gamma}=\dfrac{1}{2}$ (4) None of these

69. Equation of the latus rectum of the hyperbola $(10x-5)^2+(10y-2)^2=9(3x+4y-7)^2$ is

(1) $y-\frac{1}{5}=-\frac{3}{4}\left(x-\frac{1}{2}\right)$

(2) $x-\frac{1}{5}=-\frac{3}{4}\left(y-\frac{1}{2}\right)$

(3) $y+\frac{1}{5}=-\frac{3}{4}\left(x+\frac{1}{2}\right)$

(4) $x+\frac{1}{5}=-\frac{3}{4}\left(y+\frac{1}{2}\right)$

70. The point (2a, a) lies inside the region bounded by the parabola $x^2=4y$ and its latus rectum. Then,

(1) $0 \le a \le 1$ (2) $0 < a < 1$
(3) $a > 1$ (4) $a < 0$

71. If ΔABC is right angled at A then the inradius of the triangle is

(1) $2(a+b-2)$ (2) $2(b+c-a)$
(3) $\frac{b+c-a}{2}$ (4) $\frac{b+c-a}{4}$

72. If $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P. then $9^{ax+1}, 9^{bx+1}, 9^{cx+1}$, $x \ne 0$ are in :

(1) G.P. (2) G.P. only if $x < 0$
(3) G.P. only if $x > 0$ (4) None of these

73. If $a_k=\int_0^{\pi}\frac{\sin(2k-1)x}{\sin x}dx$, then –

(1) a_1, a_2, ... are in A.P.
(2) a_1, a_2, ... are in G.P.
(3) a_1, a_2, ... are in H.P.
(4) a_1, a_2, ... form a constant sequence

74. Let $f(x)=e^x$, $g(x)=\sin^{-1}x$ and $h(x)=f(g(x))$, then $h'(x)/h(x)=$

(1) $e^{\sin^{-1}x}$ (2) $1/\sqrt{1-x^2}$
(3) $\sin^{-1}x$ (4) $1/(1-x^2)$

75. Which of the following functions is differentiable at $x=0$?

(1) $\cos(|x|)+|x|$ (2) $\cos(|x|)-|x|$
(3) $\sin(|x|)+|x|$ (4) $\sin(|x|)-|x|$

76. The largest interval lying in $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ for which the function,

$f(x)=4^{-x^2}+\cos^{-1}\left(\frac{x}{2}-1\right)+\log(\cos x)$ is defined, is

(1) $\left[-\frac{\pi}{4},\frac{\pi}{2}\right)$ (2) $\left[0,\frac{\pi}{2}\right)$
(3) $[0,\pi]$ (4) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

77. If $f(x)=\sin x+\cos x$, $g(x)=x^2-1$, then $g(f(x))$ is invertible in the domain

(1) $\left[0,\frac{\pi}{2}\right]$ (2) $\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$
(3) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (4) $[0,\pi]$

78. Which of the following functions is NOT one-one ?

(1) $f: R \to R$ defined by $f(x)=6x-1$.
(2) $f: R \to R$ defined by $f(x)=x^2+7$.
(3) $f: R \to R$ defined by $f(x)=x^3$.
(4) $f: R-\{7\} \to R$ defined by $f(x)=\frac{2x+1}{x-7}$.

Space for Rough Work

79. $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}............+\frac{1}{n}\sec^2 1\right]$ equals

(1) $\frac{1}{2}\sec 1$ (2) $\frac{1}{2}$ cosec 1

(3) tan 1 (4) $\frac{1}{2}$ tan 1

80. The differential equation of the family of curves represented by $c(y+c)^2 = x^3$ is

(1) $y\frac{d^2y}{dx^2} - y^2\left(\frac{dy}{dx}\right)^2 = 27x$

(2) $12y\left(\frac{dy}{dx}\right)^2 = 8x\left(\frac{dy}{dx}\right)^3 - 27x$

(3) $8y\left(\frac{dy}{dx}\right)^3 = 12x\left(\frac{dy}{dx}\right)^2 - 27x$

(4) $\left(\frac{dy}{dx}\right)^3 - \left(\frac{dy}{dx}\right)^2 + \left(\frac{dy}{dx}\right) - y = 27x$.

Section - B

81. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -3\hat{i} + \hat{j} - \hat{k}$, $\vec{r}\times\vec{a} = \vec{b}\times\vec{a}, \vec{r}\times\vec{b} = \vec{a}\times\vec{b}$ and $\vec{r}\cdot\vec{a} = 30$, then magnitude of $\vec{r}$ is _____.

82. The acute angle between two lines such that the direction cosines l, m, n, of each of them satisfy the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ is (in degree) _____.

83. In a two player game that always has a winner, A beats B with probability 2/3; B beats C with probability 2/3 & C beats A with the same probability. If B plays with C and then the winner plays with A, the chance that A will be the final winner is ______.

84. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{c} = \vec{i} + \alpha\vec{j} + \beta\vec{k}$ are linearly independent vectors and $|\vec{c}| = \sqrt{3}$, then greatest value of $\alpha + \beta$ is __.

85. If the area lying between the curves y = tan x, y = cot x and x-axis, $x \in \left[0, \frac{\pi}{2}\right]$ is log k, then value of k is ________.

86. From the point (15, 12) three normals are drawn to the parabola $y^2 = 4x$. If centroid of triangle formed by three co-normal points is (m, n), then value of m + n is _____.

87. If a + b + c = 3 and a > 0, b > 0, c > 0, and the greatest value of $a^2 b^3 c^2$ is $\frac{3^p.2^q}{7^r}$, then value of (p + q + r) is _____.

88. If z_1, z_2, z_3, z_4 be the vertices of a quadrilateral taken in order such that $z_1 + z_3 = z_2 + z_4$, $|z_1 - z_3| = |z_2 - z_4|$ and $\arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = \pm k\pi$ then value of k is ______.

89. The function defined by

$$f(x) = \begin{cases} \left(x^2 + e^{\frac{1}{2-x}}\right)^{-1}, & x \neq 2, \text{ is} \\ k & x = 2 \end{cases}$$

continuous from right at the point x = 2, then k is _____.

90. Consider a rectangle whose length is increasing at the uniform rate of 2 m/sec, breadth is decreasing at the uniform rate of 3 m/sec and the area is decreasing at the uniform rate of 5 m^2/sec. If after some time the breadth of the rectangle is 2 m then the length of the rectangle is ____m.

Space for Rough Work

Mock Test 10

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

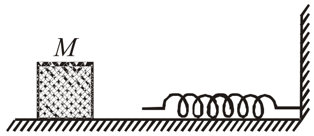
1. A uniformly tapering conical wire is made from a material of Young's modulus Y and has a normal, unextended length L. The radii, at the upper and lower ends of this conical wire, have values R and 3R, respectively. The upper end of the wire is fixed to a rigid support and a mass M is suspended from its lower end. The equilibrium extended length, of this wire, would equal :

(1) $L\left(1+\frac{2}{9}\frac{Mg}{\pi YR^2}\right)$ (2) $L\left(1+\frac{1}{9}\frac{Mg}{\pi YR^2}\right)$

(3) $L\left(1+\frac{1}{3}\frac{Mg}{\pi YR^2}\right)$ (4) $L\left(1+\frac{2}{3}\frac{Mg}{\pi YR^2}\right)$

2. The block of mass *M* moving on the frictionless horizontal surface collides with the spring of spring constant *k* and compresses it by length

L. The maximum momentum of the block after collision is



(1) $\frac{kL^2}{2M}$ (2) $\sqrt{Mk}\ L$

(3) $\frac{ML^2}{k}$ (4) Zero

3. A mass M, attached to a horizontal spring, executes S.H.M. with amplitude A_1. When the mass M passes through its mean position then a smaller mass m is placed over it and both of them move together with amplitude A_2. The ratio of $\left(\frac{A_1}{A_2}\right)$ is:

(1) $\frac{M+m}{M}$ (2) $\left(\frac{M}{M+m}\right)^{\frac{1}{2}}$

(3) $\left(\frac{M+m}{M}\right)^{\frac{1}{2}}$ (4) $\frac{M}{M+m}$

4. Three particles of equal mass m are situated at the vertices of an equilateral triangle of side L. What should be the velocity of each particle so that they move on a circular path without changing L?

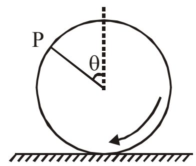
(1) $\sqrt{(GM/2L)}$ (2) $\sqrt{(GM/L)}$
(3) $\sqrt{(2GM/L)}$ (4) $\sqrt{(GM/3L)}$

5. A charge Q is distributed over two concentric hollow spheres of radii r and R (>r) such that the surface densities are equal and placed on the same axial points. Then the potential at the common centre is

(1) $\frac{Q(R^2+r^2)}{4\pi\varepsilon_0(R+r)}$ (2) $\frac{Q}{R+r}$

(3) Zero (4) $\frac{Q(R+r)}{4\pi\varepsilon_0(R^2+r^2)}$

6. A wheel is rolling straight on ground without slipping. If the axis of the wheel has speed v, the instantenous velocity of a point P on the rim, defined by angle θ, relative to the ground will be



(1) $v\cos\left(\frac{1}{2}\theta\right)$ (2) $2v\cos\left(\frac{1}{2}\theta\right)$

(3) $v(1+\sin\theta)$ (4) $v(1+\cos\theta)$

7. In the Bohr model an electron moves in a circular orbit around the proton. Considering the orbiting electron to be a circular current loop, the magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, is :

(1) $\left(\frac{e}{2m}\frac{n^2h}{2\pi}\right)$ (2) $\left(\frac{e}{m}\right)\frac{nh}{2\pi}$

(3) $\left(\frac{e}{2m}\right)\frac{nh}{2\pi}$ (4) $\left(\frac{e}{m}\right)\frac{n^2h}{2\pi}$

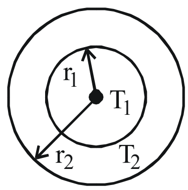
8. From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path.

The relation between H, u and n is:

(1) $2gH = n^2u^2$ (2) $gH = (n-2)^2 u^2 d$

(3) $2gH = nu^2 (n-2)$ (4) $gH = (n-2)u^2$

9. The figure shows a system of two concentric spheres of radii r_1 and r_2 are kept at temperatures T_1 and T_2, respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to



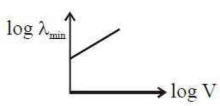
(1) $In\left(\frac{r_2}{r_1}\right)$

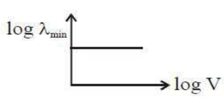
(2) $\frac{(r_2 - r_1)}{(r_1 r_2)}$

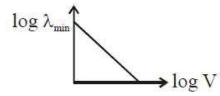
(3) $(r_2 - r_1)$

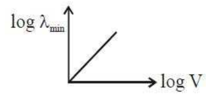
(4) $\frac{r_1 r_2}{(r_2 - r_1)}$

10. An electron beam is accelerated by a potential difference V to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-rays. If λ_{min} is the smallest possible wavelength of X-ray in the spectrum, the variation of log λ_{min} with log V is correctly represented in :

(1) 

(2) 

(3) 

(4) 

11. A cell of e.m.f. E is connected across a reistance R. The potential difference between the terminals of the cell is found to be V. The internal resistance of the cell must be

(1) [2(E–V)V]/R (2) [2(E–V)/V]R

(3) [(E–V)/V]R (4) (E–V)R

12. A cylindrical vessel of cross-section A contains water to a height h. There is a hole in the bottom of radius 'a'. The time in which it will be emptied is:

(1) $\frac{2A}{\pi a^2}\sqrt{\frac{h}{g}}$ (2) $\frac{\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$

(3) $\frac{2\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$ (4) $\frac{A}{\sqrt{2}\pi a^2}\sqrt{\frac{h}{g}}$

13. Two straight long conductors AOB and COD are perpendicular to each other and carry currents I_1 and I_2 respectively. The magnitude of the magnetic induction at a point P at a distance 'a' from the point O where the two conductors intersect, in a direction perpendicular to the plane ABCD is

(1) $\frac{\mu_0}{2\pi a}(I_1 + I_2)$ (2) $\frac{\mu_0}{2\pi a}(I_1 - I_2)$

(3) $\frac{\mu_0}{2\pi a}\left[I_1^2 + I_2^2\right]^{1/2}$ (4) $\frac{\mu_0}{2\pi a}\left(\frac{I_1 I_2}{I_1 + I_2}\right)$

Space for Rough Work

14. In a series resonant circuit, having L, C and R as its elements, the resonant current is i. The power dissipated in the circuit at resonance is

(1) $\dfrac{i^2R}{\left(\omega L - \dfrac{1}{\omega C}\right)}$ (2) zero

(3) $i^2\omega L$ (4) i^2R

15. The co-ordinates of a moving particle at any time 't' are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time 't' is given by

(1) $3t\sqrt{\alpha^2 + \beta^2}$ (2) $3t^2\sqrt{\alpha^2 + \beta^2}$

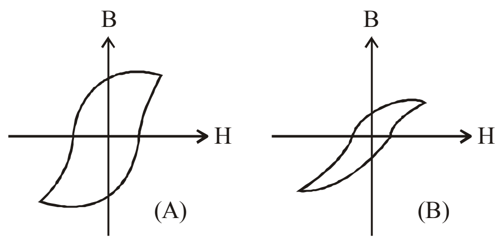
(3) $t^2\sqrt{\alpha^2 + \beta^2}$ (4) $\sqrt{\alpha^2 + \beta^2}$

16. The mutual inductance of a pair of coils, each of N turns, is M henry. If a current of I ampere in one of the coils is brought to zero in t second, the emf induced per turn in the other coil, in volt, will be

(1) $\dfrac{MI}{t}$ (2) $\dfrac{NMI}{t}$

(3) $\dfrac{MN}{It}$ (4) $\dfrac{MI}{Nt}$

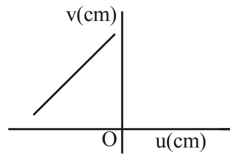
17. Hysteresis loops for two magnetic materials A and B are given below :

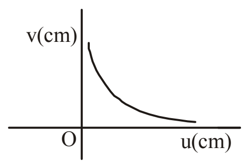


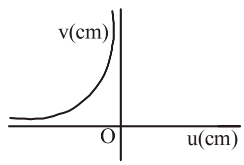
These materials are used to make magnets for elecric generators, transformer core and electromagnet core. Then it is proper to use :

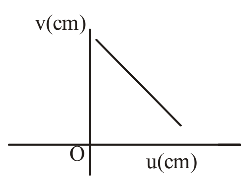
(1) A for transformers and B for electric generators.
(2) B for electromagnets and transformers.
(3) A for electric generators and trasformers.
(4) A for electromagnets and B for electric generators.

18. A student measures the focal length of a convex lens by putting an object pin at a distance 'u' from the lens and measuring the distance 'v' of the image pin. The graph between 'u' and 'v' plotted by the student should look like

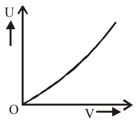
(1)


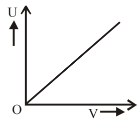
(2)


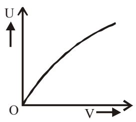
(3)


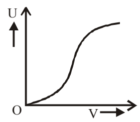
(4)


19. A battery is connected across series combination of a capacitor and a resistor, at t = 0. If at an instant t potential difference across the capacitor be 'V' and energy stored in it be U then which of the following graph is correct?

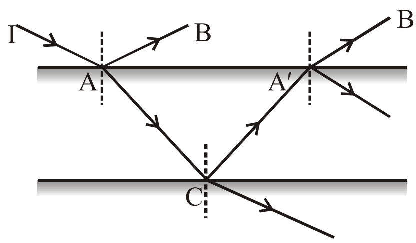
(1) 

(2) 

(3) 

(4) 

20. A ray of light of intensity I is incident on a parallel glass-slab at a point A as shown in fig. it undergoes partial reflection and refraction. At each reflection 25% of incident energy is reflected. The rays AB and A′B′ undergo interference. The ratio I_{max} / I_{min} is



(1) 4 : 1 (2) 8 : 1
(3) 7 : 1 (4) 49 : 1

Section - B

21. The time taken by a body in sliding down a rough inclined plane of angle of inclination 45° is n times the time taken by the same body in slipping down a similar frictionless plane. The coefficient of dynamic friction between the body and the plane is $\mu = 1 - \left(\frac{1}{n^x}\right)$. Find the value of x.

22. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field (in V/m) associated with the electromagnetic radiation at a distance of 5 m from the lamp will be nearly ______

23. A Carnot's engine works as a refrigerator between 250 K and 300 K. If it receives 750 calories of heat from the reservoir at the lower temperature, the amount of heat rejected at the higher temperature is ______ cal.

24. The electric field in a region of space is given by, $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$ where $E_o = 100$ N/C. The flux of the field through a circular surface of radius 0.02 m parallel to the Y-Z plane is nearly ____ Nm^2/C

25. A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0°C. If it were to go up straight without colliding with any other molecules, it rises to $h = \frac{819k_B}{xMg}$. Find the value of x. Assume that the height attained is much less than radius of the earth. (k_B is Boltzmann constant).

26. In a npn transistor 10^{10} electrons enter the emitter in 10^{-6} s. 4% of the electrons are lost in the base. The current transfer ratio will be

Space for Rough Work

27. 56 tuning forks are so arranged in series that each fork gives 4 beats per sec with the previous one. The frequency of the last fork is 3 times that of the first. The frequency of the first fork is ______ Hz.

28. For sky wave propagation of a 10 MHz signal, what should be the minimum electron density (in m^{-3}) in ionosphere?

29. In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half- a degree (= 0.5°), then the least count of the instrument is ______ minute.

30. The counting rate observed from a radioactive source at t = 0 second was 1600 counts per second and at t = 8 seconds it was 100 counts per second. The counting rate observed, as counts per second at t = 6 seconds will be

CHEMISTRY

Section - A

31. If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is :

(1) $\frac{5A}{9}$ (2) $\frac{9A}{5}$

(3) $\frac{36A}{5}$ (4) $\frac{36A}{7}$

32. The value of $\frac{t_{0.875}}{t_{0.50}}$ for n^{th} order reaction is

(1) $2^{(2n-2)}$ (2) $2^{(2n-2)-1}$

(3) $\frac{8^{n-1}-1}{2^{n-1}-1}$ (4) None of these

33. Urea and hydrazine react to form ammonia gas along with compound X which reacts with aldehydes and ketones to form specific crystalline derivatives. X is

(1) phenylhydrazine (2) semicarbazide

(3) biuret (4) acetylurea

34. The IUPAC name of $\underset{\displaystyle CN}{\underset{|}{CH_2}}-\underset{\displaystyle CN}{\underset{|}{CH}}-\underset{\displaystyle CN}{\underset{|}{CH_2}}$ is

(1) 1, 2, 3-propanetrinitrile

(2) 1, 2, 3-tricyanopropane

(3) 3-cyano-1, 5-dinitrilepentane

(4) 1, 2, 3-propanetricarbonitrile

35. The catenation tendency of C, Si and Ge is in the order Ge < Si < C. The bond energies (in $kJmol^{-1}$) of C – C, Si – Si and Ge –Ge bonds are respectively;

(1) 348, 297, 260 (2) 297, 348, 260

(3) 348, 260, 297 (4) 260, 297, 348

36. The repeating units of acrilan are

(1) $H_2C=\overset{\overset{H}{|}}{C}-COOCH_3$

(2) $H_2C=\overset{\overset{H}{|}}{C}-COOC_2H_5$

(3) $H_2C=\overset{\overset{H}{|}}{C}-CN$

(4) $H_2C=\overset{\overset{CH_3}{|}}{C}-COOCH_3$

37. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is

(1) $Mg(NO_3)_2$ (2) Mg_3N_2

(3) NH_3 (4) MgO

38. The turbidity of a polymer solution measures

(1) the light scattered by solution

(2) the light absorbed by a solution

(3) the light transmitted by a solution

(4) None of these

39. Which of the following has a shape different from others?

(1) $[Zn(NH_3)_4]^{2+}$ (2) $Ni(CO)_4$

(3) $[Cd(CN)_4]^{2+}$ (4) $[Cu(NH_3)_4]^{2+}$

40. Di-n-propyl ether and diallyl ether can be distinguished by

(1) Acetic acid

(2) Sodium Metal

(3) Cold dilute $KMnO_4$ solution

(4) PCl_5

41. In which of the following cases, the stability of two oxidation states is correctly represented

(1) $Ti^{3+} > Ti^{4+}$ (2) $Mn^{2+} > Mn^{3+}$

(3) $Fe^{2+} > Fe^{3+}$ (4) $Cu^{+} > Cu^{2+}$

42. Which one of the following oxides of chlorine is obtained by passing dry chlorine over silver chlorate at 90°C ?

(1) Cl_2O (2) ClO_3

(3) ClO_2 (4) ClO_4

43. The main product of the following reaction is

$$C_6H_5CH_2CH(OH)CH(CH_3)_2 \xrightarrow{conc.H_2SO_4} ?$$

(1) $(H_5C_6)(H)C=C(H)(CH(CH_3)_2)$

(2) $(C_6H_5CH_2)(H)C=C(CH_3)(CH_3)$

(3) $(H_5C_6CH_2CH_2)(H_3C)C=CH_2$

(4) $(C_6H_5)(H)C=C(CH(CH_3)_2)(H)$

44. Zirconium phosphate $[Zr_3(PO_4)_4]$ dissociates into three zirconium cations of charge + 4 and four phosphate anions of charge – 3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct?

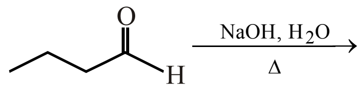
(1) $S=[K_{sp}/(6912)^{1/7}]$

(2) $S=[K_{sp}/144]^{1/7}$

(3) $S=[K_{sp}/6912]^{1/7}$

(4) $S=[K_{sp}/6912[^{7}$

45. The final product of the following sequence of reactions



$$B \xrightarrow[H_2O]{LiCu(CH_3)_2} C \xrightarrow[H_2O]{LiAlH_4} D$$

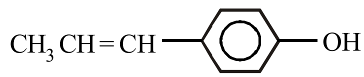
(1) 3-ethyl-2-methyl-1-hexanol
(2) 2, 3-dimethyl-1-pentanol
(3) 2-ethyl-3-methyl-1-hexanol
(4) 3, 3-dimethyl-1-pentanol

46. A solution containing As^{3+}, Cd^{2+}, Ni^{2+} and Zn^{2+} is made alkaline with dilute NH_4OH and treated with H_2S. The precipitate obtained will consist of
(1) As_2S_3 and CdS
(2) CdS, NiS and ZnS
(3) NiS and ZnS
(4) Sulphide of all ions

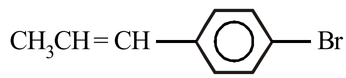
47. The intermediate formed during the addition of HCl to propene in the presence of peroxide is
(1) $CH_3\dot{C}HCH_2Cl$ (2) $CH_3\overset{+}{C}HCH_3$
(3) $CH_3CH_2\dot{C}H_2$ (4) $CH_3CH_2\overset{+}{C}H_2$

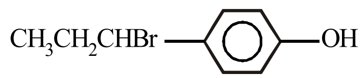
48. An organic compound 'A' having molecular formula C_2H_3N on reduction gave another compound 'B'. Upon treatment with nitrous acid, 'B' gave ethyl alcohol. On warming with chloroform and alcoholic KOH, B formed an offensive smelling compound 'C'. The compound 'C' is
(1) $CH_3CH_2NH_2$
(2) $CH_3CH_2N \stackrel{\rightarrow}{=} C$
(3) $CH_3C \equiv N$
(4) CH_2CH_2OH

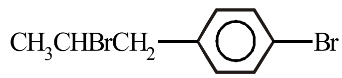
49. The reaction of

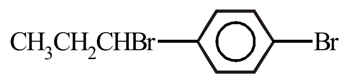


with HBr gives

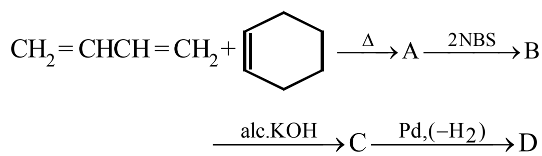
(1) 

(2) 

(3) 

(4) 

50. Final product (D) formed in the following reaction sequence is



(1) Naphthalene (2) Tetralin
(3) Benzene (4) Anthracene

Section - B

51. Determine enthalpy of formation for $H_2O_2(l)$, using listed enthalpies of reaction :

$$N_2H_4(l) + 2H_2O_2(l) \longrightarrow N_2(g) + 4H_2O(l);$$
$$\Delta_r H_1^\circ = -818 \text{ kJ/mol}$$
$$N_2H_4(l) + O_2(g) \longrightarrow N_2(g) + 2H_2O(l);$$
$$\Delta_r H_2^\circ = -622 \text{ kJ/mol}$$
$$H_2(g) + 1/2O_2(g) \longrightarrow H_2O(l);$$
$$\Delta_r H_3^\circ = -285 \text{ kJ/mol}$$

52. A metallic element exists as cubic lattice. Each edge of the unit cell is 2.88 Å. The density of the metal is 7.20 g cm^{-3}. How many unit cell will be present in 100 g of the metal in the multiple of 10^{23}?

53. The solution containing 4.0 % PVC in one litre of dioxane was found to have osmotic pressure of 6.0×10^{-4} atm at 300 K. Find the molecular mass of polymer in terms of $x \times 10^5$.

54. For a chemical reaction $X \rightarrow Y$, the rate of reaction increases by a factor of 1.837 when the concentration of X is increased by 1.5 times. Find the order of the reaction with respect to X

55. M_2SO_4 (M^+ is a monovalent metal ion) has a K_{sp} of 3.2×10^{-5} at 298 K. The maximum concentration of SO_4^{-2} ion that could be attained in a saturated solution of this solid at 298 K is $x \times 10^{-2}$ M. Find the value of x.

56. The colour of $KMnO_4$ solution is decolourised by Fe^{2+} solution, one mole of Fe^{2+} reacts with x moles of $KMnO_4$. Find x.

57. For the coagulation of 500 mL of arsenious sulphide sol, 2 mL of 1 M NaCl is required. What is the flocculation value of NaCl?

58. The standard reduction potential for Cu^{2+}/Cu is + 0.34. Calculate the reduction potential at pH = 14 for the above couple.
(K_{sp} $Cu(OH)_2 = 1 \times 10^{-19}$)

59. A current of 2.0A when passed for 5 hours through a molten metal salt deposits 22.2g of metal of atomic weight 177. Find the oxidation state of the metal in the metal salt

60. A gaseous mixture of three gases A, B and C has a pressure of 10 atm. The total number of moles of all the gases is 10. If the partial pressures of A and B are 3.0 and 1.0 atm respectively and if 'C' has mol. wt of 2.0 what is the weight of 'C' in g present in the mixture

MATHEMATICS

Section - A

61. ${}^{n+4}C_r - {}^nC_r - 3.{}^nC_{r-1} - 3.{}^nC_{r-2} - {}^nC_{r-3}$ is equal to
(1) ${}^{n+1}C_{r-1}$ (2) ${}^{n+2}C_{r-1}$
(3) ${}^{n+3}C_{r-1}$ (4) ${}^{n+4}C_{r-1}$

62. Let $\alpha/(\alpha-1)$ and $\beta/(\beta-1)$ be the roots of $x^2 + ax + b = 0$. Then $1/\alpha$ and $1/\beta$ are the roots of
(1) $bx^2 + ax + 1 = 0$
(2) $bx^2 - ax + 1 = 0$
(3) $bx^2 + (a + 2b)x + a + b + 1 = 0$
(4) $bx^2 - (a + 2b)x + a + b + 1 = 0$

Space for Rough Work

63. For any real θ, the maximum value of $\cos^2(\cos\theta) + \sin^2(\sin\theta)$ is
(1) 1 (2) $1 + \sin^2 1$
(3) $1 + \cos^2 1$ (4) 1/2

64. If tan(A/2), tan(B/2), tan(C/2) are in A.P. then sec A, sec B, sec C are in
(1) A.P. (2) G.P.
(3) H.P. (4) None

65. The equation of the common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is given by
(1) $xa^{1/3} + yb^{1/3} + a^{2/3}b^{2/3} = 0$
(2) $xb^{1/3} + ya^{1/3} + a^{2/3}b^{2/3} = 0$
(3) $xa^{1/3} + yb^{1/3} - a^{2/3}b^{2/3} = 0$
(4) None of these

66. If tangents be drawn to the cricle $x^2 + y^2 = 12$ at its points of intersection with the circle $x^2 + y^2 - 5x + 3y - 2 = 0$, then the tangents intersect at the point
(1) $\left(-6, \frac{18}{5}\right)$ (2) $\left(6, \frac{18}{5}\right)$
(3) $\left(-6, -\frac{18}{5}\right)$ (4) $\left(6, -\frac{18}{5}\right)$

67. Let $R = \{(x, y) : x, y \in N \text{ and } x^2 - 4xy + 3y^2 = 0\}$, where N is the set of all natural numbers. Then the relation R is :
(1) reflexive but neither symmetric nor transitive.
(2) symmetric and transitive.
(3) reflexive and symmetric,
(4) reflexive and transitive.

68. The value of
$$\lim_{x\to\pi/6} (4 - 3\sin x - 2\cos^2 x)^{\frac{1}{2\sin x - 1}} \text{ is}$$
(1) 1 (2) e
(3) $\sqrt{e}$ (4) $e^{-1/2}$

69. The function $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x - 3$, $(0 \le x \le \pi)$ is –
(1) Increasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$
(2) Increasing in $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$
(3) Decreasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$
(4) All of the above

70. Value of $\int_0^{\pi} \sin^n x \cos^{2m+1} x\, dx$ (where m, n $\in$ N) is
(1) $\frac{(2m+1)!}{n!}$ (2) $\frac{(2m+1)!}{n^2}$
(3) $\int_0^{\pi} \cos^{2m+1} x\, dx$ (4) None of these

71. If $f'(x) = f(x)$ and $f(0) = 2$, then
$$\int \frac{f(x)}{3 + 4f(x)} dx =$$
(1) $\log(3 + 8e^x) + C$
(2) $\frac{1}{4}\log(3 + 8e^x) + C$
(3) $\frac{1}{2}\log(3 + 8e^x) + C$
(4) None of these

Space for Rough Work

72. The area under the curve $y = |\cos x - \sin x|$, $0 \le x \le \frac{\pi}{2}$, and above x-axis is :
(1) $2\sqrt{2}$
(2) $2\sqrt{2}-2$
(3) $2\sqrt{2}+2$
(4) None of these

73. Let coordinates of a point 'p' with respect to the system of non-coplanar vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is (3, 2, 1). Then coordinates of 'p' with respect to the system of vectors $\vec{a}+\vec{b}+\vec{c}$, $\vec{a}-\vec{b}+\vec{c}$ and $\vec{a}+\vec{b}-\vec{c}$ is
(1) $\left(\frac{3}{2},\frac{1}{2},1\right)$ (2) $\left(\frac{3}{2},1,\frac{1}{2}\right)$
(3) $\left(\frac{1}{2},\frac{3}{2},1\right)$ (4) None

74. Equation of the line through the point (2,3,l) and parallel to the line of intersection of the planes $x-2y-z+5=0$ and $x+y+3z=6$ is
(1) $\frac{x-2}{-5}=\frac{y-3}{-4}=\frac{z-1}{3}$
(2) $\frac{x-2}{5}=\frac{y-3}{-4}=\frac{z-1}{3}$
(3) $\frac{x-2}{5}=\frac{y-3}{4}=\frac{z-1}{3}$
(4) $\frac{x-2}{4}=\frac{y-3}{3}=\frac{z-1}{2}$

75. If $x+2=6$, then $x=4$. So, which statement is its converse?
(1) If $x \ne 4$, then $x+2 \ne 6$
(2) If $x = 4$, then $x+2 \ne 6$
(3) If $x = 4$, then $x+2 = 6$
(4) If, $x \ne 4$ then $x+2=6$

76. If y = f(x) be a monotonically increasing or decreasing function of x and M is the median of variable x, then the median of y is
(1) f(M) (2) M/2
(3) f^{-1}(M) (4) None of these

77. A fair die is tossed eight times. The probability that a third six is observed on the eighth throw is
(1) ${}^7C_2 \frac{5^5}{6^8}$ (2) ${}^7C_3 \frac{5^3}{6^8}$
(3) ${}^7C_6 \frac{5^6}{6^8}$ (4) None of these

78. If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and λ is a real number then
$[\lambda(\vec{a}+\vec{b})\ \lambda^2\vec{b}\ \lambda\vec{c}] = [\vec{a}\quad \vec{b}+\vec{c}\quad \vec{b}]$ for
(1) exactly one value of λ
(2) no value of λ
(3) exactly three values of λ
(4) exactly two values of λ

79. $S = \tan^{-1}\left(\frac{1}{n^2+n+1}\right) + \tan^{-1}\left(\frac{1}{n^2+3n+3}\right) + \ldots + \tan^{-1}\left(\frac{1}{1+(n+19)(n+20)}\right)$, then $\tan S$ is equal to :
(1) $\frac{20}{401+20n}$ (2) $\frac{n}{n^2+20n+1}$
(3) $\frac{20}{n^2+20n+1}$ (4) $\frac{n}{401+20n}$

Space for Rough Work

80. The middle term in the expansion of $\left(1-\frac{1}{x}\right)^n (1-x)^n$ in powers of x is

(1) $-\,^{2n}C_{n-1}$ (2) $-\,^{2n}C_n$
(3) $^{2n}C_{n-1}$ (4) $^{2n}C_n$

Section - B

81. The value of $S = \frac{5}{1^2 \cdot 4^2} + \frac{11}{4^2 \cdot 7^2} + \frac{17}{7^2 \cdot 10^2} + \ldots \infty$ is ______ .

82. The line $2x + y = 3$ cuts the ellipse $4x^2 + y^2 = 5$ at P and Q. If θ be the angle between the normals at these points, then value of $\tan\theta$ is ______ .

83. Matrix M_r is defined as $M_r = \begin{pmatrix} r & r-1 \\ r-1 & r \end{pmatrix}$, $r \in \mathbf{N}$. If value of $\det(M_1) + \det(M_2) + \det(M_3) + \ldots + \det(M_{2007}) = (k)^2$, then value of k is ____ .

84. Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and

$(10)B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$.

If B is the inverse of matrix A, then value of α is ______ .

85. The derivative of $\tan^{-1}[(3x^2 - 1)/(3x - x^3)]$ with respect to $\sin^{-1}[(x^2 - 1)/(x^2 + 1)]$ is ______ .

86. If $2a + 3b + 6c = 0$ and at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval (m, n), then value of m + n is ______ .

87. If $Z_1 \neq 0$ and Z_2 be two complex numbers such that $\frac{Z_2}{Z_1}$ is a purely imaginary number, then $\left|\frac{2Z_1 + 3Z_2}{2Z_1 - 3Z_2}\right|$ is ______ .

88. If the straight lines
$x = 1 + s, y = -3 - \lambda s, z = 1 + \lambda s$
and $x = \frac{t}{2}, y = 1 + t, z = 2 - t,$ with parameters s and t respectively, are co-planar, then value of λ is ______ .

89. A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that neither of them occurs is 1/3. The most probability of occurrence of A is ______ .

90. If X and Y are independent binomial variates $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$, then P (X + Y = 3) is ______ .

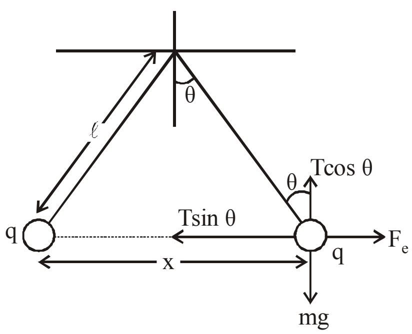
Mock Test-1

ANSWER KEY									
1	(4)	**16**	(1)	**31**	(1)	**46**	(1)	**61**	(3)	**76**	(3)
2	(4)	**17**	(3)	**32**	(1)	**47**	(1)	**62**	(2)	**77**	(1)
3	(3)	**18**	(1)	**33**	(2)	**48**	(1)	**63**	(2)	**78**	(4)
4	(2)	**19**	(1)	**34**	(2)	**49**	(2)	**64**	(2)	**79**	(4)
5	(4)	**20**	(2)	**35**	(4)	**50**	(3)	**65**	(3)	**80**	(1)
6	(1)	**21**	(170)	**36**	(3)	**51**	(1.24)	**66**	(1)	**81**	(0)
7	(3)	**22**	(6.67)	**37**	(4)	**52**	(2.25)	**67**	(3)	**82**	(0.80)
8	(4)	**23**	(6.275)	**38**	(3)	**53**	(0.5)	**68**	(1)	**83**	(0.25)
9	(1)	**24**	(20)	**39**	(3)	**54**	(1.75)	**69**	(1)	**84**	(0.33)
10	(1)	**25**	(0.25)	**40**	(3)	**55**	(5.97)	**70**	(4)	**85**	(7.00)
11	(1)	**26**	(4)	**41**	(4)	**56**	(400)	**71**	(1)	**86**	(1.00)
12	(2)	**27**	(40.05)	**42**	(4)	**57**	(210)	**72**	(2)	**87**	(3.00)
13	(2)	**28**	(5.3)	**43**	(4)	**58**	(0.04)	**73**	(1)	**88**	(1.50)
14	(2)	**29**	(23)	**44**	(2)	**59**	(6)	**74**	(4)	**89**	(14.00)
15	(2)	**30**	(2)	**45**	(3)	**60**	(0.241)	**75**	(1)	**90**	(3.20)

Solutions

PHYSICS

1. (4)



In equilibrium, $F_e = T\sin\theta$

$mg = T\cos\theta$

$$\tan\theta = \frac{F_e}{mg} = \frac{q^2}{4\pi\in_0 x^2 \times mg}$$

also $\tan\theta \approx \sin = \dfrac{x/2}{\ell}$

Hence, $\dfrac{x}{2\ell} = \dfrac{q^2}{4\pi\in_0 x^2 \times mg}$

$$\Rightarrow x^3 = \frac{2q^2\ell}{4\pi\in_0 mg}$$

$$\therefore x = \left(\frac{q^2\ell}{2\pi\in_0 mg}\right)^{1/3}$$

Therefore $x \propto \ell^{1/3}$

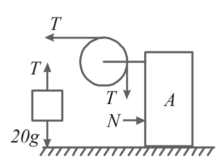
2. (4) **Case I :**

$T - N = 40a$

and $20g - T = 20a$

Also $N = 20a$

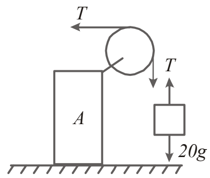
After simplifying, we get



$$a = \frac{g}{4}$$

Acceleration of block B, $= \sqrt{2}a = \frac{g}{2\sqrt{2}}$.

Case II :



$T = 40\,a$

and $20\,g - T = 20\,a$

After simplifying above equation, we get

$a = g/3$

$$\text{Ratio} = \frac{g/2\sqrt{2}}{g/3} = \frac{3}{2\sqrt{2}}.$$

3. (3) $V_{\text{in}} = \frac{-GM}{2R}\left[3 - \left(\frac{r}{R}\right)^2\right]$,

$$V_{\text{surface}} = \frac{-GM}{R}, V_{\text{out}} = \frac{-GM}{r}$$

4. (2) $T\downarrow$ (300K to 70K)

$T\downarrow$ $R_{\text{metal}}\downarrow$ $R\uparrow$ semi-conductor

(Al) (Si)

5. (4) When two rods are connected in series

$$Q = \frac{A(T_1 - T_2)t}{\frac{d_1}{K_1} + \frac{d_2}{K_1}} = \frac{A(T_1 - T_2)t}{(d_1 + d_2)/K}$$

$$\therefore \frac{d_1 + d_2}{K} = \frac{d_1}{K_1} + \frac{d_2}{K_2};$$

$$\therefore K = \frac{(d_1 + d_2)}{\frac{d_1}{K_1} + \frac{d_2}{K_2}}$$

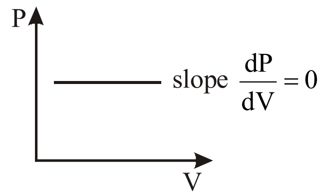
6. (1) When the ring rotates about its axis with a uniform frequency f Hz, the current flowing in the ring is

$$I = \frac{q}{T} = qf$$

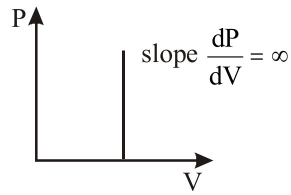
Magnetic field at the centre of the ring is

$$B = \frac{\mu_0 I}{2R} = \frac{\mu_0 qf}{2R}$$

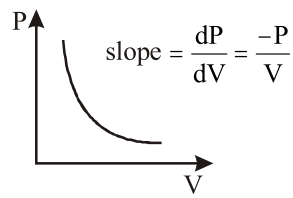
7. (3) P-V indicator diagram for isobaric



P-V indicator diagram for isochoric process



P-V indicator diagram for isothermal process



8. (4) Let 'S' be the distance between two ends 'a' be the constant acceleration

As we know $v^2 - u^2 = 2aS$

or, $aS = \frac{v^2 - u^2}{2}$

Let v_c be velocity at mid point.

Therefore, $v_c^2 - u^2 = 2a\frac{S}{2}$

$$v_c^2 = u^2 + aS$$

$$v_c^2 = u^2 + \frac{v^2 - u^2}{2}$$

$$v_c = \sqrt{\frac{u^2 + v^2}{2}}$$

9. (1) Angular retardation,

$$\alpha = \frac{\omega_2 - \omega_1}{t} = \frac{2\pi(n_2 - n_1)}{t} = \frac{2\pi(0 - 900/60)}{60} = -\frac{\pi}{2}\,\text{rad/s}^2.$$

10. (1) Given : $k_A = 300\,N/m,\ \ k_B = 400\,N/m$

Let when the combination of springs is compressed by force F. Spring A is compressed by x_A. Therefore compression in spring B

$x_B = (8.75 - x_A)$ cm

$F = 300 \times x_A = 400(8.75 - x_A)$

Solving we get, $x_A = 5$ cm

$x_B = 8.75 - 5 = 3.75$ cm

$$\frac{E_A}{E_B} = \frac{\frac{1}{2}k_A(x_A)^2}{\frac{1}{2}k_B(x_B)^2} = \frac{300 \times (5)^2}{400 \times (3.75)^2} = \frac{4}{3}$$

11. (1) From $F = \frac{R}{t^2}v(t) \Rightarrow m\frac{dv}{dt} = \frac{R}{t^2}v(t)$

Integrating both sides $\int \frac{dv}{v(t)} = \int \frac{Rdt}{mt^2}$

$\text{In } v = -\frac{R}{mt}$

$\therefore \ln v \propto \frac{1}{t}$

12. (2) In reverse bias on p-n junction when high voltage is applied, electric break down of junction takes place, resulting large increase in reverse current. This high voltage applied is called zener voltage.

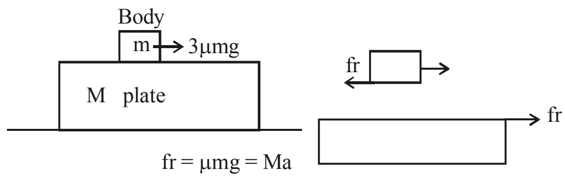
13. (2) If in nuclear reaction binding energy per nucleon increases, energy is released.

14. (2) Let initial amount be 100 gm.

		disintegrated		Left
100 gm	$\xrightarrow{5\text{ days}}$	$\frac{100 \times 10}{100}$	10	90
90	$\xrightarrow{\text{Next 5 days}}$	$\frac{90 \times 10}{100}$	9	81
81	$\xrightarrow{\text{Next 5 days}}$	$\frac{81 \times 10}{100}$	8.1	≈ 73
73	$\xrightarrow{\text{Next 5 days}}$	$\frac{73 \times 10}{100}$	7.3	≈ 66

15. (2) As $\mu_v > \mu_r$ therefore, $v_v < v_r$.

16. (1) $a = \frac{\mu mg}{M}$



fr = μmg = Ma

17. (3) Einstein equation KE_{max} = E – Work function;

$\frac{1}{2}mv^2 = E - W$

Using this concept,

$$\frac{\frac{1}{2}mV_1^2\max}{\frac{1}{2}mV_2^2\max} = \frac{1 - .5}{2.5 - .5} = \frac{1}{4} \text{ or } \frac{V_1\max}{V_2\max} = \frac{1}{2}$$

18. (1) Velocity of electron in n^{th} orbit of hydrogen atom is given by :

$$V_n = \frac{2\pi KZe^2}{nh}$$

Substituting the values we get,

$$V_n = \frac{2.2 \times 10^6}{n}\,\text{m/s} \quad \text{or} \quad V_n \propto \frac{1}{n}$$

As principal quantum number increases, velocity decreases.

19. (1) From $\eta = 1 - \frac{T_2}{T_1},\ \frac{T_2}{T_1} = 1 - \eta = 1 - \frac{1}{6} = \frac{5}{6}$ ...(i)

In 2nd case :

$$\frac{T_2 - 62}{T_1} = 1 - \eta' = 1 - \frac{2}{6} = \frac{2}{3} \quad .....(ii)$$

Using (i), $T_2 - 62 = \frac{2}{3}T_1 = \frac{2}{3} \times \frac{6}{5}T_2 = \frac{4}{5}T_2$

or $\frac{1}{5}T_2 = 62$, $T_2 = 310\,K = 310 - 273 = 37°C$

$$T_1 = \frac{6}{5}T_2 = \frac{6}{5} \times 310 = 372K$$

$$= 372 - 273 = 99°C$$

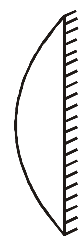
20. (2) Given,
$V_0 = 283$ volt, $\omega = 320$, $R = 5\,\Omega$, $L = 25$ mH, C $= 1000\ \mu$F
$X_L = \omega L = 320 \times 25 \times 10^{-3} = 8\,\Omega$
$$X_C = \frac{1}{\omega C} = \frac{1}{320 \times 1000 \times 10^{-6}} = 3.1\,\Omega$$
Total impedance of the circuit :
$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{25 + (4.9)^2} = 7\,\Omega$$
Phase difference between the voltage and current
$$\tan\phi = \frac{X_L - X_C}{R}$$
$$\tan\phi = \frac{4.9}{5} \approx 1 \Rightarrow \phi = 45°$$

21. (170) From Doppler's effect
$$f(\text{direct}) = f\left(\frac{340}{340-5}\right) = f_1$$
$$f(\text{by wall}) = f\left(\frac{340}{340+5}\right) = f_2$$
Beats $= (f_1 - f_2)$
$$5 = f\left(\frac{340}{340-5} - \frac{340}{340+5}\right)$$
$$\Rightarrow f = 170 \text{ Hz.}$$

22. (6.67) In balance position of bridge,
$$\frac{P}{Q} = \frac{l}{(100-l)}$$
Initially neutral position is 60 cm from A, so
$$\frac{4}{60} = \frac{Q}{40} \Rightarrow Q = \frac{16}{6} = \frac{8}{3}\,\Omega$$
Now, when unknown resistance R is connected in series to P, neutral point is 80 cm from A then,
$$\frac{4+R}{80} = \frac{Q}{20}$$
$$\frac{4+R}{80} = \frac{8}{60}$$
$$R = \frac{64}{6} - 4 = \frac{64-24}{6} = \frac{40}{6}\,\Omega$$
Hence, the value of unknown resistance R is $\frac{20}{3}\,\Omega$

23. (6.275) Pitch $= 1$ mm
Number of divisions on circular scale = 200
$$L.C = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$
$$= \frac{1\text{ mm}}{200} = 0.005\text{ mm} = 0.0005\text{ cm}$$
Diameter of the wire = (Main scale reading + Circular scale reading × L.C.) – zero error
$= 6\text{ mm} + 45 \times 0.005 - (-0.05)$
$= 6\text{ mm} + 0.225\text{ mm} + 0.05\text{ mm} = 6.275\text{ mm}$

24. (20) The silvered plano convex lens behaves as a concave mirror; whose focal length is given by
$$\frac{1}{F} = \frac{2}{f_1} + \frac{1}{f_m}$$



If plane surface is silvered
$$f_m = \frac{R_2}{2} = \frac{\infty}{2} = \infty$$
$$\therefore \frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$= (\mu - 1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \frac{\mu - 1}{R}$$
$$\therefore \frac{1}{F} = \frac{2(\mu-1)}{R} + \frac{1}{\infty} = \frac{2(\mu-1)}{R}$$
$$\Rightarrow F = \frac{R}{2(\mu-1)}$$
Here $R = 20$ cm, $\mu = 1.5$
$$\therefore F = \frac{20}{2(1.5-1)} = 20\text{cm}$$

25. (0.25) Resistors $4\,\Omega$, $6\,\Omega$ and $12\,\Omega$ are connected in parallel, its equivalent resistance (R) is given by

$$\frac{1}{R}=\frac{1}{4}+\frac{1}{6}+\frac{1}{12} \Rightarrow R=\frac{12}{6}=2\Omega$$

Again R is connected to 1.5 V battery whose internal resistance $r = 1\ \Omega$.

Equivalent resistance now,

$R' = 2\Omega + 1\Omega = 3\Omega$

Current, $I_{total}=\frac{V}{R'}=\frac{1.5}{3}=\frac{1}{2}$A

$I_{total}=\frac{1}{2}=3x+2x+x=6x \Rightarrow x=\frac{1}{12}$

$\therefore$ Current through 4Ω resistor $= 3x$

$=3\times\frac{1}{12}=\frac{1}{4}$A

Therefore, rate of Joule heating in the 4Ω resistor

$=I^2R=\left(\frac{1}{4}\right)^2\times 4=\frac{1}{4}=0.25$W

26. (4) Stress $=\frac{\text{Normal force}}{\text{Area}}=\frac{N}{A}=\frac{N}{(2\pi a)b}$

Stress = B×strain

$$\frac{N}{(2\pi a)b}=B\frac{2\pi a\Delta a\times b}{\pi a^2 b}$$

$$\Rightarrow N=B\frac{(2\pi a)^2\Delta ab^2}{\pi a^2 b}$$

Force needed to push the cork.

$f=\mu N=\mu 4\pi b\Delta aB=(4\pi\mu Bb)\Delta a$

27. (40.05) $I_m=\frac{V_m}{R_f+R_L}=\frac{25}{(10+1000)}=24.75$ mA

$I_{dc}=\frac{I_m}{\pi}=\frac{24.75}{3.14}=7.87$ mA

$I_{rms}=\frac{I_m}{2}=\frac{24.75}{2}=12.37$ mA

$P_{dc}=I_{dc}^2\times R_L=(7.87\times10^{-3})^2\times10^3=61.9$ mW

$P_{ac}=I_{rms}^2(R_f+R_L)=(12.37\times10^{-3})^2\times(10+1000)$

$=154.54$ mW

Rectifier efficiency

$$\eta=\frac{P_{dc}}{P_{ac}}\times100=\frac{61.9}{154.54}\times100=40.05\%$$

28. (5.3) This is a example of uniform circular motion.

$\omega=\frac{2\pi}{T}=2\pi n=2\pi\times\frac{7}{100}=0.44$ rad / sec;

$V=R\omega=0.44\times12=5.3$ cm/sec.

29. (23) Given, $B=4\times10^{-5}$ T

$R_E=6.4\times10^6$ m

Dipole moment of the earth M = ?

$$B=\frac{\mu_0}{4\pi}\frac{M}{d^3}$$

$$4\times10^{-5}=\frac{4\pi\times10^{-7}\times M}{4\pi\times\left(6.4\times10^6\right)^3}$$

$\therefore$ $M\cong10^{23}$ Am^2

30. (2) Total energy, $E=\frac{1}{2}m\omega^2a^2$;

$K.E.=\frac{3E}{4}=\frac{1}{2}m\omega^2(a^2-y^2)$.

So, $\frac{3}{4}=\frac{a^2-y^2}{a^2}$ or $y^2=\frac{a^2}{4}$ or $y=\frac{a}{2}$.

CHEMISTRY

31. (1) The order of stability of resonating structures: carrying no charge > carrying minimum charge and each atom having octet complete.

32. (1) Bond order in N_2 and O_2^{2+} is 3 (calculated by energy level diagram)

33. (2) In A—O—H, if EN of 'A' is 2.1 then it will be neutral, as $X_A-X_0=X_0-X_H$. (where X is EN)

34. (2) Correct order is B < Be < O < N.

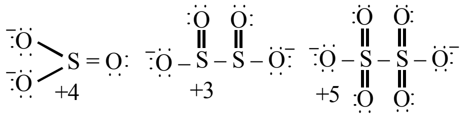
35. (4) In disproportionation reaction, one element of a compound will simultaneously get reduced and oxidised. In ClO_4^-, oxidation number of Cl is + 7 and it can not increase it further. So, ClO_4^- will not get oxidised and so will not undergo disporportionation reaction.

36. (3) $K_a = \frac{[H_3O^+][F^-]}{[HF][H_2O]}$ and

$K_b = \frac{[HF][OH^-]}{[F^-][H_2O]}$.

Therefore, $K_a \times K_b = [H_3O^+][OH^-] = K_w$.

37. (4) The chemical bond method gives the O.N.



38. (3) As Sb_2S_3 is a negative sol, so $Al_2(SO_4)_3$ will be the most effective coagulant due to higher positive charge on Al (Al^{3+}) – Hardy-Schulze rule.

39. (3) *o*-Nitrophenol is not sufficiently strong acid so as to react with $NaHCO_3$.

40. (3) Because the layer of Al_2O_3 (oxide) is inert, insoluble and impervious.

41. (4) Urotropine is used as antibiotic for urinary tract infection.

42. (4) To convert covalent compounds into ionic compounds such as NaCN, Na_2S, NaX, etc.

43. (4)

$$Zn^{2+} + 2NH_4OH \longrightarrow \underset{\text{White ppt.}}{Zn(OH)_2} + 2NH_4^+$$

$$Zn(OH)_2 + 2NH_4OH \longrightarrow \underset{\text{Soluble}}{(NH_4)_2ZnO_2} + 2H_2O$$

$$(NH_4)_2\,ZnO_2 + H_2S \longrightarrow \underset{\text{White ppt.}}{ZnS} + 2NH_4OH$$

44. (2) $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

$5NO_2^- + 2MnO_4^- + 6H^+ \longrightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

45. (3) Formaldehyde can not produce iodoform, as only those compound which contains either $CH_3-\underset{\underset{OH}{|}}{CH}-$ group or $CH_3-\underset{\underset{O}{\|}}{C}-$ group on reaction with potassium iodide and sod. hypochlorite yield iodoform.

46. (1) $2Al + 6HCl \xrightarrow{\Delta,\text{ air}} 2AlCl_3 + 3H_2$

$2Al + 3Cl_2 \longrightarrow 2AlCl_3$

$$Al_2O_3 + 3C + 3Cl_2 \xrightarrow{1000°C} \underset{\text{vapours}}{2AlCl_3} + 3CO$$

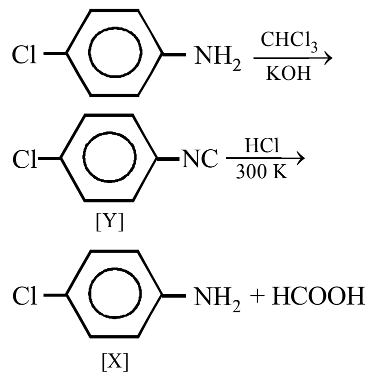
$$\downarrow \text{Cooling}$$

$$\text{Anhydrous } AlCl_3$$

$$AlCl_3.6H_2O \xrightarrow{\Delta} Al(OH)_3 + 3HCl + 3H_2O$$

Thus $AlCl_3$ cannot be obtained by this method.

47. (1)



48. (1) Let the rate law be $r = [A]^x[B]^y$

Divide (3) by (1) $\frac{0.10}{0.10} = \frac{[0.024]^x[0.035]^y}{[0.012]^x[0.035]^y}$

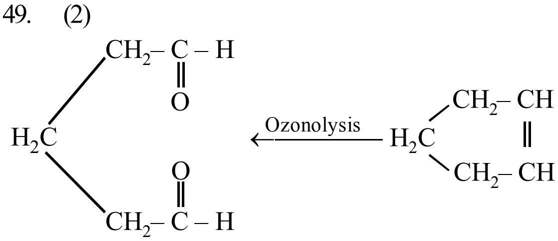
$\therefore 1 = [2]^x, x = 0$

Divide (2) by (3) $\frac{0.80}{0.10} = \frac{[0.024]^x[0.070]^y}{[0.024]^x[0.035]^y}$

$\therefore 8 = (2)^y, y = 3$

Hence rate equation, $R = K[A]^0[B]^3 = K[B]^3$

49. (2)



50. (3) $2H_2O_2 \xrightarrow{\Delta} 2H_2O + O_2$

$2KI + H_2O_2 \longrightarrow 2KOH + I_2$

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

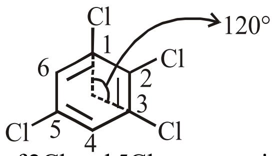
51. (1.24) Energy $= N_A h\nu$

$495.5 = 6.023 \times 10^{23} \times 6.6 \times 10^{-34} \times \nu$

$$\nu = \frac{495.5 \times 10^3}{6.023 \times 10^{23} \times 6.6 \times 10^{-34}} = 12.4 \times 10^{14}$$

$= 1.24 \times 10^{15}\ s^{-1}$.

52. (2.25)



Dipole moments of 2Cl and 5Cl are vectorically cancelled.

It is due 1 Cl and 3 Cl

$$\mu^2 = \mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\theta$$

$= (1.5)^2 + (1.5)^2 + 2 \times 1.5 \times 1.5 \cos 120°$

$= 2.25 + 2.25 + 4.5 \times -\frac{1}{2}$

$= 2.25 + 2.25 - 2.25$

$= 2.25$ D

$\therefore \mu = 2.25$ D

53. (0.5) Moles of H_2SO_4 in 98 mg of H_2SO_4

$= \frac{1}{98} \times 0.098 = 0.001$

Moles of H_2SO_4 remove

$$= \frac{3.01 \times 10^{20}}{6.02 \times 10^{23}} = 0.5 \times 10^{-3} = 0.0005$$

Moles of H_2SO_4 left $= 0.001 - 0.0005$

$= 0.5 \times 10^{-3}$

54. (1.75) According to Boyle's law

$$\frac{V_1}{V_2} = \frac{P_2}{P_1};\ \frac{750}{V_2} = \frac{360}{840}$$

$V_2 = 1750$ mL $= 1.750$ L

55. (5.97) Isoelectric point (pI)

$$= \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{2.34 + 9.60}{2} = 5.97$$

56. (400) Let bond energy of A_2 be x then bond energy of AB is also x and bond energy of B_2 is $x/2$.

Enthalpy of formation of AB is – 100 kJ/mol

$A_2 + B_2 \rightarrow 2\,AB$;

$\frac{1}{2}A_2 + \frac{1}{2}B_2 \rightarrow AB;\ \Delta_f H = -100\ kJ\ mol^{-1}$

or $-100 = \left(\frac{x}{2} + \frac{x}{4}\right) - x$

$\therefore -100 = \frac{2x + x - 4x}{4}$

$x = 400\ kJ\ mol^{-1}$

57. (210) Osmotic pressure (π) of isotonic solutions are equal. For solution of unknown substance C_1(concentration).

$$C_1 = \frac{5.25/M}{V}$$

Where M represents molar mass.

For solution of urea, C_2 (concentration)

$$= \frac{1.5/60}{V}$$

Given, $\pi_1 = \pi_2$

$\because \pi = CRT$

$\therefore C_1RT = C_2RT$ or $C_1 = C_2$ or

$$\frac{5.25/M}{V} = \frac{1.5/60}{V}$$

$\therefore M = 210$ g/mol

58. (0.04) $\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

$-n_3FE_3^\circ = -n_1FE_1^\circ - n_2FE_2^\circ$

$$E_3^\circ = \frac{n_1E_1^\circ + n_2E_2^\circ}{n_3}$$

$$E_3^\circ = \frac{1 \times 0.77 + 2(-0.44)}{3}$$

$$= \frac{0.77 - 0.88}{3} = -\frac{0.11}{3} \approx -0.04\ V$$

59. (6) $2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4 + 2H_2O$

Oxidation number of Mn in K_2MnO_4 is 6

K_2MnO_4; $2 + x - 8 = 0$

$x = +6$

60. (0.241) Solid AB crystallizes as NaCl structure, so it has coordination number 6 and its r^+/r^- ranges from 0.414 – 0.732.
For maximum radius of anion, we have to take the lower limit of the range 0.414 – 0.732. So, $\frac{r^+}{r^-} = 0.414$

$$\Rightarrow \; r^- = \frac{0.100}{0.414}\text{nm} = 0.241\text{ nm}$$

MATHEMATICS

61. (3) $\frac{2}{\cos 2\alpha} = \frac{\tan^2\beta + 1}{\tan\beta} = \frac{1}{\sin\beta\cos\beta}$

$$\Rightarrow \sin 2\beta = \cos 2\alpha = \sin(90-2\alpha)$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{4}$$

62. (2) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{\underline{|n}}{\underline{|r}\,\underline{|n-r}} \cdot \frac{\underline{|r-1}\,\underline{|n-r+1}}{\underline{|n}} = \frac{\underline{|n-r+1}}{\underline{|n-r}}$

$$= \frac{\underline{|n-r+1}}{\underline{|n-r}} = n - r + 1$$

$$\therefore \sum_{r=1}^{5} = n + (n-1) + (n-2) + (n-3) + (n-4)$$
$$= 5n - 10 = 5(n-2)$$

63. (2) ${}^{14}C_7 + \sum_{i=1}^{3} {}^{17-i}C_6$

$$= {}^{14}C_7 + {}^{14}C_6 + {}^{15}C_6 + {}^{16}C_6$$
$$= {}^{15}C_7 + {}^{15}C_6 + {}^{16}C_6$$
$$= {}^{16}C_7 + {}^{16}C_6 = {}^{17}C_7$$

64. (2) Given relation is $e^y(x+1) = 1$.
Differentiating both sides w.r.t. x, we get

$$(x+1)e^y \frac{dy}{dx} + e^y \cdot 1 = 0$$

$$\therefore \frac{dy}{dx} = -\frac{1}{x+1} \quad \text{...(i)}$$

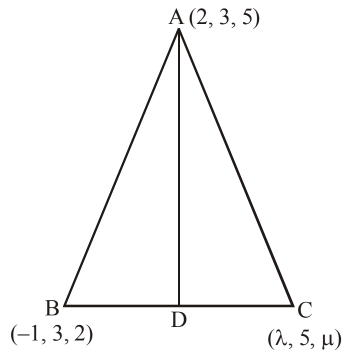
Differentiating again w.r.t. x both sides, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(-\frac{1}{x+1}\right)$$

$$\therefore \frac{d^2y}{dx^2} = \frac{1}{(x+1)^2} \quad \text{...(ii)}$$

From (i) and (ii), we get $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

65. (3) Since AD is the median



$$\therefore \; D = \left(\frac{\lambda - 1}{2}, 4, \frac{\mu + 2}{2}\right)$$

Now, dr's of AD is

$$a = \left(\frac{\lambda - 1}{2} - 2\right) = \frac{\lambda - 5}{2}$$

$$b = 4 - 3 = 1, \; c = \frac{\mu + 2}{2} - 5 = \frac{\mu - 8}{2}$$

Also, a, b, c are dr's
$\therefore \; a = kl, b = km, c = kn$ where $l = m = n$ and $l^2 + m^2 + n^2 = 1$

$$\Rightarrow l = m = n = \frac{1}{\sqrt{3}}$$

Now, $a = 1$, $b = 1$ and $c = 1$
$\Rightarrow \lambda = 7$ and $\mu = 10$

66. (1) $a_1 = 2, b_1 = 2, c_1 = -1$ and $a_2 = 1, b_2 = 2, c_2 = 2$

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{2 + 4 - 2}{\sqrt{4+4+1}\sqrt{1+4+4}} = \pm\frac{4}{9}.$$

67. (3) Contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$
$\therefore$ contrapositive of $(p \vee q) \Rightarrow r$ is
$\sim r \Rightarrow \sim (p \vee q)$ i.e. $\sim r \Rightarrow (\sim p \wedge \sim q)$

68. (1) Let the co-ordiante of other ends are (x, y, z).
The centre of sphere is C(3, 6, 1)

Therefore, $\frac{x+2}{2}=3 \Rightarrow x=4$

$\frac{y+3}{2}=6 \Rightarrow y=9$ and $\frac{z+5}{2}=1 \Rightarrow z=-3$

69. (1) $I=\int \frac{dx}{(x-\beta)\sqrt{(x-\alpha)(\beta-x)}}$

Put $x=\alpha\sin^2\theta+\beta\cos^2\theta$

$dx=2(\alpha-\beta)\sin\theta\cos\theta\,d\theta$

Also, $(x-\alpha)=(\beta-\alpha)\cos^2\theta$

$(x-\beta)=(\alpha-\beta)\sin^2\theta$

$\therefore I=\int \frac{2(\alpha-\beta)\sin\theta\cos\theta\,d\theta}{(\alpha-\beta)\sin^2\theta(\beta-\alpha)\sin\theta\cos\theta}$

$=\frac{2}{\beta-\alpha}\int\frac{d\theta}{\sin^2\theta}=\frac{2}{\beta-\alpha}\int \text{cosec}^2\theta\,d\theta$

$=\frac{2}{\beta-\alpha}(-\cot\theta)+C=\frac{2}{\alpha-\beta}\cot\theta+C$

Now, $x=\alpha\sin^2\theta+\beta\cos^2\theta$

$\Rightarrow x\,\text{cosec}^2\theta=\alpha+\beta\cot^2\theta$

$\Rightarrow x(1+\cot^2\theta)=\alpha+\beta\cot^2\theta$

$\therefore \cot\theta=\sqrt{\frac{x-\alpha}{\beta-x}}$;

$\therefore I=\frac{2}{\alpha-\beta}\sqrt{\frac{x-\alpha}{\beta-x}}+C$

70. (4) Given planes are
$P: x+y-2z+7=0$
$Q: x+y+2z+2=0$
and $R: 3x+3y-6z-11=0$
Consider Plane P and R.
Here $a_1=1, b_1=1, c_1=-2$
and $a_2=3, b_2=3, c_2=-6$
Since, $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}=\frac{1}{3}$
therefore P and R are parallel.

71. (1) $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+....\infty=\frac{\pi^4}{90}$

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+.....+\infty+\frac{1}{2^4}\left(\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+....\infty\right)$

$=\frac{\pi^4}{90}$

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\frac{1}{7^4}+....+\infty+\frac{1}{16}\times\frac{\pi^4}{90}=\frac{\pi^4}{90}$

$\therefore \frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\frac{1}{7^4}+....+\infty=\frac{\pi^4}{90}-\frac{1}{16}\left(\frac{\pi^4}{90}\right)$

$=\frac{15}{16}\left(\frac{\pi^4}{90}\right)=\frac{\pi^4}{96}.$

72. (2) We have, $f(x)=\exp\left(\sqrt{5x-3-2x^2}\right)$

i.e., $f(x)=e^{\sqrt{5x-3-2x^2}}$

For domain of $f(x)$, $5x-3-2x^2$ should be +ve.

i.e., $5x-3-2x^2\ge 0$

$\Rightarrow 2x^2-5x+3\le 0$

(By taking –ve sign common)

$\Rightarrow 2x(x-1)-3(x-1)\le 0$

$\Rightarrow (2x-3)(x-1)\le 0$

$\Rightarrow 2x-3\le 0$ or $x-1\ge 0$

$\Rightarrow x\le\frac{3}{2}$ or $x\ge 1$

$\therefore 1\le x\le\frac{3}{2}$ i.e., $x\in\left[1,\frac{3}{2}\right]$

Hence, domain of the given function is $\left[1,\frac{3}{2}\right]$.

73. (1) Given determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix}=0$$

$$\Rightarrow \begin{vmatrix} 1+a^2-2a\cos x & a & a^2 \\ 0 & \cos nx & \cos(n+1)x \\ 0 & \sin nx & \sin(n+1)x \end{vmatrix}=0$$

By applying $C_1\to C_1+C_3-2\cos x\,C_2$

By expanding

$(1 + a^2 - 2a\cos x)\,[\cos nx \sin (n+1)x - \sin nx \cos (n+1)x] = 0$

Now, $(1 + a^2 - 2a\cos x)\sin(n+1-n)x = 0$

$\Rightarrow (1+a^2-2a\cos x)\sin x = 0$

$\sin x = 0$ or $\cos x = \dfrac{1+a^2}{2a}$

As $a \neq 1 \;\therefore \left(\dfrac{1+a^2}{2a}\right) > 1$

$\Rightarrow \cos x > 1$ It is not possible.

$\therefore \sin x = 0$

74. (4) We have, $a = \cos\left(\dfrac{2\pi}{7}\right) + i\sin\left(\dfrac{2\pi}{7}\right)$

$\Rightarrow \; a^7 = \left[\cos\left(\dfrac{2\pi}{7}\right) + i\sin\left(\dfrac{2\pi}{7}\right)\right]^7$

$= \cos 2\pi + i\sin 2\pi = 1$ …(i)

Let $S = \alpha + \beta = (a + a^2 + a^4) + (a^3 + a^5 + a^6)$

$[\because \; \alpha = a + a^2 + a^4, \beta = a^3 + a^5 + a^6]$

$\Rightarrow \; S = a + a^2 + a^3 + a^4 + a^5 + a^6$

$= \dfrac{a(1-a^6)}{1-a}$

$\Rightarrow \; S = \dfrac{a-a^7}{1-a} = \dfrac{a-1}{1-a} = -1$ …(ii)

Let $P = \alpha\beta = (a + a^2 + a^4)(a^3 + a^5 + a^6)$

$= a^4 + a^6 + a^7 + a^5 + a^7 + a^8 + a^7 + a^9 + a^{10}$

$= a^4 + a^6 + 1 + a^5 + 1 + a + 1 + a^2 + a^3$ [from Eq. (i)]

$= 3 + (a + a^2 + a^3 + a^4 + a^5 + a^6) = 3 + S$

$= 3 - 1 = 2$ [from Eqn. (ii)]

Required equation is, $x^2 - Sx + P = 0$

$\Rightarrow \; x^2 + x + 2 = 0$

75. (1) We have,

$$AB = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta\cos^2\phi + \cos\theta\cos\phi\sin\theta\sin\phi & \cos^2\theta\cos\phi\sin\phi + \cos\theta\sin\theta\sin^2\phi \\ \cos\theta\sin\theta\cos^2\phi + \sin^2\theta\cos\phi\sin\phi & \cos\theta\cos\phi\sin\theta\sin\phi + \sin^2\theta\sin^2\phi \end{bmatrix}$$

$$= \cos(\theta-\phi)\begin{bmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi \\ \sin\theta\cos\phi & \sin\theta\sin\phi \end{bmatrix}$$

Since, $AB = 0$, $\therefore \; \cos(\theta-\phi) = 0$

$\therefore \; \theta - \phi$ is an odd multiple of $\dfrac{\pi}{2}$

76. (3) $f(x) = xe^{x(1-x)}, x \in R$

$f'(x) = e^{x(1-x)}.\left[1 + x - 2x^2\right]$

$= -e^{x(1-x)}.\left[2x^2 - x - 1\right]$

$= -2e^{x(1-x)}.\left[\left(x + \dfrac{1}{2}\right)(x-1)\right]$

$f'(x) = -2e^{x(1-x)}.A$

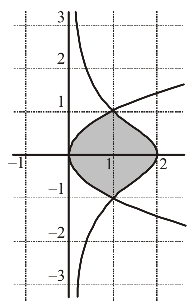
where $A = \left(x + \dfrac{1}{2}\right)(x-1)$

Now, exponential function is always +ve and $f'(x)$ will be opposite to the sign of A which is –ve in $\left[-\dfrac{1}{2}, 1\right]$

Hence, $f'(x)$ is +ve in $\left[-\dfrac{1}{2}, 1\right]$

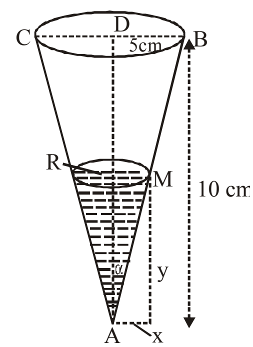
$\therefore \; f(x)$ is increasing on $\left[-\dfrac{1}{2}, 1\right]$

77. (1) Solving curves, we get point of intersections $(1, \pm 1)$. Required area



$$= 2\int_0^1 \left(\frac{2}{1+y^2} - y^2\right)dy = \pi - \frac{2}{3}.$$

78. (4) Let y be the level of water at time t and x the radius of the surface and V, the volume of water.



We know that the volume of cone

$= \frac{1}{3}\pi\ (\text{radius})^2 \times \text{height}$

$\therefore\ V = \frac{1}{3}\pi x^2 y.$ Let $\angle BAD = \alpha$

$\Rightarrow\ \tan\alpha = \frac{BD}{AD} = \frac{5}{10} = \frac{1}{2}.$

Again, from right angled ΔAMR, we have

$\tan\alpha = \frac{MR}{AR} = \frac{x}{y}; \Rightarrow \frac{1}{2} = \frac{x}{y};\quad \therefore x = \frac{y}{2}.$

$\therefore V = \frac{1}{3}\pi x^2 y = \frac{1}{3}\pi\left(\frac{y}{2}\right)^2 . y = \frac{\pi}{12}y^3$ ......(1).

By question, the rate of change of volume

$= \frac{dV}{dt} = 4$ cub.cm./min.

We have to find out the rate of increase of water-level i.e. $\frac{dy}{dt}$.

Differentiating (1) with respect to t, we get

$\frac{dV}{dt} = \frac{\pi}{12}.3y^2.\frac{dy}{dt};\quad \therefore 4 = \frac{\pi}{4}y^2.\frac{dy}{dt};\quad \therefore\ \frac{dy}{dt} = \frac{16}{\pi y^2}.$

When y = 6 cm,

$\frac{dy}{dt} = \frac{16}{\pi 6^2} = \frac{4}{9\pi}$ cub.cm./ min.

79. (4) Slope of the tangent to $4x^2 - 9y^2 = 36$ is given by

$8x - 18y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{4x}{9y}$ or $m_1 = \frac{4x}{9y}$

Slope of the straight line, $5x + 2y - 10 = 0$ is

$m_2 = -\frac{5}{2}$

Therefore, for the perpendicularity, $m_1 m_2 = -1$

Now, $\frac{4x}{9y} \times \frac{-5}{2} = -1 \Rightarrow y = \frac{10x}{9}.$

Putting $y = \frac{10x}{9}$ in $4x^2 - 9y^2 = 36$

gives imaginary roots resulting in no tangents.

80. (1) $I = \int_{\log\sqrt{\pi/2}}^{\log\sqrt{\pi}} e^{2x}\sec^2\left(\frac{1}{3}e^{2x}\right)dx$

Put $e^{2x} = t \Rightarrow 2e^{2x}dx = dt$

When $x = \log\sqrt{\pi/2}$, $t = e^{2\log\sqrt{\pi/2}}$

$= e^{\log\pi/2} = \frac{\pi}{2}$

When $x = \log\sqrt{\pi}$, $t = e^{2\log\sqrt{\pi}} = \pi$

$\therefore\ I = \int_{\frac{\pi}{2}}^{\pi}\frac{1}{2}\sec^2\left(\frac{1}{3}t\right)dt$

$= \frac{1}{2}.\frac{1}{\frac{1}{3}}\left[\tan\frac{t}{3}.\right]_{\pi/2}^{\pi}$

$= \frac{3}{2}\left[\tan\frac{\pi}{3} - \tan\frac{\pi}{6}\right]$

$= \frac{3}{2}\left[\sqrt{3} - \frac{1}{\sqrt{3}}\right] = \sqrt{3}$

81. (0) If the G.P be $a, ar, ar^2, ...$ then $a_n = ar^{n-1}$

$$D = \begin{vmatrix} \log a + (n-1)\log r & \log a + n\log r & \log a + (n+1)\log r \\ \log a + n\log r & \log a + (n+1)\log r & \log a + (n+2)\log r \\ \log a + (n+1)\log r & \log a + (n+2)\log r & \log a + (n+3)\log r \end{vmatrix}$$

$R_3 \to R_3 - R_2$ and $R_2 \to R_2 - R_1$ gives,

$$= \begin{vmatrix} \log a + (n-1)\log r & \log a + n\log r & \log a + (n+1)\log r \\ \log r & \log r & \log r \\ \log r & \log r & \log r \end{vmatrix}$$

= 0, since R_2 and R_3 are identical.

82. (0.80) Total no. of arrangements of the letters of the word UNIVERSITY is $\frac{10!}{2!}$.

No. of arrangements when both I's are together = 9!

So. the no. of ways in which 2 I's do not together $= \frac{10!}{2!} - 9!$

$\therefore$ Required probability

$= \frac{\frac{10!}{2!} - 9!}{\frac{10!}{2!}} = \frac{10! - 9!\,2!}{10!}$

$= \frac{10 \times 9! - 9!.2!}{10!} = \frac{9![10-2]}{10 \times 9!}$

$= \frac{8}{10} = \frac{4}{5} = 0.80$

83. (0.25) $n(S)$ = the area of the circle of radius r

$n(E)$ = the area of the circle of radius $\frac{r}{2}$

$\therefore$ The probability $= \frac{n(E)}{n(S)} = \frac{\pi\left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}.$

84. (0.33) $P(\bar{E}E) + P(\bar{E}\bar{E}\bar{E}\bar{E}\bar{E}E) + ...\infty$

$$= \frac{5}{6}\times\frac{1}{6} + \left(\frac{5}{6}\right)^4 \times \frac{1}{6} + \left(\frac{5}{6}\right)^8 \frac{1}{6} + ...\infty$$

$$= \frac{5}{36}\left[1 + \left(\frac{5}{6}\right)^3 + ...\infty\right] = \frac{30}{91}$$

85. (7.00) Given equation of ellipse is

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

eccentricity $= e = \sqrt{1 - \frac{b^2}{16}}$

foci: $\pm ae = \pm 4\sqrt{1 - \frac{b^2}{16}}$

Equation of hyperbola is $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$

$$\Rightarrow \frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{81}{25}} = 1$$

Eccentricity $= e = \sqrt{1 + \frac{81}{25} \times \frac{25}{144}} = \sqrt{1 + \frac{81}{144}}$

$$= \sqrt{\frac{225}{144}} = \frac{15}{12}$$

foci: $\pm ae = \pm\frac{12}{5} \times \frac{15}{12} = \pm 3$

Since, foci of ellipse and hyperbola coincide

$\therefore \pm 4\sqrt{1 - \frac{b^2}{16}} = \pm 3 \Rightarrow b^2 = 7$

86. (1.00) For $x > 10$, $f(x) = x - 2$.

Therefore, $g(x) = x - 2 - 2 = x - 4$

$\therefore g'(x) = 1.$

87. (3.00) $\because$ A.M. $\geq$ G.M. $\Rightarrow \frac{a+b+c}{3} \geq (abc)^{1/3}$

$\Rightarrow a + b + c \geq 3(abc)^{1/3}$ (1)

But given : ab^2c^3, $a^2b^3c^4$, $a^3b^4c^5$ are also in AP.

($\because abc \neq 0$)

$\Rightarrow 2abc = 1 + a^2b^2c^2 \Rightarrow (abc - 1)^2 = 0$

$\therefore abc = 1$

Now from equation (1), $a + b + c \geq 3(1)^{1/3}$

$\Rightarrow (a + b + c) \geq 3$

Hence, minimum value of is $a + b + c$ is 3.

88. (1.50) Sum of coefficients is obtained by simply putting x = 1 in the expression

So, sum of coefficients $= \left(\frac{2}{3}\right)^{199} \times \left(\frac{3}{2}\right)^{200}$

$$= \left(\frac{3}{2}\right)^{-199} \times \left(\frac{3}{2}\right)^{200} = \frac{3^{200-199}}{2^{200-199}} = \frac{3}{2} = 1.50$$

89. (14.00) Since $0 < y < x < 2y$

$\therefore y > \frac{x}{2} \Rightarrow x - y < \frac{x}{2}$

$\therefore x - y < y < x < 2x + y$

Hence median $= \frac{y + x}{2} = 10$

$\Rightarrow x + y = 20$ ...(i)

And range $= (2x + y) - (x - y) = x + 2y$

But range = 28

$\therefore x + 2y = 28$ ...(ii)

From equations (i) and (ii),

$x = 12, y = 8$

$\therefore$ Mean

$$= \frac{(x - y) + y + x + (2x + y)}{4} = \frac{4x + y}{4}$$

$$= x + \frac{y}{4} = 12 + \frac{8}{4} = 14$$

90. (3.20) $\int \frac{dx}{\cos^3 x\sqrt{4\sin x \cos x}}$

$$= \int \frac{dx}{2\cos^4 x\sqrt{\tan x}}$$

Let $\tan x = t^2 \Rightarrow \sec^2 x = 1 + t^4$

$\sec^2 x\, dx = 2t\, dt$

$$= \int \frac{\sec^4 x\, dx}{2\sqrt{\tan x}} = \int \frac{\sec^2 x(\sec^2 x\, dx)}{2\sqrt{\tan x}}$$

$$= \int \frac{(1 + t^4)2t\ dt}{2t} = \int (1 + t^4)\ dt$$

$$= t + \frac{t^5}{5} + k$$

$$= \sqrt{\tan x} + \frac{1}{5}\tan^{5/2} x + k\left[t = \sqrt{\tan x}\right]$$

$$A = \frac{1}{2},\ B = \frac{5}{2},\ C = \frac{1}{5}$$

$$A + B + C = \frac{16}{5} = 3.20$$

Mock Test-2

ANSWER KEY											
1	(1)	**16**	(1)	**31**	(3)	**46**	(3)	**61**	(2)	**76**	(2)
2	(3)	**17**	(4)	**32**	(3)	**47**	(2)	**62**	(1)	**77**	(3)
3	(4)	**18**	(1)	**33**	(2)	**48**	(2)	**63**	(3)	**78**	(4)
4	(1)	**19**	(1)	**34**	(4)	**49**	(3)	**64**	(3)	**79**	(3)
5	(1)	**20**	(1)	**35**	(1)	**50**	(2)	**65**	(3)	**80**	(2)
6	(1)	**21**	(45°)	**36**	(4)	**51**	(546)	**66**	(1)	**81**	(16)
7	(3)	**22**	(7.6)	**37**	(4)	**52**	(8)	**67**	(1)	**82**	(3)
8	(4)	**23**	(2)	**38**	(4)	**53**	(1.47)	**68**	(1)	**83**	(0.5)
9	(3)	**24**	(45.5)	**39**	(3)	**54**	(2)	**69**	(1)	**84**	(2008)
10	(1)	**25**	(3)	**40**	(2)	**55**	(0.2303)	**70**	(1)	**85**	(18)
11	(3)	**26**	(5)	**41**	(4)	**56**	(4)	**71**	(1)	**86**	(12)
12	(3)	**27**	(0.5)	**42**	(2)	**57**	(2)	**72**	(1)	**87**	(0.75)
13	(4)	**28**	(6)	**43**	(4)	**58**	(964)	**73**	(3)	**88**	(0.38)
14	(3)	**29**	$(2^{\frac{-2}{3}})$	**44**	(2)	**59**	(60)	**74**	(1)	**89**	(336.00)
15	(3)	**30**	(5)	**45**	(3)	**60**	(10.85)	**75**	(1)	**90**	(9)

Solutions

PHYSICS

1. (1) Least count of vernier callipers
= value of one division of main scale
– value of one division of vernier scale
Now, $N \times a = (N+1)\,a'$
(a' = value of one division of vernier scale)
$$a' = \frac{N}{N+1}a$$
$\therefore$ Least count $= a - a' = \dfrac{a}{N+1}$

2. (3) Let mass of smaller sphere (which has to be removed) is m
Radius $= \dfrac{R}{2}$ (from figure)
$$\frac{M}{\frac{4}{3}\pi R^3} = \frac{m}{\frac{4}{3}\pi\left(\frac{R}{2}\right)^3}$$
$$\Rightarrow m = \frac{M}{8}$$
Mass of the left over part of the sphere
$$M' = M - \frac{M}{8} = \frac{7}{8}M$$
Therefore gravitational field due to the left over part of the sphere
$$= \frac{GM'}{x^2} = \frac{7}{8}\frac{GM}{x^2}$$

3. (4)

4. (1) Charge conservation is violated in [2, 3, 4], nucleon conservation is violated in (4), (1) works.

5. (1) $C = \sin^{-1}\left(\dfrac{1}{\mu}\right)$ and $\mu \propto \dfrac{1}{\lambda}$
Yellow, orange and red have higher wavelengths than green, so μ will be less for these rays, consequently critical angle for these rays will be high, hence if green is just totally internally reflected then yellow, orange and red rays will emerge out.

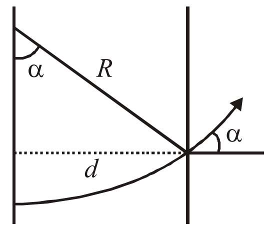
6. (1) The conductivity of semiconductor
$\sigma = e\,(\eta_e\mu_e + \eta_h\mu_h)$
$= 1.6 \times 10^{-19}(5 \times 10^{18} \times 2 + 5 \times 10^{19} \times 0.01)$
$= 1.6 \times 1.05 = 1.68$

7. (3) $E_1 = \dfrac{1}{2}kx^2,\ E_2 = \dfrac{1}{2}ky^2,$

$$E = \frac{1}{2}k(x+y)^2 = E_1 + E_2 + 2\sqrt{E_1E_2}$$

$$= 2 + 8 + 2\sqrt{16} = 18J$$

8. (4) From figure, $\sin\alpha = d/R$



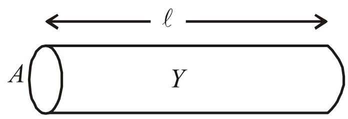
And we know, $\frac{mv^2}{R} = qvB$

$\Rightarrow \quad R = \frac{mv}{qB}$

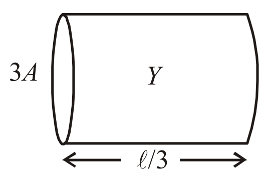
$\therefore \quad \sin\alpha = \frac{dqB}{mv}$

$\sin\alpha = Bd\sqrt{\frac{q}{2mV}} \quad \left[\because qV = \frac{1}{2}mv^2\right]$

9. (3)



Wire (1)



Wire (2)

As shown in the figure, the wires will have the same Young's modulus (same material) and the length of the wire of area of cross-section $3A$ will be $\ell/3$ (same volume as wire 1).

For wire 1, $Y = \frac{F/A}{\Delta x/\ell}$ ...(i)

For wire 2, $Y = \frac{F'/3A}{\Delta x/(\ell/3)}$ ...(ii)

From (i) and (ii), $\frac{F}{A}\times\frac{\ell}{\Delta x} = \frac{F'}{3A}\times\frac{\ell}{3\Delta x}$

$\Rightarrow F' = 9F$

10. (1) Weight of cylinder = upthrust due to both liquids

$$V \times D \times g = \left(\frac{A}{5}\times\frac{3}{4}L\right)\times d\times g + \left(\frac{A}{5}\times\frac{L}{4}\right)\times 2d\times g$$

$$\Rightarrow \left(\frac{A}{5}\times L\right)\times D\times g = \frac{A\times L\times d\times g}{4} \Rightarrow \frac{D}{5} = \frac{d}{4}$$

$$\therefore D = \frac{5}{4}d$$

11. (3) Shift $= \frac{\beta(\mu-1)}{\lambda}t_1 - \frac{\beta(\mu-1)}{\lambda}t_2$

$$= \frac{\beta(\mu-1)}{\lambda}(t_1 - t_2)$$

12. (3) $I = \frac{m\ell^2}{12}$;

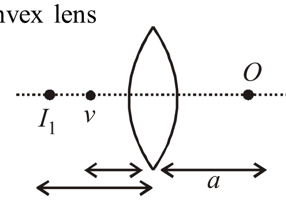
Restoring torque

$$= qE\ell\sin\theta \approx qE\ell\,\theta = \frac{d^2\theta}{dt^2}I$$

$$\therefore \omega = \sqrt{\frac{qE\ell}{I}} = \sqrt{\frac{qE\ell\times 12}{m\ell^2}} = \frac{2\pi}{T}$$

$$\therefore T = \pi\sqrt{\frac{m\ell}{3qE}}$$

13. (4) When object is kept at a distance 'a' from thin convex lens

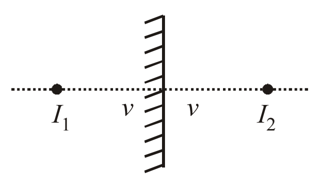


By lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

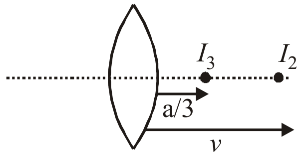
$$\frac{1}{V} - \frac{1}{(-a)} = \frac{1}{f}$$

or, $\frac{1}{v} = \frac{1}{f} - \frac{1}{a}$ ...(i)

Mirror forms image at equal distance from mirror



Now, again from lens formula



$$\frac{3}{a}-\frac{1}{V}=\frac{1}{f}$$

$$\frac{3}{a}-\frac{1}{f}+\frac{1}{a}=\frac{1}{f} \text{ [From eqn. (i)]}$$

Hence, $a = 2f$

14. (3) The equation of wave moving in negative x-direction, assuming origin of position at x = 2 and origin of time (i.e. initial time) at t = 1 sec.

$$y = 0.1 \sin(2\pi t + 4x)$$

Shifting the origin of position to left by 2m, that is, to x = 0. Also shifting the origin of time backwards by 1 sec, that is to t = 0 sec.

$$y = 0.1 \sin(2\pi(t-1)+4(x-2))$$

15. (3) Acceleration (a) $=\dfrac{v-u}{t}$

$$=\frac{(0-50)}{(10-0)}=-5 \text{ m/s}^2$$

u = 50 m/s

$\therefore$ v = u + at = 50 – 5t

Veocity in first two seconds t = 2

$$v_{(\text{at } t=2)}=40 \text{ m/s}$$

From work-energy theorem,

$$\Delta K.E. = W = \frac{1}{2}(40^2-50^2)\times 10 = -4500 \text{ J}$$

16. (1) For a particle undergoing SHM with an amplitude A and angular frequency ω, the maximum acceleration = $\omega^2 A$

Here the maximum force on the particle = QE_0

$$\therefore \text{ maximum acceleration} = \frac{QE_0}{m}=\omega^2 A$$

$$\therefore A=\frac{QE_0}{m\omega^2}$$

17. (4) $\dfrac{1}{2}mv^2=\dfrac{hc}{\lambda}-\phi$

$$\frac{1}{2}mv'^2=\frac{hc}{(3\lambda/4)}-\phi=\frac{4hc}{3\lambda}-\phi.$$

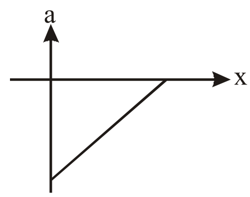
Clearly, $v' > \sqrt{\dfrac{4}{3}}v$

18. (1) The linear relationship between v and x is $v = -mx + C$ where m and C are positive constants.

$$\therefore \text{ Acceleration, } a = v\frac{dv}{dx}=-m(-mx+C)$$

$$\therefore a = m^2x - mC$$

Hence the graph relating a to x is



19. (1) Divide the ring into infinitely small lengths of mass dm_1. Even though mass distribution is non-uniform, each mass dm_1 is at same distance R from origin.

$\therefore$ MI of ring about z-axis is

$$= dm_1R^2 + dm_2R^2 + \ldots\ldots\ldots + dm_nR^2 = MR^2$$

20. (1) $E=\dfrac{-dV}{dx}$ $V = 5x^2+10x-9,$

$$E=\frac{-d}{dx}(5x^2+10x-9)=-(10x+10)$$

On putting value of x in it we get,

$$E = -(10\times 1+10) = -20 \text{ V/m}$$

21. (45°) If T is the tension i n the string , Then

$T = mg$ (For outer masses)

$2T\cos\theta=\sqrt{2}mg$ (For inner mass)

Eliminating T, we get $2(mg)\cos\theta=\sqrt{2}mg$

or $\cos\theta = 1/\sqrt{2} \Rightarrow \theta = 45°$

22. (7.6) Here I = 4A

$$\theta = 30° = \left(\frac{30\times\pi}{180}\right)=\frac{\pi}{6}$$

$$\text{Now, } k=\frac{I}{\theta}=\frac{4}{\left(\frac{\pi}{6}\right)}=\frac{4\times 6\times 7}{22}=\frac{2\times 6\times 7}{11}$$

$$=\frac{84}{11}=7.6 \text{ A/rad}$$

23. (2) Centre of mass of the rod is given by:

$$x_{cm}=\frac{\int_0^L (ax+\frac{bx^2}{L})dx}{\int_0^L (a+\frac{bx}{L})dx}=\frac{\frac{aL^2}{2}+\frac{bL^2}{3}}{aL+\frac{bL}{2}}=\frac{L\left(\frac{a}{2}+\frac{b}{3}\right)}{a+\frac{b}{2}}$$

Now $\frac{7L}{12} = \frac{\frac{a}{2}+\frac{b}{3}}{a+\frac{b}{2}}$

On solving we get, $b = 2a$

24. (45.5) Given : $h_R = 50$ m

$h_T = 32$ m

Maximum distance, $d_M = ?$

Applying, $d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$

$= \sqrt{2\times 6.4\times 10^6\times 32} + \sqrt{2\times 6.4\times 10^6\times 50}$

$= 45.5$ km

25. (3) Given : m = 0.160 kg

$\theta = 60°$; v = 10 m/s

Angular momentum $L = \vec{r}\times m\vec{v}$

$= H\,mv\cos\theta$

$= mv\frac{v^2\sin^2\theta}{2g}\cos\theta \quad \left[H = \frac{v^2\sin^2\theta}{2g}\right]$

$= \frac{10^2\times\sin^2 60°\times\cos 60°\times 0.160\times 10}{2\times 10}$

$= 3$ kg m^2/s

26. (5) Power factor $_{(old)}$

$= \frac{R}{\sqrt{R^2+X_L^2}} = \frac{R}{\sqrt{R^2+(2R)^2}} = \frac{R}{\sqrt{5}R}$

Power factor$_{(new)}$

$= \frac{R}{\sqrt{R^2+(X_L-X_C)^2}} = \frac{R}{\sqrt{R^2+(2R-R)^2}}$

$= \frac{R}{\sqrt{2}R}$

$\therefore \frac{\text{New power factor}}{\text{Old power factor}} = \frac{\frac{R}{\sqrt{2}R}}{\frac{R}{\sqrt{5}R}} = \sqrt{\frac{5}{2}}$

27. (0.5) As the surrounding is identical, vessel is identical time taken to cool both water and liquid (from 30°C to 25°C) is same 2 minutes, therefore

$\left(\frac{dQ}{dt}\right)_{water} = \left(\frac{dQ}{dt}\right)_{liquid}$

or, $\frac{(m_w C_w + W)\Delta T}{t} = \frac{(m_\ell C_\ell + W)\Delta T}{t}$

(W = water equivalent of the vessel)

or, $m_w C_w = m_\ell C_\ell$

$\therefore$ Specific heat of liquid, $C_\ell = \frac{m_W C_W}{m_\ell}$

$= \frac{50\times 1}{100} = 0.5$ kcal / kg

28. (6) From question,

$B_0 = 20$ nT $= 20\times 10^{-9}$T

$\vec{E}_0 = \vec{B}_0\times\vec{C}$

$|\vec{E}_0| = |\vec{B}_0|.|\vec{C}| = 20\times 10^{-9}\times 3\times 10^8 = 6$ V/m.

($\because$ velocity of light in vacuum $C = 3\times 10^8$ ms^{-1})

29. ($2^{\frac{-2}{3}}$) For isothermal process :

$PV = P_i.2V$

$P = 2P_i$ ...(i)

For adiabatic process

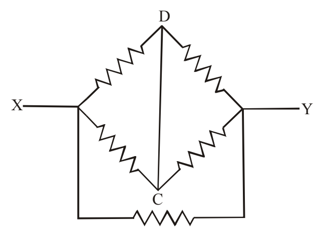
$PV^\gamma = P_a(2V)^\gamma$

($\because$ for monatomic gas $\gamma = 5/3$)

or, $2P_iV^{\frac{5}{3}} = P_a(2V)^{\frac{5}{3}}$ [From (i)]

$\Rightarrow \frac{P_a}{P_i} = \frac{2}{2^{\frac{5}{3}}} \Rightarrow \frac{P_a}{P_i} = 2^{\frac{-2}{3}}$

30. (5)



Equivalent resistance = 5Ω

CHEMISTRY

31. (3) $Cd(s) + 2Ag^+(aq) \longrightarrow Cd^{2+}(aq) + Ag(s)$,

$E_{cell} = E^\circ_{cell} - \frac{0.059}{2}\log\frac{[Cd^{2+}]}{[Ag^+]^2}$

32. (3) Gas equation is $PV = \frac{m}{M}RT$ ...(i)

Again $\frac{P}{2}V = \frac{m_1}{M}R.\frac{2}{3}T$ ...(ii)

Divide (i) by (ii)

$$2 = \frac{m}{m_1} \times \frac{3}{2}$$

$\therefore m_1 = \frac{3}{4} m$. Gas escaped is then $= \frac{1}{4} m$

33. (2)

Molecule	Hybridization
SO_3	*sp*^{2}
C_2H_2	*sp*
C_2H_4	*sp*^{2}
CH_4	*sp*^{3}
CO_2	*sp*

34. (4) (1) Tranquillizers are the substances used for the treatment of mental diseases. These act on higher centres of the central nervous system. These are also called psychotherapeutic drugs.

(2) Vitamin C, Vitamin E and β-carotene are antioxidants.

(substances that act against oxidants).

(3) Sodium or potassium salt of a long chain fatty acid is called soap.

35. (1)

$$E = E^\circ - \frac{0.0591}{2} \log \frac{1}{[Cl^-]^2}; \text{as } [Cl^-]\uparrow, E\uparrow$$

36. (4) Ester of dicarboxylic acids undergo an intramolecular version of the claisen condensation when a five or six membered ring can be formed. This reaction is an example of a Dieckmann's condensation

37. (4) (1) The change of atmospheric temperature with altitude is called the lapse rate.

(2) The gases responsible for greenhouse effects are CO_2, water vapour, CH_4 and ozone.

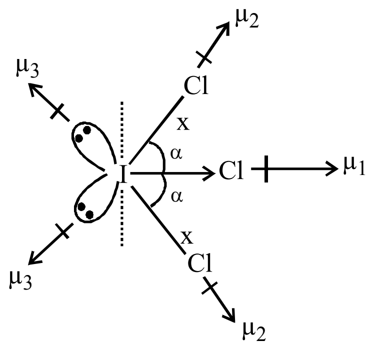
(3) The order of contribution to the acid rain is

$H_2SO_4 > HNO_3 > HCl$

38. (4) $CH_3—CH_2—\underset{\underset{Cl}{|}}{CH}—CH{=}CH_2 \xrightarrow[Pt]{H_2}$

$Et—\underset{\underset{Cl}{|}}{CH}—Et$ optically inactive.

39. (3)



($\mu_D \neq 0$)

Molecule is polar and planar.

Both ∠Cl I Cl are equal

Equatorial I – Cl bond has more *s*-character than axial I–Cl bond.

40. (2) Since the compound is formed by hydration of an alkene, to get the structure of alkene remove a molecule of water from the alcohol.

$$\underset{\text{Isopropyl alcohol}}{CH_3\underset{\underset{OH}{|}}{C}HCH_3} \xrightarrow{-H_2O} \underset{\text{Propylene}}{CH_2 = CHCH_3}$$

41. (4) Na reacts vigorously with water (exothermic process).

42. (2) (1) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

(2) $(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + H_2SO_4$

(3) $2NH_4ClO_4 \xrightarrow{\Delta} N_2 + Cl_2 + 2O_2 + 4H_2O$

(4) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

43. (4) In first case the given compounds have same anion but different cations having different charge hence they will precipitate negatively charged sol i.e. 'A'.

In second case the given compounds have similar cation but different anion with different charge. Hence they will precipitate positively charged sol. i.e. 'B'.

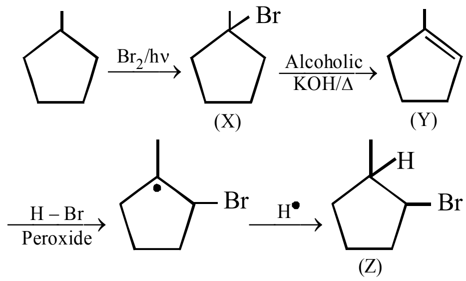
44. (2) $HC \equiv CH \xrightarrow[H_2SO_4]{HgSO_4}$

$CH_3 - CHO \xrightarrow{\text{aldol reaction}}$

$$\underset{3}{CH_3 - \overset{\overset{OH}{|}}{CH}} - \underset{2}{CH_2} - \underset{1}{CHO}$$

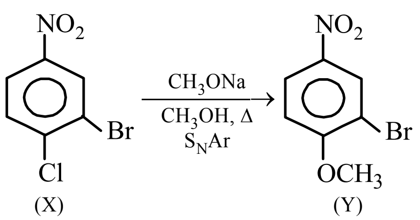
3-hydroxy butanal

45. (3)



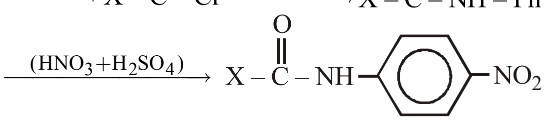
46. (3) (1) When a dilute solution of an acid is added to a dilute solution of a base, neutralization reaction takes place.
(2) In acid-base titrations, at the end point, the amount of acid becomes chemically equilvalent to the amount of base present.
(3) In the case of a strong acid and a strong base titration, at the end point the solution becomes neutral (i.e., pH = 7)
(4) In acid-base titrations the end point is determined by the hydrogen ion concentration of the solution.

47. (2)



48. (2) $X-\overset{\overset{\displaystyle O}{\|}}{C}-OH_3 \xrightarrow[H^{\oplus}]{NaOI} X-\overset{\overset{\displaystyle O}{\|}}{C}-OH$

$\xrightarrow{SOCl_2} X-\overset{\overset{\displaystyle O}{\|}}{C}-Cl \xrightarrow{Ph-NH_2} X-\overset{\overset{\displaystyle O}{\|}}{C}-NH-Ph$



49. (3)

$\underset{(X)}{CH_3-\underset{\underset{\displaystyle CH_2}{\|}}{C}-O-CH_2-CH_3} \xrightarrow{H_3O^+}$

$CH_3-\underset{\underset{\displaystyle CH_2}{\|}}{C}-OH + \underset{(Z)}{HO-CH_2-CH_3}$

$\rightleftharpoons$

$\underset{(Y)}{CH_3-\underset{\underset{\displaystyle O}{\|}}{C}-CH_3}$

50. (2) Given $k_b = x$ K kg mol^{-1}

$\Delta T_b = k_b \times m$

$\therefore \quad y = x \times m$

$m = \frac{y}{x}\frac{K}{K\ Kg\ mol^{-1}} = \frac{y}{x}$ mol Kg^{-1}

We know

$\Delta T_f = k_f \times m$

On substituting value of m,

$\Delta T_t = \frac{yz}{x}K$

51. (546) Use $d = \frac{zM}{N_A a^3}$

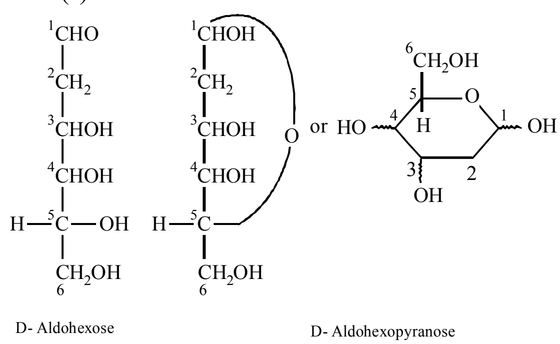
$z = 4; M = 40 + 2 \times 19 = 78$

$a^3 = \frac{zM}{N_A d}$

$= \frac{4 \times 78}{6.022 \times 10^{23} \times 3.18}$

$a = 546$ pm

52. (8)



Thus, total number of stereoisomers in pyranose form of D-configuration = $2^3 = 8$

53. (1.47) Volume of gold foil = $25 \times 40 \times 0.25$ mm^3

$= 250 \times 10^{-3}$ cm^3

Mass of gold foil = $19.32 \times 250 \times 10^{-3}$ g

= 4.83 g

No. of gold atoms = $\frac{4.83}{197} \times N_A = 1.47 \times 10^{22}$

54. (2) $A \longrightarrow B$

Initial concentration	Rate of reaction
2×10^{-3} M	2.40×10^{-4} Ms^{-1}
1×10^{-3} M	0.60×10^{-4} Ms^{-1}

rate of reaction r = $k[A]^x$

where x = order of reaction

hence

$2.40 \times 10^{-4} = k\,[2 \times 10^{-3}]^x$ ......(i)

$0.60 \times 10^{-4} = k\,[1 \times 10^{-3}]^x$ ......(ii)

On dividing eqn.(i) from eqn. (ii) we get

$4 = (2)^x$

$\therefore \; x = 2$

i.e. order of reaction = 2

55. (0.2303) $k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$;

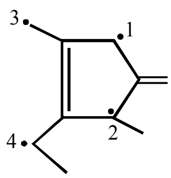
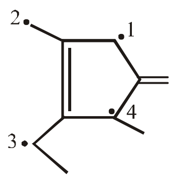
$$2.303 \times 1 = 2.303 \log\frac{[A_0]}{[A]}$$

$$\frac{[A_0]}{[A]} = 10$$

$$\therefore \; [A] \Rightarrow \frac{1}{10} \Rightarrow 0.1$$

$\therefore$ rate after 1 min $r_1 = k \cdot [A]$

$\Rightarrow$ $2.303 \times 0.1 \Rightarrow 0.2303 \text{ M min}^{-1}$

56. (4)  

57. (2) $\Delta v = \frac{0.001}{100} \times 30{,}000 = 0.3 \text{ cm sec}^{-1}$

According to uncertainty principle,

$$\Delta x \,.\, \Delta p \approx \frac{h}{4\pi}\,;\; \Delta x . \Delta v \approx \frac{h}{4\pi m}$$

$$\Delta x \times 9.1 \times 10^{-28} \times 0.3 \approx \frac{6.625 \times 10^{27} \times 7}{4 \times 22}$$

$\Delta x \approx 1.93$ cm. ≈ 2

58. (964) Given $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$;

$$\Delta H_f = -46.0 \text{ kJ/mol}$$

$$H + H \rightleftharpoons H_2;\ \Delta H_f = -436 \text{ kJ/mol}$$

$$N + N \rightleftharpoons N_2;\ \Delta H_f = -712 \text{ kJ/mol}$$

$$\Delta H_f(NH_3) = \frac{1}{2}\Delta H_{N-N} + \frac{3}{2}\Delta H_{H-H} - \Delta H_{N-H}$$

$$-46 = \frac{1}{2}(-712) + \frac{3}{2}(-436) - \Delta H_{N-H}$$

On calculation

$$\Delta H_{N-H} = -964 \text{ kJ/mol}$$

59. (60) According to Arrhenius equation

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log\frac{1.3 \times 10^{-3}}{1.3 \times 10^{-4}} = \frac{E_a}{2.303 \times 8.314}\left[\frac{1}{373} - \frac{1}{423}\right]$$

$$1 = \frac{E_a}{2.303 \times 8.314}\left[\frac{1}{373} - \frac{1}{423}\right]$$

$$E_a = 60 \text{ kJ/mol}$$

60. (10.85) For this cell, reaction is; $Zn + Fe^{2+} \rightarrow Zn^{2+} + Fe$

$$E = E^\circ - \frac{0.0591}{n}\log\frac{c_1}{c_2};\ E^\circ = E + \frac{0.0591}{n}\log\frac{c_1}{c_2}$$

$$E^\circ = 0.2905 + \frac{0.0591}{2}\log\frac{10^{-2}}{10^{-3}} = 0.32 \text{ V.}$$

$$E^\circ = \frac{0.0591}{2}\log K_{eq}$$

$$\log K_{eq} = \frac{0.32 \times 2}{0.0591} = \frac{0.32}{0.0295}$$

$$\therefore K_{eq} = 10^{\frac{0.32}{0.0295}}$$

Comparing the value of 10^x,

$$x = \frac{0.32}{0.0295} = 10.847 \approx 10.85$$

MATHEMATICS

61. (2) $f'(3^+) = \lim_{h\to0}\frac{f(3+h) - f(3)}{h}$

$$= \lim_{h\to0}\frac{(2 - e^h) - 1}{h} = -\lim_{h\to0}\left(\frac{e^h - 1}{h}\right) = -1$$

$$f'(3^-) = \lim_{h\to0}\frac{f(3-h) - f(3)}{-h}$$

$$= \lim_{h\to0}\frac{\sqrt{10 - (3-h)^2} - 1}{-h} = -\lim_{h\to0}\frac{\sqrt{1 + (6h - h^2)} - 1}{-h}$$

$$= \lim_{h\to0}\frac{6h - h^2}{-h(\sqrt{1 + 6h - h^2} + 1)} = \frac{-6}{2} = -3$$

Hence, $f'(3^+) \neq f(3^-)$

62. (1) We have $|z+1| = |z+4-3|$ ...(i)
Now $|z+4-3| \le |z+4| + |-3| \le 3+3 = 6$
[Given $|z+4| \le 3$ & $|-3| = 3$] $\therefore |z+1| \le 6$
Again $|z+1| \ge 0$ [modulus is always non-negative]
$\therefore$ Least value of $|z+1|$ may be zero, which occurs when $z=-1$, For $z=-1$, $|z+4| = |-1+4| = 3$
Which satisfies the given condition that $|z+4| \ge 3$
Hence, the least and the greatest values of $|z+1|$ are 0 and 6.

63. (3)
$$\left(x-\frac{1}{5}\right)^2+\left(y-\frac{2}{5}\right)^2=(\lambda^2-4\lambda+4)\left(\frac{3x+4y-1}{5}\right)^2$$
i.e., $$\sqrt{\left(x-\frac{1}{5}\right)^2+\left(y-\frac{2}{5}\right)^2}=|\lambda-2|\left|\frac{3x+4y-1}{\sqrt{5}}\right|$$
is an ellipse.
If $0<|\lambda-2|<1$ i.e., $\lambda \in (1,2)\cup(2,3)$

64. (3) One vertex of square is $(-4, 5)$ and equation of one diagonal is $7x - y + 8 = 0$
Diagonal of a square are perpendicular and bisect each other
Let the equation of the other diagonal be $y = mx + c$ where m is the slope of the line and c is the y-intercept.
Since this line passes through $(-4, 5)$
$\therefore$ $5 = -4m + c$ ...(i)
Since this line is at right angle to the line $7x - y + 8 = 0$ or $y = 7x + 8$, having slope $= 7$,
$\therefore$ $7 \times m = -1$ or $m = \dfrac{-1}{7}$
Putting this value of m in equation (i) we get
$C = \dfrac{31}{7}$
Hence equation of the other diagonal is
$y = -\dfrac{1}{7}x + \dfrac{31}{7}$ or $x + 7y = 31$.

65. (3) $p \Rightarrow q \equiv \sim p \vee q$ $\therefore \sim(p \Rightarrow q) \equiv p \wedge \sim q$.

66. (1) $$\int e^{\sin x}\left(\frac{x\cos^3 x-\sin x}{\cos^2 x}\right)dx$$
$$=\int e^{\sin x}\, x\cos x\,dx-\int e^{\sin x}\tan x\sec x\,dx$$
$$=\int x\,d\left(e^{\sin x}\right)-\int e^{\sin x}\,d(\sec x)$$
$$=\left\{x e^{\sin x}-\int e^{\sin x}dx\right\}-\left\{e^{\sin x}\sec x-\int e^{\sin x}\sec x\cos x\,dx\right\}$$
$$=x e^{\sin x}-e^{\sin x}\sec x+C$$

67. (1) $f(x) = x^2 - 4x + a$
$= x^2 - 4x + 4 - 4 + a$
$= (x-2)^2 + (a-4)$
So, $a \in (4, \infty)$ for domain $[2, \infty)$ and range $(0, \infty)$

68. (1) For real roots $D \ge 0$
$(k-2)^2 - 4(k^2 + 3k + 5) \ge 0$
$(k^2 + 4 - 4k) - 4k^2 - 12k - 20 \ge 0$
$-3k^2 - 16k - 16 \ge 0$; $3k^2 + 16k + 16 \le 0$
$\left(k+\dfrac{4}{3}\right)(k+4) \le 0$
Now $E = \alpha^2 + \beta^2$; $E = (\alpha+\beta)^2 - 2\alpha\beta$
$E = (k-2)^2 - 2(k^2 + 3k + 5) = -k^2 - 10k - 6$
$E = -(k^2 + 10k + 6) = -[(k+5)^2 - 19]$
$= 19 - (k+5)^2$
$\therefore$ E_{min} occurs when $k = -4/3$
$\therefore$ $E_{min} = 19 - \dfrac{121}{9} = \dfrac{171-121}{9} = \dfrac{50}{9}$
E_{max} occurs when $k = -4$
$E_{max} = 19 - 1 = 18$

69. (1) $$\frac{20!}{p!q!r!}(2x)^p(-y)^q(z)^r=\frac{20!}{p!q!r!}2^p(-1)^q x^p y^q z^r$$
$p + q + r = 20$, $q = 0$
$p + r = 20$ (p is even and r is odd).
even + odd = even (never possible)
Coefficient of such power never occur
$\therefore$ coefficient is zero

70. (1)
Starting with 1

1				

23456789
$= {}^8C_4 = 70$
Starting with 2

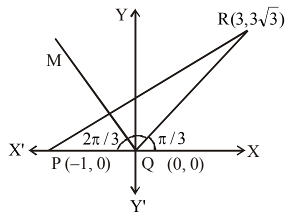
2				

3456789
$= {}^7C_4 = 35$
Total = 105
$(105)^{th}$ number 26789

71. (1) $(\hat{a}.\hat{c})\hat{b} - (\hat{a}.\hat{b})\hat{c} = \dfrac{1}{\sqrt{2}}\hat{b} + \dfrac{1}{\sqrt{2}}\hat{c}$
$\therefore$ $\hat{a}.\hat{c} = \dfrac{1}{\sqrt{2}}$ and $\hat{a}.\hat{b} = -\dfrac{1}{\sqrt{2}}$
$\Rightarrow$ angle between $\hat{a}$ and $\hat{c} = \dfrac{\pi}{4}$ and angle between $\hat{a}$ and $\hat{b} = \dfrac{3\pi}{4}$

72. (1) $|z_1+z_2| \le |z_1|+|z_2| = |24+7i|+6 = 25+6 = 31$
Also, $|z_1+z_2| = |z_1-(-z_2)| \ge ||z_1|-|z_2||$
$\Rightarrow |z_1+z_2| \ge |25-6| = 19$
Hence the least value of $|z_1+z_2|$ is 19 and the greatest value is 31.

73. (3) The coordinates of points P, Q, R are $(-1, 0), (0, 0), (3, 3\sqrt{3})$ respectively.



Slope of QR $= \sqrt{3}$

$\Rightarrow \tan\theta = \sqrt{3}$

$\Rightarrow \theta = \frac{\pi}{3}$

$\Rightarrow \angle RQX = \frac{\pi}{3}$

$\therefore \angle RQP = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

Let QM bisects the $\angle PQR$,

$\therefore$ Slope of the line QM $= \tan\frac{2\pi}{3} = -\sqrt{3}$

$\therefore$ Equation of line QM is $(y-0) = -\sqrt{3}(x-0)$

$\Rightarrow y = -\sqrt{3}x \Rightarrow \sqrt{3}x + y = 0$

74. (1) Given $\int \sqrt{2}\sqrt{1+\sin x}\, dx = -4\cos(ax+b)+C$

$$\Rightarrow \int \sqrt{2}\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)dx = -4\cos(ax+b)+C$$

$$\Rightarrow \int \sqrt{2}\,.\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin\frac{x}{2}+\frac{1}{\sqrt{2}}\cos\frac{x}{2}\right)dx = -4\cos(ax+b)+C$$

$$\Rightarrow \int 2\left(\cos\frac{\pi}{4}\sin\frac{x}{2}+\sin\frac{\pi}{4}\cos\frac{x}{2}\right)dx = -4\cos(ax+b)+C$$

$$\Rightarrow \int 2\sin\left(\frac{x}{2}+\frac{\pi}{4}\right)dx = -4\cos(ax+b)+C$$

$$\Rightarrow -4\cos\left(\frac{x}{2}+\frac{\pi}{4}\right) = -4\cos(ax+b)+C$$

$$\Rightarrow a = \frac{1}{2},\ b = \frac{\pi}{4}$$

75. (1) $4x^3 + 4y^3\,(dy/dx) = 0 \Rightarrow dy/dx = -x^3/y^3$
Equation of tangent, $Y - y = -x^3/y^3\,(X-x)$
$\Rightarrow y^3Y + x^3X = x^4 + y^4 = a^4$

$$\Rightarrow \frac{X}{a^4/x^3} + \frac{Y}{a^4/y^3} = 1$$

Here, $p = a^4/x^3$, $q = a^4/y^3$

$$\Rightarrow p^{-4/3} + q^{-4/3} = \frac{a^{-16/3}}{x^{-4}} + \frac{a^{-16/3}}{y^{-4}} = a^{-16/3}\left(x^4+y^4\right)$$

$$= a^{-16/3}(a^4) = a^{-4/3}$$

76. (2) Let the equation of curve
$y^2(2a-x) = x^3$ ...(i)
and equation of line $x = 2a$ ...(ii)
The given curve is symmetrical about x-axis and passes through origin.

From (i) we have, $y^2 = \frac{x^3}{2a-x}$

But $\frac{x^3}{2a-x} < 0$ for $x > 2a$ and $x < 0$

So, curve does not lie in the portion $x > 2a$ and $x < 0$, therefore curve lies in $0 \le x \le 2a$.

$\therefore$ Area bounded by the curve and line

$$= \int_0^{2a} y\,dx = \int_0^{2a} \frac{x^{3/2}}{\sqrt{2a-x}}\,dx$$

Put $x = 2a\sin^2\theta$ and $dx = 4a\sin\theta\cos\theta\, d\theta$

$$\therefore\ I = \int_0^{\pi/2} 8a^2\sin^4\theta\, d\theta = 8a^2\left[\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}\right]$$

$$\left(\text{using } \int_0^{\pi/2}\sin^m x\cos^n x\,dx = \frac{\Gamma\frac{m+1}{2}\Gamma\frac{n+1}{2}}{2\Gamma\frac{m+n+2}{2}}\right)$$

$$= \frac{3\pi a^2}{2} \text{ sq. unit}$$

77. (3) Rewrite the given differential equation as follows :

$\frac{dy}{dx} + \frac{2x}{x^2-1}y = \frac{1}{x^2-1}$, which is a linear form

The integrating factor I.F.

$$= e^{\int\frac{2x}{x^2-1}dx} = e^{\ell n(x^2-1)} = x^2 - 1$$

Thus multiplying the given equation by (x^2-1), we get $(x^2-1)\frac{dy}{dx}+2xy=1$

$$\Rightarrow \frac{d}{dx}[y(x^2-1)]=1$$

On integrating we get $y(x^2-1)=x+c$

78. (4) $f_2(x)=f(f(x))=f(x)=x$

$f_3(x)=f(f_2(x))=f(x)=x$

$\Rightarrow x^3-25x^2+175x-375=0$

$(x-5)(x^2-20x+75)=0$

$(x-5)^2(x-15)=0 \Rightarrow x=5, 15$

79. (3) Let $I_1=\int_0^{\sin^2 x}\sin^{-1}\sqrt{t}\,dt$

Put $t=\sin^2 u \Rightarrow dt=2\sin u\cos u\,du$

$\Rightarrow dt=\sin 2u du$

$$\therefore I_1=\int_0^x u\sin 2u\,du$$

Let $I_2=\int_0^{\cos^2 x}\cos^{-1}\sqrt{t}\,dt$

Put $t=\cos^2 v \Rightarrow dt=-2\cos v\sin v\,dv$

$\Rightarrow dt=-\sin 2v\,dv$

$$\therefore I_2=\int_{\frac{\pi}{2}}^{x} v(-\sin 2v)dv=-\int_{\frac{\pi}{2}}^{x} v\sin 2v\,dv$$

$$=-\int_{\frac{\pi}{2}}^{x} u\sin 2u\,du \text{ [change of variable]}$$

$\therefore I=I_1+I_2$

$$=\int_0^x u\sin 2u\,du-\int_{\frac{\pi}{2}}^{x} u\sin 2u\,du$$

$$=\int_0^{\frac{\pi}{2}} u\sin 2u\,du+\int_{\frac{\pi}{2}}^{x} u\sin 2u\,du-\int_{\frac{\pi}{2}}^{x} u\sin 2u\,du$$

$$=\int_0^{\frac{\pi}{2}} u\sin 2u\,du=\frac{\pi}{4} \text{ [Integrate by parts]}$$

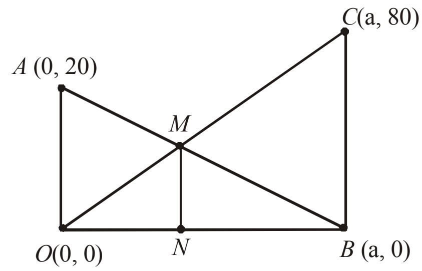
80. (2) For non-trivial solution,

$$\Delta=\begin{vmatrix}\sqrt{2}a & \sin B & \cos B\\ 1 & \cos B & \sin B\\ -1 & \sin B & -\cos B\end{vmatrix}=0$$

$$\Rightarrow a\sqrt{2}[-\cos^2 B-\sin^2 B]-\sin B[-\cos B+\sin B]$$
$$+\cos B[\sin B+\cos B]=0$$

$$\Rightarrow -a\sqrt{2}+\sin 2B+\cos 2B=0 \Rightarrow a\in[-1,1]$$

81. (16.00)



We put one pole at origin.

$BC=80$ m, $OA=20$ m

Line OC and AB intersect at M.

To find: Length of MN.

Eqn of OC: $y=\left(\frac{80-0}{a-0}\right)x$

$$\Rightarrow y=\frac{80}{a}x \quad ...(1)$$

Eqn of AB: $y=\left(\frac{20-0}{0-a}\right)(x-a)$

$$\Rightarrow y=\frac{-20}{a}(x-a) \quad ...(2)$$

At M: (1) = (2)

$$\Rightarrow \frac{80}{a}x=\frac{-20}{a}(x-a)$$

$$\Rightarrow \frac{80}{a}x=\frac{-20}{a}x+20 \Rightarrow x=\frac{a}{5}$$

$$\therefore y=\frac{80}{a}\times\frac{a}{5}=16$$

82. (3.00) $x+iy=\dfrac{3}{\cos\theta+i\sin\theta+2}$

$$\Rightarrow \frac{1}{x+iy}=\frac{\cos\theta+i\sin\theta+2}{3}$$

$$\Rightarrow \frac{x-iy}{(x+iy)(x-iy)}=\frac{1}{3}[(\cos\theta+2)+i\sin\theta]$$

$$\Rightarrow \frac{x}{x^2+y^2}=\frac{1}{3}(\cos\theta+2)$$

$$\Rightarrow \frac{x}{x^2+y^2}-\frac{2}{3}=\frac{1}{3}\cos\theta$$

and $-\dfrac{y}{x^2+y^2}=\dfrac{1}{3}\sin\theta$

Squaring and adding, we get

$$\left(\frac{x}{x^2+y^2}-\frac{2}{3}\right)^2+\left(\frac{-y}{x^2+y^2}\right)^2=\frac{1}{9}$$

$$\Rightarrow \frac{1}{x^2+y^2}(3-4x)+1=0$$

$$\Rightarrow 3-4x=-x^2-y^2$$

$$\Rightarrow 4x-x^2-y^2=3$$

83. (0.50) Let A be the event such that sum is Rs. 20 or more

$\therefore$ P (1) = 1 – P (Total value is < 20)

$$=1-\frac{{}^6C_2-{}^2C_2}{{}^8C_2}=1-\frac{14}{28}=1-\frac{1}{2}=\frac{1}{2}$$

8 — 1,1; 5,5; 10,10; 20,20

84. (2008.00) $\theta_1+\theta_2=\dfrac{\pi}{2}$

$$\therefore\ I=\int_{\theta_1}^{\theta_2}\frac{d\theta}{1+\tan\left(\frac{\pi}{2}-\theta\right)}=\int_{\theta_1}^{\theta_2}\frac{\tan\theta\,d\theta}{1+\tan\theta}$$

and also $I=\displaystyle\int_{\theta_1}^{\theta_2}\frac{d\theta}{1+\tan\theta}$

$$\therefore\ 2I=\int_{\theta_1}^{\theta_2}d\theta=\theta_2-\theta_1=\frac{1002\pi}{2008}\Rightarrow I=\frac{501\pi}{2008}$$

Hence, K = 2008.

85. (18.00) $\dfrac{{}^nC_k}{{}^nC_{k+1}}=\dfrac{1}{2}$

$$\Rightarrow \frac{n!}{k!(n-k)!}\frac{(k+1)!(n-k-1)!}{n!}=\frac{1}{2}$$

or $\dfrac{k+1}{n-k}=\dfrac{1}{2}$

$2k+2=n-k$

$n-3k=2$ ...... (1)

Similarly, $\dfrac{{}^nC_{k+1}}{{}^nC_{k+2}}=\dfrac{2}{3}$

$$\frac{n!}{(k+1)!(n-k-1)!}\cdot\frac{(k+2)!(n-k-2)!}{n!}=\frac{2}{3}$$

$\dfrac{k+2}{n-k-1}=\dfrac{2}{3}$

$3k+6=2n-2k-2$

$2n-5k=8$ ...... (2)

From (1) and (2)

$n=14$ and $k=4$

$\therefore\ n+k=18$

86. (12.00) Let the observations be x_1, x_2, x_3, x_4, x_5 and x_6, so

their mean $\bar{x}=\dfrac{\sum_{i=1}^{6}x_i}{6}=8\Rightarrow\sum_{i=1}^{6}x_i=48$

On multiplying each observation by 3, we get the new observations as $3x_1, 3x_2, 3x_3, 3x_4, 3x_5$ and $3x_6$.

Now, their mean

$$=\bar{x}=\frac{\sum_{i=1}^{6}3x_i}{6}=\frac{3\times 48}{6}=24$$

Variance of new observations

$$=\frac{\sum_{i=1}^{6}(3x_i-24)^2}{6}=\frac{3^2\sum_{i=1}^{6}(x_i-8)^2}{6}$$

= 9 × Variance of old observations

$=9\times 4^2=144$

Thus, standard deviation of new observations

$=\sqrt{\text{Variance}}=\sqrt{144}=12$

87. (0.75) $\lim_{x\to 0} \frac{(\tan x)^{3/2}[1-(\cos x)^{3/2}]}{x^{3/2}.x^2}$.

$= 1 \times \lim_{x\to 0} \frac{1-\cos^3 x}{x^2} . \frac{1}{1+(\cos x)^{3/2}}$

$= \frac{1}{2}.\frac{1}{2}(1+\cos x+\cos^2 x) = \frac{3}{4} = 0.25$

88. (0.38) n = 3, P (success) = P (HT or TH) = 1/2

$\Rightarrow \quad p = q = \frac{1}{2}$ and $r = 2$

$P(r=2) = {}^3C_2 \left(\frac{1}{2}\right)^2 . \frac{1}{2} = \frac{3}{8} = 0.38$

89. (336.00) Let $S = \{1, 2, 3\} \Rightarrow n(S) = 3$

Now, $P(S)$ = set of all subsets of S

total no. of subsets = $2^3 = 8$

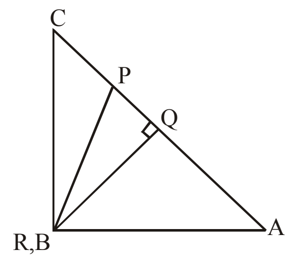
$\therefore n[P(S)] = 8$

Now, number of one-to-one functions from $S \to$

$P(S)$ is ${}^8P_3 = \frac{8!}{5!} = 8 \times 7 \times 6 = 336.$

90. (9.00) $2\sec 4C + \sin^2 2A + \sqrt{\sin B} = 0$

$A = 45°, B = 90°$ and $C = 45°$



Let $AQ = a$, then $BP = \frac{a}{2}$,

$PQ = \frac{a}{2}$ and $QR = a$

$\therefore PR = \sqrt{a^2 + \frac{a^2}{4}} = \frac{\sqrt{5}a}{2}$

$\therefore 1 : \alpha : \beta = \frac{a}{2} : a : \frac{\sqrt{5}a}{2} = 1 : 2 : \sqrt{5}$

$\therefore \alpha = 2$ and $\beta = \sqrt{5} \therefore \alpha^2 + \beta^2 = 9$

Mock Test-3

ANSWER KEY											
1.	(3)	**16.**	(4)	**31.**	(3)	**46.**	(4)	**61.**	(4)	**76.**	(4)
2.	(1)	**17.**	(2)	**32.**	(3)	**47.**	(4)	**62.**	(1)	**77.**	(1)
3.	(1)	**18.**	(1)	**33.**	(3)	**48.**	(2)	**63.**	(2)	**78.**	(3)
4.	(2)	**19.**	(1)	**34.**	(1)	**49.**	(4)	**64.**	(2)	**79.**	(1)
5.	(1)	**20.**	(2)	**35.**	(4)	**50.**	(4)	**65.**	(1)	**80.**	(4)
6.	(1)	**21.**	(1600)	**36.**	(2)	**51.**	(2)	**66.**	(4)	**81.**	(2.33)
7.	(2)	**22.**	(0.025)	**37.**	(2)	**52.**	(14.28)	**67.**	(4)	**82.**	(9.00)
8.	(3)	**23.**	(19039)	**38.**	(4)	**53.**	(2)	**68.**	(3)	**83.**	(32.00)
9.	(2)	**24.**	(0.33)	**39.**	(1)	**54.**	(80)	**69.**	(2)	**84.**	(343.00)
10.	(1)	**25.**	(1.25)	**40.**	(1)	**55.**	(5.0)	**70.**	(1)	**85.**	(3.00)
11.	(3)	**26.**	(72)	**41.**	(3)	**56.**	(1)	**71.**	(3)	**86.**	(0.53)
12.	(3)	**27.**	(4.8×10^8)	**42.**	(1)	**57.**	(395)	**72.**	(2)	**87.**	(2.00)
13.	(2)	**28.**	(0.2)	**43.**	(2)	**58.**	(3)	**73.**	(2)	**88.**	(3.00)
14.	(1)	**29.**	(8)	**44.**	(1)	**59.**	(3)	**74.**	(1)	**89.**	(0.50)
15.	(3)	**30.**	(10)	**45.**	(1)	**60.**	(1.14)	**75.**	(1)	**90.**	(0.50)

Solutions

PHYSICS

1. (3) For total internal reflection on face *AC* $\theta >$ critical angle (C) and $\sin\theta \ge \sin C$

$$\sin\theta \ge \frac{1}{{}^{w}\mu_g}$$

$$\sin\theta \ge \frac{\mu_w}{\mu_g} \Rightarrow \sin\theta \ge \frac{\frac{4}{3}}{\frac{3}{2}}$$

$$\therefore \sin\theta \ge \frac{8}{9}.$$

2. (1) As we know,

Gravitational potential energy $= \dfrac{-GMm}{r}$

and orbital velocity, $v_0 = \sqrt{GM/R+h}$

$$E_f = \frac{1}{2}mv_0^2 - \frac{GMm}{3R} = \frac{1}{2}m\frac{GM}{3R} - \frac{GMm}{3R}$$

$$= \frac{GMm}{3R}\left(\frac{1}{2}-1\right) = \frac{-GMm}{6R}$$

$$E_i = \frac{-GMm}{R} + K$$

$$E_i = E_f$$

Therefore minimum required energy,

$$K = \frac{5GMm}{6R}$$

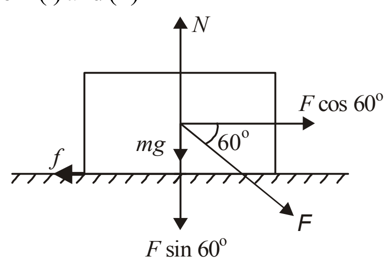
3. (1) The forces acting on the block are shown. Since the block is not moving forward for the maximum force F applied, therefore

$F\cos 60° = f = \mu N$... (i) (Horizontal Direction)

Note: For maximum force F, the frictional force is the limiting friction $= \mu N$]

and $F\sin 60° + mg = N$... (ii)

From (i) and (ii)

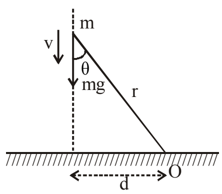


$F\cos 60° = \mu\,[F\sin 60° + mg]$

$$\Rightarrow \quad F = \frac{\mu mg}{\cos 60° - \mu \sin 60°}$$

$$= \frac{\frac{1}{2\sqrt{3}} \times \sqrt{3} \times 10}{\frac{1}{2} - \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{2}} = \frac{5}{\frac{1}{4}} = 20\text{ N}$$

4. (2) If student will use angular momentum = mvr.
He/she may conclude answer (1) as r is decreasing angular momentum must decrease hence (1) is incorrect.



But the magnitude of angular momentum of particle about O = mvd
Since speed v of particle increases, its angular momentum about O increases.
Magnitude of torque of gravitational force about O = mgd ⇒ constant
Moment of inertia of particle about O = mr^2
Hence MI of particle about O decreases.

Angular velocity of particle about $O = \frac{v\sin\theta}{r}$

∵ v and sin θ increases and r decreases.
∴ angular velocity of particle about O increases.

5. (1) As, $P = \frac{1}{3}\left(\frac{U}{V}\right)$

But $\frac{U}{V} = KT^4$

So, $P = \frac{1}{3}KT^4$

or $\frac{uRT}{V} = \frac{1}{3}KT^4$ [As PV = u RT]

$\frac{4}{3}\pi R^3 T^3 = \text{constant}$

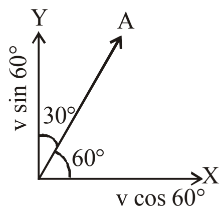
Therefore, $T \propto \frac{1}{R}$

6. (1) For particle C,
According to law of conservation of linear momentum, verticle component,
$2\,mv'\sin\theta = mv\sin 60° + mv\sin 45°$

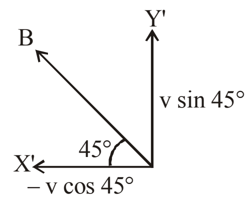
$$2mv'\sin\theta = \frac{mv}{\sqrt{2}} + \frac{mv\sqrt{3}}{2} \qquad \text{...... (i)}$$

Horizontal component,
$2\,mv'\cos\theta = mv\sin 60° - mv\cos 45°$

$$2mv'\cos\theta = \frac{mv}{2} - \frac{mv}{\sqrt{2}} \qquad \text{...... (ii)}$$



For particle A



For particle B

Dividing eq^n (i) by eq^n (ii),

$$\tan\theta = \frac{\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}}{\frac{1}{2} - \frac{1}{\sqrt{2}}} = \frac{\sqrt{2}+\sqrt{3}}{1-\sqrt{2}}$$

7. (2) $n_0 = \frac{v}{2\ell}$,

$n_1 = \frac{v}{2\,(\ell/2 - \Delta\ell)}$, $n_2 = \frac{v}{2\,(\ell/2 + \Delta\ell)}$

Beat frequency = $n_1 - n_2$

$$\Rightarrow v\left[\frac{1}{\ell - 2\Delta\ell} - \frac{1}{\ell + 2\Delta\ell}\right]$$

$$= v\left[\frac{(\ell + 2\Delta\ell) - (\ell - 2\Delta\ell)}{\ell^2 - 4\Delta\ell^2}\right]$$

$$= v\frac{4\Delta\ell}{\ell^2 - 4\Delta\ell^2} = \frac{8}{\ell}\frac{\Delta\ell v}{2\ell} = \frac{8\Delta\ell n_0}{\ell}$$

8. (3) Wavelength for which maximum obtained at the hole has the maximum intensity on passing.

So, $x = \frac{n\lambda D}{d}$

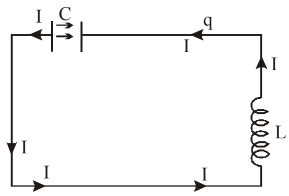
$$\lambda = \frac{xd}{nD} = \frac{1 \times 10^{-3} \times 0.5 \times 10^{-3}}{n \times 50 \times 10^{-2}}$$

$= \frac{1\times10^{-6}}{n} = \frac{1000\text{nm}}{n}$

$n = 1$, $\lambda = 1000$ nm $\rightarrow$ Not in the given range

$n = 2, \lambda = 500$nm

9. (2)



As it can be easily seen by the direction of I that Q is decreasing thus, energy of capacitor is decreasing and hence, energy of inductance is increasing or $\left(\frac{1}{2}LI^2\right)$ gives that I is increasing.

10. (1) Capacitors 2μF and 2μF are parallel, their equivalent = 4 μF

6μF and 12 μF are in series, their equivalent = 4 μF

Now 4μF (2 and 2 μF) and 8μF in series $= \frac{3}{8}\mu\text{F}$

And 4μF (12 & 6 μF) and 4μF in parallel = 4 + 4 = 8μF

8μF in series with $1\mu\text{F} = \frac{1}{8}+1 \Rightarrow \frac{8}{9}\mu\text{F}$

Now $C_{eq} = \frac{8}{9}+\frac{8}{3} = \frac{32}{9}$

With C, $\frac{1}{C_{eq}} = \frac{1}{C}+\frac{9}{32} = 1 \Rightarrow C = \frac{32}{23}\mu\text{F}$

11. (3) From question,

Horizontal velocity (initial),

$u_x = \frac{40}{2} = 20\,\text{m/s}$

Vertical velocity (initial), $50 = u_y t + \frac{1}{2}gt^2$

$\Rightarrow \quad u_y \times 2 + \frac{1}{2}(-10)\times 4$

or, $\quad 50 = 2u_y - 20$

or, $\quad u_y = \frac{70}{2} = 35\,\text{m/s}$

$\therefore \quad \tan\theta = \frac{u_y}{u_x} = \frac{35}{20} = \frac{7}{4}$

$\Rightarrow$ Angle $\theta = \tan^{-1}\frac{7}{4}$

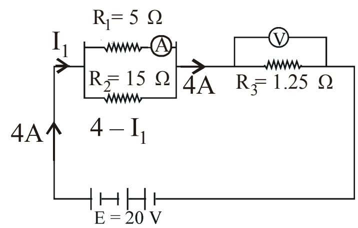
12. (3) As insect moves along a diameter, the effective mass and hence the M.I. first decreases then increases so from principle of conservation of angular momentum, angular speed, first increases then decreases.

13. (2) $R_{eq} = \frac{R_1R_2}{R_1+R_2}$ (Parallel combination);

$R_{Net} = R_{eg} + R_3$(Series combination)

R_{Net} of the circuit

$= \frac{5\times15}{5+15} + \frac{125}{100} = \frac{75}{20} + \frac{5}{4} = 5\,\Omega$



We know that $I = \frac{E}{R_{eq}} = \frac{20}{5} = 4\,\text{A}$

Potential difference across R_1 and R_2 are same (Parallel combination) $I_1R_1 = (4 - I_1)R_2$

$\Rightarrow 5I_1 = (4 - I_1)\times 15 \Rightarrow I_1 = 12 - 3I_1 \Rightarrow I_1 = 3\text{A}$

Thus reading of ammeter = 3A

Voltage across $1.25\Omega = I \times R = 4 \times 1.25 = 5\text{V}$

(Reading of voltmeter)

14. (1) Let initial temperature and volume be T_0 and V_0.

Since the process is adiabatic, the first temperature and volume is $TV^{\gamma-1} = T_0V_0^{\gamma-1}$ (γ = 5/3 for monoatomic gas)

$\therefore \; T = T_0\left(\frac{V_0}{V_0/8}\right)^{2/3} = 4T_0$

15. (3) Flux going in pyramid $= \frac{Q}{2\varepsilon_0}$

Which is divided equally among all 4 faces

$\therefore$ Flux through one face $= \frac{Q}{8\varepsilon_0}$

16. (4) From the figure it is clear that liquid 1 floats on liquid 2. The lighter liquid floats over heavier liquid. Therefore we can conclude that $\rho_1 < \rho_2$ Also $\rho_3 < \rho_2$ otherwise the ball would have sink to the bottom of the jar.
Also $\rho_3 > \rho_1$ otherwise the ball would have floated in liquid 1. From the above discussion we conclude that $\rho_1 < \rho_3 < \rho_2$.

17. (2) A = 0.1m, m = 0.1 kg, $KE_{max} = 18 \times 10^{-3}$ J, $\phi = \frac{\pi}{4}$

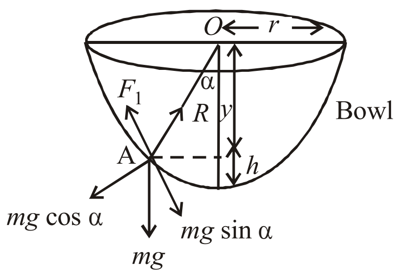
$$k = \frac{36 \times 10^{-3}}{(0.1)^2} = 3.6 \ ;$$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{3.6}{0.1}} = 6 \text{ rad/s}$$

$\therefore$ Eqn. $y = 0.1 \sin\left(6t + \frac{\pi}{4}\right)$

18. (1) First, the length of wire goes on increasing ie., area decreases and finally at breaking stress the wire breaks.

19. (1)



The insect crawls up the bowl upto a certain height h only till the component of its weight along the bowl is balanced by limiting frictional force.
For limiting condition at point A

$R = mg\cos\alpha$ ...(i)

$F_1 = mg\sin\alpha$ ...(ii)

Dividing eq. (ii) by (i)

$$\tan\alpha = \frac{1}{\cot\alpha} = \frac{F_1}{R} = \mu\left[As\ F_1 = \mu R\right]$$

$$\Rightarrow \quad \tan\alpha = \mu = \frac{1}{3}\left[\because \mu = \frac{1}{3}(\text{Given})\right]$$

$\therefore \quad \cot\alpha = 3$

20. (2) $T = 2\pi\sqrt{\frac{I}{M \times B}} = 2\pi\sqrt{\frac{I}{MB}}$

where $I = \frac{1}{12}m\ell^2$

When the magnet is cut into three pieces the pole strength will remain the same and

$$\text{M.I. } (I') = \frac{1}{12}\left(\frac{m}{3}\right)\left(\frac{\ell}{3}\right)^2 \times 3 = \frac{I}{9}$$

We have, Magnetic moment (M)
= Pole strength (m) $\times\ \ell$
$\therefore$ New magnetic moment,

$$M' = m \times \left(\frac{\ell}{3}\right) \times 3 = m\ell = M$$

$$\therefore\ T' = \frac{T}{\sqrt{9}} = \frac{2}{3}s.$$

21. (1600) Given, $R_1 = 100\,\Omega$, r' = r/2, R_2 = ?

Resistance of wire, $R = \frac{\rho l}{A}$

$\because$ Area × length = volume

Hence, $R = \frac{\rho V}{A^2}$

Since, $\rho \to$ constant, $V \to$ constant

$$R \propto \frac{1}{A^2}$$

or $R \propto \frac{1}{r^4}$ $\quad \because A = \pi r^2$

$$\frac{R_2}{R_1} = 16 \Rightarrow R_2 = 16 \times 100 = 1600\,\Omega$$

22. (0.025) At equilibrium, weight of the given block is balanced by force due to surface tension, i.e.,
2L. S = W

or $S = \frac{W}{2L} = \frac{1.5 \times 10^{-2} N}{2 \times 0.3\ m} = 0.025 Nm^{-1}$

23. (19039) Given: $\frac{dN_0}{dt}$ = 20 decays/min

$\frac{dN}{dt}$ = 2 decays/min

$T_{1/2}$ = 5730 years
As we know,

$$N = N_0 e^{-\lambda t}$$

$$\log\frac{N_0}{N} = \lambda t$$

$$\therefore \quad t = \frac{1}{\lambda}\log\frac{N_0}{N}$$

$$= \frac{2.303\times T_{1/2}}{0.693}\times\log_{10}\frac{N_0}{N}$$

But $\dfrac{\frac{dN_0}{dt}}{\frac{dN}{dt}} = \dfrac{N_0}{N} = \dfrac{20}{2} = 10$

$$\therefore \; t = \frac{2.303\times 5730}{0.693}\times 1$$

$= 19039$ years

24. (0.33) The thermal resistance is given by

$$\frac{x}{KA}+\frac{4x}{2KA}=\frac{x}{KA}+\frac{2x}{KA}=\frac{3x}{KA}$$

$$\therefore \frac{dQ}{dt}=\frac{\Delta T}{\frac{3x}{KA}}=\frac{(T_2-T_1)KA}{3x}$$

$$=\frac{1}{3}\left\{\frac{A(T_2-T_1)K}{x}\right\}$$

$$\therefore f=\frac{1}{3}$$

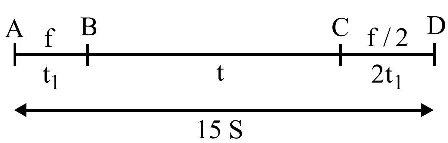
25. (1.25) $\dfrac{I_e}{I_h}=\dfrac{n_eeAv_e}{n_heAv_h}\Rightarrow\dfrac{7}{4}=\dfrac{7}{5}\times\dfrac{v_e}{v_h}\Rightarrow\dfrac{v_e}{v_h}=\dfrac{5}{4}$

26. (72) Distance from A to B $= S = \dfrac{1}{2}ft_1^{\,2}$

Distance from B to C $= (ft_1)t$

Distance from C to D $= \dfrac{u^2}{2a} = \dfrac{(ft_1)^2}{2(f/2)}$

$= ft_1^{\,2} = 2S$



$\Rightarrow \quad S + f\,t_1t + 2S = 15S$

$\Rightarrow \; f\,t_1t = 12S$ .... (i)

$\dfrac{1}{2}f\,t_1^2 = S$ .... (ii)

Dividing (i) by (ii), we get $t_1 = \dfrac{t}{6}$

$$\Rightarrow \; S=\frac{1}{2}f\left(\frac{t}{6}\right)^2=\frac{f\,t^2}{72}$$

27. (4.8×10^8) Optical source frequency

$$f=\frac{c}{\lambda}=3\times10^8/(800\times10^{-9})=3.8\times10^{14}\,\text{Hz}$$

Bandwidth of channel (1% of above)
$= 3.8 \times 10^{12}$Hz
Number of channels = (Total bandwidth of channel) / (Bandwidth needed per channel)
Number of channels for audio signal

$$=(3.8\times10^{12})/(8\times10^3)\sim 4.8\times10^8$$

28. (0.2) The current voltage relation of diode is

$$I=(e^{1000\,V/T}-1)\text{ mA (given)}$$

When, $I = 5mA,\, e^{1000\,V/T} = 6mA$

Also, $dI = (e^{1000\,V/T})\times\dfrac{1000}{T}$

(By exponential function)

$$=(6\,mA)\times\frac{1000}{300}\times(0.01)=0.2\text{ mA}$$

29. (8) Force acting on conductor B due to conductor A is given by relation

$$F=\frac{\mu_0 I_1 I_2 l}{2\pi r}$$

l-length of conductor B
r-distance between two conductors

$$\therefore F=\frac{4\pi\times10^{-7}\times10\times2\times2}{2\times\pi\times0.1}=8\times10^{-5}\text{ N}$$

30. (10) By Newton's law of cooling

$$\frac{\theta_1-\theta_2}{t}=-K\left[\frac{\theta_1+\theta_2}{2}-\theta_0\right]$$

where θ_0 is the temperature of surrounding.
Now, hot water cools from 60°C to 50°C in 10 minutes,

$$\frac{60-50}{10}=-K\left[\frac{60+50}{2}-\theta_0\right] \quad ...(i)$$

Again, it cools from 50°C to 42°C in next 10 minutes.

$$\frac{50-42}{10} = -K\left[\frac{50+42}{2} - \theta_0\right] \quad \text{...(ii)}$$

Dividing equations (i) by (ii) we get

$$\frac{1}{0.8} = \frac{55-\theta_0}{46-\theta_0}$$

$$\frac{10}{8} = \frac{55-\theta_0}{46-\theta_0}$$

$$460 - 10\theta_0 = 440 - 8\theta_0$$

$$2\theta_0 = 20$$

$$\theta_0 = 10°C$$

CHEMISTRY

31. (3) $\nu = RcZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$$\nu_1 = RcZ^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = RcZ^2$$

$$\nu_2 = RcZ^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3RcZ^2}{4}$$

$$\nu_3 = RcZ^2\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{RcZ^2}{4}$$

$\therefore\ \nu_1 - \nu_2 = \nu_3$

32. (3) $ICl_2^- \Rightarrow 2\,bp + 3\,lp$
(thus, sp^3d hybridisation) = linear

$BrF_2^+ \Rightarrow 2\,bp + 2\,lp$
(thus, sp^3 hybridisation) = pyramidal

$ClF_4^- \Rightarrow 4\,bp + 2\,lp$
(thus, sp^3d^2 hybridisation) = square planar

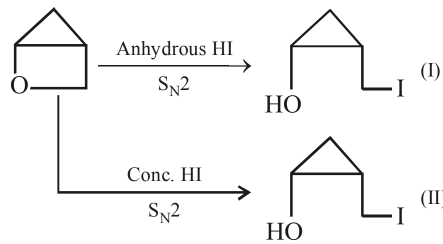
$AlCl_4^- \Rightarrow 4\,bp + 0\,lp$
(thus sp^3 hybridisation) = tetrahedral

33. (3) The correct formula of inorganic benzene is $B_3N_3H_6$ so (4) is incorrect statement.

Boric acid (H_3BO_3 or $\begin{matrix} OH \\ | \\ B - OH \\ | \\ OH \end{matrix}$) is a Lewis acid so (1) is incorrect statement.

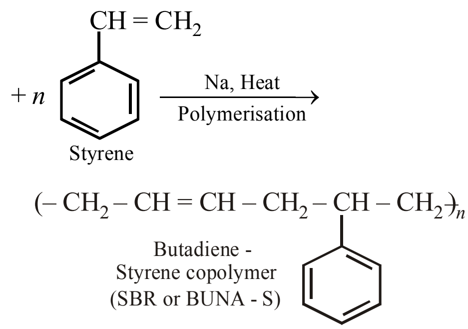
The coordination number exhibited by beryllium is 4 and not 6 so statement (2) is incorrect.
Both $BeCl_2$ and $AlCl_3$ exhibit bridged structures in solid state so (3) is correct statement.

34. (1)



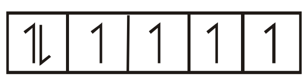
35. (4) One mole of a substance contains the number of molecules which is independent of pressure.

36. (2) $nCH_2 = CH - CH = CH_2$
1, 3-Butadiene



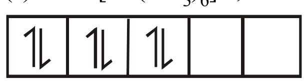
37. (2) In electrolysis of NaCl when Pt electrode is taken, then H_2 liberated at cathode, while with Hg cathode it forms sodium amalgam because more voltage is required to reduce H^+ at Hg than at Pt.

38. (4)
(1) For tetrahedral d^6 ion,



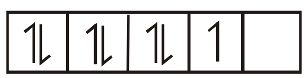
4 unpaired electrons

(2) For $[Co(NH_3)_6]^{3+}$,



0 unpaired electrons

(3) For square planar d^7 ion,



1 unpaired electrons

(4) B.M. $= \sqrt{n(n+2)}$, n = unpaired electrons

$5.92\text{ B.M.} = \sqrt{n(n+2)}$

n = 5 unpaired electrons

39. (1) (1) $SiC \Rightarrow$ Covalent carbide

(2) $Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4\uparrow$

(3) $CaC_2 + 2H_2O \longrightarrow HC{\equiv}CH\uparrow + Ca(OH)_2$

(4) $Mg_2C_3 + 2H_2O \longrightarrow 2Mg(OH)_2 + H_3C{-}C \equiv CH\uparrow$

40. (1) $\underset{(A)}{CH_3CH_2OCH(CH_3)_2} \xrightarrow{\text{Conc. HI}} \underset{(B)}{CH_3CH_2I} + \underset{(C)}{(CH_3)_2CHI}$

$\underset{(B)}{CH_3CH_2I} \xrightarrow{NaOH} CH_3CH_2OH \xrightarrow{[O]} CH_3COOH$

$\underset{(C)}{(CH_3)_2CHI} \xrightarrow{NaOH} (CH_3)_2CHOH \xrightarrow{[O]} (CH_3)_2CO$

hence the IUPAC name of compound is

$$CH_3{-}CH_2{-}O{-}\overset{2}{C}H{-}\overset{1}{C}H_3 \quad (\text{with } {}^{3}CH_3 \text{ on C-2})$$

2–ethoxy propane

41. (3)

(1) $\underset{\text{Yellow}}{CdS\downarrow} + (NH_4)_2S \longrightarrow \underset{\text{Yellow}}{CdS\downarrow} \xrightarrow{(NH_4)_2S_2} \underset{\text{Insoluble}}{CdS\downarrow}$

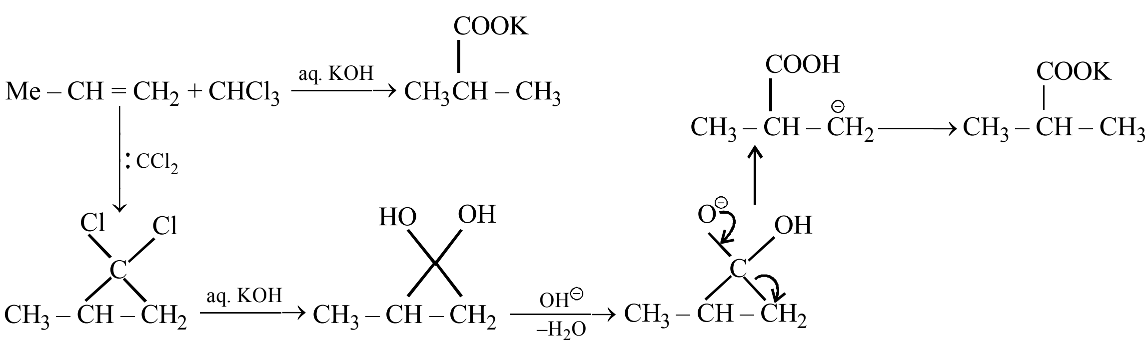
(2) $SnS_2\downarrow + (NH_4)_2S \longrightarrow \underset{\text{Yellow}}{CdS\downarrow} \xrightarrow{(NH_4)_2S_2} \underset{\text{Soluble}}{(NH_4)_2SnS_3}$

(3) $Cd^{2+} + (NH_4)_2S \longrightarrow \underset{\text{Yellow}}{CdS\downarrow} \xrightarrow{(NH_4)_2S_2} \underset{\text{Insoluble}}{CdS\downarrow}$

(4) $Sn^{2+} + (NH_4)_2S \longrightarrow \underset{\text{Brown}}{SnS\downarrow} \xrightarrow{(NH_4)_2S_2} (NH_4)_2SnS_3$
(Amm. thiostannate)
Soluble

42. (1) The compounds of the type $M(AA)_2B_2$ exhibit both geometrical and optical isomerism.

43. (2)



44. (1)

$$2N_2O_5(CCl_4) \longrightarrow 2N_2O_4(CCl_4) + O_2(g)$$

t = 0	a mol	0
t = 500 min.	a – x	$\frac{x}{2}$ mol
t = ∞	0	$\frac{a}{2}$ mol

$a = 100$
$x = 90$

$$k = \frac{1}{t}\ln\frac{a}{a-x} = \frac{2.303}{500}$$

45. (1) The order of basic character of the transition metal monoxide is TiO > VO > CrO > FeO because basic character of oxides decreases with increase in atomic number.

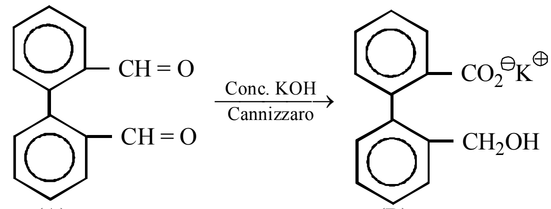
46. (4) $NH_4ClO_4 + HNO_3\,(\text{dilute}) \longrightarrow \underset{(X)}{NH_4NO_3} + HClO_4$

$$\underset{(X)}{NH_4NO_3} \xrightarrow{\text{heat}} \underset{(Y)}{N_2O} + 2H_2O$$

47. (4) $3CH_4 + 2O_3 \longrightarrow \underset{\text{Formaldehyde}}{3CH_2{=}O} + 3H_2O$

$\underset{\text{Acrolein}}{CH_2 = CHCH = O}$ $\underset{\text{Peroxyacetyl nitrate (PAN)}}{CH_3COONO_2}$

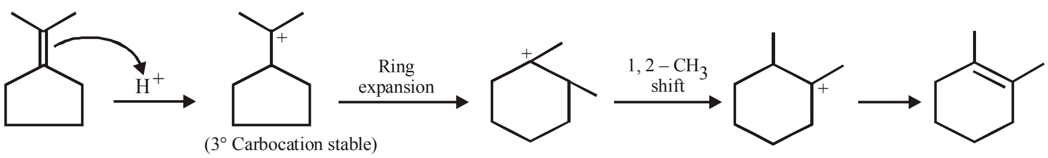
48. (2)



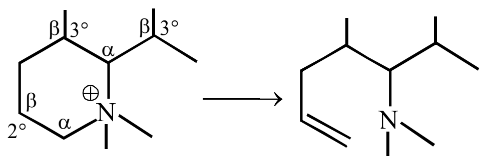
(A) (B)

49. (4) (1) The concentration of sodium thiosulphate solution should always be less than the concentration of the potassium iodide solution.
(2) Freshly prepared starch solution should be used
(3) Experiments should be performed with the fresh solutions of H_2O_2 and KI.

50. (4)



51. (2)



(2αH)

52. (14.28) $H_2(g) + CO_2(g) \rightleftharpoons CO\,(g) + H_2O\,(g)$

At eq^m 0.25–x 0.25–x x x

$$K_p = 0.16 = \frac{x^2}{(0.25-x)^2}$$

$$\Rightarrow 0.4 = \frac{x}{0.25-x} \Rightarrow 0.1 - 0.4x = x$$

$x = 0.0714$

$$\text{Mole\% of CO (g)} = \frac{0.0714}{0.50}\times 100 = 14.28$$

53. (2) No. of $A^{2+} = \frac{1}{8}\times 8 = 1$

No. of $B^{3+} = \frac{1}{2}\times 4 = 2$

No. of $O^{2-} = 8\times\frac{1}{8}\times 6\times\frac{1}{2} = 4$

(AB_2O_4)
∴ value of n = 2

54. (80) $\frac{P^\circ - P}{P^\circ} = \frac{n_2}{n_1 + n_2}$

$$\frac{640-600}{640} = \frac{2.5/m}{39/78}$$

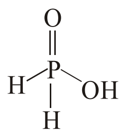
$$m = \frac{640\times 78\times 2.5}{39\times 40} = 80$$

55. (5.0) $pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

$pK_a = -\log(1.8 \times 10^{-5}) = 4.7447$
$[CH_3COO^-] = 2 \times [(CH_3COO)_2Ba] = 0.2\,M$
$[CH_3COOH] = 0.1\,M$

$pH = 4.7447 + \log \frac{0.2}{0.1} = 5.046 \approx 5.0$

56. (1) H_3PO_2 is named as hypophosphorous acid. It is monobasic as it contains only one P – OH bond, its basicity is one.



57. (395)Since, work is done against constant pressure and thus, irreversible.

Given, $\Delta V = (6-2) = 4\,L$; $P = 1$ atm

$\therefore \quad W = -1 \times 4\text{ L-atm} = -\frac{1 \times 4 \times 1.987}{0.0821}$ cal

(since 0.0821 L-atm = 1.987 cal)
$= -96.81$ cal $= -96.81 \times 4.184$ J

$(\because 1\text{ cal} = 4.184\text{ J})$

$= -405.05$ J
Now from I[st] law of thermodynamics
$q = \Delta U - W$
$800 = \Delta U + 405.05$
$\therefore \quad \Delta U = 395$ J

58. (3) Three, these are $CH_3CH_2OCH_2CH_3$ (I), $CH_3OCH_2CH_2CH_3$ (II) and $CH_3OCH(CH_3)_2$ (III). Here I and II, I and III are pairs of metamers.

59. (3) $m = \frac{E.wt \times Q}{96500}$;

$\therefore E.wt = \frac{m \times 96500}{Q} = \frac{22.2 \times 96500}{2 \times 5 \times 60 \times 60} = 60.3$

Oxidation state $= \frac{\text{At wt.}}{\text{Eq. wt.}} = \frac{177}{60.3} = 3$

60. (1.14) Given $-\frac{dMnO_4^-}{dt} = 4.56 \times 10^{-3}\,Ms^{-1}$

From the reaction given,

$-\frac{1}{2}\frac{dMnO_4^-}{dt} = \frac{4.56 \times 10^{-3}}{2} Ms^{-1}$

$-\frac{1}{2}\frac{dMnO_4^-}{dt} = \frac{1}{5}\frac{dI_2}{dt}$

$\therefore \quad -\frac{5}{2}\frac{dMnO_4^-}{dt} = \frac{dI_2}{dt}$

On substituting the given value

$\therefore \quad \frac{dI_2}{dt} = \frac{4.56 \times 10^{-3} \times 5}{2} = 1.14 \times 10^{-2}$ M/s

MATHEMATICS

61. (4) Centre of the circle is $\left(\frac{1}{2}, -\frac{3}{2}\right)$.

Its distance from the line $x + y - 1 = 0$ is $\sqrt{2}$
Let the required line be $mx - y = 0$

$\therefore \quad \left|\frac{\frac{m}{2} + \frac{3}{2}}{\sqrt{m^2+1}}\right| = \sqrt{2} \Rightarrow m = 1, -1/7$

$\therefore$ The lines are $x - y = 0$, $x + 7y = 0$

62. (1) Since α_1, α_2 and β_1, β_2 are the roots of $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ respectively, therefore

$\alpha_1 + \alpha_2 = \frac{-b}{a}, \alpha_1\alpha_2 = \frac{c}{a}$ . ..(1)

and $\beta_1 + \beta_2 = \frac{-q}{p}, \beta_1\beta_2 = \frac{r}{p}$ ...(2)

Since the given system of equations has a non-trivial solution

$\therefore \quad \begin{vmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{vmatrix} = 0$ i.e. $\alpha_1\beta_2 - \alpha_2\beta_1 = 0$

or $\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2} = \frac{\alpha_1 + \alpha_2}{\beta_1 + \beta_2} = \sqrt{\frac{\alpha_1\alpha_2}{\beta_1\beta_2}}$

$\Rightarrow \quad \frac{pb}{qa} = \sqrt{\frac{pc}{ra}} \Rightarrow \frac{b^2}{q^2} = \frac{ac}{pr}$

63. (2) Suppose, there are two points x_1 and x_2 in (a, b) such that $f'(x_1) = f'(x_2) = 0$. By Rolle's theorem applied to f' on $[x_1, x_2]$, there must be a $c \in (x_1, x_2)$ such that $f''(c) = 0$. This contradicts the given condition $f''(x) < 0, \forall x \in (a, b)$.

Hence, our assumption is wrong. Therefore, there can be atmost one point in (a, b) at which $f'(x)$ is zero.

64. (2) $I = \int_0^{16} f(t)\,dt$

Consider $g(x) = \int_0^{x^4} f(t)\,dt \Rightarrow g(0) = 0$

LMVT for g in [0, 1] gives, some $\alpha \in (0, 1)$ such that $\frac{g(1) - g(0)}{1 - 0} = g'(\alpha)$ ..... (1)

Similarly, LMVT in [1, 2] gives, some $\beta \in (1, 2)$ such that $\frac{g(2) - g(1)}{2 - 1} = g'(\beta)$ .... (2)

Eq. (1) + Eq. (2)

$g'(\alpha) + g'(\beta) = g(2) - \underbrace{g(0)}_{\text{zero}}$;

but $g'(x) = f(x^4) \cdot 4x^3$

$\therefore\ 4\left[\alpha^3 f(\alpha^4) + \beta^3 f(\beta^4)\right] = \int_0^{16} f(t)\,dt$

65. (1) $\begin{vmatrix} 3u^2 & 2u^3 & 1 \\ 3v^2 & 2v^3 & 1 \\ 3w^2 & 2w^3 & 1 \end{vmatrix} = 0$

$R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$

$\Rightarrow \begin{vmatrix} u+v & u^2+v^2+vu & 0 \\ v+w & v^2+w^2+vw & 0 \\ w^2 & w^3 & 1 \end{vmatrix} = 0$

$R_1 \to R_1 - R_2$

$\Rightarrow \begin{vmatrix} 1 & u+w+v & 0 \\ v+w & v^2+w^2+vw & 0 \\ w^2 & w^3 & 1 \end{vmatrix} = 0$

$\Rightarrow (v^2 + w^2 + vw) - (v + w)[v + w + u] = 0$

$\Rightarrow uv + vw + wu = 0$

66. (4) $x \in [-1, 0]$

$x + \frac{1 + x^2}{2} = -2x$

$x^2 + 6x + 1 = 0$

$x = 2\sqrt{2} - 3 \Rightarrow |10a| = [|20\sqrt{2} - 30|] = 30 - 20\sqrt{2}$

$x \in [0, 1]$

$x + \frac{1 + x^2}{2} = 2x$

$1 + x^2 = 2x \Rightarrow x = 1 \Rightarrow |10a| = 10$

$|10a| = 10,\ |20\sqrt{2} - 30|$

$\Rightarrow [|10a|] = 1, 10$

67. (4) Each point (x, y) has an image in line $y = 0$ as $(x, -y)$. So, replacing y by $-y$ in the given equation, we get the image as

$ax^2 - 2hxy + by^2 = 0.$

68. (3) $x^2 - 2x\cos\theta + 1 = 0,$

$x = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2} = \cos\theta \pm i\sin\theta$

Let $x = \cos\theta + i\sin\theta$

$\therefore\ x^{2n} - 2x^n \cos n\theta + 1$

$= \cos 2n\theta + i\sin 2n\theta - 2(\cos n\theta + i\sin n\theta)\cos n\theta + 1$

$= \cos 2n\theta + 1 - 2\cos^2 n\theta + i(\sin 2n\theta - 2\sin n\theta\cos n\theta)$

$= 0 + i\,0 = 0$

69. (2) $I_1 = \int_0^1 2^{x^2} dx,\ I_2 = \int_0^1 2^{x^3} dx,$

$I_3 = \int_1^2 2^{x^2} dx,\ I_4 = \int_1^2 2^{x^3} dx$

$\forall\ 0 < x < 1,\ x^2 > x^3$

$\Rightarrow \int_0^1 2^{x^2} dx > \int_0^1 2^{x^3} dx \Rightarrow I_1 > I_2.$

Also $\forall\ 1 < x < 2$

$x^2 < x^3 \Rightarrow \int_1^2 2^{x^2} dx < \int_1^2 2^{x^3} dx \Rightarrow I_3 < I_4$

70. (1)

$I = \int_0^{3\pi/4} (\sin x + \cos x)\,dx + \int_0^{3\pi/4} \underset{\mathrm{I}}{x}\,\underbrace{(\sin x - \cos x)}_{\mathrm{II}}\,dx$

$$= \int_0^{3\pi/4} (\sin x + \cos x)\, dx + \underbrace{x(-\cos x - \sin x)\big|_0^{3\pi/4}}_{\text{zero}}$$

$$+ \int_0^{3\pi/4} (\sin x + \cos x)\, dx$$

$$= 2\int_0^{3\pi/4} (\sin x + \cos x)\, dx = 2(\sqrt{2}+1)$$

71. (3) Given circles are $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y - 2 = 0$
$\therefore$ locus of centres is
$9x - 10y + 11 = 0$

72. (2) If n is odd, the required sum is

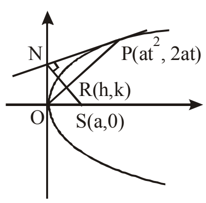
$$1^2 + 2.2^2 + 3^2 + 2.4^2 + ... + 2.(n-1)^2 + n^2$$

$$= \frac{(n-1)(n-1+1)^2}{2} + n^2 \ [\because (n-1) \text{ is even}$$

$\therefore$ using given formula for the sum of $(n-1)$ terms.]

$$= \left(\frac{n-1}{2}+1\right)n^2 = \frac{n^2(n+1)}{2}$$

73. (2)



$T : ty = x + at^2$ ..... (i)
Line perpendicular to (1) through (a, 0)
$tx + y = ta$ ..... (ii)

Equation of OP : $y - \frac{2}{t}x = 0$ ...... (iii)

From equations (ii) and (iii) eliminating t we get locus
$2x^2 + y^2 - 2ax = 0$

74. (1) $z + \frac{1}{z} = 2\cos\theta$

$\Rightarrow z^2 - 2\cos\theta z + 1 = 0$

$$\Rightarrow z = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}$$
$$= \cos\theta + i\sin\theta$$

Taking positive sign,

$$\therefore \frac{z^{2n}-1}{z^{2n}+1} = \frac{2i\sin n\theta}{2\cos n\theta} = i\tan n\theta$$

Taking negative sign, we get

$$\frac{z^{2n}-1}{z^{2n}+1} = \frac{-2i\sin n\theta}{2\cos n\theta} = -i\tan n\theta$$

$$\Rightarrow \left|\frac{z^{2n}-1}{z^{2n}+1}\right| = |\pm i\tan n\theta| = |\tan n\theta|$$

75. (1) Two curves cuts at right angle if product of their slopes is -1.
Two given curves are
$x^3 - 3xy^2 + 2 = 0$ ..... (i)
and $3x^2y - y^3 - 2 = 0$ ...... (ii)
Differentiate equ. (i),

$$\Rightarrow m_1 = \frac{dy}{dx} = \frac{3(x^2 - y^2)}{6xy}$$

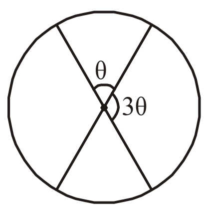
Differentiate equ. (ii),

$$\Rightarrow m_2 = \frac{dy}{dx} = \frac{-2xy}{(x^2 - y^2)}$$

$$\therefore m_1 \times m_2 = \frac{(x^2 - y^2)}{2xy} \times \frac{-2xy}{(x^2 - y^2)}$$

$$\Rightarrow m_1 \times m_2 = -1$$

76. (4)



Let one angle be θ then other = 3θ
Clearly $\theta + 3\theta = 180 \Rightarrow \theta = 45^\circ$
$\therefore$ Angle between the diameters represented by combined equation
$ax^2 + 2(a+b)xy + by^2 = 0$ is 45°

$$\therefore \text{Using } \tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$\Rightarrow 1 = \frac{2\sqrt{a^2 + b^2 + ab}}{a+b}$$

$\Rightarrow (a+b)^2 = 4(a^2+b^2+ab)$

$\Rightarrow 3a^2+3b^2+2ab=0$

77. (1)

p	q	$p \wedge q$	$p \vee q$	$\sim(p \vee q)$	$(p \wedge q) \wedge \sim(p \vee q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

$\therefore (p \wedge q) \wedge (\sim(p \vee q))$ is a contradiction.

78. (3) Since, $1, \omega, \omega^2, \ldots\ldots\ldots\ldots \omega^{n-1}$ are the n roots of unity

Consider $x^n = 1$

$\therefore (x^n - 1) = (x-1)(x-\omega)(x-\omega^2)\ldots\ldots(x-\omega^{n-1})$

$\Rightarrow (x-\omega)(x-\omega^2)\ldots\ldots(x-\omega^{n-1}) = \dfrac{x^n-1}{x-1}$

Taking $\lim\limits_{x\to 1}$ on both side

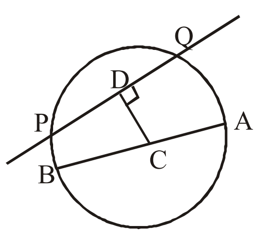
$\lim\limits_{x\to 1}(x-\omega)(x-\omega^2)\ldots\ldots(x-\omega^{n-1})$

$= \lim\limits_{x\to 1}\dfrac{x^n-1}{x-1}$

$\Rightarrow \quad (1-\omega)(1-\omega^2)\ldots\ldots(1-\omega^{n-1}) = n$

79. (1) Since, $\angle APB = \angle AQB = \dfrac{\pi}{2}$ so $y = mx + 8$ intersect the circle whose diameter is AB.

Equation of circle is $x^2 + y^2 = 16$, $CD < 4$



$\Rightarrow \dfrac{8}{\sqrt{1+m^2}} < 4 \Rightarrow 1+m^2 > 4$

$\Rightarrow m \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$

If the line passing throw the point A (–4, 0), B(4, 0)

then $\angle APB = \angle AQB = \dfrac{\pi}{2}$ does not formed.

$\therefore m \neq \pm 2$

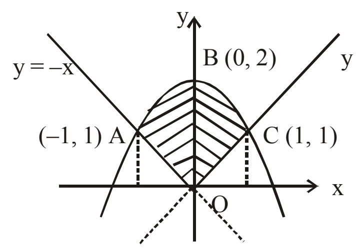
80. (4) (1) $T_r(kA) = k(a_{11}+a_{22}+a_{33}) = kT_r(A)$

(2) $T_r(A+B) = a_{11}+b_{11}+a_{22}+b_{22}+a_{33}+b_{33} = T_r(A)+T_r(B)$

(3) $T_r(I_3) = 1+1+1 = 3$

(4) $T_r(A^2) = \Sigma a_{11}^2 + \Sigma a_{12}^2 \neq (a_{11}+a_{22}+a_{33})^2$

81. (2.33)



We first draw the given curves

The first curve $x^2 - y^2 = 0 \Rightarrow y = \pm x$ represents a pair of straight lines with slopes 1 and –1 passing through origin. The second curve

$\Rightarrow x^2 + y - 2 = 0 \Rightarrow x^2 = -y + 2$

$\Rightarrow x^2 = -(y-2)$

represents a parabola with vertex (0,2) axis as y–axis and concavity downwards. Both the curves are plotted in the figure and the required area is shown by the shaded region.

The points A and C are the points of intersection of $y^2 = x^2$ with $x^2 + y - 2 = 0$.

Solving the two equations, we get

$y^2 + y - 2 = 0$ [putting value of $x^2 = y^2$]

$\Rightarrow (y+2)(y-1) = 0$

giving $y = -2$ and 1, but $y = -2$ is discarded as the required area is above the x-axis.

$\therefore y = 1 \Rightarrow x = \pm 1$

The points A and C are respectively (–1, 1) and (1, 1) now due to symmetry

Area of the bounded region OABCO

$$= 2 \times \text{Area OBCO} = 2 \times \int_0^1 \left[(2 - x^2) - x\right] dx$$

[Since $y = 2 - x^2$ is the upper curve and $y = x$ is the lower curve]

$$= 2\left[2x - \frac{x^3}{3} - \frac{x^2}{2}\right]_0^1 = 2\left[2 - \frac{1}{3} - \frac{1}{2}\right] = \frac{7}{3} = 2.33$$

82. (9.00) Number of digits are 9
Select 2 places for the digit 1 and 2 in ${}^9C_2$ ways from the remaining 7 places select any two places for 3 and 4 in ${}^7C_2$ ways and from the remaining 5 places select any two for 5 and 6 in ${}^5C_2$ ways
Now, the remaining 3 digits can be filled in 3! ways
$\therefore$ Total ways $= {}^9C_2 . {}^7C_2 . {}^5C_2 . 3!$

$$= \frac{9!}{2!.7!}.\frac{7!}{2!.5!}.\frac{5!}{2!.3!}.3! = 9.7! = k. 7!$$

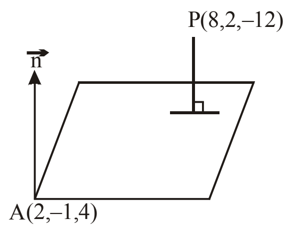
$\Rightarrow k = 9$

83. (32.00) $\vec{n} = 7\hat{i} + 2\hat{j} - \hat{k}$ is normal to plane

(Assuming $n = a\hat{i} + b\hat{j} + c\hat{k}$ and using

$\vec{n}.\overrightarrow{AB} = 0, \vec{n}.\overrightarrow{BC} = 0, \vec{n}.\overrightarrow{AC} = 0$)

$P = (8, 2, -12)$



$\overrightarrow{AP} = 6\hat{i} + 3\hat{j} - 16\hat{k}$

$\therefore$ Distance

$$d = \left|\frac{\overrightarrow{AP}.\vec{n}}{|\vec{n}|}\right| = \left|\frac{42+6+16}{\sqrt{49+4+1}}\right| = \frac{64}{\sqrt{54}} = \frac{64}{3\sqrt{6}} = \frac{64\sqrt{6}}{18} = \frac{32\sqrt{6}}{9}$$

$$= \frac{k\sqrt{6}}{9} \Rightarrow k = 32.$$

84. (343.00)

$$\vec{V} = \vec{A} \times \left[(\vec{A}.\vec{B})\vec{A} - (\vec{A}.\vec{A})\vec{B}\right].\vec{C}$$

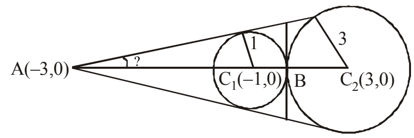
$$= \left(\underbrace{\vec{A}\times(\vec{A}.\vec{B})\vec{A}}_{\text{zero}} - (\vec{A}.\vec{A})\vec{A}\times\vec{B}\right).\vec{C} = -|\vec{A}|^2 [\vec{A}\ \vec{B}\ \vec{C}]$$

Now, $|\vec{A}|^2 = 4 + 9 + 36 = 49$

$$[\vec{A}\ \vec{B}\ \vec{C}] = \begin{vmatrix} 2 & 3 & 6 \\ 1 & 1 & -2 \\ 1 & 2 & 1 \end{vmatrix} = 7$$

$$\therefore \left|-|\vec{A}|^2 [\vec{A}\ \vec{B}\ \vec{C}]\right| = 49 \times 7 = 343$$

85. (3.00) A divides C_1C_2 externally in the ratio 1 : 3.



$\therefore$ coordinates of A are (–3, 0)
We have $\sin\theta = 1/2 \quad \therefore \theta = 30°$

Area $= 3 \times 3 \tan 30° = 3\sqrt{3} = \lambda\sqrt{3}$
$\Rightarrow \lambda = 3$

86. (0.53) Box — 6R, 5B, 4W

P(E) = P(R R B W or B B R W or W W R B)

$n(E) = {}^6C_2 \cdot {}^5C_1 \cdot {}^4C_1 + {}^5C_2 \cdot {}^6C_1 \cdot {}^4C_1 + {}^4C_2 \cdot {}^6C_1 \cdot {}^5C_1$

$n(S) = {}^{15}C_4$

$$\therefore P(E) = \frac{720 \cdot 4!}{15 \cdot 14 \cdot 13 \cdot 12} = \frac{48}{91} = 0.53$$

87. (2.00)

$$\frac{\sin\theta + \sin 2\theta}{\cos\theta + \cos 2\theta} = \frac{2\sin\left(\frac{3\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos\left(\frac{3\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)} = \tan\left(\frac{3\theta}{2}\right)$$

Hence period $= \dfrac{2\pi}{3} = k\dfrac{\pi}{3}$

$\Rightarrow k = 2.$

88. (3.00) The equation of the tangent at

$\left(4\cos\theta, \dfrac{16}{\sqrt{11}}\sin\theta\right)$ to the ellipse

$16x^2 + 11y^2 = 256$ is

$$16x\,(4\cos\theta) + 11y\left(\frac{16}{\sqrt{11}}\sin\theta\right) = 256$$

or $\quad 4x\cos\theta + \sqrt{11}y\sin\theta = 16$

This touches the circle $(x-1)^2 + y^2 = 4^2$ if

$$\left|\frac{4\cos\theta - 16}{\sqrt{16\cos^2\theta + 11\sin^2\theta}}\right| = 4$$

$\Rightarrow (\cos\theta - 4)^2$

$= 16\cos^2\theta + 11\sin^2\theta$

$\Rightarrow \quad 15\cos^2\theta + 11\sin^2\theta + 8\cos\theta - 16 = 0$

$\Rightarrow \quad 4\cos^2\theta + 8\cos\theta - 5 = 0$

$\Rightarrow \quad (2\cos\theta - 1)(2\cos\theta + 5) = 0$

$\Rightarrow \quad \cos\theta = \frac{1}{2} \Rightarrow \theta = \pm\frac{\pi}{3} = \pm\frac{\pi}{k}$

$\Rightarrow \quad k = 3$

89. (0.50) A: blood result says positive about the disease

B_1: Person suffers from the disease

$\therefore P(B_1) = \frac{1}{100}$

B_2 : person does not suffer $\therefore P(B_2) = \frac{99}{100}$

$P(A/B_1) = \frac{99}{100}$, $P(A/B_2) = \frac{1}{100}$

$$P(B_1/A) = \frac{P(B_1).P(A/B_1)}{P(B_1).P(A/B_1) + P(B_2).P(A/B_2)}$$

$$= \frac{\frac{1}{100}\cdot\frac{99}{100}}{\frac{1}{100}\cdot\frac{99}{100} + \frac{99}{100}\cdot\frac{1}{100}} = \frac{99}{2\cdot 99} = \frac{1}{2} = 0.50$$

90. (0.50) $a_n = \int_0^{\pi/2} (1-\sin t)^n \sin 2t\, dt$

Let $1 - \sin t = u \Rightarrow -\cos t\, dt = du$

$$= 2\int_0^1 u^n(1-u)\,du = 2\left(\int_0^1 u^n du - \int_0^1 u^{n+1} du\right) = 2\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

Hence, $\frac{a_n}{n} = 2\left(\frac{1}{n(n+1)} - \frac{1}{n(n+2)}\right)$

$$\lim_{n\to\infty}\sum_1^n \frac{a_n}{n} = 2\left(\sum\left(\frac{1}{n} - \frac{1}{n+1}\right) - \frac{1}{2}\sum\left(\frac{1}{n} - \frac{1}{n+2}\right)\right)$$

$$= 2\left(\sum_1^n\left(\frac{1}{n} - \frac{1}{n+1}\right)\right) - \sum_1^n\left(\frac{1}{n} - \frac{1}{n+2}\right)$$

$$= 2(1) - \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \ldots..\right] = 2 - \frac{3}{2} = \frac{1}{2}$$

$= 0.50$

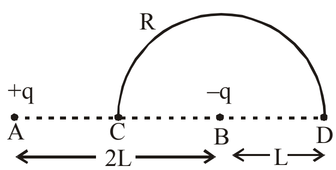
Mock Test-4

ANSWER KEY											
1	(3)	**16**	(4)	**31**	(2)	**46**	(1)	**61**	(3)	**76**	(4)
2	(2)	**17**	(4)	**32**	(2)	**47**	(2)	**62**	(2)	**77**	(2)
3	(4)	**18**	(4)	**33**	(4)	**48**	(2)	**63**	(3)	**78**	(2)
4	(2)	**19**	(1)	**34**	(4)	**49**	(2)	**64**	(2)	**79**	(2)
5	(3)	**20**	(2)	**35**	(4)	**50**	(2)	**65**	(2)	**80**	(4)
6	(3)	**21**	$(10\sqrt{2})$	**36**	(3)	**51**	(600)	**66**	(2)	**81**	(4.00)
7	(2)	**22**	(83.3)	**37**	(4)	**52**	(10.93)	**67**	(2)	**82**	(0.13)
8	(2)	**23**	(3)	**38**	(1)	**53**	(8)	**68**	(4)	**83**	(0.20)
9	(1)	**24**	(70.7)	**39**	(4)	**54**	(68.4)	**69**	(4)	**84**	(75)
10	(2)	**25**	(10.6)	**40**	(3)	**55**	(2)	**70**	(4)	**85**	(23.00)
11	(4)	**26**	(1)	**41**	(2)	**56**	(4)	**71**	(3)	**86**	(1.00)
12	(3)	**27**	(2)	**42**	(3)	**57**	(16)	**72**	(3)	**87**	(1.00)
13	(1)	**28**	(2)	**43**	(4)	**58**	(69.6)	**73**	(3)	**88**	(3.50)
14	(3)	**29**	(2)	**44**	(3)	**59**	(3.8)	**74**	(1)	**89**	(5.00)
15	(2)	**30**	$(5\sqrt{2})$	**45**	(1)	**60**	(2)	**75**	(3)	**90**	(2.25)

Solutions

PHYSICS

1. (3)



Potential at C = $V_C = 0$

Potential at D = V_D

$$= K\left(\frac{-q}{L}\right) + \frac{Kq}{3L} = -\frac{2}{3}\frac{Kq}{L}$$

Potential difference

$$V_D - V_C = \frac{-2}{3}\frac{Kq}{L} = \frac{1}{4\pi\in_0}\left(-\frac{2}{3}.\frac{q}{L}\right)$$

$\Rightarrow$ Work done = $Q(V_D - V_C)$

$$= -\frac{2}{3}\times\frac{1}{4\pi\varepsilon_0}\frac{qQ}{L} = \frac{-qQ}{6\pi\varepsilon_0 L}$$

2. (2) After 2 sec the pulses will overlap completely. The string becomes straight and therefore does not have any potential energy and it entire energy must be kinetic.

3. (4) $mg = 2TL \Rightarrow \pi r^2 Ldg = 2TL \Rightarrow \pi r^2 dg = 2T.$

This relation is independent of L.

4. (2) When either of *A* or *B* is 1 i.e. closed then lamp will glow.

In this case, Truth table

Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

This represents OR gate.

5. (3) Heat required to change the temperature of vessel by a small amount dT

$$-dQ = mC_p dT$$

Total heat required

$$-Q = m\int_{20}^{4} 32\left(\frac{T}{400}\right)^3 dT$$

$$= \frac{100\times10^{-3}\times32}{(400)^3}\left[\frac{T^4}{4}\right]_{20}^{4}$$

$\Rightarrow$ $Q = 0.001996$ kJ

Work done required to maintain the temperature of sink to T_2

$$W = Q_1 - Q_2 = \frac{Q_1 - Q_2}{Q_2}Q_2 = \left(\frac{T_1}{T_2} - 1\right)Q_2$$

$$\Rightarrow \quad W = \left(\frac{T_1 - T_2}{T_2}\right)Q_2$$

For $T_2 = 20$ K

$$W_1 = \frac{300-20}{20}\times0.001996 = 0.028 \text{ kJ}$$

For $T_2 = 4$ K

$$W_2 = \frac{300-4}{4}\times0.001996 = 0.148 \text{ kJ}$$

As temperature is changing from 20k to 4 k, work done required will be more than W_1 but less than W_2.

6. (3) As we know, time period, $T = 2\pi\sqrt{\frac{\ell}{g}}$

When additional mass M is added then

$$T_M = 2\pi\sqrt{\frac{\ell+\Delta\ell}{g}}$$

$$\frac{T_M}{T} = \sqrt{\frac{\ell+\Delta\ell}{\ell}} \quad \text{or} \quad \left(\frac{T_M}{T}\right)^2 = \frac{\ell+\Delta\ell}{\ell}$$

or, $\left(\frac{T_M}{T}\right)^2 = 1 + \frac{Mg}{AY} \quad \left[\because \Delta\ell = \frac{Mg\ell}{AY}\right]$

$$\therefore \frac{1}{Y} = \left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{A}{Mg}$$

7. (2) More the initial temperature more is the rate of cooling. Hence, $T_3 > T_2 > T_1$

or

The rate of cooling decreases with decrease in temperature difference between body and surrounding.

8. (2) We have, $F = kx$

where, F, x and k are force, length and constant respectively.

$\therefore \quad 5 = kx$ ......(1)

and $7 = ky$ ......(2)

Multiplying eq (2) by 2

$14 = 2ky$ ......(3)

Subtracting eq (1) from (3),

$14 - 5 = 2ky - kx$ or $9 = k(2y - x)$

Hence, required length $= 2y - x$

9. (1) From the graph, it is clear that for the same value of load, elongation is maximum for wire OA. Hence OA is the thinnest wire among the four wires

10. (2) Resistance between P and Q

$$r_{PQ} = r \parallel \left(\frac{r}{3} + \frac{r}{2}\right) = \frac{r\times\frac{5}{6}r}{r+\frac{5}{6}r} = \frac{5}{11}r$$

Resistance between Q and R

$$r_{QR} = \frac{r}{2} \parallel (r + \frac{r}{3}) = \frac{\frac{r}{2}\times\frac{4}{3}r}{\frac{r}{2}+\frac{4}{3}r} = \frac{4}{11}r$$

Resistance between P and R

$$r_{PR} = \frac{r}{3} \parallel \left(\frac{r}{2} + r\right) = \frac{\frac{r}{3}\times\frac{3}{2}r}{\frac{r}{3}+\frac{3}{2}r} = \frac{3}{11}r$$

Hence, it is clear that r_{PQ} is maximum

11. (4) Energy in joule (E)

= charge × potential diff. in volt

$E_{electron} = q_e V$ and $E_{proton} = q_p 4V$

de-Broglie wavelength $\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$

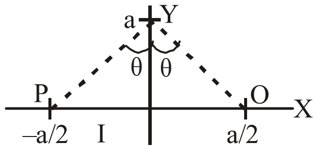
$$\lambda_e = \frac{h}{\sqrt{2m_e eV}} \text{ and } \lambda_P = \frac{h}{\sqrt{2m_P e4V}}$$

$(\because q_e = q_p)$

$$\therefore \frac{\lambda_e}{\lambda_P} = \frac{\frac{h}{\sqrt{2m_e eV}}}{\frac{h}{\sqrt{2m_P e4V}}} = \sqrt{\frac{2m_P e4V}{2m_e eV}} = 2\sqrt{\frac{m_P}{m_e}}$$

12. (3) $B = \frac{\mu_0}{4\pi}\frac{I}{a}(\sin\theta + \sin\theta) = \frac{\mu_0 I\sin\theta}{2\pi a}$

B at any point on Y axis is inversely proportional to a.



13. (1) $f_{max} = \mu mg,\ a_{max} = \mu g.$
If A is the amplitude a_{max}

$$= A\omega^2 = 4\pi^2 AV^2 = \mu g.$$

Therefore, $A = \dfrac{\mu g}{4\pi^2 V^2}$.

14. (3) Total time taken to travel distance d is :

$$\frac{d}{2n_1} + \frac{d}{2n_2} = d\left(\frac{n_1 + n_2}{2n_1 n_2}\right) = \frac{d}{n_{eff}};$$

$$n_2 = 3n_1 \Rightarrow n_{eff} = \frac{3}{2} n_1$$

15. (2) On the screen, we have four amplitudes pair wise coherent.

$$(A_1 + A_2) + (A_3 + A_4) = A_{12} + A_{34}$$

However, if A_{12} and A_{34} have equal magnitude because of random phase of A_{12} and A_{34}, no fringes will be seen.

16. (4) For good demodulation,

$$\frac{1}{f} << RC \text{ or, } RC >> \frac{1}{f}$$

17. (4) 1. $\lambda = \dfrac{0.693}{t^{1/2}}$ 2. $R = \lambda N_t$

Radioactivity at T_1 is $R_1 = \lambda N_1$,
Radioactivity at T_2 is $R_2 = \lambda N_2$
$\therefore$ Number of atoms decayed in time $(T_1 – T_2) = (N_1 – N_2)$

$$= \frac{R_1 - R_2}{\lambda} = \frac{(R_1 - R_2)T}{0.693}$$

i.e., $\propto (R_1 - R_2)T$

18. (4) In the graph given, slope of curve 2 is greater than the slope of curve 1.

$$\left(\frac{\gamma P}{V}\right)_2 > \left(\frac{\gamma P}{V}\right)_1 \Rightarrow \gamma_2 > \gamma_1$$

$$\gamma_{He} > \gamma_{O_2}$$

Since, $\gamma_{monoatomic} > \gamma_{diatomic}$
Hence, curve 2 corresponds to helium and curve 1 corresponds to oxygen.

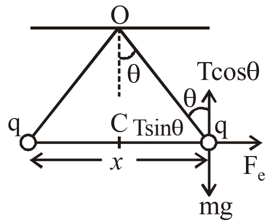
19. (1) $W = MB(\cos\theta_1 - \cos\theta_2)$

$$= MB(\cos 0° - \cos 60°)$$

$$= MB(1 - \frac{1}{2}) = \frac{MB}{2}$$

$$\therefore \tau = MB\sin\theta = MB\sin 60° = \sqrt{3}\frac{MB}{2} = \sqrt{3}W$$

20. (2)



In equilibrium, $F_e = T\sin\theta$

$$mg = T\cos\theta$$

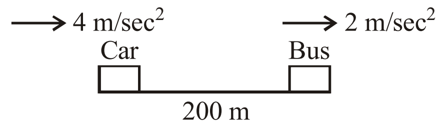
$$\tan\theta = \frac{F_e}{mg} = \frac{q^2}{4\pi\epsilon_0 x^2 \times mg}$$

$$\therefore x = \sqrt{\frac{q^2}{4\pi\epsilon_0 \tan\theta\, mg}}$$

Electric potential at the centre of the line

$$V = \frac{kq}{x/2} + \frac{kq}{x/2} = 4\sqrt{kmg\tan\theta}$$

21. $(10\sqrt{2})$



Given, $u_C = u_B = 0$, $a_C = 4\ m/s^2$, $a_B = 2\ m/s^2$
hence relative acceleration, $a_{CB} = 2\ m/sec^2$

Now, we know, $s = ut + \dfrac{1}{2}at^2$

$$200 = \frac{1}{2} \times 2t^2 \quad \because u = 0$$

Hence, the car will catch up the bus after time

$$t = 10\sqrt{2} \text{ second}$$

22. (83.3) Power of source = EI = 240 × 0.7 = 166

$$\Rightarrow \text{Efficiency} = \frac{140}{166} \Rightarrow \eta = 83.3\%$$

23. (3) In the given equation $[\rho] = [b][x]$;
$\therefore [b] = [\rho]/[x]$. But ρ is mass per unit length and x is distance, therefore $[b] = ML^{-1}/L = ML^{-2}T^0$

24. (70.7) Electric field intensity at the centre of the disc.

$E = \frac{\sigma}{2\epsilon_0}$ (given)

Electric field along the axis at any distance x from the centre of the disc

$$E' = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$$

From question, x = R (radius of disc)

$$\therefore E' = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{R}{\sqrt{R^2+R^2}}\right)$$

$$= \frac{\sigma}{2\epsilon_0}\left(\frac{\sqrt{2}R - R}{\sqrt{2}R}\right)$$

$$= \frac{4}{14}E$$

$\therefore$ % reduction in the value of electric field

$$= \frac{\left(E - \frac{4}{14}E\right)\times 100}{E} = \frac{1000}{14}\% \simeq 70.7\%$$

25. (10.6) Work done in going from a distance r_1 to a distance r_2 away from centre of the earth, by a body of mass m, is,

$W = GMm\,(1/r_1 - 1/r_2)$,

For our case we should have

$1/2\, mv^2 = GMm\,[(1/R_e) - (1/10R_e)]$

$= (GMm/R_e) \times (9/10)$

$v = \sqrt{[(2GM/R_e) \times (9/10)]}$

$= \sqrt{(9/10)} \times$ escape velocity

$= \sqrt{(9/10)} \times 11.1$ km/s $= 10.6$ km/s

26. (1) $mvr = \frac{nh}{2\pi}$, $\lambda = \frac{h}{mv}$;

Using the two concept we get, $mvr = \frac{nh}{2\pi}$ (where n = 1)

$$2\pi r = \frac{1\times h}{mv} \quad ....(1);$$

$$\lambda = \frac{h}{mv} \quad ....(2)$$

Divide (1) by (2), $\frac{2\pi r}{\lambda} = \frac{h\times mv}{mv \times h} = \frac{1}{1} = 1:1$

27. (2) The angle for which the ranges are same is complementary.

Let one angle be θ, then other is $90° - \theta$

$$T_1 = \frac{2u\sin\theta}{g}, T_2 = \frac{2u\cos\theta}{g}$$

$$T_1T_2 = \frac{4u^2\sin\theta\cos\theta}{g^2} = \frac{2R}{g} \quad (\because R = \frac{u^2\sin 2\theta}{g})$$

Hence it is proportional to R.

28. (2) On comparing the given equation to

$\vec{E} = a_0\hat{i}\cos(\omega t - kz)$

$\omega = 6\times 10^8$,

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{c}$$

$$k = \frac{\omega}{c} = \frac{6\times 10^8}{3\times 10^8} = 2\,m^{-1}$$

29. (2) $\omega_{rod} = \omega_{point} = \left(\frac{v_{rel}}{r}\right)$,

v_{rel} represents the velocity of one point w.r.t. other.

$= \frac{3v - v}{r}$ and 'r' being the distance between them.

$= \frac{2v}{r}$

30. $(5\sqrt{2})$

$$i_R = \frac{V_0}{R} = \frac{100}{20} = 5, i_L = \frac{V_0}{X_L} = \frac{100}{10} = 10$$

and $i_C = \frac{V_0}{X_C} = \frac{100}{20} = 5$

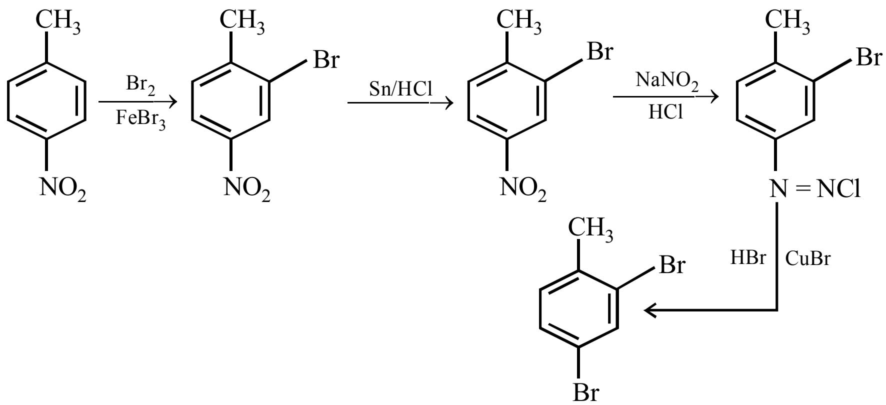
Current, $i = \sqrt{i_R^2 + (i_C - i_L)^2}$

$= \sqrt{5^2 + 5^2} = 5\sqrt{2}$ amp.

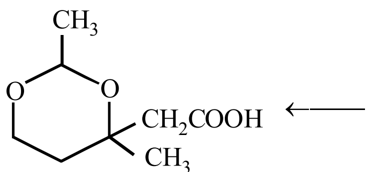
CHEMISTRY

31. (2) The molecule 2,3 - pentadiene does not have any chiral C but at the same time it does not have any mirror plane which makes the molecule chiral.

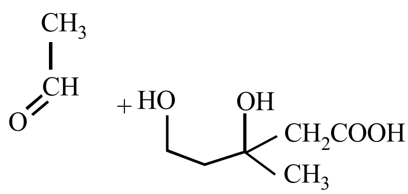
32. (2)



33. (4) The compound A is cyclic acetal, so it should have an aldehyde and a diol as the two starting compounds.



[A]



34. (4) In N_2^+, there is one unpaired electron hence it is paramagnetic.

35. (4) In Ag_2O (O.N. of Ag +1) in Ag the O.N. is O. There is gain of electrons, hence H_2O_2 is acting as reducing agent.

36. (3) $H_2C\underset{O}{-}CH_2 + RMgX \longrightarrow \underset{OMgX}{CH_2} - \underset{R}{CH_2}$

$\xrightarrow[-Mg(OH)X]{H_2O} \underset{OH}{CH_2} - \underset{R}{CH_2}$

37. (4) Total energy of a revolving electron is the sum of its kinetic and potential energy.

Total energy = K.E. + P. E.

$$= \frac{e^2}{2r} + \left(-\frac{e^2}{r}\right) = -\frac{e^2}{2r}$$

38. (1) $3Cl_2 + 2NaI \rightarrow 2NaCl + I_2$

I_2 gives violet colouration in $CHCl_3$.

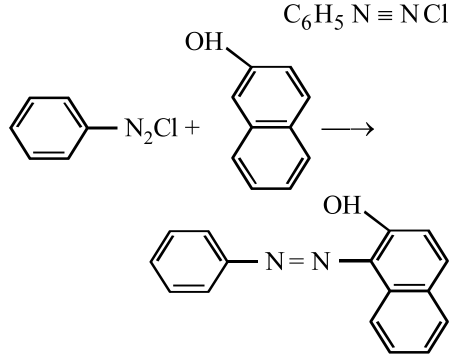
$$5Cl_2 + 6H_2O + I_2 \rightarrow \underset{\text{Colourless}}{HIO_3} + 10HCl$$

39. (4) Tertiary halides on treatment with base, such as sodium methoxide, readily undergo elimination resulting in the formation of alkenes. (Williamson's Synthesis)

$$\underset{\text{t-Butyl chloride}}{CH_3 - \overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}} - Cl} + N\overset{+}{a}\,\overset{-}{O}\,CH_2CH_3 \xrightarrow{E2} \underset{\text{Isobutylene}}{CH_3 - \underset{\underset{CH_3}{|}}{C} = CH_2}$$

40. (3) Performic acid causes hydroxylation of the double bond ; the two –OH groups add in *anti*-manner. Hence *cis*-isomer gives racemic mixture while the *trans*-isomer gives meso.

41. (2) $C_6H_5NH_2 + NaNO_2 + HCl \xrightarrow[0°C]{} C_6H_5\overset{+}{N} \equiv N\overset{-}{Cl}$



Red dye

42. (3) Multiple bonds formation tendency with carbon and nitrogen decreases from sulphur to tellurium.
CS_2 (S = C = S) is moderately stable,
CSe_2 (Se = C = Se) decomposes readily whereas,
CTe_2 (Te = C = Te) does not exist.

43. (4) Liquation is the principle based on difference in melting points.

44. (3) In NO_2^+ odd (unpaired) electron is removed. In peroxides (O_2^{2-}) no unpaired electrons are persent as the antibonding π M.O.'s acquired one more electron each for pairing. AlO_2^- containing Al^{3+} ($2s^2p^6$) configuration and 2 oxides (O^{2-}) ions each of which does not contain unpaired electron. Superoxide O_2^- has one unpaired electron in π antibonding M.O.

45. (1) The two solutions are isotonic hence there will be no movement of H_2O.

46. (1) Melamine plastic crockery is a copolymer of HCHO and melamine.

47. (2) Hydrated $CoCl_2 . 6H_2O$ is pink coloured and contains octahedral $[Co(H_2O)_6]^{2+}$ ions. If this is partially dehydrated by heating, then blue coloured tetrahedral ions $[Co(H_2O)_4]^{2+}$ are formed.

$$\underset{\text{pink}}{[Co(H_2O)_6]^{2+}} \rightleftharpoons \underset{\text{blue}}{[Co(H_2O)_4]^{2+}} + 2H_2O$$

48. (2)

49. (2)
- Li does not form peroxide or superoxide due to its small size.
- Solubility of carbonates and biocarbonates increases on moving down the group.
- The increasing order of size of hydrated ions of alkali metals is
$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
- Cesium used in photoelectric cells due to its low I.E. Hence statements (2) is correct.

50. (2) (i) $A^{3+} + e^- \longrightarrow A^{2+}$, $\Delta G_1 = -1\ F\ y_2$
(ii) $A^{2+} + 2e^- \longrightarrow A$, $\Delta G_2 = -2F(-y_1) = 2Fy_1$
Add, (i) and (ii), we get
$A^{3+} + 3e^- \longrightarrow A$;
$\Delta G_3 = \Delta G_1 + \Delta G_2$
$-3FE° = -Fy_2 + 2Fy_1$
$-3FE° = -F(y_2 - 2y_1)$
$$E° = \frac{y_2 - 2y_1}{3}$$

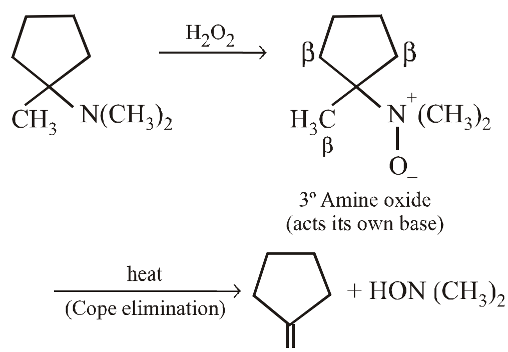
51. (600) In a *fcc* lattice, the distance between the cation and anion is equal to the sum of their radii, which is equal to half of the edge length of unit cell,
i.e. $r^+ + r^- = \frac{a}{2}$ (where a = edge length)
$r^+ = 100$ pm, $r^- = 200$ pm
Edge length $= 2r^+ + 2r^- = (2 \times 100 + 2 \times 200)$ pm
$= (200 + 400)$ pm $= 600$ pm.

52. (10.93) $\Delta H = (-40.8) \times \frac{1.8}{18} = -4.08$ kJ ;
$$\Delta S = \frac{\Delta H}{T} = \frac{-4.08 \times 10^3}{373.15} = -10.93\ JK^{-1}$$

53. (8) $n = 5$ means $l = 0, 1, 2, 3, 4$
since $m_l = +1$
hence total no. of electrons will be
$= 0(\text{from } s) + 2(\text{from } p) + 2(\text{from } d) + 2(\text{from } f) + 2(\text{from } g)$
$= 0 + 2 + 2 + 2 + 2 = 8$

54. (68.4) As we know,
$$\text{Molarity} = \frac{\text{No. of moles of sugar}}{\text{Volume of solution (in L)}}$$
$$0.1 = \frac{\text{No. of moles of sugar}}{2\,L}$$
So, no. of moles of sugar = 0.2 mol
$\therefore$ Mass of sugar = No. of moles of sugar × Molar mass of sugar
$= 0.2 \times 342 = 68.4$ g

55. (2) The reaction involves Cope eliminiation (heating of a 3° amine oxide to form an alkene with the elimination of a 2° hydroxylamine).



There are two unsaturated carbon in the product.

56. (4) According to question C – Terminal must be alanine and N – Terminal do have chiral carbon means it should not be glycine. So possible sequence is :

Val Phe Gly Ala
Val Gly Phe Ala
Phe Val Gly Ala
Phe Gly Val Ala

57. (16) $\frac{r_{O_2}}{r_{H_2}} = \frac{n_{O_2}}{n_{H_2}} = \frac{w/32}{4/2} = \frac{w}{64} = \sqrt{\frac{M_{H_2}}{M_{O_2}}}$

$= \sqrt{\frac{2}{32}} = \frac{1}{4}$

w = 64/4 = 16 g

58. (69.6) $\frac{P° - P_S}{P°} = \frac{w/m}{W/M}$

(640–600)/640 = wM / mW
40/640 = 2.175 × 78/m × 39.08
m = 2.175 × 78 × 640 / 39.08 × 40
= 69.458 ≅ 69.60

59. (3.8) $K = \frac{[H_3O^+][HCO_3^-]}{[CO_2][H_2O]^2}$

As pH = 6.0 $[H_3O]^+ = 10^{-6}$

$K = \frac{[H_3O^+][HCO_3^-]}{[CO_2][H_2O]^2}$ (H_2O is in excess, therefore its conc. remains constant)

$\frac{[HCO_3^-]}{[CO_2]} = \frac{K}{[H_3O^+]} = \frac{3.8\times10^{-6}}{10^{-6}} = 3.8$

60. (2) $t_{1/2} = \frac{0.69}{k}$, $t_{3/14} = t_{75\%}$

$t_{3/4} = \frac{2.303}{k}\log\frac{a}{\left(a - \frac{3a}{4}\right)}$

$= \frac{2.303}{k}\log 4 = \frac{2.303}{k} \times 2 \times 0.3010 = \frac{0.69 \times 2}{k}$

$\frac{t_{3/4}}{t_{1/2}} = \frac{0.69\times2}{k} \times \frac{k}{0.69} \Rightarrow t_{3/4} = 2t_{1/2}$

MATHEMATICS

61. (3) Given equation is

$(p-q)x^2 + (q-r)x + (r-p) = 0$

By using formula for finding the roots

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, we get

$x = \frac{(r-q) \pm \sqrt{(q-r)^2 - 4(r-p)(p-q)}}{2(p-q)}$

$\Rightarrow x = \frac{(r-q) \pm (q+r-2p)}{2(p-q)} = \frac{r-p}{p-q}, 1$

62. (2) $(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + b^2 + c^2 + d^2 \le 0$

$\Rightarrow (a^2p^2 - 2abp + b^2) + (b^2p^2 - 2bcp + c^2) + (c^2p^2 - 2cdp + d^2) \le 0$

$\Rightarrow (ap - b)^2 + (bp - c)^2 + (cp - d)^2 \le 0$

$\Rightarrow ap - b = 0,\ bp - c = 0\ \&\ cp - d = 0$

$\Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \Rightarrow$ a, b, c and d are in G.P

Also ad = bc

63. (3)

(1) $\log(a + 2b) = \frac{1}{2}\log(a + 2b)^2$

$= \frac{1}{2}\log(a^2 + 4b^2 + 4ab)$

$= \frac{1}{2}\log(12ab + 4ab)$

$= \frac{1}{2}\log(2^4 .ab)$

$= \frac{1}{2}(4\log 2 + \log a + \log b)$

(2) Let $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b} = k$

$\Rightarrow \log x = k(b-c),\ \log y = k(c-a),$

$\log z = k(a-b)$

$\therefore x^a . y^b . z^c = p^{k[a(b-c))+b(c-a)+c(a-b)]}$

$= p^{k(0)} = 1$

where p is any arbitrary base of the log.

(3) Given expression

$$= \log_{xyz} xy + \log_{xyz} yz + \log_{xyz} zx$$

$$= \log_{xyz}(xy\,.\,yz\,.\,zx)$$

$$= \log_{xyz}(x^2.y^2.z^2)$$

$$= 2\log_{xyz}(xyz) = 2 \times 1 = 2$$

64. (2) $\alpha+\beta+\gamma = \frac{\pi}{2} \Rightarrow \alpha+\gamma = \frac{\pi}{2} - \beta.$

so that $\cot(\alpha+\gamma) = \cot\left(\frac{\pi}{2}-\beta\right)$

$$\Rightarrow \frac{\cot\alpha\cot\gamma - 1}{\cot\alpha + \cot\gamma} = \frac{1}{\cot\beta}$$

$\Rightarrow \cot\alpha\cot\gamma - 1 = 2 \Rightarrow \cot\alpha\cot\gamma = 3.$

(since $\cot\alpha + \cot\gamma = 2\cot\beta$)

65. (2) The equation of the chord, having mid-point as (x_1, y_1), of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by

$T = S_1$ ..... (i)

where, $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$ and

$$S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$$

According to the question,

$(x_1, y_1) = (5, 3)$ and $a^2 = 16, b^2 = 25$

as $25x^2 - 16y^2 = 400$

$$\Rightarrow \frac{x^2}{16} - \frac{y^2}{25} = 1 \qquad \therefore$$

$$\frac{5x}{16} - \frac{3y}{25} = \frac{25}{16} - \frac{9}{25} \quad \text{[Using (i)]}$$

$\Rightarrow 125x - 48y = 625 - 144 \Rightarrow 125x - 48y = 481$

66. (2) $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4 = (x-\sqrt{3})^2 + 1$

$\because$ RHS ≥ 1 so, the solution exists

If and only if $x - \sqrt{3} = 0 \Rightarrow x = \sqrt{3}$

and then equation is obviously satisfied

67. (2) We have $z^5 - 1 = (z-1)(z-\alpha_1) \ldots (z-\alpha_4)$

Putting $z = \omega$ we get

$$(\omega-1)(\omega-\alpha_1)(\omega-\alpha_2)(\omega-\alpha_3)(\omega-\alpha_4) = \omega^5 - 1$$

Putting $z = \omega^2$ we get

$$(\omega^2-1)(\omega^2-\alpha_1)(\omega^2-\alpha_2)(\omega^2-\alpha_3)(\omega^2-\alpha_4) = \omega^{10} - 1$$

$$\therefore \frac{(\omega-1)(\omega-\alpha_1)(\omega-\alpha_2)(\omega-\alpha_3)(\omega-\alpha_4)}{(\omega^2-1)(\omega^2-\alpha_1)(\omega^2-\alpha_2)(\omega^2-\alpha_3)(\omega^2-\alpha_4)}$$

$$= \frac{\omega^5 - 1}{\omega^{10} - 1}$$

$$\Rightarrow \frac{(\omega-\alpha_1)(\omega-\alpha_2)(\omega-\alpha_3)(\omega-\alpha_4)}{(\omega^2-\alpha_1)(\omega^2-\alpha_2)(\omega^2-\alpha_3)(\omega^2-\alpha_4)}$$

$$= \frac{(\omega^2-1)^2}{(\omega-1)^2} = (\omega+1)^2 = \omega^4 = \omega$$

68. (4)

(1) We have $|AB| = |A||B|$

Also for a square matrix of order 3, $|kA| = k^3 |A|$ because each element of the matrix A is multiplied by k and hence in this case we will have k^3 common

$\therefore |3AB| = 3^3 |A||B| = 27(-1)(3) = -81$

(2) Since A is invertible, therefore A^{-1} exists and $AA^{-1} = I \Rightarrow \det(AA^{-1}) = \det(I)$

$\Rightarrow \det(A)\det(A^{-1}) = 1$

$$\Rightarrow \det(A^{-1}) = \frac{1}{\det(A)}$$

(3) $(A+B)^2 = (A+B)(A+B)$

$= A^2 + AB + BA + B^2$

$= A^2 + 2AB + B^2$ if $AB = BA$.

69. (4) $f(x) = |x-1| = \begin{cases} -x+1, & x < 1 \\ x-1, & x \geq 1 \end{cases}$

Consider $f(x^2) = (f(x))^2$

If it is true it should be $\forall\ x$

$\therefore$ Put $x = 2$

LHS $= f(2^2) = |4 - 1| = 3$

RHS $= (f(2))^2 = 1$

$\therefore$ (2) is not correct

Consider $f(x + y) = f(x) + f(y)$

Put $x = 2, y = 5$ we get

$f(7) = 6; f(2) + f(5) = 1 + 4 = 5$

$\therefore$ (2) is not correct

Consider $f(|x|) = |f(x)|$

Put $x = -5$ then $f(|-5|) = f(5) = 4$

$|f(-5)| = |-5-1| = 6$

$\therefore$ (3) is not correct.

70. (4) Let $a = \tan\theta$ and $b = \tan\phi$

$$\therefore\ \sin^{-1}\left[\frac{2a}{1+a^2}\right] = \sin^{-1}\left[\frac{2\tan\theta}{1+\tan^2\theta}\right]$$

$$= \sin^{-1}[\sin 2\theta] = 2\theta = 2\tan^{-1} a$$

$$\text{and } \sin^{-1}\left[\frac{2b}{1+b^2}\right] = \sin^{-1}\left[\frac{2\tan\phi}{1+\tan^2\phi}\right]$$

$$= \sin^{-1}[\sin 2\phi] = 2\phi = 2\tan^{-1} b$$

$$\text{Thus, } \sin^{-1}\left[\frac{2a}{1+a^2}\right] = 2\tan^{-1} a \text{ and}$$

$$\sin^{-1}\left[\frac{2b}{1+b^2}\right] = 2\tan^{-1} b$$

$$\therefore\ 2\tan^{-1}x = \sin^{-1}\left[\frac{2a}{1+a^2}\right] + \sin^{-1}\left[\frac{2b}{1+b^2}\right]$$

$$= 2\tan^{-1} a + 2\tan^{-1} b$$

$$\Rightarrow \tan^{-1} x = \tan^{-1} a + \tan^{-1} b$$

$$\tan^{-1} x = \tan^{-1}\frac{a+b}{1-ab}$$

$$\therefore\ x = \frac{a+b}{1-ab}$$

71. (3) Let f image of goh image, fogoh = F (x)

$= f[\text{goh}(x)]$

$= f[g(\sqrt{x+3})] = f(\cos\sqrt{x+3})$

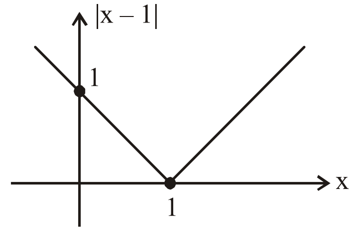
$$F(x) = \frac{2}{\cos\sqrt{x+3}+1}$$

Now $x + 3 \ge 0$ and $-1 < \cos\sqrt{x+3} \le 1$

$\sqrt{x+3} \ne (2n-1)\pi, n \in N$

So, range of fogoh is $[1, \infty)$

72. (3) Since $|x|$ is not diff. at $x = 0$



$\Rightarrow |x-1|$ is not diff at $x = 1$.

$x^n|x|$ is n times diff. at $x = 0$

$\Rightarrow (x-1)^2|x-1|$ is twice diff. at $x = 1$

but not thrice diff. at $x = 1$

73. (3) $f(x) = \dfrac{1}{x-1}$ is discontinuous at $x = 1$.

$(gof)(x) = g(f(x)) = -\dfrac{(x-1)^2}{(2x-1)(x-2)}$, which is not defined at $x = 1/2, 2$.

Hence the set of points where (gof) (x) is discontinuous is $\{1/2, 1, 2\}$

74. (1) $\sum_{r=0}^{m} {}^{n+r}C_n = \sum_{r=0}^{m} {}^{n+r}C_r$

$(\because {}^{n+r}C_n = {}^{n+r}C_{n+r-n})$

$= {}^{n}C_0 + {}^{n+1}C_1 + {}^{n+2}C_2 + {}^{n+3}C_3 + \ldots\ldots + {}^{n+m}C_m$

Using, ${}^{n}C_0 = 1 = {}^{n+1}C_0$

$= ({}^{n+1}C_0 + {}^{n+1}C_1) + {}^{n+2}C_2 + {}^{n+3}C_3 + \ldots. + {}^{n+m}C_m$

Using, ${}^{n}C_r + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}$

$= ({}^{n+2}C_1 + {}^{n+2}C_2) + {}^{n+3}C_3 + \ldots.. + {}^{n+m}C_m$

Using this again and again, we are left with

$= {}^{n+m}C_{m-1} + {}^{n+m}C_m$

$= {}^{n+m+1}C_m = {}^{n+m+1}C_{n+1}$

75. (3) $I = \int_{1/n}^{(an-1)/n} \frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}\,dx$ ....(i)

Then, $I = \int_{1/n}^{(an-1)/n} \frac{\sqrt{a-x}}{\sqrt{x}+\sqrt{a-x}}\,dx$ ....(ii)

Adding (i) and (ii), we get

$$2I = \int_{1/n}^{(an-1)/n} 1.dx = \frac{an-2}{n} \Rightarrow I = \left(\frac{an-2}{2n}\right)$$

76. (4) $I = \int_{-\pi/4}^{\pi/4} (\underbrace{x|x|}_{\text{odd f}} + \underbrace{\sin^3 x}_{\text{odd f}} + \underbrace{x\tan^2 x}_{\text{odd f}} + 1)\,dx$

$$I = \int_{-\pi/4}^{\pi/4} dx = \frac{\pi}{2}$$

$[\because \int_{-a}^{a} f(x)dx = 0,\ \text{as } f(x) \text{ is an odd function}]$

77. (2) $(1-x-2x^2)^6 = (1+x)^6(1-2x)^6$
$= 1 + a_1x + a_2x^2 + \ldots\ldots.. + a_{12}x^{12}$
Putting $x = 1/2$, we have $0 = 1 + a_1/2 + a_2/2^2$
$+ a_3/2^3 + a_4/2^4 + \ldots\ldots.. + a_{12}/2^{12}$ .....(1)
Putting $x = -1/2$, we have
$1 = 1 - a_1/2 + a_2/2^2 - a_3/2^3 + a_4/2^4 - \ldots..$
$+ a_{12}/2^{12}$ ....(2)

Adding (1) and (2), we have
$1 = 2(1 + a_2/2^2 + a_4/2^4 + \ldots..+ a_{12}/2^{12})$
$\Rightarrow a_2/2^2 + a_4/2^4 + a_6/2^6 \ldots..+ a_{12}/2^{12} = -1/2$

78. (2) $y^2 = 4x$ & $\frac{x^2}{8} + \frac{y^2}{2} = 1$

Equation of tangent to above curves are respectively.

$$y = mx + \frac{1}{m} \text{ and } y = mx + \sqrt{8m^2+2}$$

Comparing $\frac{1}{m} = \sqrt{8m^2+2}$

$\Rightarrow m^2(8m^2+2) = 1$

seeing the options

$$m = \pm\frac{1}{2} \text{ satisfy the equation}$$

$\Rightarrow y = \pm\frac{1}{2}x \pm 2 \Rightarrow 2y = \pm x \pm 4$

i.e. $2y = x + 4$ & $x + 2y + 4 = 0$

79. (2) Let the point be (x_1, y_1).
Therefore $y_1 = (x_1-3)^2$ ...(i)
$\therefore$ Now slope of the tangent at (x_1, y_1) is $2(x_1-3)$, but it is equal to 1.

Therefore, $2(x_1-3) = 1 \Rightarrow x_1 = \frac{7}{2}$

$$y_1 = \left(\frac{7}{2}-3\right)^2 = \frac{1}{4}.$$

Hence the point is $\left(\frac{7}{2}, \frac{1}{4}\right)$.

80. (4) $x = \tan A, y = \tan B, -z = \tan C.$
Then $(x+y-z) = -xyz.$
$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$
$\Rightarrow A + B + C = \pi \quad \Rightarrow 2A + 2B = 2\pi - 2C$
$\Rightarrow \tan(2A+2B) = \tan(2\pi - 2C) = -\tan 2C$
$\Rightarrow \tan 2A + \tan 2B + \tan 2C$
$= \tan 2A.\tan 2B.\tan 2C$

$$\Rightarrow \frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C}$$

$$= \frac{2\tan A}{1-\tan^2 A}.\frac{2\tan B}{1-\tan^2 B}.\frac{2\tan C}{1-\tan^2 C}$$

Put the value of $\tan A$, $\tan B$, $\tan C$, we get

$$\Rightarrow \frac{2x}{1-x^2} + \frac{2y}{1-y^2} - \frac{2z}{1-z^2}$$

$$= -\frac{8xyz}{(1-x^2)(1-y^2)(1-z^2)}$$

81. (4.00) Set $A = \{a_1, a_2, \ldots.., a_{20}\}$ has 20 distinct elements.
We have to select 5-element subset.
$\therefore$ Number of 5-element subsets $= {}^{20}C_5$

According to question

$${}^{20}C_5 = \left({}^{19}C_4\right).k$$

$$\Rightarrow \quad \frac{20!}{5!\ 15!} = k.\left(\frac{19!}{4!\ 15!}\right)$$

$$\Rightarrow \quad \frac{20}{5} = k \Rightarrow k = 4$$

82. (0.13) $\cos 36° \cos 42° \cos 78°$

$$= \cos 36° \cos(60° - 18°)\cos(60° + 18°)$$

$$= \frac{\sqrt{5}+1}{4}\left(\cos^2 60° - \sin^2 18°\right)$$

$$= \left(\frac{\sqrt{5}+1}{4}\right)\left[\frac{1}{4} - \left(\frac{\sqrt{5}-1}{4}\right)^2\right]$$

$$= \frac{\sqrt{5}+1}{16}\left[1 - \frac{\left(\sqrt{5}-1\right)^2}{4}\right]$$

$$= \frac{\sqrt{5}+1}{16}\left[\frac{4-6+2\sqrt{5}}{4}\right] = \frac{1}{8} = 0.13$$

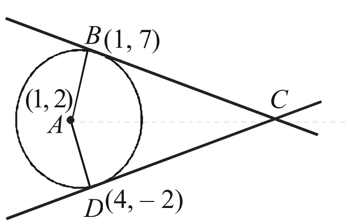
83. (0.20) The given expression is equal to

$$\left|\cos(\cos^{-1} x + \sin^{-1} x + \sin^{-1} x)\right|$$

$$= \left|\cos\left(\frac{\pi}{2} + \sin^{-1} x\right)\right| = \left|-\sin(\sin^{-1} x)\right| = |-x| = \left|-\frac{1}{5}\right|$$

$$= \frac{1}{5} = 0.20 \qquad \left[\text{Using } \cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}\right]$$

84. (75.00)



Here, centre is A (1,2), and Tangent at B (1,7) is

$$x.1 + y.7 - 1\,(x+1) - 2\,(y+7) - 20 = 0$$

or $\quad y = 7 \qquad$ ...(1)

Tangent at D (4,–2) is

$$3x - 4y - 20 = 0 \qquad ...(2)$$

Solving (1) and (2), we get C is (16, 7)

Area $ABCD = 2$ (Area of Δ ABC) $= 2 \times \frac{1}{2} AB \times BC$

$AB \times BC = 5 \times 15 = 75$ units

85. (23.00) Given system of equations can be written in matrix form as AX = B where

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{pmatrix} \text{ and } B = \begin{pmatrix} 6 \\ 9 \\ b \end{pmatrix}$$

Since, system is consistent and has infinitely many solutions

$\therefore$ (adj. A) B = 0

$$\Rightarrow \begin{pmatrix} 3a-25 & 15-2a & 1 \\ 10-a & a-6 & -2 \\ -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} 6 \\ 9 \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow -6 - 9 + b = 0 \Rightarrow b = 15$

and $6(10 - a) + 9(a - 6) - 2(b) = 0$

$\Rightarrow 60 - 6a + 9a - 54 - 30 = 0$

$\Rightarrow 3a = 24 \Rightarrow a = 8$

Hence, $a + b = 8 + 15 = 23$.

86. (1.00) Let $B = I + A + A^2 + \ldots + A^{k-1}$

Now multiply both sides by (I – A), we get

$B(I - A) = (I + A + A^2 + \ldots + A^{k-1})\,(I - A)$

$= I - A + A - A^2 + A^2 - A^3 + \ldots - A^{k-1} + A^{k-1} - A^k$

$= I - A^k = I$, since $A^k = 0 \Rightarrow B = (I - A)^{-1}$

Hence $(I - A)^{-1} = I + A + A^2 + \ldots + A^{k-1}$ Thus

$|p| = |-1| = 1$

87. (1.00) As $x \to \frac{1}{3}$; $\{x + 1\} \to \{1 + 1/3\} \to 1/3$

Similarly $\{x + 2\} \to \frac{1}{3}$ as $x \to \frac{1}{3}$

$$\Rightarrow \lim_{x \to 1/3} f(x) = \lim_{x \to 1/3} \frac{x - 1/3}{x - 1/3} = 1$$

88. (3.50) Given differential equation is

$$\frac{(2 + \sin x)}{(1 + y)} \cdot \frac{dy}{dx} = \cos x$$

which can be rewritten as

$$\frac{dy}{1 + y} = \frac{\cos x}{2 + \sin x} dx$$

Integrate both the sides, we get

$$\int \frac{dy}{1+y} = \int \frac{\cos x\, dx}{2+\sin x}$$

$\Rightarrow$ $\log(1+y) = \log(2+\sin x) + \log C$

$\Rightarrow$ $1 + y = C(2+\sin x)$

Given $y(0) = 2$

$\Rightarrow$ $1 + 2 = C[2+\sin 0] \Rightarrow C = \frac{3}{2}$

Now, $y\left(\frac{\pi}{2}\right)$ can be found as

$$1 + y = \frac{3}{2}\left(2+\sin\frac{\pi}{2}\right) \Rightarrow 1 + y = \frac{9}{2}$$

$\Rightarrow$ $y = \frac{7}{2}$

Hence, $y\left(\frac{\pi}{2}\right) = \frac{7}{2}$

89. (5.00) Differentiating w.r.t.x, we get

$$\frac{1}{f(x)} = \frac{1}{(f(x))^2} 2f(x)f'(x)$$

$$\Rightarrow f'(x) = \frac{1}{2} \Rightarrow f(x) = \frac{1}{2}x + \alpha$$

where α is a constant.

$\because$ $f(0) = 5 \Rightarrow 0 + \alpha = 5 \Rightarrow \alpha = 5$

90. (2.25) Given, in ΔABC $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$

$\Rightarrow$ $1(c^2 - ab) - a(c - a) + b(b - c) = 0$

$\Rightarrow$ $a^2 + b^2 + c^2 - ab - bc - ca = 0$

$\Rightarrow$ $2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$

$\Rightarrow$ $(a^2 + b^2 - 2ab) + (b^2 + c^2 - 2bc) + (c^2 + a^2 - 2ca) = 0$

$\Rightarrow$ $(a - b)^2 + (b - c)^2 + (c - a)^2 = 0$

Here, sum of squares of three members can be zero if and only if $a = b = c$

$\Rightarrow$ ΔABC is equilateral.

$\Rightarrow \angle A = \angle B = \angle C = 60°$

$\therefore$ $\sin^2 A + \sin^2 B + \sin^2 C$

$$= (\sin^2 60° + \sin^2 60° + \sin^2 60°)$$

$$= 3 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{9}{4} = 2.25$$

Mock Test-5

ANSWER KEY											
1	(3)	**16**	(4)	**31**	(2)	**46**	(2)	**61**	(3)	**76**	(3)
2	(2)	**17**	(1)	**32**	(1)	**47**	(3)	**62**	(3)	**77**	(4)
3	(1)	**18**	(1)	**33**	(4)	**48**	(3)	**63**	(1)	**78**	(3)
4	(3)	**19**	(4)	**34**	(2)	**49**	(2)	**64**	(1)	**79**	(4)
5	(3)	**20**	(4)	**35**	(2)	**50**	(1)	**65**	(1)	**80**	(1)
6	(3)	**21**	(4)	**36**	(4)	**51**	(19.55)	**66**	(4)	**81**	(3.00)
7	(1)	**22**	(3)	**37**	(3)	**52**	(8)	**67**	(3)	**82**	(2.00)
8	(2)	**23**	(20)	**38**	(1)	**53**	(128)	**68**	(3)	**83**	(0.60)
9	(1)	**24**	(7)	**39**	(2)	**54**	(0.32)	**69**	(4)	**84**	(1.00)
10	(3)	**25**	(9)	**40**	(4)	**55**	(0.4)	**70**	(3)	**85**	(17.67)
11	(2)	**26**	(6.5)	**41**	(2)	**56**	(68.13)	**71**	(4)	**86**	(325.00)
12	(3)	**27**	(5)	**42**	(1)	**57**	(5.19)	**72**	(2)	**87**	(0.50)
13	(4)	**28**	(8.66)	**43**	(3)	**58**	(4.0)	**73**	(3)	**88**	(1.50)
14	(2)	**29**	(2)	**44**	(2)	**59**	(4.0)	**74**	(1)	**89**	(2.00)
15	(2)	**30**	(1.25)	**45**	(2)	**60**	(100)	**75**	(3)	**90**	(6.00)

Solutions

PHYSICS

1. (3) For the man standing in the lift, the acceleration of the ball

$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - a$$

Where 'a' is the acceleration of the mass (because the acceleration of the lift is 'a')

For the man standing on the ground, the acceleration of the ball

$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - 0 = g$$

2. (2) For a diamagnetic material, the value of μ_r is less than one. For any material, the value of $\in_r$ is always greater than 1.

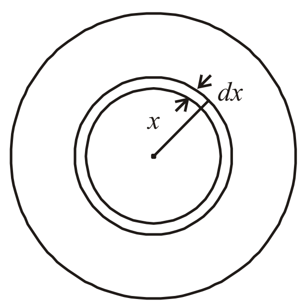
3. (1) Let us consider a spherical shell of radius x and thickness dx.

Charge on this shell

$$dq = \rho.4\pi x^2 dx = \rho_0\left(\frac{5}{4} - \frac{x}{R}\right).4\pi x^2 dx$$

$\therefore$ Total charge in the spherical region from centre to r $(r < R)$ is

$$q = \int dq = 4\pi\rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R}\right) x^2 dx$$



$$= 4\pi\rho_0\left[\frac{5}{4}.\frac{r^3}{3} - \frac{1}{R}.\frac{r^4}{4}\right] = \pi\rho_0 r^3\left(\frac{5}{3} - \frac{r}{R}\right)$$

$\therefore$ Electric field at r,

$$E = \frac{1}{4\pi \in_0}.\frac{q}{r^2}$$

$$= \frac{1}{4\pi \in_0} \cdot \frac{\pi \rho_0 r^3}{r^2}\left(\frac{5}{3} - \frac{r}{R}\right)$$

$$= \frac{\rho_0 r}{4 \in_0}\left(\frac{5}{3} - \frac{r}{R}\right)$$

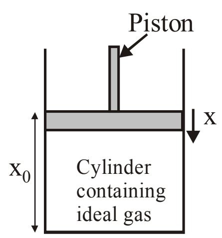
4. (3) $\frac{Mg}{A} = P_0$

$$P_0 V_0^{\gamma} = PV^{\gamma}$$

$$Mg = P_0 A$$

$$P_0 A x_0^{\gamma} = PA(x_0 - x)^{\gamma}$$

$$P = \frac{P_0 x_0^{\gamma}}{(x_0 - x)^{\gamma}}$$



Let piston is displaced by distance x

$$Mg - \left(\frac{P_0 x_0^{\gamma}}{(x_0 - x)^{\gamma}}\right) A = F_{\text{restoring}}$$

$$P_0 A\left(1 - \frac{x_0^{\gamma}}{(x_0 - x)^{\gamma}}\right) = F_{\text{restoring}}$$

$$[x_0 >>> x]$$

$$F = -\frac{\gamma P_0 A x}{x_0}$$

$\therefore$ Frequency with which piston executes SHM.

$$f = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0 A}{x_0 M}} = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0 A^2}{M V_0}}$$

5. (3) Inside the sphere, V is constant and is equal to that on the surface of the sphere. Outside the sphere it comes out to be, $V = -Gm/r$ i.e. $|V| \propto 1/r$. Hence the graph (3) is correct.

6. (3)

7. (1) Rate of cooling $\propto$ surface area and for a given mass, surface area of sphere is minimum.

8. (2) $\because$ The E.M. wave are transverse in nature i.e.,

$$= \frac{\vec{k} \times \vec{E}}{\mu\omega} = \vec{H} \qquad \ldots(i)$$

where $\vec{H} = \frac{\vec{B}}{\mu}$

and $\frac{\vec{k} \times \vec{H}}{\omega\varepsilon} = -\vec{E} \qquad \ldots(ii)$

$\vec{k}$ is $\perp$ $\vec{H}$ and $\vec{k}$ is also $\perp$ to $\vec{E}$

or In other words $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E} \times \vec{B}$

9. (1) For conservation of momentum, we have, $m_1 v_1 = (m_1 + m_2) v$ or $v = m_1/(m_1 + m_2) v_1$

Now the loss of energy

$$= \frac{1}{2} m_1 v_1^2 - \frac{1}{2}(m_1 + m_2)v^2$$

$\therefore$ Fraction of energy lost

$$= \frac{\frac{1}{2} m_1 v_1^2 - \frac{1}{2}(m_1 + m_2)v^2}{\frac{1}{2} m_1 v_1^2}$$

$$= 1 - [(m_1+m_2)/m_1] \times (v^2/v_1^2)$$

$$= 1 - [(m_1+m_2)/m_1] \times [m_1^2 v_1^2/[(m_1+m_2)^2] \times (1/v_1^2)$$

$$= 1 - [m_1/[(m_1+m_2)] = m_2/(m_1+m_2)$$

10 (3) $I = I_0 \sin(\omega t + \phi)$;

$E = E_0 \sin(\omega t + \phi)$;

$I = 4 \sin \omega t$;

$$E = 100 \sin\left(\omega t + \frac{\pi}{2} + \frac{\pi}{3}\right)$$

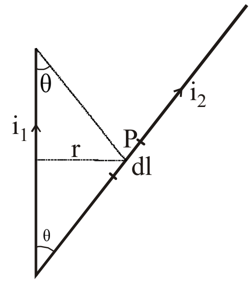
Phase diff. between I and E $= \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$

11. (2) Mutual inductance $= \frac{\phi}{I} = \frac{BA}{I}$

$$[\text{Henry}] = \frac{[MT^{-1}Q^{-1}L^2]}{[QT^{-1}]} = ML^2Q^{-2}$$

12. (3) Magnetic field due to current in wire 1 at point P distant r from the wire is

$$B = \frac{\mu_0}{4\pi}\frac{i_1}{r}[\cos\theta + \cos\theta]$$



$B = \dfrac{\mu_0}{2\pi}\dfrac{i_1 \cos\theta}{r}$ (directed perpendicular to the plane of paper, inwards)

The force exerted due to this magnetic field on current element $i_2\, dl$ is

$dF = i_2\, dl\, B \sin 90°$

$\therefore\ dF = i_2\, dl$

$$\left[\frac{\mu_0}{2\pi}\frac{i_1\cos\theta}{r}\right] = \frac{\mu_0}{2\pi r} i_1 i_2\, dl \cos\theta$$

13. (4) Let $a_1 = a, I_1 = a_1^2 = a^2$

$a_2 = 2a, I_2 = a_2^2 = 4a^2$

$I_2 = 4I_1$

$I_r = a_1^2 + a_2^2 + 2a_1a_2 \cos\phi$

$= I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$$I_r = I_1 + 4I_1 + 2\sqrt{4I_1^2}\cos\phi$$

$$\Rightarrow\quad I_r = 5I_1 + 4I_1\cos\phi \qquad \ldots(1)$$

Now, $I_{max} = (a_1 + a_2)^2 = (a + 2a)^2 = 9a^2$

$$I_{max} = 9I_1 \ \Rightarrow\ I_1 = \frac{I_{max}}{9}$$

Substituting in equation (1)

$$I_r = \frac{5I_{max}}{9} + \frac{4I_{max}}{9}\cos\phi$$

$$I_r = \frac{I_{max}}{9}[5 + 4\cos\phi]$$

$$I_r = \frac{I_{max}}{9}\left[5 + 8\cos^2\frac{\phi}{2} - 4\right]$$

$$I_r = \frac{I_{max}}{9}\left[1 + 8\cos^2\frac{\phi}{2}\right]$$

14. (2) Using conservation of momentum,

$$m_1 v_1 = m_1\frac{v_1}{3} + mv$$

$$\Rightarrow\ m_1 v_1 - \frac{m_1 v_1}{3} = mv$$

($v = \sqrt{5gl}$ to complete motion along vertical circle)

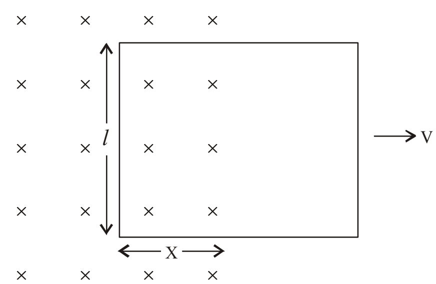
$$\frac{2m_1v_1}{3} = mv \ \Rightarrow\ v = \frac{2}{3}\frac{m_1 v_1}{m}$$

or $\quad v_1 = \dfrac{3}{2}\dfrac{m}{m_1}\sqrt{5g\ell}$

15. (2) The p-n junction diode is a half wave rectifier which produces output in forward biased mode only. Thus, there will be no output corresponding to –5V input. Hence, output will be obtained corresponding to +5V only.

16. (4) The induced emf is

$$e = \frac{-d\phi}{dt} = -\frac{d(\vec{B}.\vec{A})}{dt} = \frac{-d(BA\cos 0°)}{dt}$$



$$\therefore\ e = -B\frac{dA}{dt} = -B\frac{d(\ell \times x)}{dt}$$

$$\therefore\ e = -B\ell\frac{dx}{dt} = -B\ell v$$

17. (1) Here, $f_c = 1.5$ MHz $= 1500$ kHz, $f_m = 10$ kHz

$\therefore$ Low side band frequency

$= f_c - f_m = 1500$ kHz $- 10$ kHz $= 1490$ kHz

Upper side band frequency

$= f_c + f_m = 1500$ kHz $+ 10$ kHz $= 1510$ kHz

18. (1) $A + \delta = i + i' \Rightarrow i' = A + \delta - i = 30 + 30 - 60 = 0°$

19. (4) As λ is increased, there will be a value of λ above which photoelectrons will be cease to come out so photocurrent will become zero.

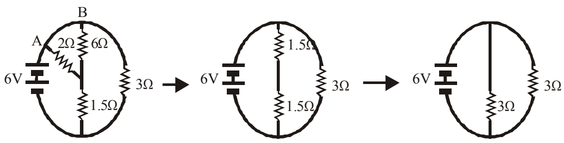
20. (4) K.E. gained by charged particle of charge q when accelerated under a pot. diff. V will be

$E_k = qV$;

For a given V, $E \propto q$.

For proton, deutron and α-particle, the ratio of charges is 1 : 1 : 2.

21. (4) 6Ω and 2Ω are in parallel combination



Hence, $R_{AB} = \frac{6 \times 2}{6+2} = 1.5\Omega$

$$V = IR \Rightarrow \frac{6}{1.5} = I \Rightarrow I = 4A$$

$$[\because R_{eq} = \frac{3 \times 3}{3+3} = \frac{9}{6} = 1.5\Omega]$$

22. (3) $x_{cm} = \frac{1 \times 0 + 1 \times PQ + 1 \times PR}{1+1+1} = \frac{PQ+PR}{3}$

and $y_{cm} = 0$

23. (20) Number of undecayed atom after time t_2;

$$\frac{N_0}{3} = N_0 e^{-\lambda t_2} \quad ...(i)$$

Number of undecayed atom after time t_1;

$$\frac{2N_0}{3} = N_0 e^{-\lambda t_1} \quad ...(ii)$$

From (i), $e^{-\lambda t_2} = \frac{1}{3}$

$$\Rightarrow -\lambda t_2 = \log_e\left(\frac{1}{3}\right) \quad ...(iii)$$

From (ii) $- e^{-\lambda t_1} = \frac{2}{3}$

$$\Rightarrow -\lambda t_1 = \log_e\left(\frac{2}{3}\right) \quad ...(iv)$$

Solving (iii) and (iv), we get

$t_2 - t_1 = 20$ min

24. (7) Apparent wt.

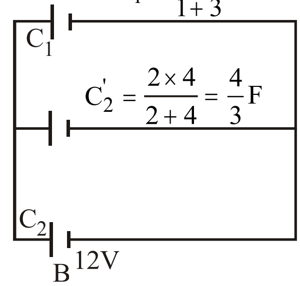
= Real wt.–Upthrust

$$= 12 - \frac{1}{2}(1000 \times 10^{-6}) \times 10^3 \times 10 = 12 - 5 = 7N$$

25. (9) As S_2 is open, hence C_1 & C_3 are in series, also C_2 & C_4 are in series combination.

$$Q = \frac{3}{4} \times V = \frac{3}{4} \times 12 = 9\mu C;$$

$$C_1' = \frac{1 \times 3}{1+3} = 3/4F$$



Charge on $C_1' = 9\mu C$;

Charge on $C_2' = \frac{4}{3} \times 12 = 16\mu C$

Hence, charge on C_1 and C_3 is $9\mu C$, as both are in series combination.

26. (6.5) Heat is extracted from the source in path DA and AB is

$$\Delta Q = \frac{3}{2}R\left(\frac{p_0 v_0}{R}\right) + \frac{5}{2}R\left(\frac{2p_0 v_0}{R}\right)$$

$$\Rightarrow \frac{3}{2}p_0 v_0 + \frac{5}{2}2p_0 v_0 = \left(\frac{13}{2}\right)p_0 v_0$$

27. (5) $E = \frac{13.6Z^2}{n^2}$; According to question,

$$\frac{13.6Z^2}{4} - \frac{13.6Z^2}{9} = 47.2$$

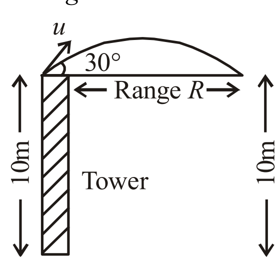
$$\Rightarrow 13.6Z^2\left(\frac{5}{36}\right) = 47.2$$

$$\Rightarrow Z^2 = 25 \Rightarrow Z = 5$$

28. (8.66)

From the figure it is clear that range is required

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{(10)^2 \sin(2 \times 30°)}{10} = 5\sqrt{3}$$



29. (2) Here, $I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos 0^o$

$$I_{min} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos 180^o$$

$$\therefore \quad I_{max} + I_{min} = 2(I_1 + I_2)$$

30. (1.25) $Y_c \times (\Delta L_c / L_c) = Y_s \times (\Delta L_s / L_s)$

$$\Rightarrow 1\times 10^{11} \times \left(\frac{1\times 10^{-3}}{1}\right) = 2\times 10^{11} \times \left(\frac{\Delta L_s}{0.5}\right)$$

$$\therefore \Delta L_s = \frac{0.5\times 10^{-3}}{2} = 0.25 \text{ mm}$$

Therefore, total extension of the composite wire

$= \Delta L_c + \Delta L_s$

$= 1\text{ mm} + 0.25\text{ mm} = 1.25\text{ mm}$

CHEMISTRY

31. (2) Correct order of increasing basic strength is $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$
32. (1) Aluminium has greater affinity for oxygen and the reaction is highly exothermic.
33. (4) Smaller the size and higher the charge more will be polarising power of cation. Since the order of the size of cation is

$$K^+ > Ca^{2+} > Mg^{2+} > Be^{2+}.$$

So the correct order of polarising power is :

$$K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$$

34. (2) The more the reduction potential, the more is the deposition of metals at cathode. Cation having E° value less than – 0.83 V (reduction potential of H_2O) will not deposit from aqueous solution.
35. (2) $\Delta H = nC_p\Delta T = 0$ means ΔH constant.

$$\Delta S = nR\ln\left(\frac{v_2}{v_1}\right) \geq 0 \qquad \Delta S \text{ increases.}$$

36. (4) Reaction between diborane and alkene are carried out in dry ether under an atmosphere of N_2 because B_2H_6 and the products are very reactive. The products further treated with alkaline H_2O_2 to convert into alcohols.

$$B_2H_6 + 6C_2H_4 \longrightarrow \underset{\text{reactive}}{B(C_2H_4)_3} \xrightarrow[H_2O_2]{\text{alkaline}} 3CH_3CH_2OH + H_3BO_3$$

37. (3) Ions I and IV are the same (trans), with mirror plane through en groups.

38. (1) $Cr\overset{2-}{O_2}\overset{-1}{Cl_2}$; Cr = +6; $Mn\overset{2-}{O_2}$; Mn = +4 (Highest oxidation state)

$$[Ni\overset{-1}{Cl_4}]^{2-}; Ni = +2; [Co\overset{-1}{Cl_4}]^{2-}; Co = +2$$

$$[Fe(\overset{-1}{CN})_6]^{3-}\ Fe = +3; [Cu(\overset{-1}{CN})_4]^{2-}\ Cu = +2$$

39. (2) $\underset{(A)}{Hg_2Cl_2} + 2NH_4OH \longrightarrow$

$$\underset{\text{(Black)}}{[H_2N - Hg - Cl + Hg]\downarrow} + NH_4Cl + 2H_2O$$

$$\underset{(B)}{HgCl_2} + 2NH_4OH \longrightarrow \underset{\text{(White)}}{Hg(NH_2)Cl} + NH_4Cl + H_2O$$

$$HgCl_2 + 2KI \longrightarrow \underset{\text{(Red)}}{HgI_2\downarrow} + 2KCl$$

$$HgI_2 + 2KI(\text{excess}) \longrightarrow K_2(HgI_4)$$

40. (4) In option (4) oxidation number changes from + 4 in NO_2 to + 3 in HNO_2 and + 5 in HNO_3.
41. (2) In neutral and alkaline medium

$$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

In acidic medium:

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

42. (1) $CH_2 = CH - CH_2 - CH = CH_2 \xrightarrow{O_3}$

$$2HCHO + OHC - CH_2 - CHO \xleftarrow[\text{Reduction}]{Zn / H_2O} HOOC - CH_2 - COOH + 2HCOOH$$

43. (3) The compound is diethyl ether $(CH_3CH_2)_2O$ which is resistant to nucleophilic attack by hydroxyl ion due to absence of double or triple bond, whereas all other compounds given are unsaturated.

$$\underset{\text{Ether}}{C_2H_5OC_2H_5} \qquad \underset{\text{Methyl acetate}}{CH_3 - \overset{\overset{O}{\|}}{C} - OCH_3}$$

$$\underset{\text{Acetonitrile}}{CH_3 - C \equiv N} \qquad \underset{\text{Acetamide}}{CH_3 - \overset{\overset{O}{\|}}{C} - NH_2}$$

44. (2) $2NaI + 2NaNO_2 + 4CH_3COOH \longrightarrow I_2 + 2NO + 4CH_3COONa + 2H_2O$

The colour of CCl_4 layer turns purple due to liberated I_2.

45. (2) Enantiomers of $C_4H_{10}O$ are

$$H_3CH_2C-\overset{\displaystyle OH}{\underset{\displaystyle H}{C}}-CH_3 \;\Big|\; H_3C-\overset{\displaystyle HO}{\underset{\displaystyle H}{C}}-CH_2CH_3$$

$$\underset{(A)}{CH_3CH_2-\overset{\displaystyle OH}{CH}-CH_3} \xrightarrow{H_2SO_4}$$

$$\underset{\text{Major (B)}}{CH_3CH=CH-CH_3} + \underset{\text{Minor (C)}}{CH_2=CHCH_2CH_3}$$

$$\underset{(B)}{CH_3CH=CH-CH_3} \xrightarrow{HBr / Peroxide}$$

$$CH_3-\overset{\displaystyle Br}{CH}-CH_2-CH_3 \xrightarrow[\text{hydrolysis}]{\text{alkaline}}$$

$$\underset{(A)}{CH_3-\overset{\displaystyle OH}{CH}-CH_2-CH_3}$$

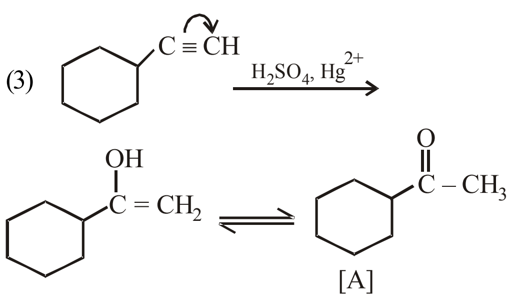
$$\underset{(C)}{CH_2=CHCH_2CH_3} \xrightarrow{HBr / Peroxide}$$

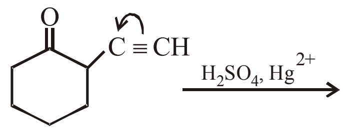
$$BrCH_2-CH_2-CH_2-CH_3 \xrightarrow[\text{hydrolysis}]{\text{alkaline}}$$

$$\underset{\text{Isomer of (A)}}{OH-CH_2-CH_2CH_2CH_3}$$

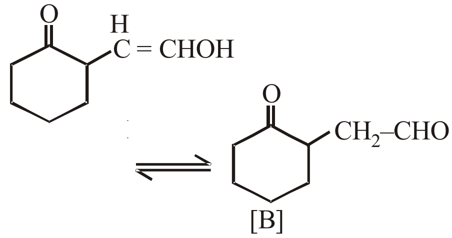
46. (2) In addition homopolymers such as teflon, empirical formula resembles with monomer.

47. (3)

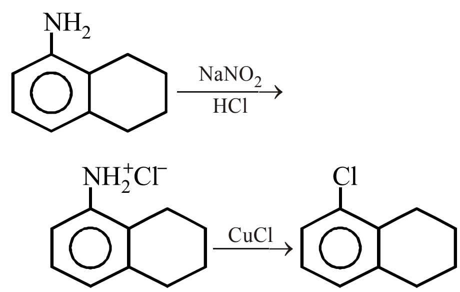




Due to electronegative nature of $>C=O$, π electrons are transferred toward benzene ring



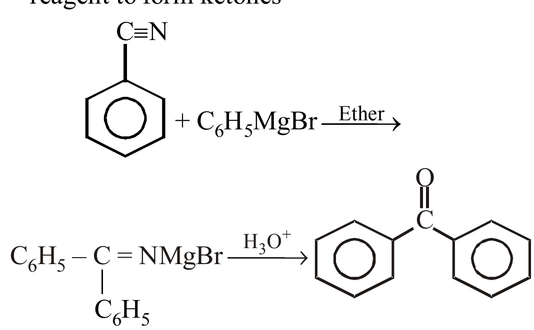
48. (3) Sandmeyer's reaction



49. (2) $C_6H_5ONa + CH_2=CH-CH_2-Br \longrightarrow C_6H_5-O-CH_2-CH=CH_2 + NaBr$

Allyl phenyl ether

50. (1) Alkyl or Aryl cyanide react with grignard reagent to form ketones



or $C_6H_5-\underset{\displaystyle C_6H_5}{C}=O + MgBrNH_2$

51. (19.55)

Energy of photon corresponding to second line of Balmer series for Li^{2+} ion

$$=(13.6)\times(3)^2\left[\frac{1}{2^2}-\frac{1}{4^2}\right]$$

$$=13.6\times\frac{27}{16}$$

Energy needed to eject electron from n = 2 level in H-atom;

$$=13.6\times1^2\times\left[\frac{1}{2^2}-\frac{1}{\infty^2}\right]\Rightarrow\frac{13.6}{4}$$

K.E. of ejected electron

$$=13.6\times\frac{9\times3}{16}-\frac{13.6}{4}=13.6\times\left(\frac{27-4}{16}\right)$$

$\Rightarrow$ 19.55 eV

52. (8) $H-O-\overset{\overset{\displaystyle O}{\|}}{\underset{\underset{\displaystyle O}{\|}}{S}}-O-H$; 6σ & 2π

53. (128) For fcc,

$$r=\frac{\sqrt{2}a}{4}=\frac{a}{2\sqrt{2}}=0.3535a$$

given $a = 361$ pm

$r = 0.3535\times361$

$= 128$ pm

54. (0.32) Cell reaction :

$$M+3Ag^+(aq)\longrightarrow M^{3+}+3Ag\,(s)$$

$$E=E^0-\frac{0.0591}{n}\log\frac{[\text{Reduced state}]}{[\text{Oxidised state}]}$$

$$0.421=E^0-\frac{0.0591}{3}\log\frac{0.001}{(0.01)^3}$$

$E^0 = 0.48$

$$E^0=E^0_{Ag^+/Ag}-E^0_{M^{3+}/M}$$

$$E^0_{M^{3+}/M}=0.8V-0.48V=0.32\text{ volt}$$

55. (0.4) 6.4 g of $CaC_2 \equiv 0.1$ mol of CaC_2

0.1 mol contains 0.1 N_A molecules of CaC_2

$\because$ 1 molecules have 4π electron

$\therefore$ 0.1 N_A molecule will have

$0.1\ N_A\times4=0.4\ N_A$

56. (68.13)

Let atomic masses of A and B be a and b amu respectively

$\therefore$ Molar mass of AB_2 = (a + 2b) g mol^{-1}

and Molar mass of AB_4 = (a + 4b) g mol^{-1}

For compound AB_2

$\Delta T_b = K_b\times W_B\times1000/\,W_A\times M_B$

$2.3 = 5.1\times1\times1000/\,20.0\times(a+2b)$ ...(i)

For compound AB_4

$1.3 = 5.1\times1\times1000/\,20.0\times(a+4b)$ ...(ii)

Solving (i) and (ii), a = 25.49 b = 42.64

Sum of atomic masses = 25.49 + 42.64 = 68.13

57. (5.19) For bcc lattice, number of atoms per unit cell = 2

Now

$$d=\frac{n\times M}{a^3\times N_A}=\frac{2\times100}{(4\times10^{-8}\,cm)^3\times6.02\times10^{23}}$$

= 200/38.528 = 5.19 g/cc

58. (4.0) Flask I:

$$1\,M\xrightarrow[t_{1/2}\,=\,4\,hr]{}0.5\,M\xrightarrow[t_{1/2}\,=\,4\,hr]{}0.25\,M$$

$t = 8$ hr

$\therefore t_{1/2} = 4$ hr.

Flask II:

$$0.6\,M\xrightarrow[t_{1/2}\,=\,4\,hr]{}0.3\,M$$

A first order reaction is independent of the initial concentration of the reactant.

59. (4.0) $pOH=pK_b+\log\frac{[\text{salt}]}{[\text{base}]}$

$$=-\log10^{-10}+\log1=10\,;\,pH=14-10=4$$

60. (100)

C (graphite) $\to$ C (diamond)

$\Delta H = \Delta U + P\,.\,\Delta V$

$$V_m(\text{diamond})=\frac{12}{3}\text{mL}$$

$$V_m(\text{graphite})=\frac{12}{2}\text{mL}$$

$\Delta H - \Delta U = (500\times10^3\times10^5\,N/m^2)$

$$\left(\frac{12}{3}-\frac{12}{2}\right)\times10^{-6}=-100\text{ kJ/mol}$$

$\Delta U - \Delta H = +100$ kJ/mol

MATHEMATICS

61. (3) (1) $x^2y^2 = 16a^4$; $L_{ST} = \left|\frac{y}{x_T}\right| \Rightarrow xy = 4a^2$

$$y + xy' = 0 \Rightarrow y' = \frac{-y}{x};$$

$$L_{ST} = \left|\frac{y}{y/x}\right| = x \Rightarrow L_{ST} = 2a$$

$\therefore$ (1) is true.

(2) $\frac{dy}{dx} = \frac{x}{2} = 1$ i.e. $x = 2$

$\therefore$ (2, 1) is the point on the curve $x^2 = 4y$ at which the normal is : $y - 1 = -1\,(x - 2)$ i.e. $x + y = 3$ $\therefore$ (2) is true

(3) $y = -4x^2$, $y = e^{-x/2}$

The curves are non-intersecting

$\therefore$ curves are not orthogonal i.e. (3) is false.

(4) $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$

$$\frac{dy}{dx} = 2x^2 - 4ax + 2 = 2(x-a)^2 + 2 - 2a^2 > 0$$

$[\because a \in (-1, 0) \Rightarrow 0 < a^2 < 1]$ $\therefore$ (4) is true

62. (3) $I = \int \frac{(x^2-1)\,dx}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}$

(Dividing Num. and Den. by x^2)

$$I = \int \frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x^2+3+\frac{1}{x^2}\right)\tan^{-1}\left(x+\frac{1}{x}\right)}$$

$$= \int \frac{dt}{\left(t^2+1\right)\tan^{-1} t}$$

$$\left(\text{where } t = x + \frac{1}{x} \Rightarrow dt = \left(1-\frac{1}{x^2}\right)dx\right)$$

$$= \log|\tan^{-1} t| + C = \log\left|\tan^{-1}\left(\frac{x^2+1}{x}\right)\right| + C$$

$$\Rightarrow f(x) = \frac{x^2+1}{x}$$

63. (1) Clearly P is $(a\cos\theta, b\sin\theta)$ and Q is $(-a\sin\theta, b\sin\theta)$ so the mid point (h, k) of PQ will be given by

$$h = \frac{a\cos\theta - a\sin\theta}{2} \text{ and } k = \frac{b\sin\theta + b\cos\theta}{2}$$

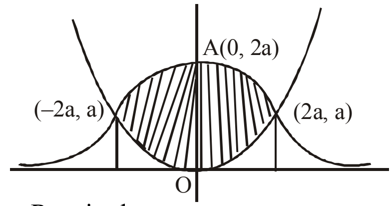
$$\therefore \frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 2 \Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = \frac{1}{2}$$

64. (1) The curve of $y(x^2+4a^2) = 8a^3$ is symmetrical about y-axis and cuts it at A (0, 2a). Tangent at A is parallel to x-axis. x-axis is asymptote. This curve meets $x^2 = 4ay$

Where,

$$\frac{x^2}{4a} = \frac{8a^3}{x^2+4a^2} \Rightarrow x^4 + 4a^2x^2 - 32a^4 = 0$$

$$\Rightarrow (x^2-4a^2)(x^2+8a^2) = 0 \Rightarrow x = \pm 2a$$



$\therefore$ Required

$$\text{area} = 2\left[\int_0^{2a} \frac{8a^3}{x^2+4a^2}\,dx - \int_0^{2a} \frac{x^2}{4a}\,dx\right]$$

$$= \frac{a^2}{3}(6\pi - 4).$$

65. (1) $p\vec{r} + (\vec{r}.\vec{b})\vec{a} = \vec{c}$ ...(1)

$$p(\vec{r}.\vec{b}) + (\vec{r}.\vec{b})(\vec{a}.\vec{b}) = \vec{c}.\vec{b}$$

$\Rightarrow \vec{r}.\vec{b} = \frac{\vec{c}.\vec{b}}{p}$, since $\vec{a}.\vec{b} = 0$, putting in (1),

$$\Rightarrow \vec{r} = \frac{\vec{c}}{p} - \frac{(\vec{c}.\vec{b})}{p^2}\vec{a} = \frac{\vec{c}}{p} - \frac{(\vec{b}.\vec{c})\vec{a}}{p^2}$$

66. (4) For $f(x)$ to be defined, we must have

$x - \sqrt{1-x^2} \geq 0$ or $x \geq \sqrt{1-x^2} > 0$

$\therefore$ $x^2 \geq 1 - x^2$ or $x^2 \geq \frac{1}{2}$.

Also, $1 - x^2 \geq 0$ or $x^2 \leq 1$.

Now, $x^2 \geq \frac{1}{2} \Rightarrow \left(x - \frac{1}{\sqrt{2}}\right)\left(x + \frac{1}{\sqrt{2}}\right) \geq 0$

$\Rightarrow x \leq -\frac{1}{\sqrt{2}}$ or $x \geq \frac{1}{\sqrt{2}}$

Also, $x^2 \leq 1 \Rightarrow (x-1)(x+1) \leq 0$

$\Rightarrow -1 \leq x \leq 1$

Thus, $x > 0,\ x^2 \ge \frac{1}{2}$ and $x^2 \le 1$

$$\Rightarrow x \in \left[\frac{1}{\sqrt{2}}, 1\right]$$

67. (3) $A = \{1, 2, 3\}, B = \{4, 5\}$

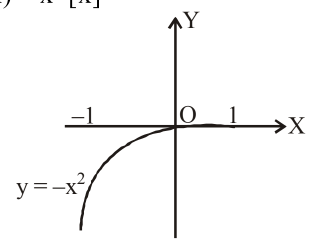
Then $A \cap B = \phi$

$A \times B = \{(1,4), (1,5), (2,4), (2,5), (3,4), (3,5)\}$

$B \times A = \{(4,1), (4,2), (4,3), (5,1), (5,2), (5,3)\}$

$(A \times B) \cap (B \times A) = \phi$

68. (3) $f(x) = x^2[x]$



$$\Rightarrow f(x) = \begin{cases} -x^2, & -1 < x < 0 \\ 0, & 0 \le x < 1 \end{cases}$$

69. (4) $y = e^x + \sin x$

$$\frac{dy}{dx} = e^x + \cos x \Rightarrow \frac{dx}{dy} = \frac{1}{e^x + \cos x}$$

Diff. w.r.t y

$$\frac{d^2x}{dy^2} = \frac{-1[e^x - \sin x]}{(e^x + \cos x)^2} \times \frac{dx}{dy} = \frac{(\sin x - e^x)}{(e^x + \cos x)^3}$$

70. (3) $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \ldots$

$$= \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{4}\right) + \left(1 - \frac{1}{8}\right) + \left(1 - \frac{1}{16}\right) + \ldots.$$

$$= n - \frac{\frac{1}{2}\left\{1 - \frac{1}{2^n}\right\}}{1 - \frac{1}{2}} = n - 1 + 2^{-n}$$

71. (4) We have $f(x) = 4x^3 - 7,\ x \in R$.

f is one-one. Let $x_1, x_2 \in R$ and $f(x_1) = f(x_2)$.

$$\Rightarrow 4x_1^3 - 7 = 4x_2^3 - 7 \Rightarrow 4x_1^3 = 4x_2^3$$

$$\Rightarrow x_1^3 = x_2^3 \Rightarrow x_1^3 - x_2^3 = 0.$$

$$\Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0.$$

$$\Rightarrow (x_1 - x_2)\left[\left(x_1 + \frac{x_2}{2}\right)^2 + \frac{3x_2^2}{4}\right] = 0.$$

$\Rightarrow x_1 - x_2 = 0$, because the other factor is non-zero.

$\Rightarrow x_1 = x_2 \therefore$ f is one-one.

f is onto. Let $k \in R$ any real number.

$$f(x) = k \Rightarrow 4x^3 - 7 = k \Rightarrow x = \left(\frac{k+7}{4}\right)^{1/3}$$

Now $\left(\frac{k+7}{4}\right)^{1/3} \in R$, because $k \in R$ and

$$f\left[\left(\frac{k+7}{4}\right)^{1/3}\right] = 4\left[\left(\frac{k+7}{4}\right)^{1/3}\right]^3 - 7$$

$$= 4\left(\frac{k+7}{4}\right) - 7 = k$$

$\therefore$ k is the image of $\left(\frac{k+7}{4}\right)^{1/3}$

$\therefore$ f is onto.

$\therefore$ f is a bijective function.

72. (2) $f : \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right] \Rightarrow [0, 4]$,

$$f(x) = \sqrt{3}\sin x - \cos x + 2$$

$$f(x) = 2\sin\left(x - \frac{\pi}{6}\right) + 2$$

$$\Rightarrow (f^{-1}(x)) = \sin^{-1}\left(\frac{x-2}{2}\right) + \frac{\pi}{6}$$

73. (3) We have $\alpha + \beta = -\frac{q}{p},\ \alpha\beta = \frac{r}{p}$

Now the given equation

$$\alpha(x - \beta)^2 + \beta(x - \alpha)^2 = 0$$

$$\Rightarrow (\alpha + \beta)x^2 - 4\alpha\beta x + \alpha\beta(\alpha + \beta) = 0$$

$$\Rightarrow \left(-\frac{q}{p}\right)x^2 - 4\frac{r}{p}x + \frac{r}{p}\left(-\frac{q}{p}\right) = 0$$

$$\Rightarrow pqx^2 + 4prx + rq = 0 \text{ ------(i)}$$

Since α and β have opposite signs, therefore p and r must have opposite signs.

$\Rightarrow$ pq and rq must have opposite signs

$\Rightarrow$ roots of equation (i) have oposite signs

74. (1) Given, $k \sin\theta + \cos 2\theta = 2k - 7$

$\Rightarrow k\sin\theta + 1 - 2\sin^2\theta = 2k - 7$

$\Rightarrow 2\sin^2\theta - k\sin\theta + (2k-8) = 0$

For the existence of real roots, discriminant ≥ 0.

$\Rightarrow k^2 - 4\times 2(2k-8) \geq 0 \Rightarrow (k-8)^2 \geq 0$, which is always true.

Roots of the quadratic equation are $\frac{k-4}{2}$, 2, but $\sin\theta \neq 2$.

$\therefore \sin\theta = \frac{k-4}{2}$ but $-1 \leq \sin\theta \leq 1$

$\Rightarrow -1 \leq \frac{k-4}{2} \leq 1 \Rightarrow -2 \leq k-4 \leq 2$

$\Rightarrow 2 \leq k \leq 6$

75. (3) $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{4x^2}{(1+x^2)^2}}}\cdot\frac{2(1+x^2)-4x^2}{(1+x^2)^2}$$

$$= \frac{2(1-x^2)}{\sqrt{(1-x^2)^2}}\cdot\frac{1}{1+x^2} = \frac{2(1-x^2)}{|1-x^2|(1+x^2)}$$

If $|x| < 1 \Rightarrow x^2 < 1$, then $1-x^2 > 0$

$\therefore |1-x^2| = 1-x^2 \Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2}$

If $|x| > 1 \Rightarrow x^2 > 1$, then $1-x^2 < 0$

$\therefore |1-x^2| = x^2 - 1 \Rightarrow \frac{dy}{dx} = -\frac{2}{1+x^2}$

Obviously $\frac{dy}{dx}$ does not exist for $x^2 = 1$ or $|x| = 1$

76. (3) Given, $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting, $y = vx$ so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$

We get, $v + x\frac{dv}{dx} = v - \cos^2 v$

$\Rightarrow \frac{dv}{\cos^2 v} = -\frac{dx}{x} \Rightarrow \sec^2 v\, dv = -\frac{dx}{x}$

Integrating, we get

$\tan v = -\ln x + \ln c$ $\tan\left(\frac{y}{x}\right) = -\ln x + \ln c$

This passes through $\left(1, \frac{\pi}{4}\right) \Rightarrow \ln c = 1$

$\therefore y = x\tan^{-1}\left(\log\frac{e}{x}\right)$

77. (4) $y = \tan^{-1}\left[\frac{2^x(2-1)}{1+2^x.2^{x+1}}\right] = \tan^{-1}\left[\frac{2^{x+1}-2^x}{1+2^x.2^{x+1}}\right]$

$= \tan^{-1}(2^{x+1}) - \tan^{-1}(2^x)$

$\Rightarrow \frac{dy}{dx} = \frac{2^{x+1}\log 2}{1+2^{2(x+1)}} - \frac{2^x\log 2}{1+2^{2x}}$

$\therefore \left(\frac{dy}{dx}\right)_{x=0} = (\log 2)\left(\frac{2}{5} - \frac{1}{2}\right) = (\log 2)\left(-\frac{1}{10}\right)$

78. (3) $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix}$

$= (1+abc)(a-b)(b-c)(c-a) = 0$ ....(1)

also, $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \neq 0$, since A, B, C are not coplanar $= (a-b)(b-c)(c-a) \neq 0$ ...(2)

From (1) & (2), $abc = -1$

79. (4) A and B will agree in a certain statement if both speak truth or both tell a lie. We define following events

E_1 = A and B both speak truth $\Rightarrow P(E_1) = xy$

E_2 = A and B both tell a lie

$\Rightarrow P(E_2) = (1-x)(1-y)$

E = A and B agree in a certain statement

Clearly, $P(E/E_1) = 1$ and $P(E/E_2) = 1$

The required probability is $P(E_1/E)$. Using Baye's theorem

$P(E_1/E)$

$$= \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2)}$$

$$= \frac{xy.1}{xy.1 + (1-x)(1-y).1} = \frac{xy}{1-x-y+2xy}$$

80. (1) $|x_1z_1 - y_1z_2|^2 + |y_1z_1 - x_1z_2|^2$
$= |x_1z_1|^2 + |y_1z_2|^2 - 2\text{Re}(x_1y_1z_1z_2)$
$+ |y_1z_1|^2 + |x_1z_2|^2 + 2\text{Re}\,(x_1y_1z_1z_2)$
$= x_1^2|z_1|^2 + y_1^2|z_2|^2 + y_1^2|z_1|^2 + x_1^2|z_2|^2$
$= x_1^2|z_1|^2 + y_1^2|z_2|^2 + y_1^2|z_1|^2 + x_1^2|z_2|^2$
$= 2(x_1^2 + y_1^2)(4^2) = 32(x_1^2 + y_1^2)$

81. (3.00)

$\mu(x) = \dfrac{1}{x-1}$, which is discontinous at $x = 1$

$f(u) = \dfrac{1}{u^2 + u - 2} = \dfrac{1}{(u+2)(u-1)}$,
which is discontinous at $u = -2, 1$

when $u = -2$, then $\dfrac{1}{x-1} = -2 \Rightarrow x = \dfrac{1}{2}$

when $u = 1$, then $\dfrac{1}{x-1} = 1 \Rightarrow x = 2$

Hence given composite function is discontinous at three points, $x = 1, \dfrac{1}{2}$ and 2.

82. (2.00) $f(x) = Pe^{2x} + Qe^{x} + Rx$

$\Rightarrow f'(x) = 2Pe^{2x} + Qe^{x} + R$

$\Rightarrow 31 = 2Pe^{2\log 2} + Qe^{\log 2} + R$

$\Rightarrow 8P + 2Q + R = 31$ .... (i)

Also, $0 = P + Q$ .... (ii)

$\& \int_0^{\log 4} (f(x) - Rx)\,dx = \dfrac{39}{2}$

$\Rightarrow \int_0^{\log 4} (Pe^{2x} + Qe^{x})\,dx = \dfrac{39}{2}$

$\Rightarrow \left[\dfrac{P}{2}e^{2x} + Qe^{x}\right]_0^{\log 4} = \dfrac{39}{2}$

$\Rightarrow \dfrac{P}{2}e^{2\log 4} + Q\,e^{\log 4} - \dfrac{P}{2} - Q = \dfrac{39}{2}$

$\Rightarrow 15P + 6Q = 39$ .... (iii)

Solving (i), (ii) and (iii), we get $P = 5, Q = -6, R = 3 \Rightarrow P + Q + R = 5 - 6 + 3 = 2$.

83. (0.60) Given, $x_1 + x_2 + ... + x_{10} = 12$

and $x_1^2 + x_2^2 + ... + x_{10}^2 = 18$

$$\sigma^2 = \frac{1}{n}\Sigma x^2 - \left(\frac{\Sigma x}{n}\right)^2$$

$$= \frac{18}{10} - \left(\frac{12}{10}\right)^2 = \frac{9}{5} - \frac{36}{25} = \frac{9}{25}$$

$\therefore$ SD $= \dfrac{3}{5} = 0.60$

84. (1.00) The given system of equations are :

$p^3x + (p+1)^3 y = (p+2)^3$ ...(1)

$px + (p+1)y = (p+2)$ ....(2)

$x + y = 1$ ....(3)

This system is consistent, if values of x and y from first two equation satisfy the third equation.

which $\Rightarrow \begin{vmatrix} p^3 & (p+1)^3 & (p+2)^3 \\ p & (p+1) & (p+2) \\ 1 & 1 & 1 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} p^3 & (p+1)^3 - p^3 & (p+2)^3 - p^3 \\ p & 1 & 2 \\ 1 & 0 & 0 \end{vmatrix} = 0$

$\Rightarrow 2(p+1)^3 - 2p^3 - (p+2)^3 + p^3 = 0$

$\Rightarrow 2(p^3 + 1 + 3p^2 + 3p) - 2p^3 - (p^3 + 8 + 12p + 6p^2) + p^3 = 0$

$\Rightarrow 2p^3 + 2 + 6p^2 + 6p - 2p^3 - p^3 - 8 - 12p - 6p^2 + p^3 = 0$

$\Rightarrow -6 - 6p = 0 \Rightarrow |p| = |-1| = 1.$

85. (17.67)

Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$

$f'(x) = 30x^9 - 56x^7 + 30x^5 - 63x^2 + 6x$

$f'(1) = 30 - 56 + 30 - 63 + 6$

$= 66 - 63 - 56 = -53$

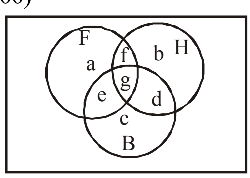
Consider $\lim\limits_{\alpha \to 0} \dfrac{f(1-\alpha) - f(1)}{\alpha^3 + 3\alpha}$

$= \lim\limits_{\alpha \to 0} \dfrac{f'(1-\alpha)(-1) - 0}{3\alpha^2 + 3}$

(By using L'hospital rule)

$= \dfrac{f'(1-0)(-1)}{3(0)^2 + 3} = \dfrac{-f'(1)}{3} = \dfrac{53}{3} = 17.67$

86. (325.00)



$a + e + f + g = 285, \quad b + d + f + g = 195$
$c + d + e + g = 115, \quad e + g = 45, f + g = 70,$
$d + g = 50$
$a + b + c + d + e + f + g = 500 - 50 = 450$

we obtain

$a + f = 240, b + d = 125, \ c + e = 65$
$a + e = 215, b + f = 145; b + c + d = 165$
$a + c + e = 255; \ a + b + f = 335$

Solving we get

$b = 95, c = 40, a = 190, \ d = 30, e = 25, f = 50$ and $g = 20$

Desired quantity $= a + b + c = 325$

87. (0.50)

$\sin(\cot^{-1}(x+1)) = \cos(\tan^{-1}x)$

$\Rightarrow \cos(\pi/2 - \cot^{-1}(x+1)) = \cos(\tan^{-1}x)$

$\Rightarrow \frac{\pi}{2} - \cot^{-1}(x+1) = 2n\pi \pm \tan^{-1}x$ Put $n = 0$

$\Rightarrow \pi/2 - \cot^{-1}(x+1) = \pm\tan^{-1}x = \tan^{-1}(\pm x)$

$\Rightarrow \pi/2 = \tan^{-1}(\pm x) + \cot^{-1}(x+1)$

$\Rightarrow x + 1 = \pm x \Rightarrow 2x + 1 = 0;$

$$|x| = \left|-\frac{1}{2}\right| = \frac{1}{2} = 0.50$$

88. (1.50)

$$\lim_{x\to0}\frac{1-\cos^3 x}{x\sin x\cos x}$$

$$= \lim_{x\to0}\frac{(1-\cos x)(1+\cos x+\cos^2 x)}{x\sin x\cos x}$$

$$= \lim_{x\to0}\frac{2\sin^2\left(\frac{x}{2}\right)}{x.2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)} \times \frac{(1+\cos x+\cos^2 x)}{\cos x}$$

$$= \lim_{x\to0}\frac{\sin\left(\frac{x}{2}\right)}{2\left(\frac{x}{2}\right)} \times \frac{1+\cos x+\cos^2 x}{\cos\left(\frac{x}{2}\right)\cos x}$$

$$= \frac{1}{2}\times3 = \frac{3}{2} = 1.50$$

89. (2.00)

$$f\{f[f(x)]\} = f\left[f\left(\frac{1}{1-x}\right)\right]$$

$$= f\left(\frac{1}{1-\frac{1}{1-x}}\right) = f\left(\frac{x-1}{x}\right)$$

$\therefore$ f (x) is not defined for x = 1; $f\left(\frac{1}{1-x}\right)$ is not defined for $x = 0$.

$\therefore$ $f\{f[f(x)]\}$ is discontinuous at $x = 0$ and 1 i.e., there are two points of discontinuity.

90. (6.00)

We have $\cos x + \cos 2x + \cos 3x = 0$

or $(\cos 3x + \cos x) + \cos 2x = 0$

or $\cos 2x(2\cos x + 1) = 0$

or $\cos 2x(2\cos x + 1) = 0$

We have, either $\cos 2x = 0$ or $2\cos x + 1 = 0$

If $\cos 2x = 0$, then $2x = (2m+1)\frac{\pi}{2}$

or $x = (2m+1)\frac{\pi}{4}, \quad m \in I$ ...(i)

If $2\cos x + 1 = 0$, then $\cos x = -\frac{1}{2} = \cos\frac{2\pi}{3}$

$\Rightarrow \quad x = 2n\pi \pm \frac{2\pi}{3}, \ n \in I$ ...(ii)

So, the required general solution are

$\therefore$ Values of x in the interval $(0 < x < 2\pi)$ in equation (i)

Put $m = 0, 1, 2, 3$

So the value is $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

In equation (ii) Put, $n = 0, 1$ $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

Hence, 6 roots of the given equation.

Mock Test-6

ANSWER KEY

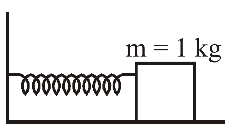
1	(3)	**16**	(4)	**31**	(3)	**46**	(2)	**61**	(1)	**76**	(4)
2	(3)	**17**	(4)	**32**	(3)	**47**	(2)	**62**	(2)	**77**	(2)
3	(2)	**18**	(4)	**33**	(3)	**48**	(3)	**63**	(1)	**78**	(4)
4	(3)	**19**	(2)	**34**	(2)	**49**	(2)	**64**	(1)	**79**	(1)
5	(4)	**20**	(4)	**35**	(2)	**50**	(3)	**65**	(3)	**80**	(3)
6	(1)	**21**	(250)	**36**	(1)	**51**	(10.43)	**66**	(4)	**81**	(0.14)
7	(1)	**22**	(4)	**37**	(1)	**52**	(1)	**67**	(2)	**82**	(10)
8	(4)	**23**	(4)	**38**	(1)	**53**	(464)	**68**	(4)	**83**	(6)
9	(1)	**24**	(2×10^{-3})	**39**	(2)	**54**	(8)	**69**	(4)	**84**	(3)
10	(4)	**25**	(1.06)	**40**	(2)	**55**	(57.5)	**70**	(3)	**85**	(80)
11	(2)	**26**	(5451)	**41**	(2)	**56**	(105.7)	**71**	(2)	**86**	(1)
12	(2)	**27**	(0.01)	**42**	(4)	**57**	(0.05)	**72**	(1)	**87**	(0)
13	(3)	**28**	(0)	**43**	(2)	**58**	(21)	**73**	(1)	**88**	(0.29)
14	(4)	**29**	(80)	**44**	(1)	**59**	(2.57)	**74**	(2)	**89**	(1)
15	(3)	**30**	(30)	**45**	(3)	**60**	(2.42)	**75**	(4)	**90**	(21.33)

Solutions

PHYSICS

1. (3) Frequency of spring $(f) = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = 1\text{ Hz}$

$\Rightarrow 4\pi^2 = \frac{k}{m}$



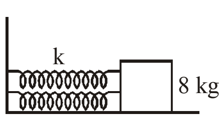
If block of mass m = 1 kg is attached then,

$k = 4\pi^2$

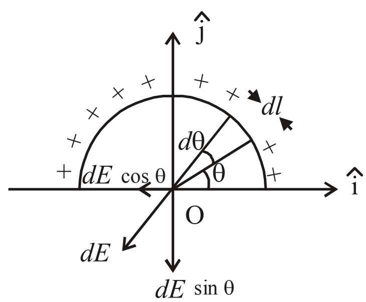
Now, identical springs are attached in parallel with mass m = 8 kg. Hence,

$k_{eq} = 2k$

$F = \frac{1}{2\pi}\sqrt{\frac{k \times 2}{m}} = \frac{1}{2}\text{Hz}$



2. (3) Let us consider a differential element dl. charge on this element.



$$dq = \left(\frac{q}{\pi r}\right) dl = \frac{q}{\pi r}(r d\theta) \qquad (\because dl = r d\theta)$$

$$= \left(\frac{q}{\pi}\right) d\theta$$

Electric field at O due to dq is

$$dE = \frac{1}{4\pi \in_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi \in_0} \cdot \frac{q}{\pi r^2} d\theta$$

The component $dE\cos\theta$ will be counter balanced by another element on left portion. Hence resultant field at O is the resultant of the component $dE \sin\theta$ only.

$$\therefore\ E = \int dE \sin\theta = \int_0^{\pi} \frac{q}{4\pi^2 r^2 \in_0} \sin\theta d\theta$$

$$= \frac{q}{4\pi^2 r^2 \in_0}[-\cos\theta]_0^{\pi}$$

$$= \frac{q}{4\pi^2 r^2 \in_0}(+1+1)$$

$$= \frac{q}{2\pi^2 r^2 \in_0}$$

The direction of E is towards negative y-axis.

$$\therefore\ \vec{E} = -\frac{q}{2\pi^2 r^2 \in_0}\hat{j}$$

3. (2) Let pole strength = m

$\therefore M = m \times \ell$ when $\ell' = \frac{\ell}{2}$

$$M' = m \times \ell' = \frac{m\ell}{2} = \frac{M}{2} = 40 \text{ units}$$

4. (3) Wave number $\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n^2}\right]$

$$\Rightarrow \lambda \propto \frac{1}{Z^2}$$

By question n = 1 and $n_1 = 2$

Then, $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

5. (4) From conservation of mechanical energy

$$\frac{1}{2}mv^2 + \frac{1}{2}I\left(\frac{v^2}{R^2}\right) = mg\left(\frac{3v^2}{4g}\right)$$

after solving $I = \frac{mR^2}{2}$ Which is for disc.

6. (1) $X = \frac{1}{\sqrt{a\omega^2 - b\omega + c}}$

At resonance X becomes maximum and $a\omega^2 - b\omega + c$ becomes minimum.

$\therefore$ Differential above term and equating it to zero.

$$\therefore\ 2a\omega - b = 0 \Rightarrow \omega = \frac{b}{2a}.$$

7. (1) $Q = KA\frac{(200 - T_1)}{L} = (2K)A\frac{(T_1 - T_2)}{L}$

$$=(1.5K)A(T_2\text{-}18) \times \frac{1}{L}$$

$200 - T_1 = 2T_1 - 2T_2 = 1.5\,T_2 - 27,$
Solving $T_1 = 116°C$ and $T_2 = 74°C$

8. (4) $4i_1 = 5i_2$

$$\therefore i_1 = \frac{5}{4}i_2$$

i_1 = current through 4 Ω
i_2 = current through Ammeter

Total $R = \frac{5 \times 4}{9} = \frac{20}{9}$

$$I = \frac{10 \times 9}{20} = \frac{9}{2} \Rightarrow \frac{5}{4}i_2 + i_2 = \frac{9}{2} \Rightarrow i_2 = 2A$$

9. (1) When positive terminal connected to A then diode D_1 is forward biased, current, $I = \frac{2}{5} = 0.4A$

When positive terminal connected to B then diode D_2 is forward biased, current,

$$I = \frac{2}{10} = 0.2A$$

10. (4) Mass per unit length of the wire = ρ
Mass of L length, M = ρL
and since the wire of length L is bent in a form of circular loop therefore $2\pi R = L \Rightarrow R = \frac{L}{2\pi}$

Moment of inertia of loop about given axis

$$= \frac{3}{2}MR^2$$

$$= \frac{3}{2}\rho L\left(\frac{L}{2\pi}\right)^2 = \frac{3\rho L^3}{8\pi^2}$$

11. (2) $\phi = \vec{B}.\vec{A}$; $\phi = BA\cos\omega t$

$$\varepsilon = -\frac{d\phi}{dt} = \omega BA\sin\omega t\ ;\ i = \frac{\omega BA}{R}\sin\omega t$$

$$P_{inst} = i^2R = \left(\frac{\omega BA}{R}\right)^2 \times R\sin^2\omega t$$

$$P_{avg} = \frac{\int_0^T P_{inst} \times dt}{\int_0^T dt}$$

$$= \frac{(\omega BA)^2}{R}\frac{\int_0^T \sin^2\omega t dt}{\int_0^T dt} = \frac{1}{2}\frac{(\omega BA)^2}{R}$$

$$\therefore P_{avg} = \frac{(\omega B\pi r^2)^2}{8R} \qquad \left[A = \frac{\pi r^2}{2}\right]$$

12. (2) Let V be volume of the load, and let its relative density be ρ then

$$Y = \frac{FL}{Al_a} = \frac{V\rho gL}{Al_a}. \quad ......(1)$$

when the load is immersed in liquid, the net weight = weight - upthrust

$$\therefore Y = \frac{F'L}{Al_w} = \frac{(V\rho g - V \times 1 \times g)L}{Al_w} \quad ......(2)$$

Equating (1) and (2), we get $\frac{\rho}{l_a} = \frac{\rho - 1}{l_w}$

which gives $\rho = \frac{l_a}{l_a - l_w}$

13. (3) **Given:** Amplitude of electric field,
$E_0 = 4$ V/m
Absolute permitivity,
$\varepsilon_0 = 8.8 \times 10^{-12}$ C²/N-m²
Average energy density u_E = ?
Applying formula,

Average energy density $u_E = \frac{1}{4}\varepsilon_0 E^2$

$$\Rightarrow \quad u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2$$
$$= 35.2 \times 10^{-12} \text{ J/m}^3$$

14. (4) From the graph it is clear that the amplitude is 1 cm and the time period is 8 second. Therefore the equation for the S.H.M. is

$$x = a\sin\left(\frac{2\pi}{T}\right) \times t = 1\sin\left(\frac{2\pi}{8}\right)t$$

$$x = \sin\frac{\pi}{4}t$$

The velocity (v) of the particle at any instant of time 't' is

$$v = \frac{dx}{dt} = \frac{d}{dt}\left[\sin\left(\frac{\pi}{4}\right)t\right] = \frac{\pi}{4}\cos\left(\frac{\pi}{4}\right)t$$

The acceleration of the particle is

$$\frac{d^2x}{dt^2} = -\left(\frac{\pi}{4}\right)^2 \sin\left(\frac{\pi}{4}\right)t$$

At $t = \frac{4}{3}s$ we get

$$\frac{d^2x}{dt^2} = -\left(\frac{\pi}{4}\right)^2 \sin\frac{\pi}{4} \times \frac{4}{3} = \frac{-\pi^2}{16}\sin\frac{\pi}{3}$$

$$= \frac{-\sqrt{3}\pi^2}{32} \text{cm}/s^2$$

15. (3) $2I_0 = 4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$ here, $\Delta\phi = \frac{\pi}{2}$

But, $\Delta\phi = \frac{2\pi}{\lambda}\Delta x$ so, $\Delta x = \frac{\lambda}{4}$

$$\frac{dy}{D} = \frac{\lambda}{4} \quad ....(i)$$

$$\frac{\lambda D}{d} = \beta \quad ....(ii)$$

Multiplying equation (i) and (ii) we get,

$$y = \frac{\beta}{4}$$

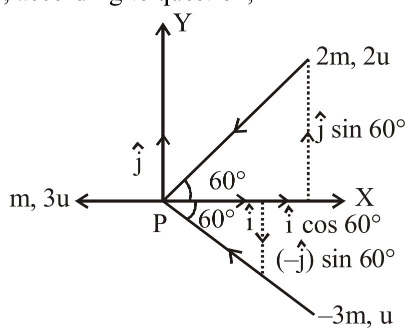
16. (4) Internal energy is a state function hence in a cyclic process change in internal energy is zero.

17. (4) From the law of conservation of momentum we know that,

$$m_1u_1 + m_2u_2 + ... = m_1v_1 + m_2v_2 + ...$$

Given $m_1 = m$, $m_2 = 2m$ and $m_3 = 3m$
and $u_1 = 3u$, $u_2 = 2u$ and $u_3 = u$

Let the velocity when they stick = $\vec{v}$
Then, according to question,



$$m \times 3u\left(\hat{i}\right) + 2m \times 2u$$

$$\left(-\hat{i}\cos 60° - \hat{j}\sin 60°\right) + 3\,m \times u$$

$$\left(-\hat{i}\cos 60° + \hat{j}\sin 60°\right) = (m + 2m + 3m)\,\vec{v}$$

$$\Rightarrow\ 3mu\hat{i} - 4mu\frac{\hat{i}}{2} - 4mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) - 3mu\frac{\hat{i}}{2}$$

$$+3mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) = 6m\vec{v}$$

$$\Rightarrow\ mu\hat{i} - \frac{3}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} = 6m\vec{v}$$

$$\Rightarrow\ -\frac{1}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} = 6m\vec{v}$$

$$\Rightarrow\ \vec{v} = \frac{u}{12}\left(-\hat{i} - \sqrt{3}\hat{j}\right)$$

18. (4) Gravitational force provides the necessary centripetal force.

$$\therefore \frac{mv^2}{(R+x)} = \frac{GmM}{(R+x)^2} \text{ also } g = \frac{GM}{R^2}$$

$$\therefore \frac{mv^2}{(R+x)} = m\left(\frac{GM}{R^2}\right)\frac{R^2}{(R+x)^2}\frac{n!}{r!(n-r)!}$$

$$\therefore \frac{mv^2}{(R+x)} = mg\frac{R^2}{(R+x)^2}$$

$$\therefore v^2 = \frac{gR^2}{R+x} \Rightarrow v = \left(\frac{gR^2}{R+x}\right)^{1/2}$$

19. (2) According to Wien's law $\lambda_m \propto \frac{1}{T}$ and from the figure $(\lambda_m)_1 < (\lambda_m)_3 < (\lambda_m)_2$ therefore $T_1 > T_3 > T_2$.

20. (4) Here, $R = 4\ k\Omega = 4 \times 10^3\ \Omega$

$V_i = 60$ V

Zener voltage $V_z = 10$ V

$R_L = 2\ k\Omega = 2 \times 10^3\ \Omega$

Load current, $I_L = \frac{V_Z}{R_L} = \frac{10}{2\times10^3} = 5$ mA

Current through R, $I = \frac{V_i - V_Z}{R}$

$$= \frac{60-10}{4\times10^3} = \frac{50}{4\times10^3} = 12.5 \text{ mA}$$

Fom circuit diagram,

$I = I_Z + I_L$

$\Rightarrow\ 12.5 = I_Z + 5$

$\Rightarrow\ I_Z = 12.5 - 5 = 7.5$ mA

21. (250) The magnetic field at a point on the axis of a circular loop at a distance x from centre is,

$$B = \frac{\mu_0 i\, a^2}{2(x^2+a^2)^{3/2}}$$

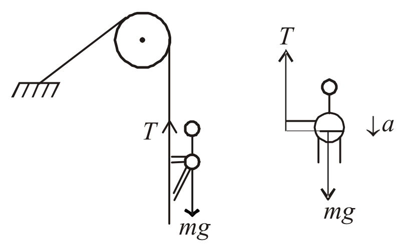
$$B' = \frac{\mu_0 i}{2a}$$

$$\therefore\ B' = \frac{B.(x^2+a^2)^{3/2}}{a^3}$$

Put $x = 4$ & $a = 3 \Rightarrow B' = \frac{54(5^3)}{3\times3\times3} = 250\ \mu T$

22. (4) $mg - T = ma$

$$\therefore\ a = g - \frac{T}{m} = 10 - \frac{360}{60} = 4m/s^2$$



23. (4) $H = \frac{u^2\sin^2\theta}{2g}$ or $H \propto u^2$

$$\frac{\Delta H}{H} = 2.\frac{\Delta u}{u} = 2(+2\%) = 4\% \text{ increased}$$

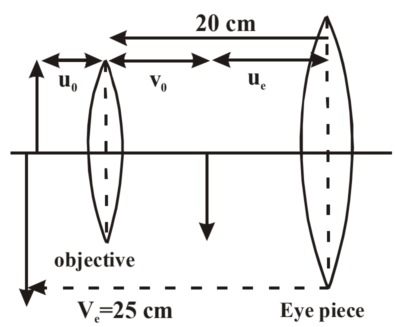
24. (2×10^{-3}) Electric Intensity $= \frac{1}{4\pi\epsilon_0}\times\frac{q}{r^2}$

or $q_{max} = \frac{1}{9\times10^9}r^2E$

$$= \frac{1}{9\times10^9}\times2.5\times2.5\times3\times10^6$$

or $q_{max} = 2\times10^{-3}$ coulomb

25. (1.06)



Final image distance $L = v_0 + u_e = 20$

for objective $\frac{1}{f_0} = \frac{1}{v_0} + \frac{1}{u_0}$

for eye piece $\frac{1}{f_e} = -\frac{1}{25} + \frac{1}{u_e}$

or $\frac{1}{5} + \frac{1}{25} = \frac{1}{u_e}$

$\therefore u_e = \frac{25}{6}$ cm

$\therefore v_o = 20 - u_e$

$v_0 = 20 - \frac{25}{6} = \frac{95}{6}$

$\therefore \frac{1}{u_0} = 1 - \frac{1}{v_0} = 1 - \frac{95}{6} = -\frac{89}{95}$

$\therefore u_0 = -\frac{95}{89} \simeq 1.06$

26. (5451) $V = \frac{1}{e}\left\{\frac{hc}{\lambda} - \phi\right\}$

Put $\lambda_1 = 3 \times 10^{-7}$ m, $V_1 = 1.85$,

$\lambda_2 = 4 \times 10^{-7}$ m, $V_2 = 0.82$

in the above equation respectively

$\phi = 1.216 \times 10^{-19}\,J = \frac{hc}{\lambda_0} \Rightarrow \lambda_0 = 5451Å$

27. (0.01) Pitch $= \frac{\text{Distance travelled on pitch scale}}{\text{Number of rotation}}$

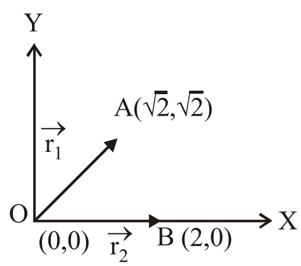
$= \frac{2mm}{4} = 0.5\,mm$

Least count

$= \frac{\text{Pitch}}{\text{Number of division on circular scale}}$

$= \frac{0.5mm}{50} = 0.01\,mm$

28. (0)



The distance of point $A(\sqrt{2}, \sqrt{2})$ from the origin,

$$OA = |\vec{r_1}| = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{4} = 2\text{units.}$$

The distance of point $B(2, 0)$ from the origin,

$$OB = |\vec{r_2}| = \sqrt{(2)^2 + (0)^2} = 2 \text{ units.}$$

Now, potential at A, $V_A = \frac{1}{4\pi \in_0} . \frac{Q}{(OA)}$

Potential at B, $V_B = \frac{1}{4\pi \in_0} . \frac{Q}{(OB)}$

$\therefore$ Potential difference between the points A and B is given by

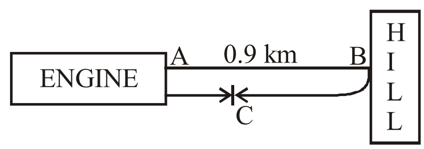
$$V_A - V_B = \frac{1}{4\pi \in_0} . \frac{Q}{OA} - \frac{1}{4\pi \in_0} . \frac{Q}{OB}$$

$$= \frac{Q}{4\pi \in_0}\left(\frac{1}{OA} - \frac{1}{OB}\right) = \frac{Q}{4\pi \in_0}\left(\frac{1}{2} - \frac{1}{2}\right)$$

$$= \frac{Q}{4\pi \in_0} \times 0 = 0.$$

29. (80) 9.25 days is equal to three half-lives for Tl-201. The fraction remaining is then :
$1/2 \times 1/2 \times 1/2 = 1/8$. Thus 1/8 of 80 mCi remains.

30. (30)



Let after 5 sec engine at point C

$t = \frac{AB}{330} + \frac{BC}{330}$

$5 = \frac{0.9 \times 1000}{330} + \frac{BC}{330}$

$\therefore BC = 750\,m$

Distance travelled by engine in 5 sec

$= 900\,m - 750\,m = 150\,m$

Therefore velocity of engine

$= \frac{150\,m}{5\,sec} = 30$ m/s .

CHEMISTRY

31. (3) The correct order of acidic strength of the given species is :

$$\underset{(iv)}{HSO_3F} > \underset{(ii)}{H_3O^+} > \underset{(iii)}{HSO_4^-} > \underset{(i)}{HCO_3^-}$$

(i) < (iii) < (ii) < (iv)

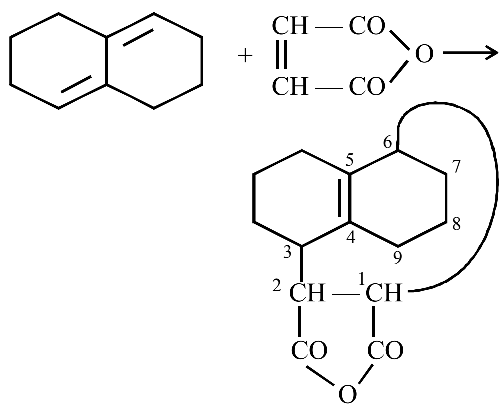
It corresponds to choice (3) which is correct answer.

32. (3) Solar energy is not responsible for green house effect instead it is a source of energy for the plants and animals.

33. (3) CH_3OCH_2Cl is a 1° halide but still hydrolysis takes place through S_N1 mechanism because $CH_3O-\overset{+}{C}H_2$ is stabilized by resonance

$$CH_3OCH_2Cl \leftrightarrow CH_3-\ddot{O}-\overset{+}{C}H_2 \leftrightarrow CH_3-\overset{+}{O}=CH_2$$

34. (2) It is an example of Diel's-Alder reaction in which a dienophile (maleic anhydride, here) reacts with a conjugated diene to form cyclic adduct. Here although structures (i), (iii) and (iv) have conjugated system of double bonds, hence theoretically all the three can undergo Diel's-Alder reaction, but structure (iii) does not undergo this reaction because this will lead to larger ring (9-membered) which is unstable.



35. (2) $Ag_2S + 2NaCN \rightleftharpoons Na_2S + 2AgCN$

$AgCN + NaCN \longrightarrow Na[Ag(CN)_2]$

36. (1) M.P. $\propto \frac{1}{\phi}$ (HCl having more covalent character)

M.P. $\propto$ Lattice energy (Ionic compd.)

LiF is ionic having highest Lattice energy.

37. (1) $AgX + 2\underset{\text{thiosulphate}}{Na_2S_2O_3} \rightarrow \underset{\text{Sodium argento (soluble complex)}}{Na_3[Ag(S_2O_3)_2]} + NaX$

$\because x + 2(-2) = -3$

$x = +1$

38. (1) $\underset{\substack{\text{Ethanol}\\(A)}}{2C_2H_5OH} + 2Na \longrightarrow \underset{\substack{\text{Sod. ethoxide}\\(B)}}{2C_2H_5ONa} + H_2\uparrow$

$$2C_2H_5OH \xrightarrow[-H_2O]{\text{conc. } H_2SO_4} \underset{\text{Diethyl ether}}{C_2H_5OC_2H_5}$$

Thus, (A) is ethanol and (B) is sodium ethoxide.

39. (2) Claisen self condensation is given by esters which contain at least one hydrogen on an alpha carbon atom (e.g., $CH_3COOC_2H_5$). Compound, $C_6H_5COOC_2H_5$ will not give this reaction. All remaining compounds will give this reaction.

40. (2) The step involved is a precipitation step. Increasing the Cl^- concentration will reduce the concentration of Ag^+ in solution. Remember $K_{sp} = [Ag^+][Cl^-]$ and the source of the Cl^- is irrelevant. Thus increased concentration of Cl^- must result in decreased concentration of Ag^+ in order to maintain the solubility product constant. Addition of Ag_2SO_4 would probably be counterproductive, since additional Ag^+ is being added, and probably all will not be recovered.

41. (2) $2F_2 + 4KOH \rightarrow 4KF + O_2 + 2H_2O$ for 1 mole of F_2 the molar ratio.

F_2	KOH	KF	O_2	H_2O
1	2	2	$\frac{1}{2}$	1

42. (4) $CH_2 = CH_2 \xrightarrow{HOCl} ClCH_2CH_2OH \xrightarrow{\text{aq. } NaHCO_3} HOCH_2CH_2OH$

43. (2) All alkali metals dissolve in liquid ammonia giving deep blue solution.

44. (1) According to van der Waal's equation for one mole of gas

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

at very high pressure $P >> \frac{a}{V^2}$

So, $\frac{a}{V^2}$ is negligible.

$P(V-b)=RT$

$PV-Pb=RT$

on dividing RT on both sides.

$\therefore Z=1+\frac{Pb}{RT}$ compressibility factor.

45. (3) For electrophilic substitution reaction, the order of reactivity among the given compounds is as follows :

$C_6H_5OCH_3 >$ benzene $> C_6H_5NO_2$

46. (2) $KI + AgNO_3$ (slight excess) $\longrightarrow AgI + KNO_3$

$AgNO_3 \longrightarrow Ag^+ + NO_3^-$;

$AgI(s) + Ag^+ \longrightarrow [AgI]Ag^+$

47. (2) Saffron is an artificial edible colour.

48. (3) Hydroxylamine and hydrazine, both do not have carbon, hence NaCN will not be formed in Lassaigne's extract leading to negative test for nitrogen.

49. (2) Both egg yolk and mustard contain large quantity of sulphur (as its compounds) which react with Ag.

$2Ag + S \rightarrow Ag_2S$ (Black colour)

50. (3) In (c), sulphate ion is present outside the coordination sphere so it can form white ppt of $BaSO_4$ with $BaCl_2$ (aq).

51. (10.43) Mass of H_2SO_4 in 100ml of 93% H_2SO_4 solution = 93g

$\therefore$ Mass of H_2SO_4 in 1000 ml of the H_2SO_4 solution = 930g

Mass of 1000 ml H_2SO_4 solution

$= 1000 \times 1.84 = 1840g$

Mass of water in 1000 ml of solution

$= 1840 - 930 = 910$ g

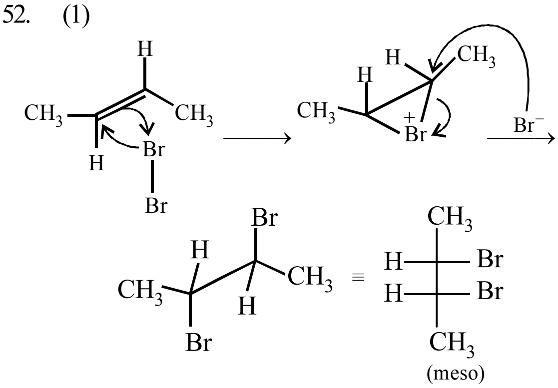
Moles of $H_2SO_4 = \frac{\text{Wt. of } H_2SO_4}{\text{Mol Wt. of } H_2SO_4} = \frac{930}{98}$

$\therefore$ Moles of H_2SO_4 in 1 kg of water

$= \frac{930}{98} \times \frac{1000}{910} = 10.43 \text{ mol kg}^{-1}$

$\therefore$ Molality of 1 litre solution = 10.43

52. (1)



53. (464) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ $\Delta H = -249$

Let the bond enthalpy of O - H is x. Then

$\Delta H = \Sigma$B.E. of reactant - Σ B.E. of product

$-249 = 433 + \frac{1}{2} \times 492 - 2x$

$\Rightarrow x = 464 \text{ kJ mol}^{-1}$.

54. (8) $KE = h\nu - h\nu_0$

$h\nu_1 - h\nu_0 = 2(h\nu_2 - h\nu_0)$

$\nu_0 = 2\,\nu_2 - \nu_1$

$= 2\,(2.0 \times 10^{16}) - (3.2 \times 10^{16})$

$= 8 \times 10^{15}\ s^{-1} = 8 \times 10^{15}$ Hz

55. (57.5) Percentage by mass of copper in malachite

$= \frac{2 \times 63.5}{221} = 57.5\%$

56. (105.7) Molar mass of solute $= \frac{K_f \times w \times 1000}{\Delta T_f \times W}$

Given : $K_f = 1.86$, $w = 1.25$ g, $W = 20$ g,

$\Delta T_f = 273 - 271.9 = 1.1K$

Therefore, molar mass of solute

$= \frac{1.86 \times 1.25 \times 1000}{1.1 \times 20} = 105.7$

57. (0.05) According to the Faraday's law of electrolysis, nF of current is required for the deposition of 1 mole.

According to the reaction,

$Ni(NO_3)_2 \longrightarrow Ni^{2+} + 2NO_3^-$

2 F of current deposits = 1 mol

$\therefore$ 0.1 F of current deposits = = 0.05 mol

58. (21) Molecular orbital electronic configuration of these species are :

$O_2^-(17e^-):\sigma 1s^2,\sigma^*1s^2,\sigma 2s^2,\sigma^*2s^2,\sigma 2p_z^2,$
$\pi 2p_x^2=\pi 2p_y^2,\pi^*2p_x^2=\pi^*2p_y^1$

$O_2(16e^-):\sigma 1s^2,\sigma^*1s^2,\sigma 2s^2,\sigma^*2s^2,\sigma 2p_z^2,$
$\pi 2p_x^2=\pi 2p_y^2,\pi^*2p_x^1=\pi^*2p_y^1$

$O_2^{2-}(18e^-):\sigma 1s^2,\sigma^*1s^2,\sigma 2s^2,\sigma^*2s^2,\sigma 2p_z^2,$
$\pi 2p_x^2:\pi 2p_y^2,\pi^*2p_x^2=\pi^*2p_y^2$

Hence number of antibonding electrons are 7, 6 and 8 respectively.

59. (2.57) Since in NaCl type of structure 4 formula units form a cell.
58.5 g of NaCl = 6.023×10^{23} atoms

1 g of NaCl = $\frac{6.023\times10^{23}}{58.5}$ atoms

4 atoms constitute 1 unit cell

$\therefore \frac{6.023\times10^{23}}{58.5}$ atoms constitute $=\frac{6.023\times10^{23}}{58.5\times4}$

$=2.57\times10^{21}$ unit cells.

60. (2.42) Given : $E^\circ_{Mg^{2+}/Mg}=-2.36V$

$Mg^{2+}(aq)+2e^-\rightarrow Mg(s)$

$$E_{Mg^{2+}/Mg}=E^\circ_{Mg^{2+}/Mg}+\frac{0.059}{2}\log\frac{[Mg^{2+}]}{[Mg]}$$

Given $[Mg^{2+}]=0.01$ M; $[Mg]=1$

Hence, $E_{Mg^{2+}/Mg}=-2.36+\frac{0.059}{2}\log(0.01)$
$=-2.42V$

MATHEMATICS

61. (1) $x=\int_0^y\frac{dt}{\sqrt{1+t^2}}\Rightarrow 1=\frac{1}{\sqrt{1+y^2}}\cdot\frac{dy}{dx}$

$$\left[\because \text{If } I(x)=\int_{\phi(x)}^{\psi(x)}f(t)\,dt,\text{ then }\frac{dI(x)}{dx}=f\{\psi(x)\}.\left\{\frac{d}{dx}\psi(x)\right\}-f\{\phi(x)\}.\left\{\frac{d}{dx}\phi(x)\right\}\right]$$

$$\frac{dy}{dx}=\sqrt{1+y^2}$$

$$\Rightarrow\frac{d^2y}{dx^2}=\frac{1}{2\sqrt{1+y^2}}.2y.\frac{dy}{dx}=\frac{y}{\sqrt{1+y^2}}.\sqrt{1+y^2}=y$$

62. (2) $|a_1x+a_2x^2+\ldots+a_nx^n|\le|a_1||x|+|a_2||x|^2+|a_3||x|^3+\ldots\ldots+|a_n||x|^n$

$\le 2[|x|+|x|^2+\ldots..+|x|^n]\ [\because|a_n|<2]$

$$=\frac{2|x|}{1-|x|}[1-|x|^n]<\frac{2|x|}{1-|x|}<\frac{2}{3}\frac{1}{1-|x|}$$

Therefore, $|a_1x+\ldots..+a_nx^n|<2.\frac{1}{3}.\frac{3}{2}=1$

$\therefore\ |a_1x+\ldots..+a_nx^n|<1$ for all n .
$-1<a_1x+\ldots..+a_nx^n<1$ for all n

$\Rightarrow 1+a_1x+a_2x^2+\ldots+a_nx^n>0$ for all n

$\Rightarrow 1+a_1x+..+a_nx^n\neq0$ for all n

63. (1) Since, $e^{-\frac{\pi}{2}}<\theta<\frac{\pi}{2}$

$\therefore \log e^{-\frac{\pi}{2}}<\log\theta<\log\frac{\pi}{2}$

i.e., $-\frac{\pi}{2}<\log\theta<\log\frac{\pi}{2}<1<\frac{\pi}{2}$

$\left[\because\frac{\pi}{2}<e\ \therefore\log\frac{\pi}{2}<\log e=1\text{ and }1<\frac{\pi}{2}\right]$

$\therefore -\frac{\pi}{2}<\log\theta<\frac{\pi}{2}\ \therefore\cos(\log\theta)>0$

But $0<\cos\theta<1$, $\therefore\log(\cos\theta)<\log1=0$

i.e., $\log(\cos\theta)<0$.

Hence, $\cos(\log\theta)>\log(\cos\theta)$

64. (1) $\sum_{\substack{r=0\\r\le s}}^{s}\sum_{s=1}^{n}{}^nC_s\,{}^sC_r$

$$=\sum_{s=1}^{n}{}^nC_s\left({}^sC_0+{}^sC_1+{}^sC_2+\ldots+{}^sC_s\right)$$

$$=\sum_{s=1}^{n}{}^nC_s2^s=\sum_{s=0}^{n}{}^nC_s2^s-{}^nC_02^0=(1+2)^n-1$$

$=3^n-1$

65. (3) Number of ways of making 3 sets of 10 balls having 2, 3 and 5 balls $\frac{10!}{2!.3!.5!}$ and further they can be given to 3 persons one each in $3!$ ways.

Total number of ways $=\frac{10!}{2!3!5!}\times 3!$

66. (4) Given, $\sin 2x\left(\frac{dy}{dx}-\sqrt{\tan x}\right)-y=0$

or, $\frac{dy}{dx}=\frac{y}{\sin 2x}+\sqrt{\tan x}$

or, $\frac{dy}{dx}-y\operatorname{cosec}2x=\sqrt{\tan x}$ ...(1)

Now, integrating factor (I.F) $= e^{\int -\operatorname{cosec}2x}$

or, I.F $= e^{-\frac{1}{2}\log|\tan x|}=e^{\log\left(\sqrt{\tan x}\right)^{-1}}$

$=\frac{1}{\sqrt{\tan x}}=\sqrt{\cot x}$

Now, general solution of eq. (1) is written as

$$y\,(\text{I. F.})=\int Q(\text{I.F.})\,dx+c$$

$$\therefore\ y\sqrt{\cot x}=\int\sqrt{\tan x}.\sqrt{\cot x}\,dx+c$$

$$\therefore\ y\sqrt{\cot x}=\int 1.dx+c$$

$$\boxed{\therefore\ y\sqrt{\cot x}=x+c}$$

67. (2) From cosine and sine formula, we have

$\cos A=\frac{b^2+c^2-a^2}{2bc}$ and

$\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}=k.$

Given, in any ΔABC,

$\cos A=\frac{\sin B}{2\sin C}$

From the above given formula, we have

$\sin B=bk$, $\sin C=ck$

$\therefore\ \cos A=\frac{bk}{2ck}=\frac{b}{2c}$

Put the value of cos A in the formula, which gives

$$\frac{b}{2c}=\frac{b^2+c^2-a^2}{2bc}$$

$\Rightarrow c^2=a^2\ \Rightarrow\ c=a$

68. (4) Operate $C_1+C_2+C_3$, we get,

$$\begin{vmatrix} x+1+\omega+\omega^2 & \omega & \omega^2 \\ x+1+\omega+\omega^2 & x+\omega^2 & 1 \\ x+1+\omega+\omega^2 & 1 & x+\omega \end{vmatrix}=0$$

$$\Rightarrow\begin{vmatrix} x & \omega & \omega^2 \\ x & x+\omega^2 & 1 \\ x & 1 & x+\omega \end{vmatrix}=0$$

$[\because\ 1+\omega+\omega^2=0]$

$$\Rightarrow x\begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & x+\omega^2 & 1 \\ 1 & 1 & x+\omega \end{vmatrix}=0\Rightarrow x=0$$

69. (4) Lines are concurrent if $\begin{vmatrix} \ell & m & n \\ m & n & \ell \\ n & \ell & m \end{vmatrix}=0$

$$\Rightarrow\begin{vmatrix} \ell+m+n & m & n \\ \ell+m+n & n & \ell \\ \ell+m+n & \ell & m \end{vmatrix}=0\quad (C_1\to C_1+C_2+C_3)$$

$$\Rightarrow(\ell+m+n)\begin{vmatrix} 1 & m & n \\ 1 & n & \ell \\ 1 & \ell & m \end{vmatrix}=0$$

$\Rightarrow(\ell+m+n)(mn+n\ell+\ell m-\ell^2-m^2-n^2)=0$

$\Rightarrow \ell+m+n=0;\ \ell^2+m^2+n^2-\ell m-mn-n\ell=0$

$\therefore\ \ell^2+m^2+n^2=\ell m+mn+n\ell$

70. (3) $\begin{vmatrix} 3 & 4 & 2 \\ 5 & 8 & 2 \\ x & y & 2 \end{vmatrix}=\begin{vmatrix} 3 & 4 & 2 \\ 2 & 4 & 0 \\ x-5 & y-8 & 0 \end{vmatrix}$

$(R_2\to R_2-R_1,\ R_3\to R_3-R_2)$

$=2\,(2y-16-4x+20)=2\,(2y-4x+4)$

$\therefore$ given determinant $=0\Rightarrow 2y-4x+4=0$

$\Rightarrow 2x-y-2=0$ which represents st. line.

71. (2) Centre of the given circle is (1,2) and its radius $=\sqrt{1+4+20}=5$. Since the radii of the two circles are equal, therefore these will touch externally and the point of contact will lie mid-way between the two centres. If (h,k) is the centre of the circle, then

$\frac{h+1}{2}=5,\ \frac{k+2}{2}=5\ \therefore h=9, k=8$

$\therefore$ its equation is $(x-9)^2+(y-8)^2=25$

i.e., $x^2+y^2-18x-16y+120=0$

72. (1) Ellipse is $\frac{x^2}{16}+\frac{y^2}{3}=1$

Now, equation of normal at (2, 3/2) is

$$\frac{16x}{2}-\frac{3y}{3/2}=16-3$$

$\Rightarrow\ 8x-2y=13 \Rightarrow y=4x-\frac{13}{2}$

Let $y=4x-\frac{13}{2}$ touches a parabola

$y^2=4ax$.

We know, a straight line $y=mx+c$ touches a parabola $y^2=4ax$ if $a-mc=0$

$\therefore\ a-(4)\left(-\frac{13}{2}\right)=0 \Rightarrow a=-26$

Hence, required equation of parabola is

$y^2=4(-26)x=-104x$

73. (1) I study or I fail = $p\vee q$

Now, $\sim(p\vee q)\equiv\sim p\wedge(\sim q)$

Hence, negation of 'I study or I fail' is I do not study and I do not fail.

74. (2) $\sin x-\cos x=1\Rightarrow \sin\left(x-\frac{\pi}{4}\right)=\sin\frac{\pi}{4}\Rightarrow x-\frac{\pi}{4}=n\pi+(-1)^n\frac{\pi}{4}\Rightarrow x=n\pi+(-1)^n\frac{\pi}{4}+\frac{\pi}{4}$

75. (4) Vector $\vec{A}=\hat{i}+\hat{j}+\hat{k},\ \vec{B}=4\hat{i}+3\hat{j}+4\hat{k}$

And $\vec{C}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$ are linearly independent,

if $\begin{vmatrix}1 & 1 & 1\\ 4 & 3 & 4\\ 1 & \alpha & \beta\end{vmatrix}=0$

Operate $C_3\to C_3-C_1$; $C_2\to C_2-C_1$

we get, $\begin{vmatrix}1 & 0 & 0\\ 4 & -1 & 0\\ 1 & \alpha-1 & \beta-1\end{vmatrix}=0$

$\Rightarrow-(\beta-1)=0\Rightarrow\beta=1$

Again, $|\vec{C}|=\sqrt{1+\alpha^2+\beta^2}=\sqrt{3}$

$\Rightarrow 1+\alpha^2+\beta^2=3 \Rightarrow 1+\alpha^2+1=3$

$\therefore \alpha^2=1\Rightarrow\alpha=\pm1$

Hence, $\alpha=\pm1,\ \beta=1$

76. (4) We know that every quadratic equation has exactly two roots which are either, both real or both are imaginary. So, any quadratic equation has neither exactly one real root nor booth roots are always real.

77. (2) We have a theorem that if a square matrix A satisfies the equation;

$a_0+a_1x+a_2x^2+.......+a_nx^n=0,$

where $a_0\neq 0$ then A is invertible.

Since A, B and C are $n\times n$ matrices and A satisfies the equation $x^3+2x^2+3x+5=0$ as $A^3+2A^2+3A+5I=0$, therefore, A is invertible.

78. (4) $\int_{\pi/2}^{3\pi/2}[2\sin x]dx=\int_{\pi/2}^{5\pi/6}[2\sin x]dx+\int_{5\pi/6}^{\pi}[2\sin x]dx+\int_{\pi}^{7\pi/6}[2\sin x]dx+\int_{7\pi/6}^{3\pi/2}[2\sin x]dx$

$=\int_{\pi/2}^{5\pi/6}1\,dx+\int_{5\pi/6}^{\pi}0\,dx+\int_{\pi}^{7\pi/6}(-1)\,dx+\int_{7\pi/6}^{3\pi/2}(-2)\,dx$

$=\left(\frac{5\pi}{6}-\frac{\pi}{2}\right)+0+\left(\pi-\frac{7\pi}{6}\right)+2\left(\frac{7\pi}{6}-\frac{3\pi}{2}\right)=-\frac{\pi}{2}$

79. (1) $Lf'(1)$

$=\underset{x\to1}{Lt}\frac{f(x)-f(1)}{x-1}=\underset{x\to1}{Lt}\frac{ax^2+b-a-b}{x-1}$

$=\underset{x\to1}{Lt}\frac{a(x^2-1)}{x-1}=\underset{x\to1}{Lt}a(x+1)=2a$

$Rf'(1)$

$=\underset{x\to1}{Lt}\frac{f(x)-f(1)}{x-1}=\underset{x\to1}{Lt}\frac{bx^2+ax+c-a-b}{x-1}$

$=\underset{x\to1}{Lt}[b(x+1)+a]+\frac{c}{x-1}$

$=2b+a+\frac{c}{x-1}=2b+a$ if $c=0$

Since, f is diff. at $x=1\ \therefore 2a=2b+a$

$\Rightarrow a=2b$. Thus, result holds if $a=2b, c=0$.

80. (3) We have $\alpha+\beta=-3$ and $\alpha\beta=a/2$

Now $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}<2 \Rightarrow \frac{\alpha^2+\beta^2}{\alpha\beta}<2$

$\Rightarrow \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}<2 \Rightarrow \frac{9-a}{a/2}<2$

$\Rightarrow \frac{9-a}{a}<1$

$\Rightarrow \frac{9-a}{a}-1<0 \Rightarrow \frac{9-2a}{a}<0 \Rightarrow \frac{2a-9}{a}>0$

$\Rightarrow a<0$ or $a>\frac{9}{2}$

81. (0.14) The probability of A winning in a game

$=P(A)=\frac{6}{12}=\frac{1}{2}$

The probability of B winning in a game = P (B)

$=\frac{4}{12}=\frac{1}{3}$

$\therefore$ Reqd. probability $=P(A\cap B\cap A)+P(B\cap A\cap B)$

$=P(A).P(B).P(A)+P(B).P(A).P(B)$

$=\frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{2}+\frac{1}{3}\cdot\frac{1}{2}\cdot\frac{1}{3}=\frac{5}{36}=0.14$

82. (10.00) Period of $\sin x=2\pi$

$\Rightarrow$ period of $\sin^3 x=2\pi$

period of $|\sin^3 x|=\pi$

$\Rightarrow$ period of $\left|\sin^3\frac{x}{2}\right|=2\pi$

period of $\cos^5 x=2\pi$

$\Rightarrow$ period of $|\cos^5 x|=\pi$

$\Rightarrow$ period of $\left|\cos^5\frac{x}{5}\right|=5\pi$

Thus required period

$=$ LCM of 2π & $5\pi=10\pi=k\pi \Rightarrow k=10$

83. (6.00) We have,

$4\cos x(2-3\sin^2 x)+(\cos 2x+1)=0$

$\Rightarrow 4\cos x(3\cos^2 x-1)+2\cos^2 x=0$

$\Rightarrow 2\cos x(6\cos^2 x+\cos x-2)=0$

$\Rightarrow 2\cos x(3\cos x+2)(2\cos x-1)=0$

$\Rightarrow$ either $\cos x=0$ which gives $x=\pi/2$ or $\cos x=-2/3$

Which gives no value of x for which $0\le x\le\pi/2$

or $\cos x=1/2$, which gives $x=\pi/3$

So, the required difference $=\pi/2-\pi/3=\pi/6$

$=\frac{\pi}{n}\Rightarrow n=6.$

84. (3.00) Since $1^2x-1<[1^2x]\le 1^2.x$

$2^2x-1<[2^2x]\le 2^2x$

$3^2x-1<[3^2x]\le 3^2x$ ....

$n^2x-1<[n^2x]\le n^2x$

Adding all terms, we get,

$$x\sum n^2-n<\{[1^2x]+[2^2x]+[3^2x]+\ldots+[n^2x]\}\le x\sum n^2$$

Dividing each term by n^3,

we get , $\frac{x\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6}-\frac{1}{n^2}<\frac{[1^2x]+[2^2x]+.....+[n^2x]}{n^3}\le\frac{x\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6}$

Let $n\to\infty$, we get,

$$\frac{x}{6}(2)-0<\underset{n\to\infty}{\mathrm{Lt}}\frac{[1^2x]+[2^2x]+....+[n^2x]}{n^3}\le\frac{x}{3}$$

$$\Rightarrow \lim_{n\to\infty}\frac{[1^2x]+[2^2x]+.....+[n^2x]}{n^3}=\frac{x}{3}$$

$=\frac{x}{n}\Rightarrow n=3.$

85. (80.00) $(1-y)^m(1+y)^n$

$=[1-{}^mC_1y+{}^mC_2y^2-......][1+{}^nC_1y+{}^nC_2y^2+.....]$

$=1+(n-m)y+\left\{\frac{m(m-1)}{2}+\frac{n(n-1)}{2}-mn\right\}y^2+.....$

By comparing coefficients with the given expression, we get

$\therefore\ a_1 = n - m = 10$ and $a_2 = \dfrac{m^2+n^2-m-n-2mn}{2} = 10$

So, $n - m = 10$ and $(m-n)^2 - (m+n) = 20$

$\Rightarrow\ m + n = 80$

86. (1.00) $\sin^4\left(\dfrac{3\pi}{2}-\alpha\right) = \cos^4\alpha,$

$\sin^4(3\pi+\alpha) = \sin^4\alpha,$

$\sin^6\left(\dfrac{\pi}{2}+\alpha\right) = \cos^6\alpha,\ \sin^6(5\pi-\alpha) = \sin^6\alpha$

$\therefore$ given quantity
$= 3[\cos^4\alpha + \sin^4\alpha] - 2[\cos^6\alpha + \sin^6\alpha]$
$= 3[1 - 2\sin^2\alpha\cos^2\alpha] - 2[1 - 3\sin^2\alpha\cos^2\alpha] = 1$

87. (0) $[\vec{A}-\vec{B}\ \ \vec{B}-\vec{C}\ \ \vec{C}-\vec{A}]$

$= [(\vec{A}-\vec{B})\times(\vec{B}-\vec{C})].(\vec{C}-\vec{A})$

$= [\vec{A}\times\vec{B} - \vec{A}\times\vec{C} - \vec{0} + \vec{B}\times\vec{C}].(\vec{C}-\vec{A})$

$= [\vec{A}\times\vec{B}].\vec{C} - [\vec{B}\times\vec{C}].\vec{A}$

$= [\vec{A}\,\vec{B}\,\vec{C}] - [\vec{A}\,\vec{B}\,\vec{C}] = 0$ $(\because [ABC]=[BCA])$

88. (0.29) $\lim\limits_{x\to 3}[f(x)+g(x)+h(x)]$

$= \lim\limits_{x\to 3}\left[\dfrac{2}{x-3}+\dfrac{x-3}{x+4}-\dfrac{2(2x+1)}{x^2+x-12}\right]$

$= \lim\limits_{x\to 3}\left[\dfrac{2x+8+x^2-6x+9-4x-2}{x^2+x-12}\right]$

$= \lim\limits_{x\to 3}\dfrac{x^2-8x+15}{x^2+x-12} = \lim\limits_{x\to 3}\dfrac{(x-3)(x-5)}{(x+4)(x-3)}$

$= \lim\limits_{x\to 3}\dfrac{x-5}{x+4} = \dfrac{3-5}{3+4} = -\dfrac{2}{7} = k$

$\Rightarrow |k| = \dfrac{2}{7} = 0.29.$

89. (1.00) Let $P(x_1, y_1)$, $Q(x_2, y_2)$ be two points on the curve $y = x + \dfrac{1}{x}$. Let $\hat{i}$ be unit vector along x-axis and let $\hat{j}$ be a unit vector along y-axis. Then

$$\overrightarrow{OP} = x_1\hat{i} + y_1\hat{j},\ \overrightarrow{OQ} = x_2\hat{i} + y_2\hat{j}$$

Since, $\overrightarrow{OP}\cdot\hat{i} = 1$ and $\overrightarrow{OQ}.\hat{i} = -1$

$\therefore\ x_1 = 1$ and $x_2 = -1 \Rightarrow y_1 = 2$ and $y_2 = -2$;

$\therefore\ \overrightarrow{OP} = \hat{i} + 2\hat{j};\ OQ = -\hat{i} - 2\hat{j}$

$\therefore\ 2\overrightarrow{OP} + 3\overrightarrow{OQ} = 2i + 4j - 3i - 6j = -i - 2j$;

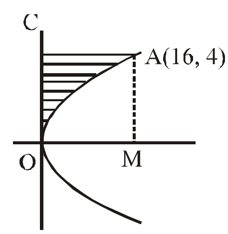
Hence, $\left|2\overrightarrow{OP} + 3\overrightarrow{OQ}\right| = \sqrt{1+4} = \sqrt{5}$

$= n\sqrt{5} \Rightarrow n = 1$

90. (21.33) $y = 4$ meets the parabola $y^2 = x$ at A, if $16 = x$

$\therefore$ A is (16, 4)

Reqd. area = Area of rect. OMAC – Area OMA



$= 4\times 16 - \int_0^{16}\sqrt{x}\,dx = 64 - \left|\dfrac{x^{3/2}}{\frac{3}{2}}\right|_0^{16}.$

$= 64 - \dfrac{2}{3}(4)^3 = 64 - \dfrac{128}{3} = \dfrac{64}{3} = 21.33$ sq.units

Mock Test-7

ANSWER KEY											
1	(3)	**16**	(3)	**31**	(4)	**46**	(3)	**61**	(4)	**76**	(1)
2	(2)	**17**	(1)	**32**	(1)	**47**	(2)	**62**	(4)	**77**	(3)
3	(3)	**18**	(1)	**33**	(3)	**48**	(3)	**63**	(3)	**78**	(3)
4	(1)	**19**	(1)	**34**	(3)	**49**	(2)	**64**	(2)	**79**	(1)
5	(2)	**20**	(1)	**35**	(1)	**50**	(3)	**65**	(1)	**80**	(3)
6	(3)	**21**	(242)	**36**	(2)	**51**	(6.5)	**66**	(1)	**81**	(1)
7	(2)	**22**	(499)	**37**	(2)	**52**	(6.24)	**67**	(2)	**82**	(4)
8	(1)	**23**	(80)	**38**	(3)	**53**	(3.6)	**68**	(2)	**83**	(0.57)
9	(1)	**24**	(0.4)	**39**	(4)	**54**	(6.7)	**69**	(1)	**84**	(34)
10	(1)	**25**	(0.1)	**40**	(2)	**55**	(0.107)	**70**	(3)	**85**	(0.63)
11	(2)	**26**	(11.2)	**41**	(2)	**56**	(10)	**71**	(3)	**86**	(5)
12	(1)	**27**	(0.67)	**42**	(3)	**57**	(6.56)	**72**	(2)	**87**	(3)
13	(1)	**28**	(1.2×10^{-7})	**43**	(3)	**58**	(–6.04)	**73**	(3)	**88**	(0.33)
14	(4)	**29**	(7.8×10^{-7})	**44**	(4)	**59**	(2)	**74**	(1)	**89**	(7.26)
15	(3)	**30**	(4)	**45**	(3)	**60**	(316)	**75**	(4)	**90**	(1.33)

Solutions

PHYSICS

1. (3) Force, $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \Rightarrow \varepsilon_0 = \frac{q_1 \cdot q_2}{4\pi F r^2}$

So dimension of ε_0

$$= \frac{[AT]^2}{[MLT^{-2}][L^2]} = [M^{-1}L^{-3}T^4A^2]$$

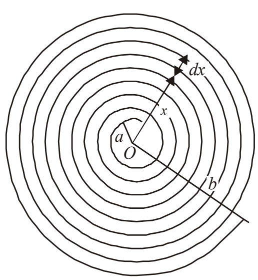
2. (2) $y = 2\sin\left(\frac{\pi t}{2} + \phi\right)$

velocity of particle $\frac{dy}{dt} = 2 \times \frac{\pi}{2}\cos\left(\frac{\pi t}{2} + \phi\right)$

acceleration $\frac{d^2y}{dt} = -\frac{\pi^2}{2}\sin\left(\frac{\pi t}{2} + \phi\right)$

Thus, $a_{max} = \frac{\pi^2}{2}$

3. (3) Let us consider a thickness dx of wire. Let it be at a distance x from the centre O. Number of turns per unit length $= \frac{N}{b-a}$



$\therefore$ Number of turns in thickness $dx = \frac{N}{b-a}dx$

Small amount of magnetic field is produced at O due to thickness dx of the wire.

$$\therefore dB = \frac{\mu_0}{2}\frac{NI}{(b-a)}\frac{dx}{x}$$

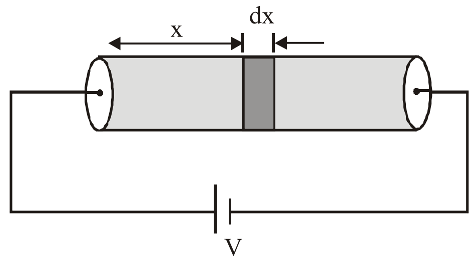
On integrating, we get,

$$B = \int_a^b \frac{\mu_0}{2}\frac{NI}{b-a}\frac{dx}{x} = \frac{\mu_0}{2}\frac{NI}{(b-a)}$$

$$\int_a^b \frac{dx}{x} = \frac{\mu_0}{2}\frac{NI}{(b-a)}[\log_e x]_a^b$$

$$B = \frac{\mu_0}{2}\frac{NI}{(b-a)}\log_e\frac{b}{a}$$

4. (1) Consider an element part of solid at a distance x from left end of width dx.



Resistance of this elemental part is,

$$dR = \frac{\rho dx}{\pi a^2} = \frac{\rho_0 x dx}{\pi a^2}$$

$$R = \int dR = \int_0^L \frac{\rho_0 x dx}{\pi a^2} = \frac{\rho_0 L^2}{2\pi a^2}$$

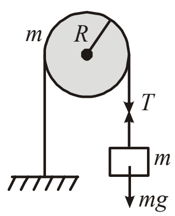
Current through cylinder is,

$$I = \frac{V}{R} = \frac{V \times 2\pi a^2}{\rho_0 L^2}$$

Potential drop across element is, $dV = I\, dR = \frac{2V}{L^2} x\, dx$

$$E(x) = \frac{dV}{dx} = \frac{2V}{L^2} x$$

5. (2) For translational motion,
$mg - T = ma$ .....(1)
For rotational motion,
$T.R = I\alpha = I\frac{a}{R}$ ....(2)



Solving (1) & (2),

$$a = \frac{mg}{\left(m + \frac{I}{R^2}\right)} = \frac{mg}{m + \frac{mR^2}{2R^2}} = \frac{2mg}{3m} = \frac{2g}{3}$$

6. (3) We know that V/T = constant

$$\therefore \quad \frac{V + \Delta V}{T + \Delta T} = \frac{V}{T} \quad \text{or} \quad \frac{\Delta V}{V \Delta T} = \frac{1}{T}$$

7. (2) $l = \frac{nh}{2\pi}, |E| \propto Z^2/n^2; n = 3$

$\Rightarrow l_H = l_{Li}$ and $|E_H| < |E_{Li}|$

8. (1) $\eta = \left(1 - \frac{T_c}{T_H}\right) \times 100 \Rightarrow 70 = \left(1 - \frac{T_c}{1000}\right) \times 100$

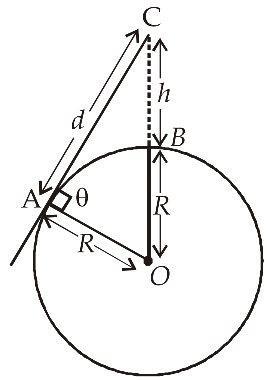
$$0.7 = 1 - \frac{T_c}{1000}$$

$$\therefore \frac{T_c}{1000} = 0.3 \text{ or } T_c = 300\text{K}.$$

9. (1) Let d is the maximum distance, upto it the objects

From ΔAOC

$$OC^2 = AC^2 + AO^2$$



$$(h+R)^2 = d^2 + R^2$$

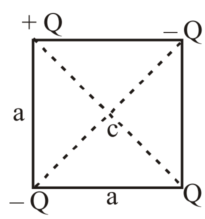
$$\Rightarrow \quad d^2 = (h+R)^2 - R^2$$

$$d = \sqrt{(h+R)^2 - R^2}$$

$$d = \sqrt{h^2 + 2hR}$$

$$d = \sqrt{500^2 + 2 \times 6.4 \times 10^6} = 80\,\text{km}$$

10. (1)



Electric potential at centre

$$= \frac{1}{4\pi\varepsilon_0}\frac{Q}{\frac{a}{\sqrt{2}}} + \frac{1}{4\pi\varepsilon_0}\frac{Q}{\frac{a}{\sqrt{2}}} - \frac{1}{4\pi\varepsilon_0}\frac{Q}{\frac{a}{\sqrt{2}}} - \frac{1}{4\pi\varepsilon_0}\frac{Q}{\frac{a}{\sqrt{2}}} = 0$$

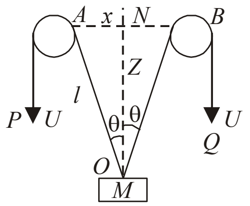
11. (2) This is a problem based on constraint motion. The motion of mass M is constraint with the motion of P and Q. Let $AN = x$, $NO = z$. Then velocity of mass is $\frac{dz}{dt}$. Also, let $OA = \ell$. then

$$\frac{d\ell}{dt} = U$$

From ΔANO, using pythagorous theorem

$\therefore \quad x^2 + z^2 = \ell^2$

Here x is a constant.
Differentiating the above



equation w.r.t to t

$$0+2z\frac{dz}{dt}=2\ell\frac{d\ell}{dt} \Rightarrow \qquad zv_M=\ell U$$

$$\Rightarrow \; v_M=\frac{\ell}{z}U=\frac{U}{z/\ell}=\frac{U}{\cos\theta}$$

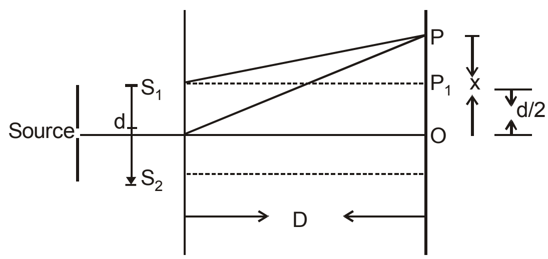
$$\left(\because \cos\theta=\frac{z}{\ell}\right)$$

12. (1) Path difference $=\dfrac{xd}{D}$

For missing wavelengths (i.e., for dark fringe)

path difference $=(2n-1)\dfrac{\lambda}{2}=\dfrac{xd}{D}$

$$\Rightarrow \lambda=\frac{2xd}{D(2n-1)}$$



but x = d/2, because point is P_1, where we want missing wave length.

$$\therefore \quad \lambda=\frac{d^2}{D(2n-1)}$$

For n = 1, 2 .... missing wavelengths are

$$\frac{d^2}{D},\frac{d^2}{3D},\frac{d^2}{5D}...............$$

13. (1) Let the body be depressed by distance x from its equilibrium position. The extra upthrust created is x ρAg which applies to whole body. If a be acceleration created then,

$$x\rho Ag=ma \Rightarrow a=\frac{\rho A}{m}xg$$

Since, acceleration a x. So it is equation of S.H.M.

$$\text{So, } \omega^2=\frac{\rho Ag}{m} \Rightarrow T=2\pi\sqrt{\frac{m}{\rho Ag}}\,;\; T \text{ a } \frac{1}{\sqrt{A}}$$

14. (4) Young's modulus, $Y=\dfrac{\text{Stress}}{\text{Strain}}$

or stress = Y. strain

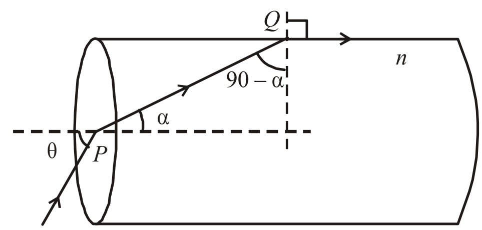
or strain = Stress / Y

$$\text{or } \Delta l=\frac{Fl}{YA}\;;\; \frac{\Delta l_1}{\Delta l_2}=\frac{F_1 l_1}{A_1Y_1}\cdot\frac{A_2Y_2}{F_1 l_1}$$

$l_1=l_2$ & $Y_1=Y_2$, $F_1=F_2$

$$\Rightarrow \frac{\Delta l_1}{\Delta l_2}=\frac{\pi r_2^{\,2}}{\pi r_1^{\,2}}=\frac{4r^2}{r^2}=4$$

15. (3)



Applying Snell's law at Q

$$\frac{1}{n}=\frac{\sin(90-\alpha)}{\sin 90^{\circ}}$$

$$\therefore \cos\alpha=\frac{1}{n}$$

$$\therefore \sin\alpha=\sqrt{1-\cos^2\alpha}=\sqrt{1-\frac{1}{n^2}}=\frac{\sqrt{n^2-1}}{n} \qquad ...(1)$$

Applying Snell's Law at P

$$n=\frac{\sin\theta}{\sin\alpha} \Rightarrow \sin\theta=n\times\sin\alpha=\sqrt{n^2-1}\,;\ \text{from (1)}$$

$$\therefore \sin\theta=\sqrt{\left(\frac{2}{\sqrt{3}}\right)^2-1}=\sqrt{\frac{4}{3}-1}=\frac{1}{\sqrt{3}}$$

$$\text{or} \qquad \theta=\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

16. (3)

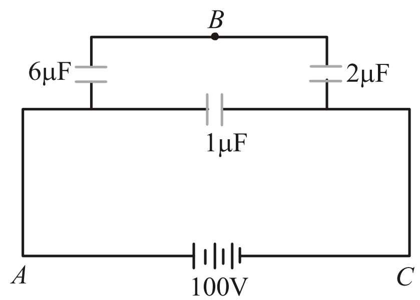
$$C_{eq}=\frac{(3+3)\times(1+1)}{(3+3)+(1+1)}+1=\left(\frac{6\times 2}{6+2}\right)+1=\frac{5}{2}\mu F$$

$\therefore Q = C \times V = \frac{5}{2} \times 100 = 250\mu C$

Charge in 6μF branch

$= VC = \left(\frac{6 \times 2}{6+2}\right)100 = 150\mu C$

$V_{AB} = \frac{150}{6} = 25V$ and $V_{BC} = 100 - V_{AB} = 75V$



17. (1) When springs are in parallel, then

$T = 2\pi\sqrt{\frac{m}{K_1 + K_2}} \Rightarrow \frac{2\pi}{T} = \omega = \sqrt{\frac{K_1 + K_2}{m}}$

18. (1) Let u be the initial velocity of the bullet of mass m.
After passing through a plank of width x, its velocity decreases to v.

$\therefore \quad u - v = \frac{u}{n}$ or, $v = u - \frac{u}{n} = \frac{u(n-1)}{n}$

If F be the retarding force applied by each plank, then using work – energy theorem,

$Fx = \frac{1}{2}mu^2 - \frac{1}{2}mv^2 = \frac{1}{2}mu^2 - \frac{1}{2}mu^2$

$\frac{(n-1)^2}{n^2} = \frac{1}{2}mu^2\left[1 - \frac{(n-1)^2}{n^2}\right]$

$Fx = \frac{1}{2}mu^2\left(\frac{2n-1}{n^2}\right)$

Let P be the number of planks required to stop the bullet.
Total distance travelled by the bullet before coming to rest = Px
Using work-energy theorem again,

$F(Px) = \frac{1}{2}mu^2 - 0$

or, $P(Fx) = P\left[\frac{1}{2}mu^2\frac{(2n-1)}{n^2}\right] = \frac{1}{2}mu^2$

$\therefore \; P = \frac{n^2}{2n-1}$

19. (1) $I = I_o\left(1 - e^{-\frac{R}{L}t}\right)$

(When current is in growth in LR circuit)

$= \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right) = \frac{5}{5}\left(1 - e^{-\frac{5}{10}\times 2}\right)$

$= (1 - e^{-1})$

20. (1) Total momentum will be conserved.
Initial momentum = Final momentum

$M.v = m \times 0 + (M - m)v'$

$\therefore v' = \frac{Mv}{M - m}$

21. (242) When capacitance is taken out, the circuit is LR.

$\therefore \; \tan\phi = \frac{\omega L}{R}$

$\Rightarrow \omega L = R\tan\phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$

Again, when inductor is taken out, the circuit is CR.

$\therefore \quad \tan\phi = \frac{1}{\omega CR}$

$\Rightarrow \quad \frac{1}{\omega c} = R\tan\phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$

Now, $Z = \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}$

$= \sqrt{(200)^2 + \left(\frac{200}{\sqrt{3}} - \frac{200}{\sqrt{3}}\right)^2} = 200\,\Omega$

Power dissipated = $V_{rms}I_{rms}\cos\phi$

$= V_{rms}.\frac{V_{rms}}{Z}.\frac{R}{Z}\left(\because \cos\phi = \frac{R}{Z}\right)$

$= \frac{V^2{}_{rms}R}{Z^2} = \frac{(220)^2 \times 200}{(200)^2} = \frac{220 \times 220}{200}$

$= 242$ W

22. (499) $v_{rms} = \sqrt{\frac{3pv}{\text{mass of the gas}}}$

23. (80) Speed, $u = 60 \times \frac{5}{18}$ m/s $= \frac{50}{3}$ m/s

$d = 20$m, $u' = 120 \times \frac{5}{18} = \frac{100}{3}$ m/s

Let declaration be a then $(0)^2 - u^2 = -2ad$

or $u^2 = 2ad$ ... (i)

and $(0)^2 - u'^2 = -2ad'$ or $u'^2 = 2ad'$ ...(ii)

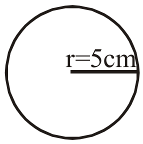
(ii) divided by (i) gives,

$$4 = \frac{d'}{d} \Rightarrow d' = 4 \times 20 = 80\text{m}$$

24. (0.4) $r = 5$ cm. $= 5 \times 10^{-2}$m

$B_E = 0.5 \times 10^{-5}$ W/m^2

we know that field due to coil at



centre $B = \dfrac{\mu_0 I}{2r}$

it annuals the earth's megnetic field

So, $\dfrac{\mu_0 I}{2r} = 0.5 \times 10^{-5}$

$$I = \frac{2r \times 0.5 \times 10^{-5}}{\mu_0} = \frac{5}{4\pi}\text{A} = 0.4\text{A}$$

25. (0.1) **Given:** Radius of air bubble,

$r = 0.1$ cm $= 10^{-3}$ m

Surface tension of liquid,

$S = 0.06$ N/m $= 6 \times 10^{-2}$ N/m

Density of liquid, $\rho = 10^3$ kg/m^3

Excess pressure inside the bubble,

$P_{exe} = 1100$ Nm^{-2}

Depth of bubble below the liquid surface,

$h = ?$

As we know, $P_{Excess} = h\rho g + \dfrac{2s}{r}$

$\Rightarrow \quad 1100 = h \times 10^3 \times 9.8 + \dfrac{2 \times 6 \times 10^{-2}}{10^{-3}}$

$\Rightarrow \quad 1100 = 9800\,h + 120$

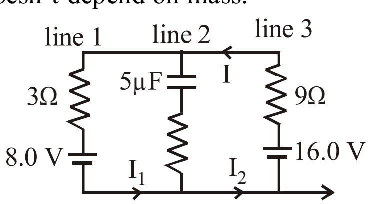
$\Rightarrow \quad 9800h = 1100 - 120$

$\Rightarrow \quad h = \dfrac{980}{9800} = 0.1$ m

26. (11.2) Escape velocity $v_e = \sqrt{2gR}$

thus, it doesn't depend on mass.

27. (0.67)



In steady state capacitor is fully charged hence no current will flow through line 2.

By simplyfing the circuit

Hence resultant potential difference across resistances will be 8.0 V.

Thus current $I = \dfrac{V}{R} = \dfrac{8.0}{3+9} = \dfrac{8}{12}$

or, $\quad I = \dfrac{2}{3} = 0.67$ A

28. (1.2×10^{-7}) Pressure of light on totally reflecting surface

$P = \dfrac{2I}{C}$ (C = velocity of light)

$P = \dfrac{F}{A} = \dfrac{2I}{C}$

$\Rightarrow F = \dfrac{2IA}{C} = \dfrac{2 \times 12 \times 1.5 \times 10^{-4}}{10^{-4} \times 3 \times 10^8}$

$= \dfrac{8 \times 15 \times 10^{-8}}{10} = 12 \times 10^{-8} = 1.2 \times 10^{-7}$ N

29. (7.8×10^{-7}) $F = qE = mg$ $(q = 6e = 6 \times 1.6 \times 10^{-19})$

Density (d) $= \dfrac{\text{mass}}{\text{volume}} = \dfrac{m}{\frac{4}{3}\pi r^3}$

or $r^3 = \dfrac{m}{\frac{4}{3}\pi d}$

Putting the value of d and m $\left(= \dfrac{qE}{g}\right)$ and solving we get $r = 7.8 \times 10^{-7}$m

30. (4) Velocity of wave $v = n\lambda$

where n = frequency of wave $\Rightarrow n = \dfrac{v}{\lambda}$

$$n_1 = \frac{v_1}{\lambda_1} = \frac{396}{99 \times 10^{-2}} = 400\,\text{Hz}$$

$$n_2 = \frac{v_2}{\lambda_2} = \frac{396}{100 \times 10^{-2}} = 396\,\text{Hz}$$

no. of beats $= n_1 - n_2 = 4$

CHEMISTRY

31. (4) (1) $\begin{matrix} CH_2Cl \\ | \\ CH_2Cl \end{matrix} \xrightarrow[\text{alcohol}]{KOH} \begin{matrix} CH \\ ||| \\ CH \end{matrix}$;

Ethylene Dichloride — Acetylene

$\begin{matrix} CH_3 \\ | \\ CHCl_2 \end{matrix} \xrightarrow[\text{alcohol}]{KOH} \begin{matrix} CH \\ ||| \\ CH \end{matrix}$; hence true

Ethylidine Chloride — Acetylene

(2) Both are position isomers

(3) Since, they are isomers, precentage of C, H and Cl in both will be same.

(4) $\begin{matrix} CH_2Cl \\ | \\ CH_2Cl \end{matrix} \xrightarrow{\text{hydrolysis}} \begin{matrix} CH_2OH \\ | \\ CHOH; \end{matrix}$

$\begin{matrix} CH_3 \\ | \\ CHCl_2 \end{matrix} \xrightarrow{\text{hydrolysis}} \begin{matrix} CH_3 \\ | \\ CH(OH)_2 \end{matrix}$

$\xrightarrow{-H_2O} \begin{matrix} CH_3 \\ | \\ CHO \end{matrix}$

Hence, statement (4) is wrong.

32. (1) According to the kinetic theory of gases, the average velocity of the molecules in the gas is given by the expression $v = \sqrt{\frac{8RT}{\pi M}}$ where T is the absolute temperature and R is the gas constant. Thus the average velocity can be taken as proportional to the square root of the absolute temperature. Hence the ratio of the average velocity at 200 ºC to that at 50 ºC will be equal to $\sqrt{\frac{273+200}{273+50}}$ which is 1.21.

33. (3) The Friedal-crafts alkylation reaction will give propyl phenyl ketone which further on Clemmenson's reduction will give butyl benzene

$$C_6H_6 + CH_3CH_2CH_2COCl \xrightarrow{AlCl_3}$$

$$C_6H_5COCH_2CH_2CH_3 \xrightarrow{\text{Zn-Hg/HCl}}$$

$$\underset{\text{Butyl benzene}}{C_6H_5CH_2CH_2CH_2CH_3}$$

34. (3) Moissan boron, which is an amorphous low purity boron, is made by reduction of boric oxide with magnesium. Boron obtained has 86% purity and is amorphous.

35. (1) Addition of a catalyst to a reaction mixture has the effect of lowering the activation energy of the reaction by changing the path or mechanism of the reaction. The reaction rate increases manifold. However, the equilibrium constant and the enthalpy (ΔH) of the reaction are unaffected.

36. (2) Since the gas B turns $CuSO_4$ solution blue, it can be NH_3.
Since formula of the given compound 'A' is M_3N, 'A' is either lithium or sodium nitride. Of the two, Li_3N is most likely since it is a stable, very high melting compound.

37. (2) He is less soluble in blood.

38. (3) (i)

$$CH_3-\overset{\overset{\displaystyle O}{||}}{\underset{\underset{\displaystyle H}{|}}{C}} \xrightarrow[\text{(ii)}H_3O^+]{\text{(i)}CH_3MgBr} \underset{\text{2-Propanol}}{CH_3-\overset{\overset{\displaystyle H}{|}}{\underset{\underset{\displaystyle CH_3}{|}}{C}}-OH}$$

(ii) $CH_3-\overset{\overset{\displaystyle H}{|}}{C}=O + C_2H_5OH \xrightarrow{HCl}$

$$\underset{\text{Hemiacetal}}{CH_3-\overset{\overset{\displaystyle H}{|}}{\underset{\underset{\displaystyle OH}{|}}{C}}-OC_2H_5} \xrightarrow[C_2H_5OH]{HCl}$$

$$\underset{\text{Acetal}}{CH_3-\overset{\overset{\displaystyle H}{|}}{C}(OC_2H_5)_2}$$

39. (4) I^- ion reduces Fe(III) to Fe(II).

$$FeI_3 \rightarrow FeI_2 + \frac{1}{2} I_2$$

40. (2) Since each of (i), (ii) and (iii) are hexa-coordinated, in the case of (ii), one of the chlorines (chloride ions) is coordinated to the central cobalt ion and in (iii), two such chlorides are coordinately linked. Thus, the ionisable chlorides in (i) is three in (ii) it is two and in (iii) it is only one.
Primary valency means the valency of the complex cation.

41. (2) Polyethylene or Polyethene,
$+CH_2 - CH_2+_n$,
is made from a single monomer, it is a homopolymer.

42. (3) Enzyme inhibition can be either reversible or irreversible. In case of irreversible inhibition the inhibitor dissociates very slowly from its target enzyme because it has become tightly bound to the enzyme either covalently or non-covalently. Some irreversible inhibitors are important drugs like Penicilline and Aspirin. Thus "drug induced poisoning" may bound irreversibly to the active site of the enzyme.

43. (3) More the *s* - character, more is the stability of the carbanion. hence the correct order is $sp > sp^2 > sp^3$.

44. (4) Amongst the sets, it is (4) that is not arranged in the order of decreasing reactivity towards electrophilic substitution. Carboxylic acid group in the case of benzoic acid deactivates the benzene nucleus towards electrophilic substitution reaction and thus it can not have higher reactivity than phenol which has a highly activating phenolic OH group.

45. (3) Deviation from ideal gas behaviour is greater, when the pressure is higher and the gas is closed to its liquefaction point or its critical temperature. Thus, the conditions of –100 ºC and 4 atm pressure among the sets given causes maximum deviation.

46. (3) $Be(OH)_2$ is amphoteric that means it can react with both acids and alkalies.

47. (2) Ionization energy increases along a period but reduces down a group. Electron affinity increases along a period from left to right and deceases on moving down in a group.

48. (3) The secondary amines react with HNO_2 to give the oily nitroso derivative. Amongst the options, (3) is the secondary amine.

49. (2) NaCl is a salt of strong acid and strong base hence on dissolution will give neutral solution.
$NaCl + AgNO_3 \rightarrow AgCl\downarrow + NaNO_3$

50. (3) Higher is the stability of a species, more likely it is formed, hence lower will be the E_{act} for its precursor.

51. (6.5) $HCOONH_4$ is a salt of weak acid and weak base;

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$\therefore pH = \frac{1}{2}\times 14 + \frac{1}{2}\times 3.8 - \frac{1}{2}\times 4.8\,; pH = 6.5$$

52. (6.24) Charge (Coulombs) passed per second $= 10^{-6}$
Number of electrons passed per second
$$= \frac{10^{-6}}{1.602\times 10^{-19}} = 6.24\times 10^{12}$$

53. (3.6) No. of atoms of hydrogen in 0.046 g of alcohol
$$= \frac{0.046}{46}\times 6.022\times 10^{23}\times 6$$
$= 1 \times 10^{-3} \times 6.022 \times 10^{23} \times 6 = 3.6 \times 10^{21}$

54. (6.7) For *bcc*
$$\Rightarrow R = \frac{\sqrt{3}\,a}{4}$$
$$\therefore \text{Empty space at edge} = a - 2R = a - \frac{\sqrt{3}\,a}{2}$$
$$\therefore \quad r_{sphere} = \frac{a - \frac{\sqrt{3}}{2}a}{2} = \left(\frac{2-\sqrt{3}}{4}\right)a$$
= diameter of sphere.
$= 0.067\,a$ (Given a = 100 pm)
$= 0.067 \times 100 = 6.7$ pm

55. (0.107) $N_1V_1 = N_2V_2$
$N_{NaOH} = M_{NaOH} = 0.164$
$\Rightarrow 25 \times N = 32.63 \times 0.164$
$$N = \frac{32.63\times .164}{25} = 0.214\ N$$
But $N_{H_2SO_4} = 2 \times M_{H_2SO_4}$
$$\Rightarrow M = \frac{\text{Normality}}{2} = \frac{0.214}{2} = 0.107\ M$$

56. (10) By the law of equivalents :
$$\frac{E + \text{Eq. mass of } OH^-}{E + \text{Eq. mass of } O}$$
$$= \frac{\text{mass of metal hydroxide}}{\text{mass of metal oxide}}$$ (E = Eq. mass of metal)

or $\dfrac{E+17}{E+8}=\dfrac{1.5}{1}\Rightarrow E=10$

57. (6.56) In order to calculate the enthalpy change for H_2O at 5 °C to ice at –5 °C, we need to calculate the enthalpy change of all the transformation involved in the process.
(i) Energy change of 1 mol, H_2O (l), at 5 °C → 1 mol, H_2O (l), 0 °C
(ii) Energy change of 1 mol, H_2O (l), at 0 °C → 1 mol, H_2O (s) (ice), 0 °C
(iii) Energy change of 1 mol, Ice (s), at 0 °C → 1 mol, Ice (s), –5 °C
Total ΔH
$= C_P[H_2O\,(l)]\,\Delta T + \Delta H$ freezing $+ C_P[H_2O\,(s)]\,\Delta T$
$=(75.3\ \text{J mol}^{-1}\text{K}^{-1})(-5)\text{K}+(-6\times10^3\ \text{J mol}^{-1})$
$+(36.8\ \text{J mol}^{-1}\text{K}^{-1})(-5)\ \text{K}$
$\Delta H = -6.56\ \text{kJ mol}^{-1}$ (exothermic process)
So, $\Delta H = 6.56\ \text{kJ mol}^{-1}$.

58. (–6.04) According to Bohr's model energy in n^{th} state

$$= -13.6\times\frac{Z^2}{n^2}\ \text{eV}$$

For second excited state, of He^+, $n = 3$

$$\therefore\ E_3(He^+) = -13.6\times\frac{2^2}{3^2}\ \text{eV} = -6.04\ \text{eV}$$

59. (2) The basic groups in the given form of lysine is NH_2 (not $\overset{+}{N}H_3$) and CO_2^-.

60. (316) $\dfrac{V\times45}{100}+\dfrac{(800-V)\times20}{100}=\dfrac{800\times29.875}{100}$

$$\Rightarrow\frac{9V}{20}+160-\frac{V}{5}=239\Rightarrow V=316\ \text{mL}$$

MATHEMATICS

61. (4) Given $f(x)=\tan^{-1}(\sin x+\cos x)$

$$f'(x)=\frac{1}{1+(\sin x+\cos x)^2}.(\cos x-\sin x)$$

$$=\frac{\sqrt{2}.\left(\frac{1}{\sqrt{2}}\cos x-\frac{1}{\sqrt{2}}\sin x\right)}{1+(\sin x+\cos x)^2}$$

$$\therefore\ f'(x)=\frac{\sqrt{2}\cos\left(x+\frac{\pi}{4}\right)}{1+(\sin x+\cos x)^2}$$

if $f'(x)>0$ then f(x) is increasing function.
Hence f(x) is increasing, if

$$-\frac{\pi}{2}<x+\frac{\pi}{4}<\frac{\pi}{2}\ \Rightarrow\ -\frac{3\pi}{4}<x<\frac{\pi}{4}$$

Hence, f(x) is increasing when

$$x\in\left(-\frac{\pi}{2},\frac{\pi}{4}\right)$$

62. (4) Equation of the tangent at the point 'θ' is

$$\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$$

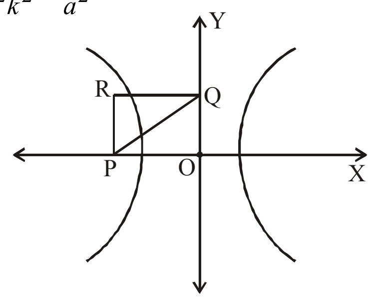
$\Rightarrow P=(a\cos\theta,\ 0)$ and $Q=(0,\ -b\cot\theta)$
Let R be (h, k) $\Rightarrow h=a\cos\theta,\ k=-b\cot\theta$

$$\Rightarrow\frac{k}{h}=\frac{-b}{a\sin\theta}\Rightarrow\sin\theta=\frac{-bh}{ak}\ \text{and}$$

$$\cos\theta=\frac{h}{a}$$

By squaring and adding,

$$\frac{b^2h^2}{a^2k^2}+\frac{h^2}{a^2}=1$$



$$\Rightarrow\frac{b^2}{k^2}+1=\frac{a^2}{h^2}\ \Rightarrow\frac{a^2}{h^2}-\frac{b^2}{k^2}=1$$

Now, given eq^{n} of hyperbola is

$$\frac{x^2}{4}-\frac{y^2}{2}=1\Rightarrow a^2=4,\ b^2=2$$

$$\therefore\ \text{R lies on }\frac{a^2}{x^2}-\frac{b^2}{y^2}=1\ \text{i.e., }\frac{4}{x^2}-\frac{2}{y^2}=1$$

63. (3) $I=\int_1^2[x^2]dx-\int_1^2[x]^2dx$

$$=\int_1^{\sqrt{2}}dx+\int_{\sqrt{2}}^{\sqrt{3}}2\,dx+\int_{\sqrt{3}}^{2}3\,dx-\int_1^2 1\,dx$$

$$=4-\sqrt{2}-\sqrt{3}$$

64. (2) Let $y=\dfrac{ax+b}{c}\Rightarrow y=\dfrac{a}{c}x+\dfrac{b}{c}$

$$\Rightarrow\ y=Ax+B,\ \text{where }A=\frac{a}{c}\ \text{and }B=\frac{b}{c}$$

So, $\bar{y} = A\bar{x} + B$ and hence

$y - \bar{y} = Ax + B - (A\bar{x} + B) = A(x - \bar{x})$

$\Rightarrow (y - \bar{y})^2 = A^2 (x - \bar{x})^2$

$\Rightarrow \sum (y - \bar{y})^2 = A^2 \sum (x - \bar{x})^2$

$\Rightarrow n\sigma_y^2 = A^2 (n\sigma_x^2) \Rightarrow \sigma_y = |A| \sigma_x$

65. (1) Consider the example: Let A = {1, 2, 3}, R = {(1, 1), (1,2)} and S = {(2, 2), (2, 3)}
Clearly R and S are transitive relations on A.
$R \cup S$ = {(1, 1), (2, 2), (1, 2), (2, 3)}
$R \cup S$ is not transitive as $(1,3) \notin R \cup S$.

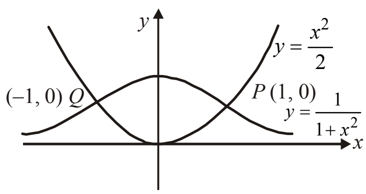
66. (1) $I = \int_0^a f(x) g(x) dx$

$= \int_0^a f(a - x) g(a - x) dx$

$= \int_0^a f(x)[2 - g(x)] dx = 2\int_0^a f(x) dx - I$

$\therefore 2I = 2\int_0^a f(x) dx \Rightarrow I = \int_0^a f(x) dx$

67. (2) Area bounded by $y = \dfrac{1}{1+x^2}$



and x-axis is $\int_{-\infty}^{\infty} \dfrac{1}{1+x^2} dx = \pi$

Area bounded by two curves is

$\int_{-1}^{1} \left(\dfrac{1}{1+x^2} - \dfrac{x^2}{2} \right) dx = \dfrac{\pi}{2} - 1$

68. (2) $f'(x) > 0$ if $x \geq 0$ and $g'(x) < 0$ if $x \geq 0$

Let $h(x) = f(g(x))$ then

$h'(x) = f'(g(x)).g'(x) < 0$ if $x \geq 0$

$\therefore$ h(x) is decreasing function

$\therefore$ $h(x) \leq h(0)$ if $x \geq 0$

$\therefore$ $f(g(x)) \leq f(g(0)) = 0$

But codomain of each function is $[0, \infty)$

$\therefore f(g(x)) = 0$ for all $x \geq 0$ $\therefore f(g(x)) = 0$

Also $g(f(x)) \leq g(f(0))$ [as above]

69. (1) It can be also solved by comparing with the linear equation $\dfrac{dy}{dx} + Py = Q$

The integrating factor, I.F. = $e^{\int 1.dx} = e^x$

Therefore, y I.F = $\int 2e^{2x}.I.F + C$

$y.e^x = \int 2e^{2x}.e^x + C$

$y.e^x = 2\int e^{3x} + C = \dfrac{2}{3} e^{3x} + C \Rightarrow y = \dfrac{2e^{2x}}{3} + ce^{-x}$

70. (3) $\Delta = \begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$

$\Rightarrow \Delta = \begin{vmatrix} 3a-x & a-x & a-x \\ 3a-x & a+x & a-x \\ 3a-x & a-x & a+x \end{vmatrix}$,

$C_1 \to C_1 + C_2 + C_3$

$= (3a - x) \begin{vmatrix} 1 & a-x & a-x \\ 1 & a+x & a-x \\ 1 & a-x & a+x \end{vmatrix} = 0$

Using $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

$\Rightarrow \Delta = (3a - x) \begin{vmatrix} 1 & a-x & a-x \\ 0 & 2x & 0 \\ 0 & 0 & 2x \end{vmatrix} = 0$

or, $4x^2(3a - x) = 0 \Rightarrow x = 0$ or $3a$

71. (3) We know that centroid divides the median in the ratio 2 : 1.

Radius of the circle $= \dfrac{2}{3} \times$ length of median

$= \dfrac{2}{3} \times 3a = 2a$

Centre of the (given) circle is C(0, 0). Therefore the equation of the circle
$(x - 0)^2 + (y - 0)^2 = (2a)^2 \Rightarrow x^2 + y^2 = 4a^2$

72. (2) Its contropositive is 'sum of digits of n is not divisible by 9'
$\Rightarrow$ n is not divisible by 9

73. (3) $n = 7$
Prob. of getting any no. out of 1, 2, 3, … 9 is p = 9/15
$\therefore$ $q = 6/5$

$P(x=7) = {}^7C_7 p^7 q^0$ [Binomial distribution]

$= \left(\frac{9}{15}\right)^7 = \left(\frac{3}{5}\right)^7$

74. (1) For $x \geq a$, the equation becomes
$x^2 - 2a(x-a) - 3a^2 = 0$

$\Rightarrow x = (1+\sqrt{2})a,\ (1-\sqrt{2})a$

for $x \leq a$, the equation becomes

$x^2 - 2a[-(x-a)] - 3a^2 = 0 \Rightarrow x^2 + 2ax - 5a^2 = 0$

$\Rightarrow x = -(1+\sqrt{6})a,\ (-1+\sqrt{6})a$

This shows $(-1+\sqrt{6})\, a$ is one of the roots.

75. (4) $n(A) = 1000, n(B) = 500, n(A \cap B) \geq 1$,
$n(A \cup B) = p$
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$p = 1000 + 500 - n\,(A \cap B)$
$1 \leq n(A \cap B) \leq 500$
Hence $p \leq 1499$ and $p \geq 1000$
$1000 \leq p \leq 1499$

76. (1) $y = \log_2 \{\log_2 (x)\} = \log_2 \{\log_e x . \log_2 e\}$

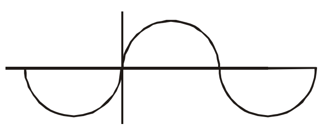
$\Rightarrow y = \log_e\{\log_e x.\log_2 e\}.\log_2 e$

$\Rightarrow \frac{dy}{dx} = \log_2 e \frac{d}{dx}[\log_e \{\log_e x.\log_2 e\}]$

$\Rightarrow \frac{dy}{dx} = \log_2 e . \frac{1}{\log_e x . \log_2 e} \frac{d}{dx}(\log_e x . \log_2 e)$

$\Rightarrow \frac{dy}{dx} = \frac{1}{\log_e x}(\log_2 e)\frac{1}{x} = \frac{\log_2 e}{x \ln x}$

77. (3) The function breaks at x = 0 and multiples of x. Hence the function is differentiable at all other points as the function is continuous at all these pts.



At x = 0, for f (x) to be continous

$\lim_{x \to 0} f(0^-) = f(x=0) = \lim_{x \to 0} f(0^+)$

$f(x) = 0$ at $x = 0$

RHL $= \lim_{x \to 0} \sin(x+h) = \sin\ h > 0$

L. H. L. $= \lim_{x \to 0} \sin(x-h) = \sin\ (-h) < 0$

Hence, not differentiable at x = 0

Similarly, f (x) is not differentiable at all multiples of π, i.e, $n\pi$ where n = 0, 1, 2.....

78. (3) The equation of the pair of tangents is given by $SS_1 = T^2$

$(3x^2 + 2y^2 - 5)(3.1^2 + 2.2^2 - 5) = (3x.1 + 2y.2 - 5)^2$

$9x^2 - 4y^2 - 24xy + 40y + 30x - 55 = 0$

further angle, θ between them can be found by using

$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b} = \frac{2\sqrt{(12)^2 - (9)(-4)}}{9 + (-4)}$

$= \frac{2\sqrt{180}}{5} = \frac{12\sqrt{5}}{5},\ \therefore \theta = \tan^{-1}\frac{12}{\sqrt{5}}$

79. (1) $\frac{x-5}{x^2+5x-14} > 1 \Rightarrow x^2 + 5x - 14 < x - 5$

$\Rightarrow x^2 + 4x - 9 < 0$

$\Rightarrow \alpha = -5, -4, -3, -2, -1, 0, 1$

$\alpha = -5$ does not satisfy any of the options

$\alpha = -4$ satisfy the option (1) $\alpha^2 + 3\alpha - 4 = 0$

80. (3) $\sin^{-1}(1-x) = \left(\frac{\pi}{2} - \sin^{-1} x\right) - \sin^{-1} x$

$(\because \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x)$

$\sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1} x$

Taking sum of both sides

$1 - x = \sin\left(\frac{\pi}{2} - 2\sin^{-1} x\right) = \cos(2\sin^{-1} x)$

$= \cos 2\theta$, where $\sin^{-1} x = \theta$

$1 - x = 1 - 2\sin^2\theta = 1 - 2x^2$ or $x(1-2x) = 0$ or

$x = 0, \frac{1}{2}$

81. (1.00)

$\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda(x\sqrt{1+y^2} - y\sqrt{1+x^2})$

$\Rightarrow \sqrt{1+x^2}(1+\lambda y) = \sqrt{1+y^2}(\lambda x - 1)$

$\Rightarrow \frac{\sqrt{1+x^2}}{\sqrt{1+y^2}} = \frac{\lambda x - 1}{\lambda y + 1}$

$\Rightarrow \frac{x^2+1}{y^2+1} = \frac{\lambda^2 x^2 - 2\lambda x + 1}{\lambda^2 y^2 + 2\lambda y + 1}$

$\Rightarrow (y^2+1)(\lambda^2 x^2 - 2\lambda x + 1)$

$= (x^2+1)(\lambda^2 y^2 + 2\lambda y + 1)$

$\Rightarrow \lambda^2 x^2 y^2 - 2\lambda x y^2 + y^2 + \lambda^2 x^2 - 2\lambda x + 1$

$= \lambda^2x^2y^2 + 2\lambda x^2y + x^2 + \lambda^2y^2 + 2\lambda y + 1$

$\Rightarrow \lambda^2(x^2 - y^2) - 2\lambda(xy^2 + x^2y + x + y) = 0$

$\Rightarrow \lambda^2(x+y)(x-y) - 2\lambda\left[xy(x+y) + (x+y)\right] = 0$

$\Rightarrow \lambda(x+y)\left[\lambda(x-y) - 2xy - 2\right] = 0$

$\Rightarrow (x+y)\left[\lambda(x-y) - 2xy - 2\right] = 0$

$\Rightarrow \lambda(x-y) - 2xy - 2 = 0$

$\Rightarrow \frac{2xy+2}{x-y} = \lambda \Rightarrow \frac{xy+1}{x-y} = \frac{\lambda}{2}$

$\Rightarrow \frac{\left(x\frac{dy}{dx} + y\right)(x-y) - (xy+1)\left(1 - \frac{dy}{dx}\right)}{(x-y)^2} = 0$

This is the first order differential equation and clearly degree of $\frac{dy}{dx}$ is 1. Hence degree of the differential equation is 1.

82. (4.00) We have $T_2 = 14\, a^{\frac{5}{2}}$

$\Rightarrow {}^nC_1\left(a^{\frac{1}{13}}\right)^{n-1}\left(a^{\frac{3}{2}}\right) = 14a^{\frac{5}{2}}$

$\Rightarrow na^{\frac{n-1}{13}+\frac{3}{2}} = 14a^{\frac{5}{2}}$

$\Rightarrow n = 14 \Rightarrow \frac{{}^nC_3}{{}^nC_2} = \frac{{}^{14}C_3}{{}^{14}C_2} = \frac{12}{3} = 4$

83. (0.57) Let the probability of occurrence of first event A, be 'a'

i.e., P(A) = a

$\therefore$ P(not A) = $1 - a$

And also suppose that probability of occurrence of second event B, P(B) = b,

$\therefore$ P(not B) = $1 - b$

Now, P(A and not B) + P(not A and B)

$= \frac{26}{49}$

$\Rightarrow P(A) \times P(\text{not } B) + P(\text{not } A) \times P(B) = \frac{26}{49}$

$\Rightarrow a \times (1 - b) + (1 - a)\, b = \frac{26}{49}$

$\Rightarrow a + b - 2ab = \frac{26}{49}$ ...(i)

And P(not A and not B) $= \frac{15}{49}$

$\Rightarrow$ P (not A) × P(not B) $= \frac{15}{49}$

$\Rightarrow (1 - a) \times (1 - b) = \frac{15}{49}$

$\Rightarrow 1 - b - a + ab = \frac{15}{49}$

$\Rightarrow a + b - ab = \frac{34}{49}$ ...(ii)

From (i) and (ii),

$a + b = \frac{6}{7}$ ...(iii)

and $ab = \frac{8}{49}$

$(a - b)^2 = (a + b)^2 - 4ab = \frac{42}{49}\times\frac{42}{49} - \frac{4\times 8}{49} = \frac{196}{2401}$ $\therefore$ $a - b = \frac{14}{49}$ ...(iv)

From (iii) and (iv),

$a = \frac{4}{7}, b = \frac{2}{7}$

Hence probability of more probable of the two events $= \frac{4}{7} = 0.57$

84. (34.00) $\alpha + \beta = 3; \alpha\beta = a$;

$\gamma + \delta = +12$; $\gamma\delta = b$

$\alpha, \beta, \gamma, \delta$ are in increasing G.P.

$\beta = \alpha x, \gamma = \alpha x^2, \delta = \alpha x^3$

$\alpha + \beta = \alpha + \alpha x = 3 = \alpha(1 + x)$ .....(1)

$\gamma + \delta = \alpha x^2 + \alpha x^3 = 12 = \alpha x^2(1 + x)$ .....(2)

Divding $\frac{3}{12} = \frac{\alpha(1+x)}{\alpha x^2(1+x)}$

or $\frac{1}{4} = \frac{1}{x^2}$ or $x = 2$

$\Rightarrow \beta = 2\alpha$ and $\alpha + 2\alpha = 3 \Rightarrow \alpha = 1$ and $\beta = 2$

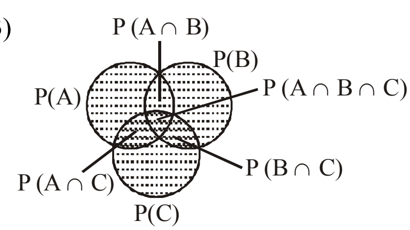
$\therefore a = \alpha\beta = 2$

$\gamma = \alpha x^2 = 1 \times 2^2 = 4; \delta = \alpha x^3 = 1 \times 2^3 = 8$

$\therefore b = \gamma\,\delta = 4 \times 8 = 32$

$\Rightarrow$ $a + b = 2 + 32 = 34.$

85. (0.63)



$\therefore$ Probability that atleast one of the events A, B, C exists is given by the shaded region.

Req. prob. $= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

$$= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} - 0 - 0 - \frac{1}{8} + 0 = \frac{5}{8} = 0.63$$

86. (5.00) $\sum_{r=0}^{n} \frac{r+2}{r+1} {}^nC_r = \sum_{r=0}^{n} \frac{r+1+1}{r+1} {}^nC_r$

$$= \sum_{r=0}^{n} {}^nC_r + \sum_{r=0}^{n} \frac{{}^nC_r}{r+1}$$

$$= 2^n + \sum_{r=0}^{n} \frac{{}^{n+1}C_{r+1}}{n+1}$$

$$= 2^n + \frac{1}{n+1} \sum_{r=0}^{n} {}^{n+1}C_{r+1}$$

$$= 2^n + \frac{1}{n+1}(2^{n+1} - 1)$$

$$= \frac{1}{n+1}[(n+1)2^n + 2^{n+1} - 1]$$

$$= \frac{1}{n+1}[2^n(n+3) - 1]$$

Given, $\frac{(n+3)2^n - 1}{n+1} = \frac{2^8 - 1}{6} = \frac{(5+3).2^5 - 1}{5+1}$

$\Rightarrow \quad n = 5$

87. (3.00) For f(x) to be continuous,

$$\lim_{x \to 0} f(x) = f(0)$$

$f(0) = k$

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin 3x}{\sin x} = \lim_{x \to 0} \frac{3 \cdot \frac{\sin 3x}{3x}}{\frac{\sin x}{x}} = 3$$

$$\left[\because \lim_{x \to 0} \frac{\sin\theta}{\theta} = 1\right]$$

$\Rightarrow \quad k = 3$

88. (0.33) $\int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{1 + \sin 2x}} dx$

$$= \int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{\sin^2 x + \cos^2 x + 2\sin x \cos x}} dx$$

$$= \int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{(\cos x + \sin x)^2}} dx = \int_0^{\pi/3} dx = \frac{\pi}{3} = k\pi$$

$\Rightarrow \quad k = \frac{1}{3} = 0.33$

89. (7.26) Let us assume an arbitary mean a = 155. Following table is constructed :

X_i	f_i	$u_i = \frac{X_i - 155}{5}$	u_i^2	f_iu_i	$f_iu_i^2$
140	4	–3	9	–12	36
145	6	–2	4	–12	24
150	15	–1	1	–15	15
155	30	0	0	0	0
160	36	1	1	36	36
165	24	2	4	48	96
170	8	3	9	24	72
175	2	4	16	8	32
Total	125			77	311

$$\therefore \text{Variance}, \sigma^2 = \left(\frac{\sum f_i u_i^2}{n} - \left(\frac{\sum f_i u_i}{n}\right)^2\right)$$

$\therefore$ S.D. $= \sqrt{\text{variance}} = 7.26$

90. (1.33) Line is $\perp$ to $3x + y = 3$

$\therefore$ Slope of line, $m = \frac{1}{3}$

Equation is, $y = mx + c = \frac{x}{3} + c$

It passes through (2, 2) $\Rightarrow 2 = \frac{2}{3} + c$

$\Rightarrow \quad c = \frac{4}{3}$

$\Rightarrow y - \frac{x}{3} = 4/3 \Rightarrow 3y - x = 4$

$\therefore$ y-intercept $= 4/3 = 1.33$

Mock Test-8

ANSWER KEY											
1	(3)	**16**	(3)	**31**	(1)	**46**	(3)	**61**	(3)	**76**	(4)
2	(3)	**17**	(1)	**32**	(4)	**47**	(3)	**62**	(3)	**77**	(3)
3	(4)	**18**	(2)	**33**	(4)	**48**	(3)	**63**	(3)	**78**	(3)
4	(3)	**19**	(3)	**34**	(1)	**49**	(3)	**64**	(1)	**79**	(2)
5	(2)	**20**	(3)	**35**	(3)	**50**	(3)	**65**	(1)	**80**	(4)
6	(1)	**21**	(4)	**36**	(1)	**51**	(112)	**66**	(2)	**81**	(0.25)
7	(3)	**22**	(264)	**37**	(3)	**52**	(3)	**67**	(1)	**82**	(1.33)
8	(1)	**23**	(4.5 × 104)	**38**	(2)	**53**	(2.1)	**68**	(2)	**83**	(32.00)
9	(3)	**24**	(48)	**39**	(4)	**54**	(2)	**69**	(2)	**84**	(0.40)
10	(1)	**25**	(0.16)	**40**	(4)	**55**	(0.177)	**70**	(2)	**85**	(0.60)
11	(4)	**26**	(10)	**41**	(2)	**56**	(9.25)	**71**	(2)	**86**	(10.00)
12	(2)	**27**	(2)	**42**	(2)	**57**	(9.6)	**72**	(1)	**87**	(0)
13	(3)	**28**	(2)	**43**	(4)	**58**	(107.2)	**73**	(4)	**88**	(0)
14	(3)	**29**	(6.52)	**44**	(3)	**59**	(–0.9)	**74**	(4)	**89**	(1.00)
15	(1)	**30**	(90)	**45**	(3)	**60**	(0.125)	**75**	(2)	**90**	(0.50)

Solutions

PHYSICS

1. (3) $TV^{\gamma-1} = \text{constant}$

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\Rightarrow T(V)^{1/2} = T_2\left(\frac{V}{4}\right)^{1/2}$$

$$\left[\because \gamma = 1.5, T_1 = T, V_1 = V \text{ and } V_2 = \frac{V}{4}\right]$$

$$\therefore \quad T_2 = \left(\frac{4V}{V}\right)^{1/2} T = 2T$$

2. (3) Initially, ϕ_B increases as magnet approaches the solenoid

$\therefore$ $\varepsilon = -$ ve and increasing in magnitude. When magnet is moving inside the solenoid, increase in ϕ_B slow down and finally ϕ_B starts decreasing

$\therefore$ emf is positive and increasing.

Only graph (3) shows these characteristic.

3. (4) Acceleration

$$a = \frac{F}{m} = \frac{\frac{\rho}{3\in_0}\left(\frac{R}{2}\right)q_0}{m} = \frac{\rho q_0 R}{6m\in_0}$$

$$\therefore \; v^2 = 2a(R/2) = 2.\frac{\rho q_0 R}{6m\in_0}.\frac{R}{2}$$

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{12}\frac{q_0\rho R^2}{\in_0}$$

4. (3) Both fall with equal acceleration g, have equal displacements in time t; therefore extension $= 0$.

5. (2) Momentum

$$Mu = \frac{E}{c} = \frac{h\nu}{c}$$

Recoil energy

$$\frac{1}{2}Mu^2 = \frac{1}{2}\frac{M^2u^2}{M} = \frac{1}{2M}\left(\frac{h\nu}{c}\right)^2$$

$$= \frac{h^2\nu^2}{2Mc^2}$$

6. (1) $I_1 = 4I_0$

$I_2 = A_{average}$;

$= [I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\phi]_{ev} = 2I_0$

$\therefore \quad \frac{I_1}{I_2} = \frac{4I_0}{2I_0} = 2$

7. (3) According to the question,

$2\pi r = n\lambda = \frac{nh}{p} = \frac{nh}{mv}$

or $mvr = \frac{nh}{2\pi}$ or $mv = \frac{nh}{2\pi r}$

$F = qvB = \frac{mv^2}{r}$

or, $qB = \frac{mv}{r} = \frac{nh}{2\pi r.r}$

or, $r^2 = \frac{nh}{2\pi qB}$

or, $r = \sqrt{\frac{nh}{2\pi qB}}$

i.e., $r \propto n^{1/2}$

8. (1) Volume of removed sphere

$V_{remo} = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{4}{3}\pi R^3\left(\frac{1}{8}\right)$

Volume of the sphere (remaining)

$V_{remain} = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi R^3\left(\frac{1}{8}\right)$

$= \frac{4}{3}\pi R^3\left(\frac{7}{8}\right)$

Therefore mass of sphere carved and remaining sphere are at respectively $\frac{1}{8}$ M and $\frac{7}{8}$ M.

Therefore, gravitational force between these two sphere,

$F = \frac{GMm}{r^2}$

$= \frac{G\frac{7M}{8}\times\frac{1}{8}M}{(3R)^2} = \frac{7}{64\times 9}\frac{GM^2}{R^2}$

$\simeq \frac{41}{3600}\frac{GM^2}{R^2}$

9. (3) In parallel combination, the equivalent thermal conductivity is given by

$K = \frac{K_1A_1 + K_2A_2 + K_3A_3 + ... + K_nA_n}{A_1 + A_2 + A_3 + ... + A_n}$

For two rods of equal area,

$K = \frac{(K_1 + K_2)A}{2A}$ (if $A_1 = A_2 = A$)

$\Rightarrow K = \frac{K_1 + K_2}{2}$

10. (1) From Einstein's photoelectric equation

$K.E._\lambda = \frac{hc}{\lambda} - \phi$ ...(i)

(for monochromatic light of wavelength λ)
where ϕ is work function

$K.E._{\lambda/2} = \frac{hc}{\lambda/2} - \phi$ ...(ii)

(for monochromatic light of wavelength $\lambda/2$)
From question,

$K.E._{\lambda/2} = 3(K.E._\lambda) \Rightarrow \frac{hc}{\lambda/2} - \phi = 3\left(\frac{hc}{\lambda} - \phi\right)$

$\frac{2hc}{\lambda} - \phi = 3\frac{hc}{\lambda} - 3\phi$

$\Rightarrow 2\phi = \frac{hc}{\lambda} \therefore \phi = \frac{hc}{2\lambda}$

11. (4) For a p-type semiconductor, the acceptor energy level, as shown in the diagram, is slightly above the top E_v of the volume band. With very small supply of energy an electron from the valence band can jump to the level E_A and ionise acceptor negatively

12. (2) The figure of merit of a galvanometer is the current required to produce unit deflection in the galvanometer. Mathematically,

$K = I/\theta$, where k is the figure of merit of the galvanometer.

13. (3) Left amount $= \dfrac{\text{Initial amount}}{(2)^n}$; $T = n \times t_{½}$

$9 = n_1 \times t_{½}$ ......(1)
$18 = n_2 \times t_{½}$ ......(2)

In case (1) $\dfrac{I_O}{3} = \dfrac{I_O}{(2)^{9/t_{1/2}}}$ ......(3);

In case (2), $I = \dfrac{I_O}{(2)^{18/t_{1/2}}} = \dfrac{I_O}{\left((2)^{9/t_{1/2}}\right)^2}$ ....(4)

From eq. (3) & (4), we get $I = \dfrac{I_0}{(3)^2} = \dfrac{I_0}{9}$

14. (3) We know that, $v = \dfrac{dx}{dt} \Rightarrow dx = v\,dt$

Integrating, $\int_0^x dx = \int_0^t v\,dt$

or $x = \int_0^t (v_0 + gt + ft^2)\,dt$

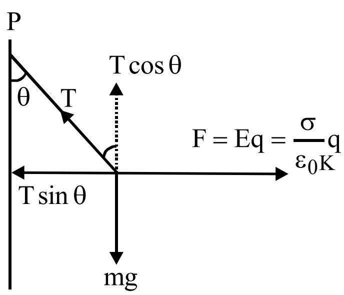
$= \left[v_0 t + \dfrac{gt^2}{2} + \dfrac{ft^3}{3}\right]_0^t$

or, $x = v_0 t + \dfrac{gt^2}{2} + \dfrac{ft^3}{3}$

At $t = 1$, $x = v_0 + \dfrac{g}{2} + \dfrac{f}{3}$.

15. (1) The correct plotting of the base current vs base voltage (i.e., input characteristics of a common emitter transistor) is shown by (1)

16. (3)



$T\sin\theta = \dfrac{\sigma}{\varepsilon_0 K}.q$ .... (i)

$T\cos\theta = mg$ .... (ii)

Dividing (i) by (ii),

$\tan\theta = \dfrac{\sigma q}{\varepsilon_0 K.mg}$

$\therefore \sigma \propto \tan\theta$

17. (1) Initial and final condition is same for all process
$\Delta U_1 = \Delta U_2 = \Delta U_3$
from first law of thermodynamics
$\Delta Q = \Delta U + \Delta W$
Work done
$\Delta W_1 > \Delta W_2 > \Delta W_3$ (Area of P.V. graph)
So $\Delta Q_1 > \Delta Q_2 > \Delta Q_3$

18. (2) Velocity of the tennis ball on the surface of the earth or ground

$v = \sqrt{\dfrac{2gh}{1+\dfrac{k^2}{R^2}}}$ (where k = radius of gyration of spherical shell $= \sqrt{\dfrac{2}{3}}R$)

Horizontal range AB $= \dfrac{v^2 \sin 2\theta}{g}$

$= \dfrac{\left(\sqrt{\dfrac{2gh}{1+k^2/R^2}}\right)^2 \sin(2\times 30°)}{g} = 2.08\text{ m}$

19. (3) Let ℓ be length of the organ pipe. Difference between successive resonating frequencies.

$= 595 - 425 = 170 = \dfrac{V}{2\ell}$

(for both open and closed pipes);

Now, $\dfrac{V}{2\ell} = 170 \Rightarrow \ell = \dfrac{340}{2\times 170} = 1\text{ m}$

20. (3) Yes, the person can catch the ball when horizontal velocity is equal to the horizontal component of ball's velocity, the motion of ball will be only in vertical direction with respect to person for that,

$\dfrac{v_o}{2} = v_o \cos\theta$ or $\theta = 60°$

21. (4) $\dfrac{1}{u} + \dfrac{1}{v} = \dfrac{1}{f} \Rightarrow v = \dfrac{uf}{u-f} = \dfrac{60\times 24}{36} = 40\text{cm}$

Further differentiation gives,

$$\frac{1}{u^2}\frac{du}{dt}+\frac{1}{v^2}\frac{dv}{dt}=0$$

$$\Rightarrow \frac{dv}{dt}=-\frac{v^2}{u^2}\frac{du}{dt}=-\left(\frac{40}{60}\right)^2\times 9=-4\,cm/s$$

22. (264) $\varepsilon\sigma\left(T_1^4-x^4\right)=\varepsilon\sigma\left(x^4-T_2^4\right)$;

$T_1^4-x^4=x^4-T_2^4$

or $2x^4=T_1^4+T_2^4$;

$$x=\left[\frac{T_1^4+T_2^4}{2}\right]^{1/4}=\left[\frac{300^4+200^4}{2}\right]^{1/4}=263.8\,K$$

23. (4.5×104) The potential difference along slabs

$V_0=E_1d_1+E_2d_2+E_3d_3$

$$=\frac{\sigma d_1}{\epsilon_0\epsilon_1}+\frac{\sigma d_2}{\epsilon_0\epsilon_2}+\frac{\sigma d_3}{\epsilon_0\epsilon_3}$$

$$=\frac{\sigma}{\epsilon_0}\left[\frac{5\times10^{-3}}{2}+\frac{3\times10^{-3}}{3}+\frac{2\times10^{-3}}{5}\right]$$

$$=\frac{\sigma}{\epsilon_0}(3.9\times10^{-3})=351$$

$$\frac{\sigma}{\epsilon_0}=\frac{351}{3.9\times10^{-3}}=9\times10^4\ V/m$$

Electric field intensity in slab A

$$E_1=\frac{\sigma}{\epsilon_0\epsilon_1}=\frac{9\times10^4}{2}=4.5\times10^4\ V/m$$

24. (48)

$$d_m=\sqrt{2\times64\times10^5\times40}+\sqrt{2\times64\times10^5\times50}$$

$$=10^3\sqrt{2\times64\times4}+10^3\sqrt{2\times64\times5}$$

$$=8\times10^3(\sqrt{8}+\sqrt{10})=48\ km$$

25. (0.16) Given: $F=100\,kN=10^5\,N$

$Y=2\times10^{11}\,Nm^{-2}$

$\ell_0=1.0\,m$

radius $r=10\,mm=10^{-2}\,m$

From formula, $Y=\frac{\text{Stress}}{\text{Strain}}$

$$\Rightarrow \text{Strain}=\frac{\text{Stress}}{Y}=\frac{F}{AY}$$

$$=\frac{10^5}{\pi r^2Y}=\frac{10^5}{3.14\times10^{-4}\times2\times10^{11}}$$

$$=\frac{1}{628}$$

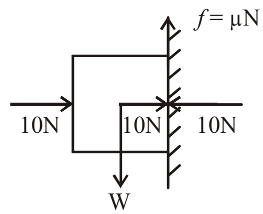
Therefore % strain $=\frac{1}{628}\times100=0.16\%$

26. (10) Concept involved $\lambda=\frac{12345}{V}$Å; From above $\lambda\propto\frac{1}{V}$

Hence, X rays of largest wavelength can be generated by applying lowest potential i.e.,10kV.

27. (2) For the block to remain stationary with the wall

$f=W$



$\mu N=W$

$0.2\times10=W\Rightarrow W=2N$

28. (2) $R_{eq}=1+\frac{2R_{eq}}{2+R_{eq}}=\frac{2+3R_{eq}}{2+R_{eq}}$;

$2R_{eq}+R_{eq}^2=2+3R_{eq}$;

$R_{eq}^2-R_{eq}-2=0 \qquad \Rightarrow R_{eq}=2\Omega$

29. (6.52) Length of the rod = observed reading - zero error

= (Main scale division + Vernier scale division x L.C.) - Zero error

$=(6.4+8\times0.01)-(-0.04)$

$=6.4+0.08+0.04=6.52$ cm

30. (90) Fractional decrease in kinetic energy of mass m

$$=1-\left(\frac{m_2-m_1}{m_2+m_1}\right)^2=1-\left(\frac{2-1}{2+1}\right)^2$$

$$=1-\left(\frac{1}{3}\right)^2=1-\frac{1}{9}=\frac{8}{9}$$

Percentage loss in energy

$$=\frac{8}{9}\times100\simeq90\%$$

CHEMISTRY

31. (1) A - Silica gel packets are used to absorb moisture and keep things dry i.e. as drying agent.

B - Silicon is a semiconductor and is used in transistors.

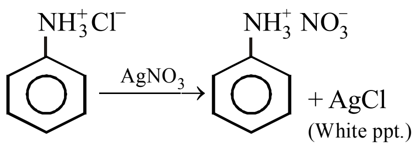
C - Silicone is used as sealant.
D - Silicates are widely used in ion-exchange beds in domestic and commercial water purification, softening and other applications.

32. (4) H_2O. Because of hydrogen-bonding the intermolecular forces increase. Hence boiling point increases.

33. (4) $\underset{+2}{Cu^{2+}} + Fe \longrightarrow Fe^{2+} + \underset{0}{Cu}$

$$\underset{+2}{2Cu^{2+}} + 2KI \longrightarrow \underset{+1}{2Cu^{+}} + 2K^{+} + I_2$$

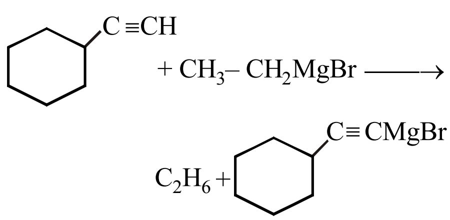
34. (1) $R-\overset{\overset{\displaystyle O}{\|}}{C}-OR' \xrightarrow{Na/alcohol} RCH_2OH + R'OH$

35. (3) 

p-Chloroaniline does not give white ppt. with $AgNO_3$.

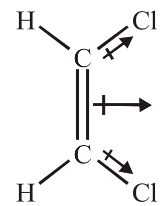
36. (1) Bauxite is not chemically reduced to Al, as aluminium is fairly electropostive and reactive metal, hence it may react with the reducing agent.

37. (3) All ortho para directors other than Halogens activate the ring to electrophilic substitution while all meta directors deactivate the ring to further electrophillic substitution, because halogen can both withdraw and release electrons through inductive effect and resonance respectively.

38. (2) 

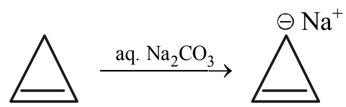
39. (4) $^{-}O-\overset{\overset{\displaystyle O}{\|}}{\underset{\underset{\displaystyle O}{\|}}{S}}-CF_3$ is the best leaving group as it is the weakest base

40. (4) $9O_3 + 2I_2 \longrightarrow I_4O_9 + 9O_2$

41. (2) 

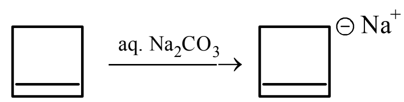
42. (2) According to Le Chatlier's principle increase in pressure will increase the reaction in the direction in which lesser no. of moles are formed to offset the increase in pressure. In reverse reaction, one gaseous mole is formed as compared to two in forward direction. Hence reverse direction is favoured.

43. (4) (1) Antiaromatic ($4\pi e^-$)



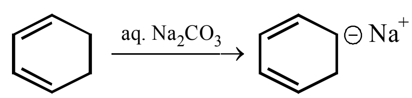
Unstable.

(2) Nonaromatic ($4\pi e^-$)



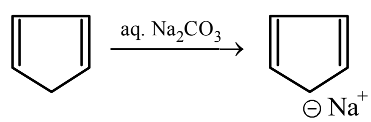
Unstable.

(3) Homoaromatic ($6\pi e^-$)



No delocalization of electron in whole ring **Unstable.**

(4) Aromatic ($6\pi e^-$)



Stable.

44. (3) $2AgBr + Na_2S_2O_3 \longrightarrow Ag_2S_2O_3 + 2NaBr$

$$Ag_2S_2O_3 \xrightarrow{Boil} \underset{\text{Black ppt.}}{Ag_2S} + SO_3$$

45. (3) Only chlorides of group I are neutral because they do not undergo hydrolysis

46. (3) Presence of electron -withdrawing group reduces efficiency of Perkin reaction.

47. (3) The nitration of aniline is difficult to carry out with nitrating mixture since $-NH_2$ group get oxidised which is not required. So the amino group is first protected by acylation to form acetanilide which is then nitrated to give *p*-nitro acetanilide as a major product.

48. (3) The inert pair effect increases down the group and stability of lower oxidation state increases.

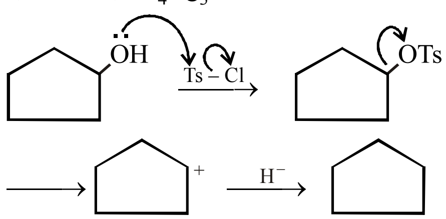
49. (3) Each cell, Cu atoms $\frac{8}{8}+\frac{6}{2}=4$

Ag atom $=\frac{12}{4}=3$

Au atom $=\frac{1}{1}=1.$

Hence, formula, Cu_4Ag_3Au

50. (3)



51. (112) Density $(\rho)=\frac{Pm}{RT}$

(1 bar = 0.987 atm)

$$\rho_{N_2}=\frac{4\times 0.987\ \text{atm}\times 28\ \text{g/mol}}{R\times 300\ \text{K}}$$

Let the molar mass of gas be x

$$\rho_{gas}=\frac{2\times 0.987\ \text{atm}\times x}{R\times 300\ \text{K}}$$

Given $\rho_{gas}=\rho_{N_2}\times 2$

$$\frac{2\times 0.987\ \text{atm}\times x}{R\times 300\ \text{K}}=\frac{4\times 0.987\ \text{atm}\times 28\ \text{g/mol}}{R\times 300\ \text{K}}\times 2$$

$\therefore x = 112$ g/mol.

52. (3)
(i) *cis*-1, 3-dichlorocyclopentane
(ii) *trans*-1, 3-dichlorocyclopentane (enantiomeric pair)

53. (2.1) $E=h\nu=\frac{hc}{\lambda}$

$$E=\frac{6.626\times 10^{-34}\times 3\times 10^{8}}{4000\times 10^{-10}\times 1.6\times 10^{-19}}=3.1\text{ eV}$$

$$=\frac{1}{2}mv^2=\frac{1}{2}\times 9\times 10^{-31}\times 36\times 10^{10}\text{ J}$$

$= 1.62\times 10^{-19}$ J

$= 1$ eV

According to photoelectric effect

K.E. $= h\nu - h\nu_0$

$h\nu_0 = h\nu -$ K.E.

Work function $(W_0) = E -$ K.E.

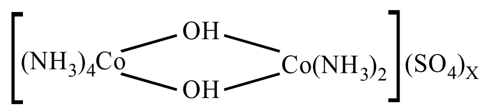
$= 3.1 - 1 = 2.1$ eV

54. (2) In $K_3[Fe(CN)_6]$

$[Fe(CN)_6]^{3-}$

$\therefore \{y-6=-3,\ y=3\}$ the oxidation state of Fe is 3

$\therefore$ in



$4\times0+1\times3+1\times(-1)+1\times(-1)+1\times3+2\times0-2x=0$

or $6-2-2x=0$; $2x=4$; or $x=2$

55. (0.177) $E=0.0591\times \log\left(\frac{1}{[H^+]}\right)$;

pH = 3 $\therefore [H^+]=10^{-3}$ $\therefore$ E = 0.177 volts

56. (9.25) $NH_3 + HCl \longrightarrow NH_4Cl$

moles of HCl = 0.2 M × 25 × 10^{-3} L = 0.005 moles HCl (total consumed)

moles of NH_3 = 0.2 M × 50 × 10^{-3} L = 0.01 moles HCl

excess NH_3 = 0.01 – 0.005 = 0.005 moles

1 mole ammonia = 1 mole NH_4Cl

0.005 NH_3 = 0.005 NH_4Cl

Total volume

$= V_{HCl}+V_{NH_3}=25+50=75$ mL

$$[NH_3]=[NH_4Cl]=\frac{0.005\text{ mole}}{75\times 10^{-3}\text{ L}}=0.066\text{ M}$$

$$pOH = pK_b + \log\frac{[NH_4Cl]}{[NH_3]}$$

$$pOH = 4.75 + \log\frac{[0.066]}{[0.066]}$$

$$pOH = 4.75$$

$$pH = 14 - pOH \Rightarrow pH = 9.25$$

57. (9.6) Mass of water present in the solution at the F. pt. $= 1000 - 200 = 800$ g

$$\Delta T_f = 0.372 = \frac{1000\ K_f W}{60\times 800} = \frac{1000\times 1.86\times W}{60\times 800}$$

$W = 9.6$ g

58. (107.2)

$$\ln\frac{K_2}{K_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln 4 = \frac{E_a}{8.314}\left(\frac{310-300}{310\times 300}\right)$$

$$2\ln 2 = \frac{E_a}{8.314}\left(\frac{310-300}{310\times 300}\right)$$

$$E_a = \frac{0.693\times 2\times 8.314\times 300\times 310}{10}$$

$$= 107.2 \text{ kJ/mol}$$

59. (–0.9) $w = -P\Delta V$

$= -(1 \text{ bar}) \times (9 \text{ L})$

$= -(10^5 \text{ Pa}) \times (9 \times 10^{-3})\ m^3$

$= -9 \times 10^2$ N m

$= -900$ J $= -0.9$ kJ

60. (0.125)

$$\underset{\substack{2 \text{ moles} \\ 1 \text{ mole}}}{2H_3AsO_4 + 5H_2S} \xrightarrow{\text{Conc. HCl}} \underset{\substack{1 \text{ mole} \\ ½ \text{ mole}}}{As_2S_5 + 8H_2O}$$

$\therefore$ number of moles of H_3AsO_4

$= \frac{35.5}{142} = 0.25$.

$\therefore$ number of moles of As_2S_5

$= \frac{0.25}{2} = 0.125$ mole.

MATHEMATICS

61. (3) $L f'(0) = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$

$$= \lim_{h\to 0}\frac{\sqrt{1-\sqrt{1-h^2}}}{-h}$$

$$= \lim_{h\to 0}\frac{\sqrt{1-\sqrt{1-h^2}}}{-h}\times\frac{1}{\sqrt{1+\sqrt{1-h^2}}}$$

$$= \lim_{h\to 0}\frac{-1}{\sqrt{1+\sqrt{1-h^2}}} = \frac{-1}{\sqrt{2}}$$

$$Rf'(0) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h\to 0}\frac{\sqrt{1-\sqrt{1-h^2}}}{h}$$

$$= \lim_{h\to 0}\frac{1}{\sqrt{1+\sqrt{1-h^2}}} = \frac{1}{\sqrt{2}}$$

Therefore, f(x) is not differentiable at $x = 0$.

Since $L f'(0)$ and $R f'(0)$ are finite therefore, f(x) is continuous at $x = 0$.

Hence $f(x)$ is continuous but not differentiable at $x = 0$.

62. (3)

$$\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)} \Rightarrow \frac{\sin(B+C)}{\sin(A+B)} = \frac{\sin(A-B)}{\sin(B-C)}$$

$\Rightarrow \sin^2 B - \sin^2 C = \sin^2 A - \sin^2 B$

$\Rightarrow \sin^2 A, \sin^2 B, \sin^2 C$ and hence a^2, b^2, c^2 are in A.P.

63. (3) $\sum_{x=0}^{50} {}^{50}C_r (2x-3)^r (2-x)^{n-r}$

$= {}^{50}C_0 (2x-3)^0 (2-x)^n + {}^{50}C_1 (2x-3)^1 (2-x)^{n-1} + {}^{50}C_2 (2x-3)^2 (2-x)^{n-2} ...... + {}^{50}C_{50} (2x-3)^n$

$= (x-1)^{50} = {}^{50}C_0 x^{50} - {}^{50}C_1 x^{49} + ...$

Coeff. of $x^{25} = -{}^{50}C_{25}$.

64. (1) The common root must satisfy

$(x^3 + 5x^2 + px + q) - (x^3 + 7x^2 + px + r) = 0$

i.e. $-2x^2+q-r=0$

Sum of common roots = 0

$\Rightarrow x_1=-5,\ x_2=-7$.

65. (1) We can decide denomination in ${}^{13}C_1$ ways.

Then 3 cards out of this denomination in ${}^{4}C_3$ ways and remaining card in 48 ways.

So no. of ways

$= {}^{13}C_1 \times {}^{4}C_3 \times 48 = 52\times 48$ ways.

66. (2) We have, $f(x)=\exp\left(\sqrt{5x-3-2x^2}\right)$

i.e., $f(x)=e^{\sqrt{5x-3-2x^2}}$

For Domain of $f(x)$, $5x-3-2x^2$ should be +ve.

i.e., $5x-3-2x^2\ge 0$

$\Rightarrow 2x^2-5x+3\le 0$

(By taking –ve sign common)

$\Rightarrow 2x(x-1)-3(x-1)\le 0$

$\Rightarrow (2x-3)(x-1)\le 0$

$\Rightarrow 2x-3\le 0$ or $x-1\ge 0$

$\Rightarrow x\le \frac{3}{2}$ or $x\ge 1$

$\therefore 1\le x\le \frac{3}{2}$ i.e., $x\in\left[1,\frac{3}{2}\right]$

Hence, domain of the given function is $[1,\frac{3}{2}]$.

67. (1) $\frac{dy}{dx}=\sin x+2x$

On integrating,

$y=-\cos x+\frac{2x^2}{2}+c=x^2-\cos x+c$.

68. (2) $\left|\frac{1-iz}{z-i}\right|=|w|=1$ (given)

$\Rightarrow \left|\frac{1-iz}{z-i}\right|=1 \quad \because \left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|}$

$\Rightarrow |1-iz|=|z-i|$

$\Rightarrow |1-i(x+iy)|=|(x+iy)-i|$

$\Rightarrow |(1+y)-ix|=|x+i(y-1)|$

$\Rightarrow \sqrt{[(1+y)^2+(-x)^2]}=\sqrt{[x^2+(y-1)^2]}$

$\Rightarrow (1+y)^2+x^2=x^2+(y-1)^2$

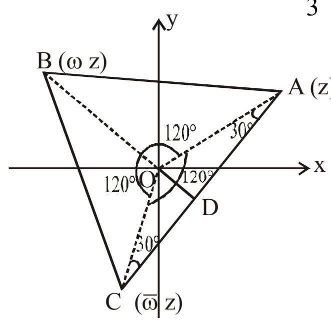
$\Rightarrow 1+2y+y^2+x^2=x^2+y^2-2y+1$

or, $4y=0 \ \therefore y=0$

From this it is known that the locus of z, is y = 0, which is a real axis. Therefore, z is situated on real axis.

69. (2) Let the point A represents the complex number z, B represents ωz and C represents $\overline{\omega}z$. ω & $\overline{\omega}$ are complex cube roots of unity clearly

ωz means rotation of z by $\frac{2\pi}{3}$ and $\omega^2 z$ $(=\overline{\omega}z)$ means rotation of ωz by $\frac{2\pi}{3}$.



$\therefore \angle AOB=\angle BOC=\angle COA=\frac{2\pi}{3}$

also OA = OB = OC = |z|.

That is the Δ ABC is equilateral.

Now AC = 2AD = 2 (OA cos 30°)

$=2|z|\frac{\sqrt{3}}{2}=\sqrt{3}|z|$

Area of $\Delta ABC=\frac{\sqrt{3}}{2}(\text{side})^2=\frac{3\sqrt{3}}{2}|z|^2$

70. (2) Equation of tangent to circle $x^2+y^2=1$ is $y=mx\pm\sqrt{1+m^2}$. This also touches the circle $(x-2)^2+y^2=4$.

$\therefore \left|\frac{2m\pm\sqrt{1+m^2}}{\sqrt{1+m^2}}\right|=2\Rightarrow m=\pm\sqrt{\frac{1}{3}}$.

Therefore common tangents are

$y = \frac{x}{\sqrt{3}} + \frac{2}{\sqrt{3}}$ & $y = \frac{-x}{\sqrt{3}} - \frac{2}{\sqrt{3}}$

On putting $y = 0$, from both equations we get $x = -2 \Rightarrow h = -2$.

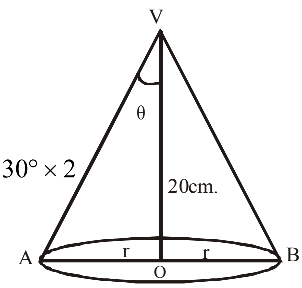
71. (2) Let θ be the semi-vertical angle and r be the radius of the cone at time t.

Then, $r = 20 \tan\theta$

$\frac{dr}{dt} = 20 \sec^2\theta \frac{d\theta}{dt}$

$\Rightarrow \frac{dr}{dt} = 20 \times \sec^2 30° \times 2$

$= \frac{160}{3}$ cm/sec



72. (1) Let $I = \int \frac{1}{1+\sin x} dx = \int \frac{dx}{1 + \frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$

$\int \frac{\left(1+\tan^2\frac{x}{2}\right)dx}{1+\tan^2\frac{x}{2} + 2\tan\frac{x}{2}} = \int \frac{\sec^2\frac{x}{2} dx}{1+\tan^2\frac{x}{2}+2\tan\frac{x}{2}}$

Substitute $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2} dx = dt$

$\Rightarrow \sec^2\frac{x}{2} dx = 2dt$.

Then

$I = \int \frac{2dt}{1+t^2+2t} = 2\int \frac{dt}{(1+t)^2} = 2\frac{-1}{(1+t)} + c$

$= \frac{-2}{1+\tan\frac{x}{2}} + c$

$= 1 - \frac{2}{1+\tan\frac{x}{2}} + (c-1) = \frac{\tan\frac{x}{2} - 1}{\tan\frac{x}{2}+1} + b$

Where b = c – 1, a new constant

$= -\frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}} + b = -\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) + b$

$= \tan\left(\frac{x}{2} - \frac{\pi}{4}\right) + b$. Clearly

$a = -\frac{\pi}{4}$ and $b \in \mathbf{R}$

73. (4) $f(x) = \begin{cases} \frac{1-[x]}{1+x}, & x \neq -1 \\ 1, & x = -1 \end{cases}$ and $f(x)$

$= \begin{cases} 1, & x < 0 \\ \frac{1-x}{1+x}, & x \geq 0 \end{cases}$

$f(2x) = \begin{cases} 1, & x < 0 \\ \frac{1-[2x]}{1+[2x]}, & x > 0 \end{cases} \Rightarrow f(2x) = \begin{cases} 1, & x < 0 \\ 1, & 0 \leq x < \frac{1}{2} \\ 0, & \frac{1}{2} \leq x \leq 1 \\ -\frac{1}{3}, & 1 \leq x < \frac{3}{2} \end{cases}$

$\Rightarrow f(x)$, for all values of x where $x < \frac{1}{2}$a, is continuous function and for $x = \frac{1}{2}$ and $x = 1$ $f(x)$ be a discontinous function.

74. (4) Let the variable circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ....(1)

Since this circle is passing through A(p, q)

$\therefore p^2 + q^2 + 2gp + 2fq + c = 0$ ....(2)

Circle (1) touches x-axis,

$\therefore g^2 - c = 0 \Rightarrow c = g^2$.

From (2), we have

$p^2 + q^2 + 2gp + 2fq + g^2 = 0$ ....(3)

Let the other end of diameter through (p, q) be (h, k), then

$\frac{h+p}{2}=-g$ and $\frac{k+q}{2}=-f$

Put in (3)

$$p^2+q^2+2p\left(-\frac{h+p}{2}\right)+2q\left(-\frac{k+q}{2}\right)+\left(\frac{h+p}{2}\right)^2=0$$

$$\Rightarrow h^2+p^2-2hp-4kq=0$$

$\therefore$ locus of (h, k) is $x^2+p^2-2xp-4yq=0$

$$\Rightarrow (x-p)^2=4qy$$

75. (2) Let $\vec{R}=x\hat{i}+y\hat{j}+z\hat{k}$. Then

$$\vec{R}\times\vec{B}=\vec{C}\times\vec{B}\Rightarrow(\vec{R}-\vec{C})\times\vec{B}=\vec{0}$$

$$\Rightarrow\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ x-4 & y+3 & z-7\\ 1 & 1 & 1\end{vmatrix}=\vec{0}$$

$$\Rightarrow (y-z+10)\hat{i}+(z-x-3)\hat{j}+(x-y-7)\hat{k}=\vec{0}$$

$$\Rightarrow y-z=-10,\ z-x=3,\ x-y=7$$

Also

$$\vec{R}\,.\,\vec{A}=0\Rightarrow 2x+0\,.\,y+z=0\Rightarrow z=-2x$$

Solving, we obtain

$x=-1, y=-8, z=2$. Hence $\vec{R}=-\hat{i}-8\hat{j}+2\hat{k}$.

76. (4) $\sum_{k=33}^{65}\left(\sin\frac{2k\pi}{8}-i\cos\frac{2k\pi}{8}\right)$

$$=\left[\sin\frac{33\pi}{4}+\sin\frac{34\pi}{4}+\ldots+\sin\frac{65\pi}{4}\right]$$

$$-i\left[\cos\frac{33\pi}{4}+\cos\frac{34\pi}{4}+\ldots+\cos\frac{65\pi}{4}\right]$$

$$=\sin\frac{\pi}{4}-i\cos\frac{\pi}{4}$$

$\because$ $\sin\alpha+\sin(\alpha+\beta)+\sin(\alpha+2\beta)+\ldots+\sin[\alpha+(n-1)\beta]$

$$=\frac{\sin\left\{\alpha+(n-1)\frac{\beta}{2}\right\}\cdot\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}$$

and $\cos(\alpha)+\cos(\alpha+\beta)+\ldots+\cos(\alpha+(n-1)\beta)$

$$=\frac{\cos\left\{\alpha+(n-1)\frac{\beta}{2}\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\frac{\beta}{2}}$$

$$=-\left(\frac{1+i}{\sqrt{2}}\right)=\frac{1-i}{\sqrt{2}}$$

77. (3) $-\frac{\pi}{2}\le\sin^{-1}x\le\frac{\pi}{2}$

$$-\frac{\pi}{2}\le 3\sin^{-1}a\le\frac{\pi}{2}\Rightarrow-\frac{\pi}{6}\le\sin^{-1}a\le\frac{\pi}{6}$$

$$\sin\left(\frac{-\pi}{6}\right)\le a\le\sin\frac{\pi}{6}\quad\Rightarrow\quad\frac{-1}{2}\le a\le\frac{1}{2}.$$

78. (3) Equation of a plane through the line of intersection of given plane is

$$ax+by+cz+d+\lambda(a'x+b'y+c'z+d')=0$$

$$\Rightarrow(a+\lambda a')x+(b+\lambda b')y+(c+\lambda c')z+(d+\lambda d')=0$$

It is parallel to $y=0, z=0$

i.e., x-axis whose direction ratios are 1, 0, 0

$$\therefore 1(a+\lambda a')+0(b+\lambda b')+0(c+\lambda c')=0$$

$$\Rightarrow\lambda=-\frac{a}{a'}.$$

Hence, the required plane is

$$y(a'b-ab')+z(a'c-ac')+(a'd-ad')=0$$

or $y(ab'-a'b)+z(ac'-a'c)+(ad'-a'd)=0$

79. (2) $\frac{x^2-y^2}{x^2+y^2}=\tan e^a=k$ (a const.)

$$\Rightarrow kx^2-x^2=ky^2+y^2$$

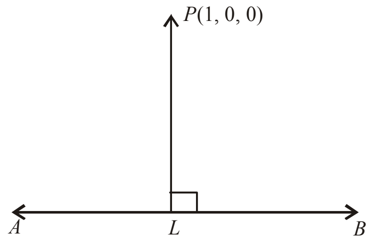
$$\Rightarrow(1-k)x^2=(1+k)y^2\Rightarrow y^2=\frac{(1-k)}{(1+k)}x^2$$

$$\Rightarrow\frac{dy}{dx}=\left(\frac{1-k}{1+k}\right)\frac{x}{y}=\frac{y^2}{x^2}.\frac{x}{y}=\frac{y}{x}$$

80. (4) Let the equation of AB is

$$\frac{x-1}{2}=\frac{y-(-1)}{-3}=\frac{z-(-10)}{8}=k$$

Let L be the foot of the perpendicular drawn from $P(1, 0, 0)$.



$\therefore \quad L = (2k+1, -3k-1, 8k-10).$

Now, direction ratio of $PL = (2k, -3k-1, 8k-10)$ and direction ratio of $AB = (2, -3, 8)$

Since, PL is perpendicular to AB

$\therefore \quad 2(2k) - 3(-3k-1) + 8(8k-10) = 0$

$$\text{Now, } k = \frac{2(1-1)+(-3)(0+1)+8(0+10)}{(2)^2+(-3)^2+(8)^2}$$

$$= \frac{0-3+80}{4+9+64} = \frac{77}{77} = 1$$

$\therefore$ Required co-ordinates

$= L = (2+1, -3-1, 8-10) = (3, -4, -2).$

81. (0.25) We have the equation

$y^2 + xy + px^2 - x - 2y + p = 0$

We know any general equation

$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ ... (1)

represents two straight lines if

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ ... (2)

On comparing given equation with (1), we get

$a = p, b = 1, h = \frac{1}{2}, g = -\frac{1}{2}, f = -1, c = p$

Put these value in equation (2)

$$(p \times 1 \times p) + \left(2 \times -1 \times -\frac{1}{2} \times \frac{1}{2}\right) - p \times (-1)^2$$

$$-1 \times \left(-\frac{1}{2}\right)^2 - p \times \left(\frac{1}{2}\right)^2 = 0$$

$$\Rightarrow \quad p^2 + \frac{1}{2} - p - \frac{1}{4} - \frac{p}{4} = 0$$

$$\Rightarrow p^2 - \frac{5p}{4} + \frac{1}{4} = 0$$

$$\Rightarrow \quad 4p^2 - 5p + 1 = 0 \Rightarrow (4p-1)(p-1) = 0$$

$$\Rightarrow \quad p = 1, \frac{1}{4}$$

82. (1.33) Length of focal chord will be

$$t^2 + 1 + \frac{1}{t^2} + 1 = \frac{25}{4} \quad \Rightarrow t + \frac{1}{t} = \pm\frac{5}{2} \Rightarrow t = \pm 2.$$

Since slope is positive. Therefore t = 2.

The end points will be (4, 4), $\left(\frac{1}{4}, -1\right)$.

$$\text{Slope} = \frac{4}{3} = 1.33$$

83. (32.00) Let the roots be $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$

$$\Rightarrow a\left(\frac{1}{r^2} + \frac{1}{r} + 1 + r + r^2\right) = 40$$

$$\text{Also,} \quad \frac{1}{a}\left(\frac{1}{r^2} + \frac{1}{r} + 1 + r + r^2\right) = 10$$

Now product of roots = – s

$a^5 = -s \Rightarrow |s| = |a|^5 = 32$

84. (0.40)

$$(\vec{a} + k\vec{b}).(\vec{a} - k\vec{b}) = 0 \Rightarrow |\vec{a}|^2 - k^2|\vec{b}|^2 = 0$$

$$k^2 = \frac{|\vec{a}|^2}{|\vec{b}|^2} \Rightarrow k = \pm\frac{|\vec{a}|}{|\vec{b}|} = \pm\frac{2}{5}$$

85. (0.60) Total no. of ways in which 2 cards can be chosen = ${}^{52}C_2 = 1326$.

Total no. of ways in which 2 cards of spades can be drawn (13 in no.) = ${}^{13}C_2 = 78$. Therefore required probability

$$= \frac{78}{1326} = \frac{1}{17} = 0.06$$

86. (10.00) f(1). = 3, f′(1). = –2, g(1). = –1, g′(1). = 4

$$\lim_{x \to a} \frac{g(x)f(a) - g(a)f(x)}{x - a}$$

$$= \lim_{x \to a} \frac{g'(x)f(a) - g(a)f'(x)}{1}$$

$$= g'(a)f(a) - g(a)f'(a) = 4.3 - (-1)(-2) = 10$$

87. (0) $\Delta'(x) = \begin{vmatrix} e^x & 2\cos 2x & 2x\sec^2 x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix}$

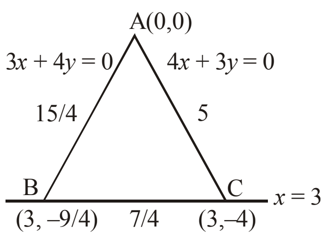
$+\begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \dfrac{1}{1+x} & -\sin x & \cos x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix}$

$+\begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ -2x\sin x^2 & e^x & 2x\cos x^2 \end{vmatrix}$

$= B + 2Cx + \ldots.$

Put $x = 0$, $B = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0$

88. (0)



$$h + k = \frac{a\,(x_1 + y_1) + b\,(x_2 + y_2) + c\,(x_3 + y_3)}{a + b + c}$$

$$= \frac{\frac{7}{4}(0) + 5\left(3 - \frac{9}{4}\right) + \frac{15}{4}(3-4)}{5 + \frac{15}{4} + \frac{7}{4}} = \frac{\frac{15}{4} - \frac{15}{4}}{5 + \frac{22}{4}} = 0$$

89. (1.00) Given that $A = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 8 & 10 \\ -6 & -12 & -15 \end{bmatrix}$

$|A| = \begin{vmatrix} 2 & 4 & 5 \\ 4 & 8 & 10 \\ -6 & -12 & -15 \end{vmatrix}$

Order of given matrix = 3

and $|A| = 2(-120 + 120) - 4(-60 + 60)$
$+ 5(-48 + 48) = 0$

$\therefore$ Rank of A < 3

i.e. Rank of A either 0, 1 or 2

Now, Applying and

$R_3 \to R_3 + 3R_1$

$\Rightarrow A = \begin{bmatrix} 2 & 4 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Since the equivalent matrix in echelon form has only one non-zero row, therefore, rank of the matrix A = 1, i.e., r (A) = 1

90. (0.50) Now there are only two possibilities :-
Either head appears odd no. of times = event A
Or head appears even no. of times = event B
Since both are mutually exclusive and equally likely. Therefore P(1). = P(2). = $\frac{1}{2}$. Therefore required probability = $\frac{1}{2}$ = 0.50.

Mock Test-9

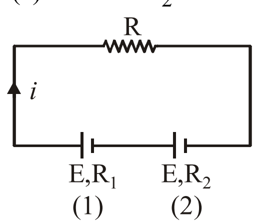
ANSWER KEY											
1	(4)	**16**	(4)	**31**	(3)	**46**	(4)	**61**	(2)	**76**	(2)
2	(4)	**17**	(3)	**32**	(2)	**47**	(1)	**62**	(2)	**77**	(2)
3	(2)	**18**	(4)	**33**	(4)	**48**	(4)	**63**	(3)	**78**	(2)
4	(3)	**19**	(3)	**34**	(3)	**49**	(2)	**64**	(4)	**79**	(4)
5	(4)	**20**	(2)	**35**	(3)	**50**	(3)	**65**	(3)	**80**	(2)
6	(2)	**21**	(7)	**36**	(4)	**51**	(610)	**66**	(3)	**81**	(1.67)
7	(2)	**22**	(2.4)	**37**	(4)	**52**	(41.5)	**67**	(3)	**82**	(60)
8	(3)	**23**	(1000)	**38**	(2)	**53**	(608)	**68**	(1)	**83**	(0.56)
9	(1)	**24**	(41.4)	**39**	(1)	**54**	(141.4)	**69**	(1)	**84**	(2)
10	(3)	**25**	(1.2)	**40**	(1)	**55**	(112)	**70**	(2)	**85**	(2)
11	(3)	**26**	(1.67×10^5)	**41**	(1)	**56**	(22.05)	**71**	(3)	**86**	(8.67)
12	(2)	**27**	(2.12)	**42**	(1)	**57**	(3)	**72**	(1)	**87**	(21)
13	(1)	**28**	(1.1)	**43**	(2)	**58**	(30)	**73**	(4)	**88**	(0.5)
14	(1)	**29**	(6.2)	**44**	(3)	**59**	(4)	**74**	(2)	**89**	(0.25)
15	(1)	**30**	(6000)	**45**	(1)	**60**	(30)	**75**	(4)	**90**	(3)

Solutions

PHYSICS

1. (4) $i = \dfrac{2E}{R + R_1 + R_2}$

From cell (2) $E = V + iR_2 = 0 + iR_2$



$$\Rightarrow E = \frac{2E}{R + R_1 + R_2} \times R_2 \Rightarrow R = R_2 - R_1$$

2. (4) $mg \sin \theta = ma$

$\therefore a = g \sin \theta$

where a is along the inclined plane

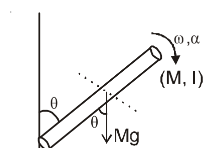
$\therefore$ vertical component of acceleration is $g \sin^2 \theta$

$\therefore$ relative vertical acceleration of A with respect to B is

$$g(\sin^2 60 - \sin^2 30] = \frac{g}{2} = 4.9 \text{ m/s}^2$$

3. (2) $h = \dfrac{2T \cos \theta}{rdg}$, Height in capillary depends upon the surface tension of the liquid and surface tension of soap water solution is less than water. So, height will be less in second case.

4. (3) Torque at angle θ



$\tau = Mg \sin \theta . \dfrac{l}{2}$

Also $\tau = I\alpha$

$\therefore \quad I\alpha = Mg \sin \theta \dfrac{l}{2}$

$\dfrac{Ml^2}{3} . \alpha = Mg \sin \theta \dfrac{l}{2} \qquad \left[\because I_{rod} = \dfrac{Ml^2}{3} \right]$

$\Rightarrow \quad \dfrac{l\alpha}{3} = g \dfrac{\sin \theta}{2} \qquad \therefore \quad \alpha = \dfrac{3g \sin \theta}{2l}$

5. (4) $E = \sigma \times \text{area} \times T^4$; T increases by a factor $\frac{3}{2}$.

Area increses by a factor $\frac{1}{4}$.

6. (2) $\because \theta_1 < \theta_2 \Rightarrow \tan\theta_1 < \tan\theta_2$

$\Rightarrow \left(\frac{V}{T}\right)_1 < \left(\frac{V}{T}\right)_2$

from $PV = \mu RT$; $\frac{V}{T} \propto \frac{1}{P}$

Hence $\left(\frac{1}{P}\right)_1 < \left(\frac{1}{P}\right)_2 \Rightarrow P_1 > P_2$.

7. (2) Core of acceptance angle

$\theta = \sin^{-1}\sqrt{n_1^2 - n_2^2}$

8. (3) Nuclear fission equation

$$_{92}U^{135} + {}_0n^1 \longrightarrow {}_{56}Ba^{141} + {}_{36}Kr^{92} + 3\,{}_0n^1 + Q(\text{energy})$$

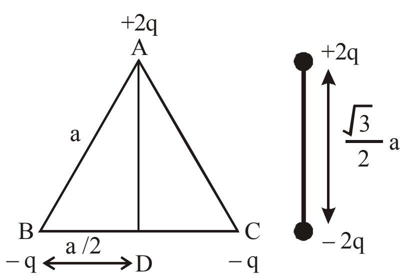
Hence particle x is neutron.

9. (1) Surface tension, $T = \frac{F}{\ell} = \frac{F}{\ell}.\frac{\ell}{\ell}.\frac{T^2}{T^2}$

(As, $F.\ell = K$ (energy); $\frac{T^2}{\ell^2} = V^{-2}$)

Therefore, surface tension = $[KV^{-2}T^{-2}]$

10. (3) This figure can be shown as



Hence $q \times r_1 = 2q \times \frac{\sqrt{3}}{2}a = \sqrt{3}qa$

11. (3) Charge on the capacitor at any time t is given by

$q = CV(1 - e^{t/\tau})$

at $t = 2\tau$

$q = CV(1 - e^{-2})$

12. (2) $\tau = MB\sin\theta$ $(\theta = 90°)$

$\tau = MB \Rightarrow \frac{B_1}{B_2} = \frac{\tau_1}{\tau_2}$

(since magnetic moment is same)

13. (1) Given wave equation is $y(x,t)$

$= e^{-\left(ax^2+bt^2+2\sqrt{ab}\,xt\right)}$

$= e^{-[(\sqrt{ax})^2+(\sqrt{b}\,t)^2+2\sqrt{a}\,x.\sqrt{b}\,t]}$

Increase cover on "n"

$= e^{-(\sqrt{a}x+\sqrt{b}t)^2}$

$= e^{-\left(x+\sqrt{\frac{b}{a}}t\right)^2}$

It is a function of type $y = f(x + vt)$

$\Rightarrow$ Speed of wave = $\sqrt{\frac{b}{a}}$

14. (1) In the given system all four gate is NOR gate

Truth Table

A	B	$(y' = \overline{A+B})$	$y'' = \overline{(A+y')}$	$y''' = \overline{(B+y')}$	$y = \overline{y''+y'''}$
0	0	1	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

i.e.,

A	B	y
0	0	1
0	1	0
1	0	0
1	1	1

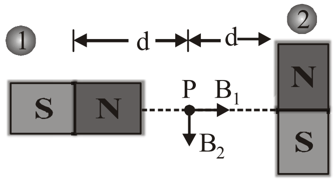
15. (1) T in general = $2\pi\sqrt{(m/k)}$; Half the cycle on one side has period $\pi\sqrt{(m/k)}$ and the other half $\pi\sqrt{(m/2k)}$

16. (4) At point P net magnetic field

$B_{net} = \sqrt{B_1^2 + B_2^2}$

where $B_1 = \frac{\mu_0}{4\pi}.\frac{2M}{d^3}$ and $B_2 = \frac{\mu_0}{4\pi}.\frac{M}{d^3}$

$\Rightarrow B_{net} = \frac{\mu_0}{4\pi}.\frac{\sqrt{5}M}{d^3}$



17. (3) At equilibrium, pressure is same, temp. and molecular weight do not change (given).

Using ideal gas eq. $P_1V_1 = n_1RT_1$

From above, $\frac{V}{m}$ is constant, $\frac{V_1}{V_2} = \frac{m_1}{m_2}$;

Add 1 on both sides, we get $\frac{V_1+V_2}{V_2} = \frac{m_1+m_2}{m_2}$

$$\Rightarrow \frac{V_2}{V_1+V_2} = \frac{m_2}{m_1+m_2} = \frac{2m}{m+2m} = \frac{2}{3}$$

18. (4) In first case :

$$f_1 = \frac{1}{\lambda_1} = RZ^2\left(\frac{1}{4}-\frac{1}{16}\right) = RZ^2\left(\frac{4-1}{16}\right)$$

$$= \frac{3}{16}RZ^2 = .18RZ^2 \qquad \text{(green light)}$$

In second case : $f_2 = \frac{1}{\lambda_2} = RZ^2\left(\frac{1}{4}-\frac{1}{25}\right)$ or

$$RZ^2 \times \frac{21}{100} = .21RZ^2$$

Hence $f_2 > f_1$ Among the options, only violet is one which has frequency more than green. Hence answer is violet.

19. (3) Since aluminium is a metal, therefore field inside this will be zero. Hence it would not affect the field in between the two plates, so capacity

$= \frac{q}{V} = \frac{q}{Ed}$ remains unchanged.

20. (2) $\frac{\Delta\phi}{\Delta t} = \frac{(W_2 - W_1)}{t}$

$$R_{tot} = (R+4R)\Omega = 5R\Omega$$

$$i = \frac{nd\phi}{R_{tot}dt} = \frac{-n(W_2-W_1)}{5Rt}$$

($\because W_2$ & W_1 are magnetic flux)

21. (7) Given, $\lambda_A = 8\lambda,\ \lambda_B = \lambda$

$$N_A = \frac{N_B}{e}$$

$$\Rightarrow \quad N_o e^{-\lambda_A t} = N_o \frac{e^{-\lambda_B t}}{e}$$

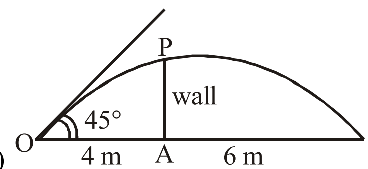
$$e^{-8\lambda t} = e^{-\lambda t}e^{-1}$$

Comparing both side powers

$$-8\lambda t = -\lambda t - 1$$

$$t = \frac{1}{7\lambda}$$

The best answer is $t = \frac{1}{7\lambda}$



22. (2.4)

As ball is projected at an angle 45° to the horizontal therefore Range = 4H

or $\quad 10 = 4H \Rightarrow H = \frac{10}{4} = 2.5\,m$

($\because$ Range = 4 m + 6 m = 10m)

Maximum height, $H = \frac{u^2\sin^2\theta}{2g}$

$$\therefore \quad u^2 = \frac{H\times 2g}{\sin^2\theta} = \frac{2.5\times2\times10}{\left(\frac{1}{\sqrt{2}}\right)^2} = 100$$

or, $\quad u = \sqrt{100} = 10\ ms^{-1}$

Height of wall PA

$$= OA\tan\theta - \frac{1}{2}\frac{g(OA)^2}{u^2\cos^2\theta}$$

$$= 4 - \frac{1}{2}\times\frac{10\times16}{10\times10\times\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}} = 2.4\,m$$

23. (1000) Since particle is moving undeflected

So, $qE = qvB$

$$\Rightarrow B = \frac{E}{V} = \frac{10^4}{10} = 10^3\ wb/m^2$$

24. (41.4) We know that escape velocity v_e and orbital velocity v_o are related as, $v_e = \sqrt{2}.v_o$ or $v_e = 1.414\, v_o$.

Thus for the satellite to escape from its orbit, % increase in the velocity required is

$= [(v_e - v_o) / v_o] \times 100 = ((\sqrt{2}\, v_o - v_o) / v_o] \times 100 = 41.4\%$

25. (1.2) $eV = \frac{hc}{\lambda} - W$

According to question,

$eV = \frac{hc}{\lambda} - 5.01$

$\Rightarrow eV = \frac{12400}{2000} - 5.01,$

$\Rightarrow eV = 6.2 - 5.01$ eV $= 1.2$ eV $= 1.2$ Volt

26. (1.67×10^5) $\Delta H = mL = 5 \times 336 \times 10^3 = Q_{sink}$

$$\frac{Q_{sink}}{Q_{source}} = \frac{T_{sink}}{T_{source}}$$

$$\therefore Q_{source} = \frac{T_{source}}{T_{sink}} \times Q_{sink}$$

Energy consumed by freezer

$$\therefore w_{output} = Q_{source} - Q_{sink}$$

$$= Q_{sink}\left(\frac{T_{source}}{T_{sink}} - 1\right)$$

Given: $T_{source} = 27°C + 273 = 300K,$

$T_{sink} = 0°C + 273 = 273\,k$

$$W_{output} = 5 \times 336 \times 10^3 \left(\frac{300}{273} - 1\right) = 1.67 \times 10^5 J$$

27. (2.12) Velocity is maximum when K.E. is maximum For minimum. P.E.,

$$\frac{dV}{dx} = 0 \Rightarrow x^3 - x = 0 \Rightarrow x = \pm 1$$

$$\Rightarrow \text{Min. P.E.} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} J$$

$K.E._{(max.)} + P.E._{(min.)} = 2$ (Given)

$$\therefore K.E._{(max.)} = 2 + \frac{1}{4} = \frac{9}{4}$$

$$K.E._{max.} = \frac{1}{2} m v_{max.}^2$$

$$\Rightarrow \frac{1}{2} \times 1 \times v_{max.}^2 = \frac{9}{4} \Rightarrow v_{max.} = \frac{3}{\sqrt{2}}$$

28. (1.1) Radius of circular path followed by electron is given by,

$$r = \frac{m\upsilon}{qB} = \frac{\sqrt{2meV}}{eB} = \frac{1}{B}\sqrt{\frac{2m}{e}V}$$

$$\Rightarrow \quad V = \frac{B^2 r^2 e}{2m} = 0.8V$$

For transition between 3 to 2.

$$E = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{13.6 \times 5}{36} = 1.88 eV$$

Work function = 1.88 eV – 0.8 eV = 1.08 eV ≈ 1.1eV

29. (6.2) Time taken for travelling vertical distance of 4.9m with zero initial velocity $= \sqrt{(2h/g)} = \sqrt{2 \times 4.9/9.8} = 1$ sec. In this time, distance travelled with horizontal velocity v should be at least 6.2 m. Thus minimum velocity v is $v \times 1 = 6.2$

or $v = 6.2$m/s.

30. (6000) $S = (\mu - 1)t\left(\frac{D}{2d}\right)$; $\beta = \frac{\lambda D}{2d}$ or $\frac{D}{2d} = \frac{\beta}{\lambda}$

or $S = (\mu - 1)t\left(\frac{\beta}{\lambda}\right)$ or 5β

$$= (1.5 - 1) \times 6 \times \frac{10^{-6} \times \beta}{\lambda}$$

$$\Rightarrow \lambda = \frac{.5}{5} \times 10^{-6} \times 6 = 6 \times 10^{-7} \text{ m} = 6000Å$$

CHEMISTRY

31. (3) It is 2-pentyne, $CH_3C \equiv CCH_2CH_3$, which being a non-terminal alkyne, will not give ppt. with ammonical Cu_2Cl_2 solution.

32. (2)

$$\frac{(K.E.)_1}{(K.E.)_2} = \frac{\nu_1 - \nu_0}{\nu_2 - \nu_0} = \frac{(1.5 \times 10^{15} - 1 \times 10^{15})}{(2.0 \times 10^{15} - 1 \times 10^{15})} = \frac{1}{2}$$

33. (4) In presence of HCl

$$\left.\begin{array}{l} ZnS \\ MnS \end{array}\right\} \xrightarrow{HCl} \begin{array}{l} ZnCl_2 \\ MnCl_2 \end{array}$$

$$\left.\begin{array}{l} CoS \\ NiS \end{array}\right\} \xrightarrow{HCl} \text{No reaction}$$

$$ZnS \xrightarrow{HCl} ZnCl_2 \xrightarrow{NaOH} \underset{\text{white ppt}}{Zn(OH)_2}$$

$$ZnCl_2 \xrightarrow{K_4[Fe(CN)_6]} \underset{\text{bluish white ppt}}{Zn_2[Fe(CN)_6]}$$

$$MnS \xrightarrow{HCl} MnCl_2 \xrightarrow{K_4[Fe(CN)]_6} \text{No reaction}$$

$$MnCl_2 \xrightarrow{NaOH} Mn(OH)_2 \xrightarrow{[O]} \underset{\text{(Brown colour)}}{MnO(OH)_2}$$

The above reaction path is confirming the presence of Zn^{2+} ion.

34. (3) The thermal stability of carbonates is in the following order.

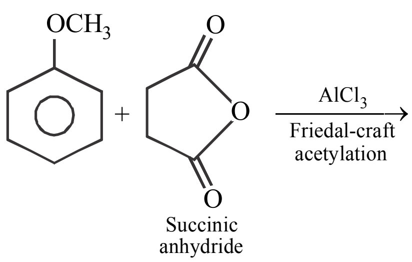
$$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$$

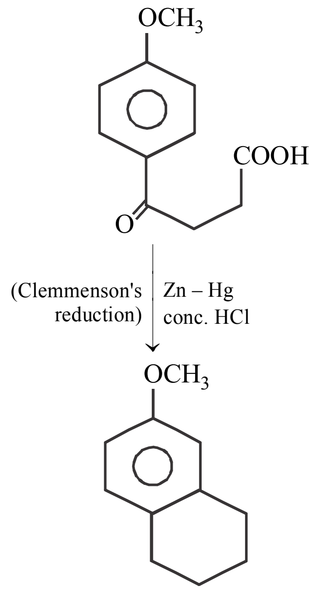
Li_2CO_3 is considerably less stable and decompose readily.

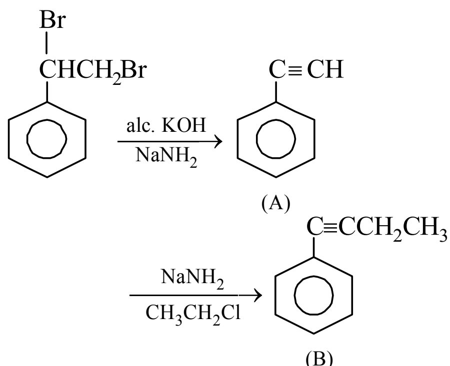
$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$

35. (3) Only secondary amines give nitroso amines on reaction with nitrous acid. This reaction is known as Libermann's nitroso reaction.

36. (4) For the given reaction $K = [CO_2]$ because concentrations of solid $CaCO_3$ and CaO have been taken as unity, since $[CaCO_3]$ or [CaO] do not come in the expression there will be no effect of addition of $CaCO_3$.

37. (4) 



38. (2) 

39. (1) Aldehydes containing no α-hydrogen atom on warming with 50% NaOH or KOH undergo disproportionation i.e. self oxidation - reduction known as Cannizzaro's reaction.

$$2C_6H_5CHO + NaOH \xrightarrow{50\%\ NaOH} C_6H_5COONa + C_6H_5CH_2OH$$

40. (1) Octahedral complexes result from d^2sp^3 (inner orbital) and sp^3d^2 (outer orbital) hybridisation eg. $[Ni(NH_3)_6]^{2+}$.
Tetrahedral complexes are formed by sp^3 hybridisation. e.g. $Ni(CO)_4$
and square planar complexes are formed by dsp^2 hybridisation. e.g. $[Ni(CN)_4]^{2-}$.

41. (1) sp-hybridized carbon atom is more electronegaive than sp^2 hybridized carbon which in turn is more electronegative than sp^3 hybridized carbon. More electronegative atom accommodates the negative charge more easily.

42. (1) Hall-Heroult process is used in extraction of Al from alumina.

43. (2)

$$\Delta H = \Delta G + T\Delta S = -nFE_{\text{cell}} - T\left[\frac{d(\Delta G)}{dT}\right]_P$$

$$= -nFE_{\text{cell}} - T\left[\frac{d(-nFE_{\text{cell}})}{dT}\right]_P$$

$$= nF\left[T\left(\frac{dE_{\text{cell}}}{dT}\right)_P - E_{\text{cell}}\right]$$

44. (3)

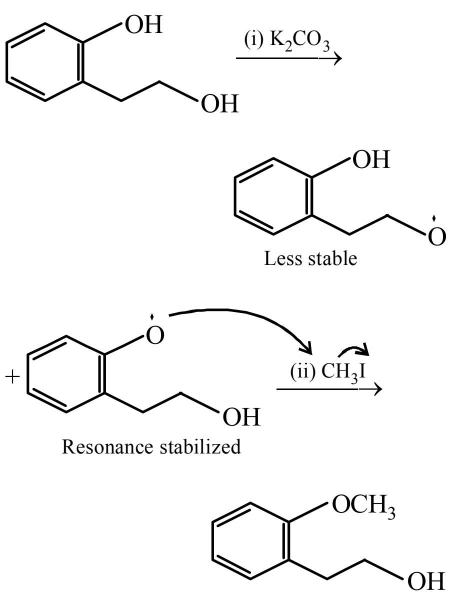
$$2Au^{3+} + 6e^- \longrightarrow 2Au,\ \Delta G^\circ = -6\times 1.52F \quad \text{......(i)}$$

$$3Au^{3+} + 6e^- \longrightarrow 3Au^+,\ \Delta G^\circ = -6\times 1.36F \quad \text{......(ii)}$$

For the reaction

$$3Au^+ \longrightarrow 2Au + Au^{3+}, \Delta G^\circ = \text{(i)} - \text{(ii)}$$
$$= 8.16F - 9.12F = -0.96F \text{(negative)}$$

45. (1)

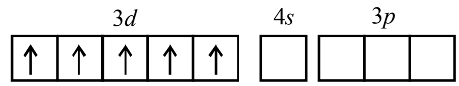


46. (4) In $Fe(CN)_6^{3-}$ the oxidation number of Fe = +3 and so it has a $3d^5$ configuration

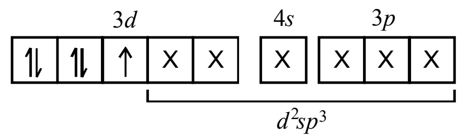
In $Fe(CN)_6^{4-}$ the oxidation number of Fe = +2 and so it has a $3d^6$ configuration.

In $Co(NO_2)_6^{3-}$ the oxidation number of Co = +3 and so it has a $3d^6$ configuration.

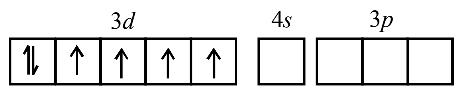
Fe(III)



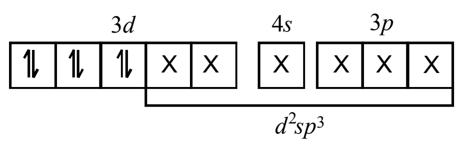
$Fe(CN)_6^{3-}$



Fe (II) and Co (III)



$Fe(CN)_6^{4-}$ and $Co(NO_2)_6^{3-}$



47. (1) sp-hybrid C is more electronegative than sp^2 hybrid C which is more electronegative than sp^3 hybrid C. As the hybrid state of α-C changes from sp^3 to sp, the acid strength tends to increase from X to Z.

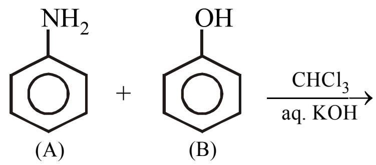
48. (4) In equation (i) $Fe_2(SO_4)_3$ and in equation (ii) $Fe_2(SO_4)_3$ on decomposing will form oxide instead of Fe

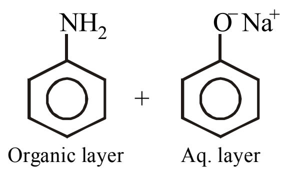
The correct sequence of reactions is

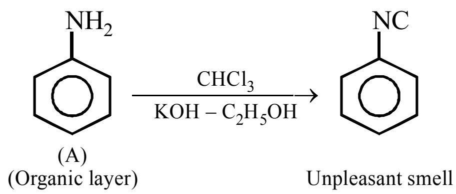
$$Fe \xrightarrow{O_2,\text{heat}} Fe_3O_4 \xrightarrow{CO,\ 600\ ^\circ C}$$
$$FeO \xrightarrow{CO,\ 700\ ^\circ C} Fe$$

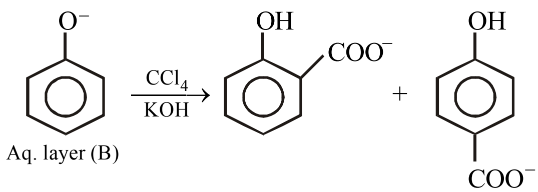
49. (2) The cubic form of BN with ZnS structure is as hard as diamond.

50. (3)









51. (610)The energy involved in the conversion of

$\frac{1}{2}Cl_2(g)$ to Cl^- (aq) is given by

$$\Delta H = \frac{1}{2}\Delta_{diss}H^{\circ}_{Cl_2} + \Delta_{eg}H^{\circ}_{Cl} + \Delta_{hyd}H^{\circ}_{Cl}$$

Substituting various values from given data, we get

$$\Delta H = \left(\frac{1}{2}\times 240\right) + (-349) + (-381)\,\text{kJmol}^{-1}$$

$= (120 - 349 - 381)\,\text{kJ mol}^{-1} = -610\,\text{kJ mol}^{-1}$

52. (41.5) Given $e = 1.60 \times 10^{-19}$ C

$d = 9.17 \times 10^{-11}$ m

From $\mu = e \times d$

$\mu = 1.60 \times 10^{-19} \times 9.17 \times 10^{-11}$

$= 14.672 \times 10^{-30}$

% ionic character

$$= \frac{\text{Observed dipole moment}}{\text{Dipole moment for 100\% ionic bond}}$$

$$= \frac{6.104\times 10^{-30}}{14.672\times 10^{-30}}\times 100$$

$= 41.5\%$

53. (608) Moles of solute $= \frac{28}{28} = 1$;

moles of water $= \frac{100-28}{18} = 4$

V.P. of solution

$$= P^{\circ}_{H_2O} \times X_{H_2O} = 760 \times \frac{4}{5} = 608 \text{ torr}$$

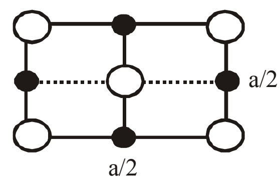
54. (141.4) For a cube as given, the Cl^- ions are at the corners and one each in the face centre i.e. it is a ccp structure.

For a ccp structure $4r^- = \sqrt{2}$ a, The face diagonal $= \sqrt{2}$ a

On the face diagonal there are only Cl^- ions

$\therefore 4r^- = \sqrt{2}$ a or $r^- = 1.414 \times 400/4$

$= 141.4$ pm



55. (112)Density $(\rho) = \frac{PM}{RT}$ (1 bar = 0.987 atm)

$$\rho_{N_2} = \frac{4\times 0.987\times 28}{R\times 300}$$

Let the molar mass of gas be x

$$\rho_{gas} = \frac{2\times 0.987\times x}{R\times 300}$$

Given $\rho_{gas} = \rho_{N_2} \times 2$

$$\frac{2\times 0.987\times x}{R\times 300} = \frac{4\times 0.987\times 28}{R\times 300}\times 2$$

$\therefore$ x = 112 g/mol

56. (22.05) $\because$ 143.5 g of AgCl = 36.5 g HCl

$0.4305 \text{ g AgCl} = \frac{36.5 \times 0.4305}{143.5} \text{ g HCl}$

$\text{geq of HCl} = \frac{36.5 \times 0.4305}{143.5 \times 36.5} = \frac{0.4305}{143.5}$

$\text{Normality of HCl} = \frac{0.4305}{143.5} \times \frac{1000}{20} = 0.15$

Normality of $H_2SO_4 = 0.6 - 0.15 = 0.45$

Strength of $H_2SO_4 = 49 \times 0.45 = 22.05$ g/L

57. (3) Total cationic charge = Total anionic charge

$2n + 6 + 24 = 36$

$n = 3$

58. (30) $a = \frac{a_0}{2^n}$, $2^n = \frac{a_0}{a} = \frac{0.08}{0.01}$, Thus $2^n = 8$ or

$n = 3$; $T = 3 \times t_{1/2} = 3 \times 10 = 30$ min

59. (4) CMC of soap

10^{-4} M to 10^{-3} M

x = 4

60. (30) $\text{pH} = 14 - \text{pK}_b - \log\frac{[B^+]}{[BOH]}$

V = volume of acid required for the equivalance point.

(i) $(\text{pH})_1 = 9 = 14 - \text{pK}_b - \log\frac{10}{V-10}$

(ii) $(\text{pH})_2 = 8 = 14 - \text{pK}_b - \log\frac{25}{V-25}$

(i) – (ii),

$1 = \log\frac{25}{V-25} - \log\frac{10}{V-10} = \log\frac{25(V-10)}{10(V-25)}$

$\Rightarrow V = 30$ mL

MATHEMATICS

61. (2) We have, $P \cup Q$

$= \{x \in \mathbf{R} : f(x) = 0 \text{ or } g(x) = 0\}$

$= \{x \in \mathbf{R} : f(x)\, g(x) = 0\}$

(1) represents the set for which either $f(x) = 0$ and $g(x) = 0$ or $f(x) = -g(x)$

(3) represents the set for which f (x) and g (x) both zero.

62. (2) $\operatorname{cosec}\theta = \frac{p+q}{p-q}$, $\sin\theta = \frac{p-q}{p+q}$

$\cos\theta = \pm\sqrt{1-\sin^2\theta}$

$= \sqrt{1-\left(\frac{p-q}{p+q}\right)^2} = \frac{2\sqrt{pq}}{(p+q)}$

$\left|\cot\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right|$

$= \left|\frac{\cot\frac{\pi}{4}\cot\frac{\theta}{2}-1}{\cot\frac{\pi}{4}+\cot\frac{\theta}{2}}\right| = \left|\frac{\cot\frac{\theta}{2}-1}{\cot\frac{\theta}{2}+1}\right|$

$= \left|\frac{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}\right|$

On rationalizing denominator, we get

$\left|\left(\frac{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}\right)\left(\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}\right)\right|$

$= \left|\frac{\cos\theta}{\sin^2\frac{\theta}{2}+\cos^2\frac{\theta}{2}+2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right|$

$= \left|\frac{\cos\theta}{1+\sin\theta}\right| = \left|\frac{2\sqrt{pq}/(p+q)}{1+\frac{(p-q)}{p+q}}\right| = \frac{\sqrt{pq}}{p}$

$= \sqrt{\frac{q}{p}}$

63. (3) Rewrite the given differential equation as follows :

$\frac{dy}{dx} + \frac{2x}{x^2-1}y = \frac{1}{x^2-1}$, which is a linear form

The integrating factor I.F.

$= e^{\int \frac{2x}{x^2-1}dx} = e^{\ell n(x^2-1)} = x^2 - 1$

Thus multiplying the given equation by (x^2-1), we get $(x^2-1)\frac{dy}{dx} + 2xy = 1$

$\Rightarrow \frac{d}{dx}[y(x^2-1)] = 1$

On integrating we get $y(x^2-1) = x + c$

64. (4) Seeing the options we can assume that

$$\int e^{(\log x + ax^2)} \cos(bx^2 + c)dx$$

$$= \frac{K}{\sqrt{a^2+b^2}} e^{ax^2} \cos\left(bx^2 + c - \tan^{-1}\frac{b}{a}\right) + A$$

Put $b = 0$ & $a = 1$

$$\Rightarrow \int xe^{x^2} \cos c\, dx = Ke^{x^2} \cos(c) + A$$

$$\Rightarrow \frac{\cos c}{2}\int 2xe^{x^2} dx = Ke^{x^2} \cos c + A$$

$$\Rightarrow \frac{\cos c}{2} e^{x^2} = Ke^{x^2} \cos c + A$$

Comparing coeff. of $e^{x^2} \cos c$, we get $K = \frac{1}{2}$

65. (3) $\int \sqrt{1+\sec x}\, dx = \int \frac{\sec x . \tan x}{\sec x . \sqrt{\sec x - 1}} dx$

(Put $\sec x = t \Rightarrow \sec x \tan x\, dx = dt$)

$$= \int \frac{dt}{t\sqrt{t-1}} = 2\int \frac{y\, dy}{(y^2+1)y} \text{ (where } y^2 = t-1)$$

$$= 2\tan^{-1} y + C = 2\tan^{-1}\sqrt{\sec x - 1} + C$$

$\therefore f(x) = \tan^{-1} x$ and $g(x) = \sqrt{\sec x - 1}.$

66. (3) $y^2 = P(x) \Rightarrow 2yy_1 = P'(x) \Rightarrow y_1 = \frac{P'(x)}{2y}$

$$\therefore y_2 = \frac{2yP''(x) - P'(x).2y_1}{4y^2}$$

$$= \frac{2yP''(x) - P'(x).\frac{P'(x)}{y}}{4y^2}$$

$$\Rightarrow y_2 = \frac{2y^2P''(x) - [P'(x)]^2}{4y^3}$$

$$= \frac{2P(x)P''(x) - [P'(x)]^2}{4y^3}$$

$$\Rightarrow 2y^3y_2 = \frac{1}{2}\left[2P(x)P''(x) - \{P'(x)\}^2\right]$$

$$\Rightarrow 2\frac{d}{dx}(y^3y_2) = \frac{1}{2}\left[\begin{matrix} 2P(x)P'''(x) + 2P'(x)P''(x) \\ - 2P'(x)P''(x) \end{matrix}\right]$$

$$\Rightarrow 2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right) = P(x)P'''(x)$$

67. (3)

(1) $f(x) = \log\left(\frac{1-x}{1+x}\right)$

$$\therefore f(-x) = \log\left(\frac{1+x}{1-x}\right) = -\log\left(\frac{1-x}{1+x}\right) = -f(x)$$

$\therefore$ It is an odd function

(2) $f(x) + f(-x) = \log(x + \sqrt{x^2+1}) + \log(-x + \sqrt{x^2+1})$

$$= \log\left\{(x + \sqrt{x^2+1})(-x + \sqrt{x^2+1}\right\} = \log\left(x^2 + 1 - x^2\right)$$

$= \log(1) = 0$ $\qquad \therefore f(-x) = -f(x)$

$\therefore$ It is an odd function

(3) $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1;$

$$f(-x)=\frac{-x}{e^{-x}-1}-\frac{x}{2}+1=\frac{-xe^x}{1-e^x}-\frac{x}{2}+1$$

$$f(x)-f(-x)=\left\{\frac{x}{e^x-1}+\frac{x}{2}+1\right\}-\left\{\frac{-xe^x}{1-e^x}-\frac{x}{2}+1\right\}$$

$$\frac{x(1-e^x)}{e^x-1}+x=-x+x=0$$

$$\therefore f(-x)=f(x)$$

Hence, it is an even function

(4) $f(x)=e^{2x}+\sin x;$

$f(-x)=e^{-2x}+\sin(-x)=e^{-2x}-\sin x\neq \pm f(x)$

68. (1) Let the equation of the required plane be

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ ...(i)

It meets co-ordinate axes in points A (a, 0, 0), B(0, b, 0), C(0, 0, c).

The centroid of ΔABC is $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$

$\Rightarrow \frac{a}{3}=\alpha,\ \frac{b}{3}=\beta,\ \frac{c}{3}=\gamma$

$\Rightarrow a=3\alpha,\ b=3\beta,\ c=3\gamma$

Hence the required plane is

$\frac{x}{3\alpha}+\frac{y}{3\beta}+\frac{z}{3\gamma}=1$ i.e., $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$.

69. (1) Given, hyperbola is

$(10x-5)^2+(10y-2)^2=9(3x+4y-7)^2$

$\Rightarrow \left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{5}\right)^2=\frac{9}{4}\left(\frac{3x+4y-7}{5}\right)^2$

$\Rightarrow$ Given curve is a hyperbola where focus is $\left(\frac{1}{2},\frac{1}{5}\right)$ and directrix is $3x+4y-7=0$. Latus rectum is a line passing through the focus and parallel to the directrix.

$\Rightarrow$ Eq. of the latus rectum is

$y-\frac{1}{5}=-\frac{3}{4}\left(x-\frac{1}{2}\right)$.

70. (2) Let $S\equiv x^2-4y$

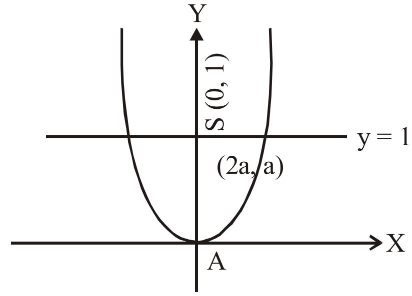
Since the point (2a, a) lies inside the parabola,

$\therefore$ $S(2a, a)=4a^2-4a<0$

or $a(a-1)<0$ ...(1)

Also, the vertex A (0, 0) and the point (2a, a) are on the same side of the line y = 1 (the equation of latus rectum)

So, $a-1<0$ i.e. $a<1$ ...(2)



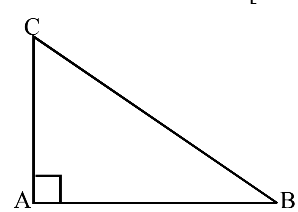
From (1) and (2), we have $a(a-1)<0$

or $0<a<1$

71. (3) $r=\frac{\Delta}{s}=\frac{bc}{a+b+c}=\frac{bc(b+c-a)}{(b+c+a)(b+c-a)}$

$$=\frac{bc(b+c-a)}{\{(b+c)^2-a^2\}}=\frac{bc(b+c-a)}{2bc}$$

[Since $a^2=b^2+c^2$]



$\therefore r=\frac{b+c-a}{2}$

72. (1) $\frac{2}{\sqrt{c}+\sqrt{a}}=\frac{1}{\sqrt{b}+\sqrt{c}}+\frac{1}{\sqrt{a}+\sqrt{b}}$

$$=\frac{2\sqrt{b}+\sqrt{a}+\sqrt{c}}{(\sqrt{b}+\sqrt{c})(\sqrt{a}+\sqrt{b})}$$

$\Rightarrow 2\sqrt{ab}+2b+2\sqrt{ac}+2\sqrt{bc}$

$=2\sqrt{bc}+2\sqrt{ac}+c+2\sqrt{ab}+a$

$\Rightarrow 2b = a + c$

$\therefore$ a, b, c, are in A.P.

$\Rightarrow$ ax, bx, cx, are in A.P.

$\Rightarrow$ ax + 1, bx + 1, cx + 1, are in A.P.

$\Rightarrow 9^{ax+1}, 9^{bx+1}, 9^{cx+1}$ are in G.P.

73. (4) $a_k - a_{k-1} = \int_0^{\pi} \frac{\sin(2k-1)x - \sin(2k-3)x}{\sin x} dx$

$= \int_0^{\pi} 2\cos 2(k-1)x \, dx = \left[\frac{2\sin 2(k-1)x}{2(k-1)}\right]_0^{\pi} = 0$

for $k = 2, 3, 4$

$\Rightarrow a_1 = a_2 = a_3 = \ldots\ldots\ldots\ldots$

$\Rightarrow$ the sequence is a constant sequence.

74. (2) $f(x) = e^x$ and $g(x) = \sin^{-1}x$ and $h(x) = f(g(x))$

$\Rightarrow h(x) = f(\sin^{-1} x) = e^{\sin^{-1} x}$

$\Rightarrow h'(x) = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$

$\Rightarrow \frac{h'(x)}{h(x)} = \frac{1}{\sqrt{1-x^2}}$

75. (4) $f(x) = \sin|x| - |x|$

$f(x)\big|_{x=0^-} = -\sin x + x \Rightarrow f'(x) = -\cos x + 1$

$f(x)\big|_{x=0^+} = \sin x - x \Rightarrow f'(x) = \cos x - 1$

$\Rightarrow \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = 0$

76. (2) $f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$

$f(x)$ is defined if $-1 \le \left(\frac{x}{2} - 1\right) \le 1$ and

$\cos x > 0$

or $0 \le \frac{x}{2} \le 2$ and $-\frac{\pi}{2} < x < \frac{\pi}{2}$

or $0 \le x \le 4$ and $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$\therefore \quad x \in \left[0, \frac{\pi}{2}\right)$

77. (2) $f(x) = \sin x + \cos x,\ g(x) = x^2 - 1$

$\Rightarrow g(f(x)) = (\sin x + \cos x)^2 - 1 = \sin 2x$

Clearly $g(f(x))$ is invertible in $-\frac{\pi}{2} \le 2x \le \frac{\pi}{2}$

$\Rightarrow -\frac{\pi}{4} \le x \le \frac{\pi}{4}$

78. (2)

(1) We have

$f(x) = 6x - 1,\ x \in R$.

Let $f(x_1) = f(x_2),\ x_1, x_2 \in R$

$\Rightarrow 6x_1 - 1 = 6x_2 - 1 \Rightarrow 6x_1 = 6x_2 \Rightarrow x_1 = x_2$.

$\therefore$ 'f' is one-one.

(2) We have $f(x) = x^2 + 7,\ x \in R$.

$f(-2) = (-2)^2 + 7 = 11,\ f(2) = (2)^2 + 7 = 11$

$\therefore$ The images of distinct elements –2 and 2 of R are equal.

$\therefore$ 'f' cannot be one-one.

(3) $f(x) = x^3,\ x \in R$.

Let $f(x_1) = f(x_2),\ x_1, x_2 \in R$

$\Rightarrow x_1^3 = x_2^3 \quad \Rightarrow x_1^3 - x_2^3 = 0$

$\Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0$

$\Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2$,

because the other factor cannot be zero

$\therefore$ 'f' is one-one

(4) We have $f(x) = \frac{2x+1}{x-7},\ x \in R - \{7\}$.

Let $f(x_1) = f(x_2),\ x_1, x_2 \in R - \{7\}$.

$\Rightarrow \frac{2x_1 + 1}{x_1 - 7} = \frac{2x_2 + 1}{x_2 - 7}$

$\Rightarrow -15x_1 = -15x_2 \Rightarrow x_1 = x_2$.

$\therefore f$ is one-one.

79. (4) $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2} + \frac{3}{n^2}\sec^2\frac{9}{n^2} + \ldots + \frac{1}{n}\sec^2 1\right]$

is equal to $\lim_{n\to\infty} \frac{r}{n^2}\sec^2\frac{r^2}{n^2}$

$= \lim_{n\to\infty} \frac{1}{n}.\frac{r}{n}\sec^2\frac{r^2}{n^2}$

$\Rightarrow$ Given limit is equal to value of integral $\int_0^1 x\sec^2 x^2 dx$

or $\frac{1}{2}\int_0^1 2x\sec^2 x^2 dx = \frac{1}{2}\int_0^1 \sec^2 t\, dt$

[put $x^2 = t$
$\Rightarrow$ $2xdx = dt$]

$= \frac{1}{2}(\tan t)\Big|_0^1 = \frac{1}{2}\tan 1.$

80. (2) The equation $c(y+c)^2 = x^3$ ...(i) has one arbitrary constant, so the equation must be of the first order. Differentiating (i) we get

$$2c(y+c)\frac{dy}{dx} = 3x^2 \quad ...(ii)$$

$$\Rightarrow 4c^2(y+c)^2\left(\frac{dy}{dx}\right)^2 = 9x^4 \quad ...(iii)$$

Divide (iii) by (i), we get

$$4c\left(\frac{dy}{dx}\right)^2 = 9x \Rightarrow c = \frac{9}{4}\frac{x}{\left(\frac{dy}{dx}\right)^2}$$

Put the value of c in (ii), we get

$$2\cdot\frac{9}{4}\frac{x}{\left(\frac{dy}{dx}\right)^2}\left[y+\frac{9}{4}\frac{x}{\left(\frac{dy}{dx}\right)^2}\right]\left(\frac{dy}{dx}\right) = 3x^2$$

$$\Rightarrow 9x\left[y+\frac{9}{4}\frac{x}{\left(\frac{dy}{dx}\right)^2}\right] = 6x^2\left(\frac{dy}{dx}\right)$$

$$\Rightarrow 3y\left(\frac{dy}{dx}\right)^2 + \frac{27}{4}x = 2x\left(\frac{dy}{dx}\right)^3$$

$$\Rightarrow 12y\left(\frac{dy}{dx}\right)^2 = 8x\left(\frac{dy}{dx}\right)^3 - 27x$$

81. (1.67) We have

$\vec{r}\times\vec{a} + \vec{r}\times\vec{b} = \vec{a}\times\vec{b} + \vec{b}\times\vec{a} = \vec{0}$

$\Rightarrow \vec{r}\times(\vec{a}+\vec{b}) = \vec{0}$

$\Rightarrow \vec{r} = \lambda(\vec{a}+\vec{b}) = \lambda(-2\hat{i}-\hat{j}+2\hat{k})$

$\Rightarrow \hat{r} = \frac{1}{3}(-2\hat{i}-\hat{j}+2\hat{k})$

$\vec{r}\cdot\vec{a} = 6\lambda = 30 \Rightarrow \lambda = 5$

$\therefore\ \vec{r} = 5(-2\hat{i}-\hat{j}+2\hat{k})$

$\therefore\ |\vec{r}| = 5/3 = 1.67$

82. (60) Let l_1, m_1, n_1 and l_2, m_2, n_2 be the d.c of line 1 and 2 respectively, then as given

$l_1 + m_1 + n_1 = 0$
and $l_2 + m_2 + n_2 = 0$
and $l_1^2 + m_1^2 - n_1^2 = 0$ and
$l_2^2 + m_2^2 - n_2^2 = 0$
($\because\ l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$)
Angle between lines, θ is
$\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$ ...(1)
As given $l^2 + m^2 = n^2$ and $l + m = -n$
$\Rightarrow (-n)^2 - 2lm = n^2 \Rightarrow\ 2lm = 0$ or $lm = 0$
So $l_1m_1 = 0, l_2m_2 = 0$
If $l_1 = 0, m_1 \neq 0$ then $l_1m_2 = 0$
If $m_1 = 0, l_1 \neq 0$ then $l_2m_1 = 0$
If $l_2 = 0, m_2 \neq 0$ then $l_2m_1 = 0$
If $m_2 = 0, l_2 \neq 0$ then $l_1m_2 = 0$
Also $l_1l_2 = 0$ and $m_1m_2 = 0$
$l^2 + m^2 - n^2 = l^2 + m^2 + n^2 - 2n^2 = 0$

$\Rightarrow\ 1 - 2n^2 = 0\ \Rightarrow\ n = \pm\frac{1}{\sqrt{2}}$

$\therefore\ n_1 = \pm\frac{1}{\sqrt{2}}, n_2 = \pm\frac{1}{\sqrt{2}}$

$\therefore\ \cos\theta = \frac{1}{2}\ \theta = 60°$ (acute angle)

83. (0.56) If A will be final winner

$\Rightarrow$ (A beats B) (B beats C)

+ (A beats C) (C beats B)

$$=\frac{2}{3}\times\frac{2}{3}+\left(1-\frac{2}{3}\right)\left(1-\frac{2}{3}\right)=\frac{4}{9}+\frac{1}{9}=\frac{5}{9}\quad =0.56$$

84. (2.00) If $\vec{a}, \vec{b}, \vec{c}$ are linearly independent vectors, then $\vec{c}$ should be a linear combination of $\vec{a}$ and $\vec{b}$.

Let $\vec{c}=p\vec{a}+q\vec{b}$

i.e. $\hat{i}+\alpha\hat{j}+\beta\hat{k}=p\left(\hat{i}+\hat{j}+\hat{k}\right)+q\left(4\hat{i}+3\hat{j}+4\hat{k}\right)$

Equating the coefficient $\vec{i}, \vec{j}, \vec{k}$, we get $1=p+4q$,

$\alpha=p+3q$ and $\beta=p+4q$.

From first and third, $\beta=1$.

Now $|\vec{c}|=\sqrt{3}\Rightarrow 1+\alpha^2+\beta^2=3$

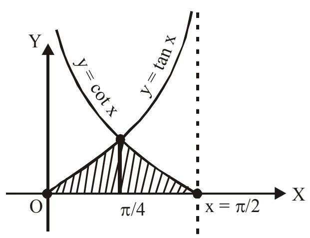
$\Rightarrow\quad \alpha^2=1,\quad \therefore\ \alpha=\pm 1.$

Hence, $\alpha=\pm 1,\ \beta=1.$

$\Rightarrow \alpha+\beta=2$ or 0.

85. (2) Required area

$$=\int_0^{\pi/4}\tan x\ dx+\int_{\pi/4}^{\pi/2}\cot x\ dx$$



$$=\left[\log\sec x\right]_0^{\pi/4}+\left[\log\sin x\right]_{\pi/4}^{\pi/2}$$

$$=\log\sqrt{2}+\log\sqrt{2}=2\log\sqrt{2}=\log 2$$

$=\log k\Rightarrow k=2$

86. (8.67) Let the equation of any normal be $y=-tx+2t+t^3$

Since it passes through the point (15, 12)

$\therefore\ 12=-15t+2t+t^3$ i.e., $t^3-13t-12=0$

$\therefore\ t=-1,\ -3,\ 4$

$\therefore$ The points are $(1,-2), (9,-6), (16, 8)$

$\therefore$ Centroid is $\left(\frac{26}{3},0\right)=(m, n).$

$$\Rightarrow m+n=\frac{26}{3}=8.67$$

87. (21.00) Taking A.M. and G.M. of 7 numbers

$\frac{a}{2},\frac{a}{2},\frac{b}{3},\frac{b}{3},\frac{b}{3},\frac{c}{2},\frac{c}{2}$, we get

$$\frac{2.\frac{a}{2}+3.\frac{b}{3}+2.\frac{c}{2}}{7}\geq\left\{\left(\frac{a}{2}\right)^2\left(\frac{b}{3}\right)^3\left(\frac{c}{2}\right)^2\right\}^{\frac{1}{7}}$$

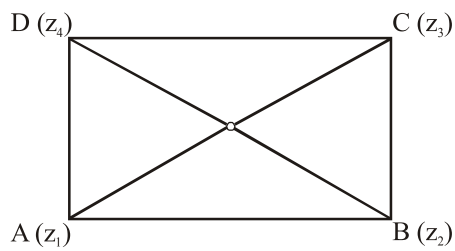
$$\Rightarrow\quad \frac{3^7}{7^7}\geq\frac{a^2b^3c^2}{2^2.3^3.2^2}\Rightarrow a^2b^3c^2\leq\frac{3^{10}.2^4}{7^7}$$

$\therefore\quad$ greatest value of $a^2b^3c^2=\frac{3^{10}.2^4}{7^7}=\frac{3^p.q^q}{7^r}$

$\Rightarrow\quad P+q+r=10+4+7=21$

88. (0.50) Since $z_1+z_3=z_2+z_4$

$$\Rightarrow\quad \frac{z_1+z_3}{2}=\frac{z_2+z_4}{2}$$



$$\Rightarrow\quad\left(\begin{matrix}\text{Midpoint of}\\ z_1\text{ and }z_3\end{matrix}\right)=\left(\begin{matrix}\text{Midpoint of}\\ z_2\text{ and }z_4\end{matrix}\right)$$

$\Rightarrow$ Quadrilateral ABCD is a parallelogram.

Also, $|z_1-z_3|=|z_2-z_4|$

$$\text{i.e.}\quad\left(\begin{matrix}\text{Distance}\\ \text{Between}\\ z_1\text{ and }z_3\end{matrix}\right)=\left(\begin{matrix}\text{Distance}\\ \text{Between}\\ z_2\text{ and }z_4\end{matrix}\right)$$

$\Rightarrow$ Length of diagonals are equal.

$\therefore$ The parallelogram ABCD is a rectangle

Now, $\arg\left(\dfrac{z_1 - z_2}{z_3 - z_2}\right)$ i.e. angle between z_1, z_2 and z_3 (taken in order) is $\pm\dfrac{\pi}{2}$ which is angle between sides AB and BC (see figure)

$$\therefore \pm k\pi = \pm\frac{\pi}{2} \Rightarrow k = \frac{1}{2} = 0.50$$

89. (0.25) $f(x) = \left[x^2 + e^{\frac{1}{2-x}}\right]^{-1}$ and $f(2) = k$

If f(x) is continuous from right at x =2 then

$$\lim_{x\to 2^+} f(x) = f(2) = k$$

$$\Rightarrow \lim_{x\to 2^+}\left[x^2 + e^{\frac{1}{2-x}}\right]^{-1} = k$$

$$\Rightarrow k = \lim_{h\to 0} f(2+h)$$

$$\Rightarrow k = \lim_{h\to 0}\left[(2+h)^2 + e^{\frac{1}{2-(2+h)}}\right]^{-1}$$

$$\Rightarrow k = \lim_{h\to 0}\left[4 + h^2 + 4h + e^{-1/h}\right]^{-1}$$

$$\Rightarrow k = \left[4 + 0 + 0 + e^{-\infty}\right]^{-1} \Rightarrow k = \frac{1}{4} = 0.25$$

90. (3) Let A be the area, b be the breadth and ℓ be the length of the rectangle.

Given : $\dfrac{dA}{dt} = -5$, $\dfrac{d\ell}{dt} = 2$, $\dfrac{db}{dt} = -3$

We know, $A = \ell \times b$

$$\Rightarrow \frac{dA}{dt} = \ell.\frac{db}{dt} + b.\frac{d\ell}{dt} = -3\ell + 2b$$

$\Rightarrow -5 = -3\ell + 2b.$

When $b = 2$, we have

$$-5 = -3\ell + 4 \Rightarrow \ell = \frac{9}{3} = 3m$$

Mock Test-10

ANSWER KEY

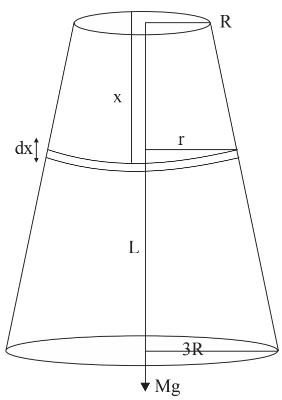
1	(3)	**16**	(2)	**31**	(4)	**46**	(4)	**61**	(3)	**76**	(1)
2	(2)	**17**	(2)	**32**	(3)	**47**	(2)	**62**	(4)	**77**	(1)
3	(3)	**18**	(3)	**33**	(2)	**48**	(2)	**63**	(2)	**78**	(2)
4	(2)	**19**	(1)	**34**	(4)	**49**	(2)	**64**	(3)	**79**	(3)
5	(4)	**20**	(4)	**35**	(1)	**50**	(1)	**65**	(1)	**80**	(4)
6	(2)	**21**	(2)	**36**	(3)	**51**	(187)	**66**	(4)	**81**	(0.33)
7	(3)	**22**	(2.68)	**37**	(2)	**52**	(5.82)	**67**	(4)	**82**	(0.60)
8	(3)	**23**	(900)	**38**	(1)	**53**	(1.6)	**68**	(4)	**83**	(2007)
9	(4)	**24**	(0.125)	**39**	(4)	**54**	(1.5)	**69**	(1)	**84**	(5.00)
10	(3)	**25**	(2)	**40**	(3)	**55**	(2)	**70**	(4)	**85**	(1.50)
11	(3)	**26**	(0.96)	**41**	(2)	**56**	(0.2)	**71**	(2)	**86**	(1.00)
12	(2)	**27**	(110)	**42**	(3)	**57**	(4)	**72**	(2)	**87**	(1.00)
13	(3)	**28**	(1.2×10^{12})	**43**	(1)	**58**	(0.22)	**73**	(1)	**88**	(2.00)
14	(4)	**29**	(1)	**44**	(3)	**59**	(3)	**74**	(1)	**89**	(0.50)
15	(2)	**30**	(200)	**45**	(3)	**60**	(12)	**75**	(3)	**90**	(0.05)

Solutions

PHYSICS

1. (3) Consider a small element dx of radius r,

$$r = \frac{2R}{L}x + R$$



At equilibrium change in length of the wire

$$\int_0^L dL = \int \frac{Mg\,dx}{\pi\left[\frac{2R}{L}x + R\right]^2 Y}$$

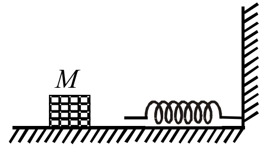
Taking limit from 0 to L

$$\Delta L = \frac{Mg}{\pi Y}\left[-\frac{1}{\left[\frac{2Rx}{L} + R\right]_0^L} \times \frac{L}{2R}\right] = \frac{MgL}{3\pi R^2 Y}$$

The equilibrium extended length of wire = L + ΔL

$$= L + \frac{MgL}{3\pi R^2 Y} = L\left(1 + \frac{1}{3}\frac{Mg}{\pi Y R^2}\right)$$

2. (2)



$$\frac{1}{2}Mv^2 = \frac{1}{2}k\,L^2 \Rightarrow v = \sqrt{\frac{k}{M}}\,.L$$

$$\text{Momentum} = M \times v = M \times \sqrt{\frac{k}{M}}\,.L = \sqrt{kM}\,.L$$

3. (3) The net force becomes zero at the mean point.

Therefore, linear momentum must be conserved.

$\therefore Mv_1 = (M+m)v_2$

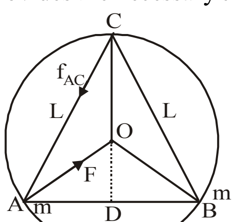
$MA_1\sqrt{\frac{k}{M}} = (M+m)A_2\sqrt{\frac{k}{m+M}}$

$\therefore \left(V = A\sqrt{\frac{k}{M}}\right)$

$\because A_1\sqrt{M} = A_2\sqrt{M+m}$

$\therefore \frac{A_1}{A_2} = \sqrt{\frac{m+M}{M}}$

4. (2) The resultant gravitational force on each particle provides the necessary centripetal force.



$\therefore$ $mv^2/r = \sqrt{(F^2 + F^2 + 2F^2\cos 60°)} = \sqrt{3}\ F$

But $r = (\sqrt{3}\ L/2) \times (2/3) = L/\sqrt{3}$

$\therefore$ $mv^2\sqrt{3}/L = \sqrt{3}GM^2/L^2 \Rightarrow v = \sqrt{(GM/L)}$

5. (4) Let the charges on spheres of radii r and R be q_1 and q_2 respectively. Then, $Q = q_1 + q_2$. The potential at the centre will be :

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{R}\right)$$

As surface densities are equal, hence

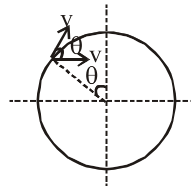
$$\sigma = \frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2} \text{ or } q_2 = q_1\left(\frac{R^2}{r^2}\right)$$

$$V = \frac{1}{4\pi\varepsilon_o}\left(\frac{q_1}{r} + q_1\frac{R}{r^2}\right) = \frac{1}{4\pi\varepsilon_o} \times \frac{q_1}{r^2}(r+R)$$

Now, $q_1 + q_2 = Q$ or $q_1 + q_1\frac{R^2}{r^2} = Q$

$$\Rightarrow \frac{q_1}{r^2} = \frac{Q}{R^2 + r^2};\ V = \frac{Q}{4\pi\varepsilon_0}\left(\frac{R+r}{R^2+r^2}\right)$$

6. (2) $v_R = \sqrt{v^2 + v^2 + 2v^2\cos\theta}$



$$= \sqrt{2v^2(1+\cos\theta)} = 2v\cos\frac{\theta}{2}$$

7. (3) Magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, i.e., $n' = (n+1)$

As magnetic moment $M_n = I_n A = i_n(\pi r_n^2)$

$$i_n = eV_n = \frac{mz^2e^5}{4\varepsilon_0^2 n^3 h^3}$$

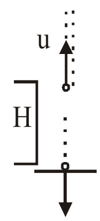
$$r_n = \frac{n^2h^2}{4\pi^2 kzme^2}\left(k = \frac{1}{4\pi\in_0}\right)$$

Solving we get magnetic moment of the hydrogen atom for n^{th} excited state

$$M_{n'} = \left(\frac{e}{2m}\right)\frac{nh}{2\pi}$$

8. (3) Speed on reaching ground

$v = \sqrt{u^2 + 2gh}$



Now, $v = u + at$

$\Rightarrow \sqrt{u^2 + 2gh} = -u + gt$

Time taken to reach highest point is $t = \frac{u}{g}$,

$\Rightarrow t = \frac{u + \sqrt{u^2 + 2gH}}{g} = \frac{nu}{g}$ (from question)

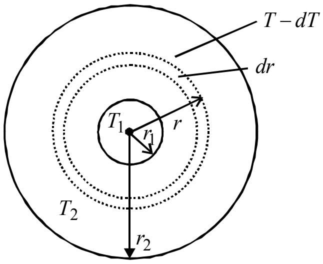
$\Rightarrow 2gH = n(n-2)u^2$

9. (4) Consider a shell of thickness (dr) and of radii (r) and the temperature of inner and outer surfaces of this shell be T, $(T - dT)$

$\frac{dQ}{dt}$ = rate of flow of heat through it

$$= \frac{KA[(T-dT)-T]}{dr} = \frac{-KAdT}{dr}$$

$$= -4\pi Kr^2 \frac{dT}{dr} \quad (\because A = 4\pi r^2)$$



To measure the radial rate of heat flow, integration technique is used, since the area of the surface through which heat will flow is not constant.

Then, $\left(\frac{dQ}{dt}\right)\int_{r_1}^{r_2} \frac{1}{r^2} dr = -4\pi K \int_{T_1}^{T_2} dT$

$$\frac{dQ}{dt}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = -4\pi K[T_2 - T_1]$$

or $\frac{dQ}{dt} = \frac{-4\pi K r_1 r_2 (T_2 - T_1)}{(r_2 - r_1)}$

$\therefore \quad \frac{dQ}{dt} \propto \frac{r_1 r_2}{(r_2 - r_1)}$

10. (3) In X-ray tube, $\lambda_{min} = \frac{hc}{eV}$

In $\lambda_{min} = In\left(\frac{hc}{e}\right) - InV$

Clearly, log λ_{min} versus log V graph slope is negative hence option (3) correctly depicts.

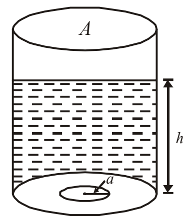
11. (3) Let r be the internal resistance of the cell. Terminal potential difference (V) < EMF of a cell (E).

$V = E - Ir$ or $r = \frac{E-V}{I} = \left(\frac{E-V}{V}\right)R$

12. (2) Let the rate of falling water level be $-\frac{dh}{dt}$

Initially at $t = 0\,;\, h = h$

$t = t\,;\, h = 0$



Then, $A\left(-\frac{dh}{dt}\right) = \pi a^2 . v$

$$dt = -\frac{A}{\pi a^2\sqrt{2gh}}dh$$

[$\because$ velocity of efflux of liquid $v = \sqrt{2gh}$]

Integrating both sides

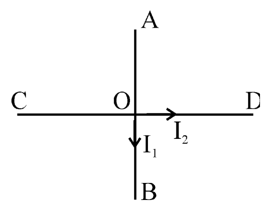
$$\int_0^t dt = -\frac{A}{\sqrt{2g}\pi a^2}\int_h^0 h^{-1/2} dh$$

$$[t]_0^t = -\frac{A}{\sqrt{2g}\pi a^2}.\left[\frac{h^{1/2}}{1/2}\right]_h^0$$

$$t = \frac{\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$$

13. (3) $\vec{B}$ due to AOB and COD are $\perp$ to each other. Hence, net $\vec{B}^2 = B_1^2 + B_2^2$

$B = \frac{\mu_0}{2\pi a}(I_1^2 + I_2^2)^{1/2}$ ($\perp$ to plane ABCD)



14. (4) At resonance $L\omega = \frac{1}{C\omega}$, $\omega = \frac{1}{\sqrt{LC}}$

Current through circuit $i = \frac{E}{R}$

Power dissipated at Resonance $= i^2R$

15. (2) $x = \alpha t^3$ and $y = \beta t^3$

$v_x = \frac{dx}{dt} = 3\alpha t^2$ and $v_y = \frac{dy}{dt} = 3\beta t^2$

$\therefore \; v = \sqrt{v_x^2 + v_y^2} = \sqrt{9\alpha^2 t^4 + 9\beta^2 t^4}$

$= 3t^2\sqrt{\alpha^2+\beta^2}$

16. (2) $\beta = \frac{\lambda D}{2d} = \frac{6.5\times10^{-7}}{10^{-3}}\times 1 = 0.65\times10^{-3}$

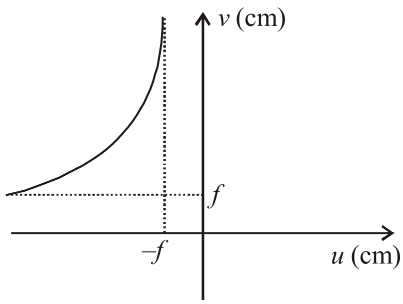
$= 0.65$ mm

The distance between fifth bright fringe from third dark fringe $= 2.5\,\beta = 2.5 \times .65$
$= 1.63$ mm.

17. (2) Graph [A] is for material used for making permanent magnets (high coercivity)

Graph [B] is for making electromagnets and transformers.

18. (3) This graph suggest that when
$u = -f, v = +\infty$

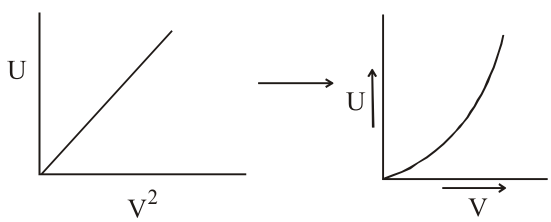


When the object is moved further away from the lens, v decreases but remains positive. When u is at $-\infty$, $v = f$.

This is how image formation takes place for different positions of the object in case of a convex lens.

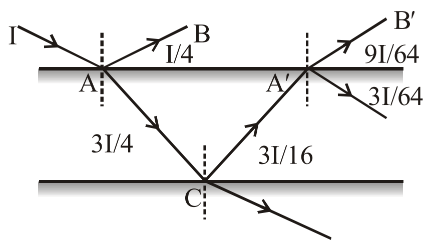
19. (1) $U = \frac{1}{2}CV^2$

From this, we get $U \propto V^2$



20. (4) From figure $I_1 = \frac{I}{4}$ and

$I_2 = \frac{9I}{64} \Rightarrow \frac{I_2}{I_1} = \frac{9}{16}$



By using

$$\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{\frac{I_2}{I_1}}+1}{\sqrt{\frac{I_2}{I_1}}-1}\right)^2 = \left(\frac{\sqrt{\frac{9}{16}}+1}{\sqrt{\frac{9}{16}}-1}\right)^2 = \frac{49}{1}$$

21. (2) Acceleration when there is no friction is a = gsinθ and acceleration when friction is there is
$a' = (g\sin\theta - \mu g\cos\theta)$

We know that $s = ut + \frac{1}{2}at^2$, under condition of $u = 0$, gives, $t = \sqrt{(2s/a)}$.
Therefore, $t'/t = \sqrt{(a/a')}$.
For our case we get $n = \sqrt{[g\sin\theta/(g\sin\theta - \mu g\cos\theta)]}$
$\Rightarrow n^2 = g\sin\theta/(g\sin\theta - \mu g\cos\theta) = 1/(1-\mu\cot\theta)$
But $\theta = 45°$
$\Rightarrow \cot\theta = 1 \therefore n^2 = 1/(1-\mu)$ or $\mu = 1-(1/n^2)$

22. (2.68) Wavelength of monochromatic green light $= 5.5\times10^{-5}$ cm

Intensity $I = \frac{\text{Power}}{\text{Area}}$

$= \frac{100\times(3/100)}{4\pi(5)^2} = \frac{3}{100\pi}\,\text{Wm}^{-2}$

Now, half of this intensity (I) belongs to electric field and half of that to magnetic field, therefore,

$\frac{I}{2} = \frac{1}{4}\varepsilon_0 E_0^2 C$ or $E_0 = \sqrt{\frac{2I}{\varepsilon_0 C}}$

$$= \sqrt{\frac{2 \times \left(\frac{3}{100}\pi\right)}{\left(\frac{1}{4\pi \times 9 \times 10^9}\right) \times \left(3 \times 10^8\right)}}$$

$$= \sqrt{\frac{6}{25} \times 30} = \sqrt{7.2}$$

$\therefore \quad E_0 = 2.68 \text{ V/m}$

23. (900) We know that

$$\eta = 1 - \frac{T_2}{T_1} = \frac{300-250}{300} = \frac{50}{300} = \frac{1}{6}$$

or $\eta = 1 - \frac{Q_2}{Q_1} \Rightarrow \frac{1}{6} = \frac{Q_1 - 750}{Q_1}$

$Q_1 = 6Q_1 - 4500$

$\Rightarrow -5Q_1 = -4500 \Rightarrow Q_1 = 900 \text{ Cal}$

24. (0.125) $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$

Given, $E_0 = 100 \text{N/c}$

So, $\vec{E} = 100\hat{i} + 200\hat{j}$

Radius of circular surface = 0.02 m

Area = $\pi r^2 = \frac{22}{7} \times 0.02 \times 0.02$

$= 1.25 \times 10^{-3}\,\hat{i}\text{ m}^2$ [Loop is parallel to Y-Z plane]

Now, flux (ϕ) = EA cosθ

$= \left(100\hat{i} + 200\hat{j}\right).1.25 \times 10^{-3}\,\hat{i}\cos\theta°$ [$\theta = 0°$]

$= 125 \times 10^{-3}$ Nm^2/c

$= 0.125$ Nm^2/c

25. (2) Kinetic energy of each molecule,

$$\text{K.E.} = \frac{3}{2}K_BT$$

In the given problem,

Temperature, T = 0°C = 273 K

Height attained by the gas molecule, h = ?

$$\text{K.E.} = \frac{3}{2}K_B(273) = \frac{819K_B}{2}$$

K.E. = P.E.

$$\Rightarrow \frac{819K_B}{2} = Mgh$$

or $h = \frac{819K_B}{2Mg}$

26. (0.96) No. of electrons reaching the collector,

$$n_C = \frac{96}{100} \times 10^{10} = 0.96 \times 10^{10}$$

Emitter current, $I_E = \frac{n_E \times e}{t}$

Collector current, $I_C = \frac{n_C \times e}{t}$

$\therefore$ Current transfer ratio,

$$\alpha = \frac{I_C}{I_E} = \frac{n_C}{n_E} = \frac{0.96 \times 10^{10}}{10^{10}} = 0.96$$

27. (110) $n_{last} = n_{first} + (n-1) \times d$

$\Rightarrow 3n = n + (56-1) \times 4$

or $2n = 55 \times 4 \Rightarrow n = 110\text{Hz}$

28. (1.2×10^{12}) The critical frequency of a sky wave for reflection from a layer of atmosphere is given by $f_c = 9(N_{max})^{1/2}$

$\Rightarrow 10 \times 10^6 = 9(N_{max})^{1/2}$

$$\Rightarrow N_{max} = \left(\frac{10 \times 10^6}{9}\right)^2 \simeq 1.2 \times 10^{12}\text{m}^{-3}$$

29. (1) 30 Divisions of vernier scale coincide with 29 divisions of main scales

Therefore 1 V.S.D = $\frac{29}{30}$ MSD

Least count = 1 MSD – 1VSD

$= 1 \text{ MSD} - \frac{29}{30} \text{ MSD}$

$= \frac{1}{30}$ MSD

$= \frac{1}{30} \times 0.5° = 1$ minute.

30. (200) After 8 sec., the counting rate falls to

$$\frac{100}{1600} = \frac{1}{16} = \left(\frac{1}{2}\right)^4 \text{ th Part ;}$$

Time period = $\frac{8}{4}$ sec. = 2 sec.

Therefore, after 6 seconds, the counting rate should be $\left(\frac{1}{2}\right)^3$ th part of 1600, i.e. $\frac{1}{8}\times 1600 = 200$.

CHEMISTRY

31. (4) For Lyman series (shortest wavelength)

$n_1 = 1, n_2 = \infty$

$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\Rightarrow \frac{1}{A} = 1^2 R\left(\frac{1}{1} - \frac{1}{\infty}\right) \Rightarrow \frac{1}{A} = R$$

Longest wavelength = 1st line

$n_1 = 3, n_2 = 4$

$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{3^2} - \frac{1}{4^2}\right) \Rightarrow \frac{1}{\lambda} = \frac{R7}{36}$$

$$R = \frac{1}{A}$$

$$\frac{1}{\lambda} = \frac{\frac{1}{A}\times 7}{36} \Rightarrow \frac{1}{\lambda} = \frac{7}{36A} \Rightarrow \lambda = \frac{36A}{7}$$

32. (3) $kt = \frac{1}{n-1}\left[\frac{1}{A_t^{n-1}} - \frac{1}{A_0^{n-1}}\right]$

$$kt_{0.5} = \frac{1}{n-1}\left[\frac{1}{\left(\frac{A_0}{2}\right)^{n-1}} - \frac{1}{A_0^{n-1}}\right] = \frac{1}{n-1}\left[\frac{2^{n-1}-1}{A_0^{n-1}}\right]$$

$$kt_{0.875} = \frac{1}{n-1}\left[\frac{1}{\left(\frac{A_0}{8}\right)^{n-1}} - \frac{1}{A_0^{n-1}}\right] = \frac{1}{n-1}\left[\frac{8^{n-1}-1}{A_0^{n-1}}\right]$$

$$\frac{t_{0.875}}{t_{0.5}} = \frac{8^{n-1}-1}{2^{n-1}-1}$$

33. (2) $NH_2CONH_2 + NH_2{-}NH_2 \longrightarrow NH_2NHCONH_2 + NH_3$

(Semicarbazide)

34. (4) $\overset{1}{C}H_2(CN)-\overset{2}{C}H(CN)-\overset{3}{C}H_2(CN)$

1,2,3-propanetricarbonitrile

35. (1) The linking of identical atoms with each other to form long chains is called catenation. However, this property decreases from carbon to lead. Decrease of this property is associated with M-M bond energy which decreases from carbon to lead.

36. (3) Polyacrylonitrile (PAN), acrilan or orlon

$$n\,CH_2{=}CH(CN) \longrightarrow {-}\!\left(CH_2-CH(CN)\right)\!{-}_n$$

Acrylonitrile Orlon

It is hard used in preparing cloths, carpets etc.

37. (2) $\underset{X}{3Mg} + N_2 \xrightarrow{\Delta} \underset{Y}{Mg_3N_2}$;

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + \underset{\text{Colourless}}{2NH_3\uparrow}$$

$$CuSO_4 + 4NH_3 \longrightarrow \underset{\text{Blue complex}}{[Cu(NH_3)_4]SO_4}$$

38. (1) Molecules of a polymer, being large in size, scatter light.

39. (4) $[Zn(NH_3)_4]^{2+}$, $Ni(CO)_4$ and $[Cd(CN)_4]^{2-}$ all form tetrahedral structures where the central atom uses sp^3 hybridisation.

Whereas $[Cu(NH_3)_4]^{2+}$ uses sp^2d hybridisation and forms square planar structure. Although Cu should use both tetrahedral (sp^3) and square planar (sp^2d) hybridisations but the later one is confirmed through X-ray analysis.

40. (3) Diallyl ether has double bond which shows addition reaction with $KMnO_4$ (change its colour) while di-n-propyl ether does not react with $KMnO_4$.

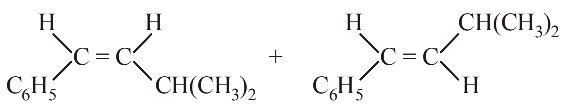
41. (2) Mn^{2+} ($3d^5$) is more stable then Mn^{3+} ($3d^4$). due to half filled orbitals.

42. (3) Pure ClO_2 is obtained by passing dry Cl_2 over $AgClO_3$ at 90°C.

$$2AgClO_3 + Cl_2(dry) \xrightarrow{90°C} 2AgCl + 2ClO_2 + O_2$$

43. (1) Whenever dehydration can produce two different alkenes, major product is formed according to **Saytzeff rule** *i.e.* more substituted alkene (alkene having lesser number of hydrogen atoms on the two doubly bonded carbon atoms) is the major product.

$$C_6H_5-CH_2-\underset{\substack{|\\OH}}{CH}-\underset{\substack{|\\CH_3}}{CH}-CH_3 \xrightarrow{\text{Conc. } H_2SO_4}$$



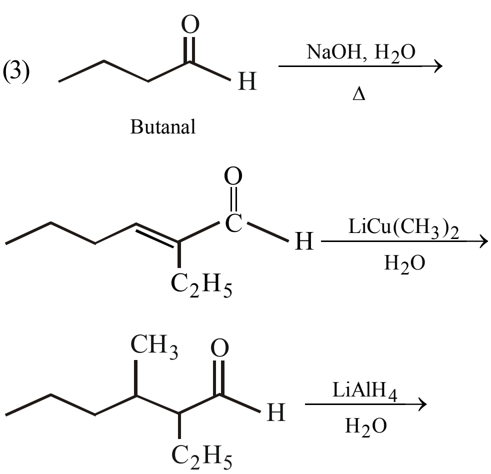
cis (minor) *trans* (major)

44. (3) $\left[Zr_3(PO_4)_4\right] \rightleftharpoons \underset{3S}{3Zr^{4+}} + \underset{4S}{4PO_4^{3-}}$

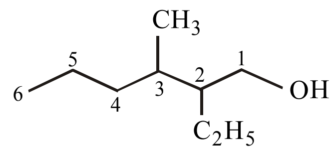
$K_{sp} = (3S)^3(4S)^4$
$= 27S^3 \times 256\,S^4$
$= 6912\,S^7.$

$\therefore \quad S = \left(\dfrac{K_{sp}}{6912}\right)^{1/7}$

45. (3)



Butanal



2-Ethyl-3 methyl-1-hexanol

46. (4) As^{3+} and Cd^{2+} are the radicals of group II, whereas Ni^{2+} and Zn^{2+} are the radicals of group IV. The solubility product of group IV radicals is higher as compared to group II. NH_4OH increases the ionisation of H_2S by removing H^+ of H_2S as unionisable water.

$$H_2S \rightleftharpoons 2H^+ + S^{2-};$$
$$H^+ + OH^- \longrightarrow H_2O$$

Thus, excess of sulphide ions are present which leads to the precipitation of all the four ions.
Note : HCl decreases ionisation of H_2S whereas NH_4OH increases the ionisation of H_2S.

47. (2) The reaction proceed through Markownikoff's rule by ionic mechanism. The initial carbonium ion $(CH_3CH_2\overset{+}{C}H_2)$ rearrange into more stable $(CH_3\overset{+}{C}HCH_3)$ carbonium ion.

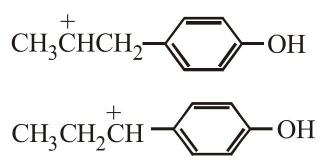
48. (2)

$$\underset{(C_2H_3N)}{A} \xrightarrow{\text{reduction}} B \xrightarrow{HNO_2} C_2H_5OH$$
$$B \xrightarrow{\text{Carbylamine reaction}} \text{Offensive smell (C)}$$

Given reactions indicate that B has 1° NH_2 group, and thus A, C_2H_3N, should be $CH_3C\equiv N$. Hence C should be CH_3CH_2NC

$$\underset{A}{CH_3C\equiv N} \xrightarrow{\text{reduction}} \underset{B}{CH_3CH_2NH_2} \xrightarrow[KOH]{CHCl_3} \underset{C}{CH_3CH_2N \overset{\rightarrow}{=} C}$$

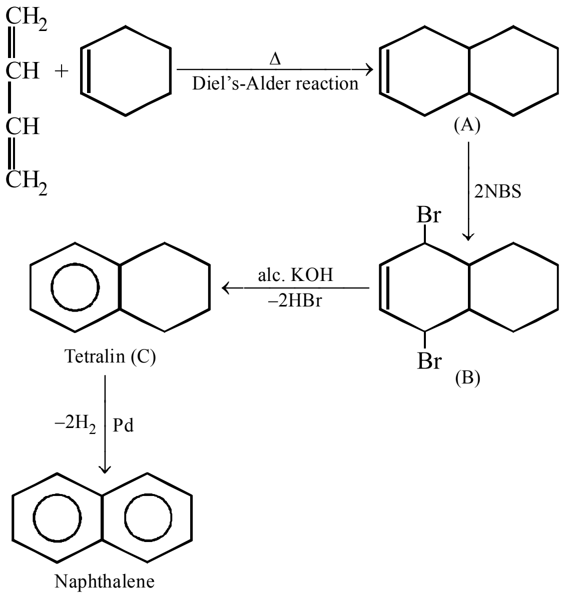
49. (2) Of the two possible carbocations, benzylic type (II) is more stable than the I.



......(I)
......(II)

hence bromination occurs on 3rd C atom

50. (1)



51. (187) For $H_2(g) + O_2(g) \longrightarrow H_2O_2(l)$

$$\Delta_f H^\circ(H_2O_2, l) = \Delta_r H_3^\circ + \frac{\Delta_r H_2^\circ}{2} - \frac{\Delta_r H_1^\circ}{2}$$

$$= (-285) + \left(\frac{-622}{2}\right) - \left(\frac{-818}{2}\right)$$

$= -187$ kJ/mol

52. (5.82) The volume of the unit cell $= (2.88\ \text{Å})^3 = 23.9 \times 10^{-24}\ \text{cm}^3$.

The volume of 100 g of the metal

$$= \frac{m}{\rho} = \frac{100}{7.20} = 13.9\ \text{cm}^3$$

Number of unit cells in this volume

$$= \frac{13.9\ \text{cm}^3}{23.9 \times 10^{-24}\ \text{cm}^3} = 5.82 \times 10^{23}$$

53. (1.6) $W_2 = 4$g, $V = 1000$ mL $= 1$ L,

$\pi = 6.0 \times 10^{-4}$ atm

$T = 300$K, $R = 0.0821$ Latm. K^{-1} mol^{-1}

$\pi = W_2 \times RT/(M_2 V)$

$\therefore M_2 = 4 \times 0.0821 \times 300/(6 \times 10^{-14} \times 1)$

$= 16.42 \times 10^4 = 1.6 \times 10^5$

54. (1.5) $(1.5)^n = 1.837$, $n = \dfrac{\log 1.837}{\log 1.5} = 1.5$

55. (2) The maximum concentration of SO_4^{2-} will be same as the solubility of M_2SO_4;

If solubility is x mol L^{-1}

$M_2SO_4 \rightleftharpoons 2M^+ + SO_4^{2-}$

$K_{sp} = 4x^3$, therefore $x = (K_{sp}/4)^{1/3}$

$[SO_4^{-2}] = x = (K_{sp}/4)^{1/3} = [(3.2 \times 10^{-5})/4]^{1/3}$

$= 2 \times 10^{-2}$ M

56. (0.2) $MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

In the above reaction 1 mole of MnO_4^- reacts with 5 mole of Fe^{2+}.

or 1 mole of Fe^{2+} reacts with 0.2 mole of MnO_4^-.

57. (4) 2 mL of 1 M NaCl contains

$NaCl = \dfrac{2}{1000} = 2$ m mole Thus 500 mL of As_2S_3 sol require NaCl for complete coagulation = 2 m mole

Hence 1 L, i.e., 1000 mL of the sol require NaCl for complete coagulation = 4 m mole

Therefore, flocculation value of NaCl = 4.

58. (0.22) When pH = 14

$[H^+] = 10^{-14}$

$$[OH^-] = \frac{Kw}{[H^+]} = \frac{10^{-14} m}{10^{-14} m} = 1\ m$$

$$K_{sp} = [Cu^{2+}][OH^-]^2 = 10^{-19}$$

$$\therefore\ [Cu^{2+}] = \frac{10^{-19}}{[OH^-]^2} = 10^{-19}$$

The half cell reaction

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

$$E = E^\circ - \frac{0.059}{2} \log \frac{1}{[Cu^{2+}]}$$

$$= 0.34 - \frac{0.059}{2} \log \frac{1}{10^{-19}} = -0.22\ V$$

59. (3) F_{metal} = (Wt of metal × 96500)/No of coulombs

$$= \frac{(22.2 \times 96500)}{(2 \times 5 \times 60 \times 60)} \approx 59.5$$

Oxidation no. of metal $= \dfrac{177}{59.5} = +3$

60. (12) Total pressure of mixture of gases A,B & C = 10 atm.

Partial pressure of

A = (No. of moles of A × 10)/ 10

or (No. of moles of A × 10)/ 10 = 3

Hence, no. of moles of A = 3

∴ Partial pressure of

B = (No. of moles of B × 10)/ 10

(since partial pressure of B = 1atm)

∴ No.of moles of B = 1

Now the no. of moles of C = 10 – (3+1) = 6;

1 mole of C weighs = 2 g.

∴ 6 mole of C will weigh = 2 × 6 = 12

MATHEMATICS

61. (3)

$^{n+4}C_r - (^nC_r + 3.^nC_{r-1} + 3.^nC_{r-2} + ^nC_{r-3})$

$= ^{n+4}C_r - (^nC_r + ^nC_{r-1}) + (^nC_{r-1} + ^nC_{r-2})$

$+ (^nC_{r-1} + ^nC_{r-2}) + (^nC_{r-2} + ^nC_{r-3})$

$= ^{n+4}C_r - ^{n+1}C_r + ^{n+1}C_{r-1} + ^{n+1}C_{r-1} + ^{n+1}C_{r-2}$

$= ^{n+4}C_r - ^{n+2}C_r + ^{n+2}C_{r-1}$

$= ^{n+4}C_r - ^{n+3}C_r = ^{n+3}C_{r-1}$

62. (4) Let $\alpha' = \alpha/(\alpha - 1)$, $\beta' = \beta/(\beta - 1)$

$\Rightarrow \alpha = \alpha'/(\alpha' - 1)$, $\beta = \beta'/(\beta' - 1)$

$\Rightarrow 1/\alpha = (\alpha' - 1)/\alpha'$, $1/\beta = (\beta' - 1)/\beta'$

Then equation whose roots are $1/\alpha$, $1/\beta$, is

$x^2 - (1/\alpha + 1/\beta)x + 1/\alpha\beta = 0$

$\Rightarrow \quad x^2 - [(\alpha' - 1)/\alpha' + (\beta' - 1)/\beta']x$

$+ [(\alpha' - 1)(\beta' - 1)] / \alpha'\beta' = 0$

$\Rightarrow \alpha'\beta'x^2 - [2\alpha'\beta' - (\alpha' + \beta')]x + \alpha'\beta' - (\alpha' + \beta') + 1 = 0$

$\Rightarrow bx^2 - (a + 2b)x + a + b + 1 = 0$

63. (2) $f(\theta) = \cos^2(\cos\theta) + \sin^2(\sin\theta)$

Now, put $\theta = \pi/2$ & 0

$f(0) = \cos^2 1 + 0 = \cos^2 1$

$f(\pi/2) = 1 + \sin^2 1$

Since, $\sin^2 1 > \cos^2 1$

$[\because \sin\theta > \cos\theta \ \forall \ \frac{\pi}{4} < \theta < \frac{\pi}{2} \ \& \ 1 > \frac{\pi}{4}]$

Now, seeing the options $1 + \sin^2 1$ is greater than all other options.

64. (3) tanA/2, tanB/2, tanC/2 are in A.P.

$\Rightarrow \tan A/2 - \tan B/2 = \tan B/2 - \tan C/2$

$$\Rightarrow \frac{\sin A/2}{\cos A/2} - \frac{\sin B/2}{\cos B/2} = \frac{\sin B/2}{\cos B/2} - \frac{\sin C/2}{\cos C/2}$$

$\Rightarrow \cos(C/2)\sin(A/2 - B/2)$

$= \cos A/2 \sin(B/2 - C/2)$

$\Rightarrow \cos B - \cos A = \cos C - \cos B$

$\Rightarrow 2\cos B = \cos A + \cos C$

$\Rightarrow$ secA, secB, secC are in H.P.

65. (1) Any tangent to the parabola $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \qquad ...(1)$$

meets $x^2 = 4by$ where $x^2 = 4b\left(mx + \frac{a}{m}\right)$

or $mx^2 - 4bm^2x - 4ab = 0$ ...(2)

If the line (1) be a tangent to the second parabola, then roots of (2) must be equal. The condition for this is

$$16b^2m^4 + 16abm = 0 \quad \text{or} \quad m^3 = -\frac{a}{b}$$

$$\therefore \quad m = -\left(\frac{a}{b}\right)^{1/3} = -\frac{a^{1/3}}{b^{1/3}}$$

Substituting this value of m in (1), we get

$$y = -\frac{a^{1/3}}{b^{1/3}}x + a\left(-\frac{b^{1/3}}{a^{1/3}}\right)$$

i.e. $xa^{1/3} + yb^{1/3} + a^{2/3}b^{2/3} = 0$

which is the required equation of the common tangent.

66. (4) Let the point of contact be (h,k); equation of chord of contact is T = 0

$\Rightarrow xh + yk - 12 = 0$ ……….(1)

Equation of common chord $C_1 - C_2 = 0$

$\Rightarrow 5x - 3y - 10 = 0$ ………(2)

(1) and (2) represent the same line

$\Rightarrow$ h/5 = k/(–3) = –12/(–10)

$\Rightarrow$ h = 6, k = –18/5

67. (4) $R = \{(x, y) : x, y \in N \text{ and } x^2 - 4xy + 3y^2 = 0\}$

Now, $x^2 - 4xy + 3y^2 = 0$

$\Rightarrow (x - y)(x - 3y) = 0$

$\therefore\ x = y$ or $x = 3y$

$\therefore$ R = {(1, 1), (3, 1), (2, 2), (6, 2), (3, 3), (9, 3),......}

Since (1, 1), (2, 2), (3, 3),...... are present in the relation, therefore R is reflexive.

Since (3, 1) is an element of R but (1, 3) is not the element of R, therefore R is not symmetric

Here $(3, 1) \in R$ and $(1, 1) \in R \Rightarrow (3, 1) \in R$

$(6, 2) \in R$ and $(2, 2) \in R \Rightarrow (6, 2) \in R$

For all such $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$

Hence R is transitive.

68. (4) $\lim\limits_{x\to\pi/6}\left(4 - 3\sin x - 2\cos^2 x\right)^{1/(2\sin x - 1)}$

$$\lim_{x\to\pi/6}\left[1 + (2\sin x - 1)(\sin x - 1)\right]^{\frac{\sin x - 1}{(2\sin x - 1)(\sin x - 1)}} = e^{-1/2}$$

69. (1) $f'(x) = -12\cos^3 x \sin x - 30\cos^2 x \sin x - 12\cos x \sin x = -6\sin x \cos x(\cos x + 2)(2\cos x + 1)$

$f'(x) = 0$, for $x = 0, \frac{\pi}{2}, \frac{2\pi}{3}, \pi$

Clearly, $f'(x) > 0$ for $\frac{\pi}{2} < x < \frac{2\pi}{3}$

And $f'(x) < 0$; for $0 < x < \frac{\pi}{2}$ or $\frac{2\pi}{3} < x < \pi$

70. (4) $I = \int_0^\pi \sin^n x \cos^{2m+1} x\,dx$

$= \int_0^\pi \sin^n(\pi - x).\cos^{2m+1}(\pi - x)\,dx$

$= -\int_0^\pi \sin^n x \cos^{2m+1} x\,dx = -I \Rightarrow I = 0$

Similarly, one can show that

$\int_0^\pi \cos^{2m+1} x\,dx = 0$

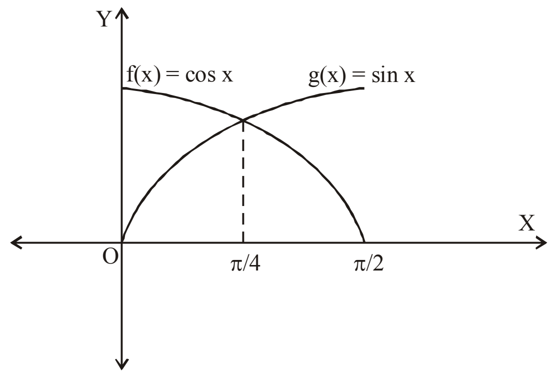
71. (2) Since, $f'(x) = f(x)$, therefore, $f(x) = \alpha e^x$. Since, $f(0) = 2$, therefore, $f(x) = 2e^x$.

$\therefore I = 2\int \frac{e^x}{3 + 8e^x}dx$

Put $e^x = t$, $\therefore dt = e^x dx$

$\therefore\ I = 2\int \frac{dt}{3 + 8t} \Rightarrow I = \frac{1}{4}\log(3 + 8e^x) + C$

72. (2) $y = |\cos x - \sin x|$



Required area $= 2\int_0^{\pi/4}(\cos x - \sin x)\,dx$

$= 2\left[\sin x + \cos x\right]_0^{\pi/4}$

$= 2\left[\frac{2}{\sqrt{2}} - 1\right] = (2\sqrt{2} - 2)$ sq. units

73. (1) Any vector (x, y, z) can be represented as a linear combination of three non coplanar vectors.

$\overrightarrow{OP} = x\vec{a} + y\vec{b} + z\vec{c}$

But $\vec{P}$ is $(3,2,1) \Rightarrow \overrightarrow{OP} = 3\vec{a} + 2\vec{b} + \vec{c}$

Now if coordinate axis changes then the vector is represented by

$x(\vec{a} + \vec{b} + \vec{c}) + y(\vec{a} - \vec{b} + \vec{c}) + 2(\vec{a} + \vec{b} - \vec{c})$

$= 3\vec{a} + 2\vec{b} + \vec{c}$

Comparing coeff of $\vec{a}, \vec{b}$ & $\vec{c}$, we get

$x + y + z = 3,\ x - y + z = 2,\ x + y - z = 1$

On solving, we get

$x = \frac{3}{2}, y = \frac{1}{2}, z = 1$

Hence the new co-ordinates of point P are $\left(\frac{3}{2},\frac{1}{2},1\right)$

74. (1) Let DR of the line of intersection of the planes be a, b, c

$\Rightarrow \; a-2b-c=0$ ....(1)

and $a+b+3c=0$ ....(2)

on solving, we get $\frac{a}{-5}=\frac{b}{-4}=\frac{c}{3}=k$

$\Rightarrow \; a=-5k, b=-4k, c=3k$

Thus required eq^n of line is

$$\frac{x-2}{-5}=\frac{y-3}{-4}=\frac{z-1}{3}$$

75. (3) $x+2=6 \Rightarrow x=4$, Its converse is $x=4 \Rightarrow x+2=6$ i.e. if $x=4$ then $x+2=6$

76. (1) Since, $x_1<x_2 \Rightarrow f(x_1)<f(x_2)$ if

$x_1,x_2,---,x_n$ are in ascending order

$\Rightarrow f(x_1),f(x_2),---f(x_n)$ are also in ascending order.

If M is the median along x, then f (M) is the median along y

77. (1) The required event occurs if two sixes are observed in the first seven throws and a six is observed on the eighth throw. If p is the probability that a six shows on the die, the number of throws n is 7, and X is the number of times a six is observed, then $X \sim B(7,p)$. Therefore the required probability equals P(X = 2) times the probability of getting a six on the eighth throw, i.e., it equals

$$({}^7C_2p^2q^5)(p)=({}^7C_2)\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^5\left(\frac{1}{6}\right)$$

$$=\frac{{}^7C_2(5^5)}{6^8}$$

78. (2) Let us consider

$\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k} \quad \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$

then as per question

$$\left[\lambda(\vec{a}+\vec{b})\ \lambda^2\vec{b}\ \lambda\vec{c}\right]=\left[\vec{a}\ \vec{b}+\vec{c}\ \vec{b}\right]$$

$$\Rightarrow \begin{vmatrix}\lambda(a_1+b_1) & \lambda(a_2+b_2) & \lambda(a_3+b_3)\\ \lambda^2b_1 & \lambda^2b_2 & \lambda^2b_3\\ \lambda c_1 & \lambda c_2 & \lambda c_3\end{vmatrix}$$

$$=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1+c_1 & b_2+c_2 & b_3+c_3\\ b_1 & b_2 & b_3\end{vmatrix}$$

$$\Rightarrow \lambda^4\begin{vmatrix}a_1+b_1 & a_2+b_2 & a_3+b_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

$$=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1+c_1 & b_2+c_2 & b_3+c_3\\ b_1 & b_2 & b_3\end{vmatrix}$$

$$\Rightarrow \lambda^4\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}=\begin{vmatrix}a_1 & a_2 & a_3\\ c_1 & c_2 & c_3\\ b_1 & b_2 & b_3\end{vmatrix}$$

$$\Rightarrow \lambda^4=-1$$

Hence λ has no real values.

79. (3)

$$\tan^{-1}\frac{1}{1+n(n+1)}+\tan^{-1}\frac{1}{1+(n+1)(n+2)}+......+$$

$$\tan^{-1}\frac{1}{1+(n+19)\,(n+20)}$$

$$=\tan^{-1}\frac{n+19}{n+21}-\tan^{-1}\frac{n-1}{n+1}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{n^2+n+1}\right)+\tan^{-1}\left(\frac{1}{n^2+3n+3}\right)+......+$$

$$\tan^{-1}\frac{1}{1+(n+19)\,(n+20)}$$

$$= \tan^{-1}\left(\frac{\frac{n+19}{n+21}-\frac{n-1}{n+1}}{1+\frac{n+19}{n+21}\times\frac{n-1}{n+1}}\right)$$

$$= \tan^{-1}\frac{20}{n^2+20n+1} = S$$

$$\therefore \tan S = \frac{20}{n^2+20n+1}$$

80. (4) Given expansion can be written as

$\left(\frac{x-1}{x}\right)^n .(1-x)^n = (-1)^n x^{-n} (1-x)^{2n}$

Total number of terms will be $2n+1$ which is odd ($\because 2n$ is always even)

$\therefore$ Middle term $= \frac{2n+1+1}{2} = (n+1)$ th

Now, $T_{r+1} = {}^nC_r (1)^r x^{n-r}$

So, $\frac{{}^{2n}C_n \cdot x^{2n-n}}{x^n.(-1)^n} = {}^{2n}C_n .(-1)^n$

Middle term is an odd term. So, $n+1$ will be odd.

So, n will be even.

$\therefore$ Required answer is ${}^{2n}C_n$.

81. (0.33) $S = \frac{5}{1^2 \cdot 4^2} + \frac{11}{4^2 \cdot 7^2} + \frac{17}{7^2 \cdot 10^2} + \ldots \infty.$

$3S = \frac{3\cdot 5}{1^2\cdot 4^2} + \frac{3\cdot 11}{4^2\cdot 7^2} + \frac{3\cdot 17}{7^2\cdot 10^2} + \ldots\infty$

$\Rightarrow 3S = \frac{(4-1)(4+1)}{1^2\cdot 4^2} + \frac{(7-4)(7+4)}{4^2\cdot 7^2} + \frac{(10-7)(10+7)}{7^2\cdot 10^2} + \ldots\infty$

$\Rightarrow S = \frac{1}{3} = 0.33$

82. (0.60) Solving $2x + y = 3$ i.e. $y = 3 - 2x$ and $4x^2 + y^2 = 5$, we get

$4x^2 + (3-2x)^2 = 5 \Rightarrow 8x^2 - 12x + 4 = 0$

$\Rightarrow 2x^2 - 3x + 1 = 0 \Rightarrow (2x-1)(x-1) = 0$

$\therefore x = \frac{1}{2}, 1$

When $x = 1$, $y = 3 - 2 = 1$ and when $x = \frac{1}{2}$, $y = 3 - 1 = 2$

$\therefore$ Points of intersection are $P\left(\frac{1}{2}, 2\right)$ and $Q(1, 1)$.

Equation of tangent at $P\left(\frac{1}{2}, 2\right)$ is

$4x\left(\frac{1}{2}\right) + y(2) = 5$ i.e. $2x + 2y = 5$

Its slope $= -1$ $\therefore$ Slope of the normal $= 1$

Again, equation of tangent at $Q(1, 1)$ is

$4x(1) + y(1) = 5$ i.e. $4x + y = 5$

Its slope $= -4$ $\therefore$ Slope of the normal $= \frac{1}{4}$

$$\therefore \tan\theta = \frac{1-\frac{1}{4}}{1+(1)\left(\frac{1}{4}\right)} = \frac{\frac{3}{4}}{\frac{5}{4}} = \frac{3}{5} = 0.6$$

83. (2007) $\det(M_r) = \begin{vmatrix} r & r-1 \\ r-1 & r \end{vmatrix} = 2r - 1$

$$\sum_{r=1}^{2007} \det(M_r) = 2\sum_{r=1}^{2007} r - 2007$$

$$= 2 \times \frac{2007 \times 2008}{2} - 2007 = (2007)^2$$

$= k^2 \Rightarrow k = 2007$

84. (5.00) $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$

Cofactors of various entries are

$4, -5, 1; 2, 0, -2; 2, 5, 3$

$|A| = 1\times 4 + (-1)\times -5 + 1\times 1 = 10$

Cofactor Matrix $C = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$

$$\therefore \text{Adj } A = C^T = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{10}\begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Comparing, we get $\alpha = 5$

85. (1.50) $y = \tan^{-1}\left(\frac{3x^2-1}{3x-x^3}\right)$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{3x-x^3}{3x^2-1}\right) = \frac{\pi}{2} + 3\tan^{-1} x$$

$$\frac{dy}{dx} = \frac{3}{1+x^2}$$

$$u = \sin^{-1}\left(\frac{x^2-1}{x^2+1}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right)$$

$$= -\frac{\pi}{2} + 2\tan^{-1} x$$

$$\frac{du}{dx} = \frac{2}{1+x^2}$$

$$\frac{dy}{dx} = \frac{dy/du}{du/dx} = \frac{3}{2} = 1.50$$

86. (1.00) Let $f(x) = ax^2 + bx + c$

$$g(x) = \int f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx$$

$$\therefore\ g(1) = \frac{a}{3} + \frac{b}{2} + c = 2a + 3b + c = 0$$

$g(0) = 0 = g(1)$

$\therefore$ g(x) is continuous apply Rolle's theorem

$\Rightarrow g(x) = 0$ for some value $x \in (0,1)$

$\Rightarrow g'(x) = f(x) = ax^2 + bx + c = 0$

has a root in $(0, 1) = (m, n)$

$\Rightarrow\ m + n = 1.$

87. (1.00) Let $z_1 = 1 + i$ and $z_2 = 1 - i$

$$\frac{z_2}{z_1} = \frac{1-i}{1+i} = \frac{(1-i)(1-i)}{(1+i)(1-i)} = -i$$

$$\frac{2z_1+3z_2}{2z_1-3z_2} = \frac{2+3\left(\frac{z_2}{z_1}\right)}{2-3\left(\frac{z_2}{z_1}\right)} = \frac{2-3i}{2+3i}$$

$$\left|\frac{2z_1+3z_2}{2z_1-3z_2}\right| = \left|\frac{2-3i}{2+3i}\right| = \left|\frac{2-3i}{2+3i}\right|$$

$$\left[\because \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}\right]$$

$$= \frac{\sqrt{4+9}}{\sqrt{4+9}} = 1$$

88. (2.00) The given lines are

$$x-1 = \frac{y+3}{-\lambda} = \frac{z-1}{\lambda} = s \qquad \text{......(1)}$$

and $2x = y - 1 = \dfrac{z-2}{-1} = t$ ......(2)

The lines are coplanar, if

$$\begin{vmatrix} 0-(-1) & -1-3 & -2-(-1) \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix} = 0$$

$$c_2 \to c_2 + c_3;\ \begin{vmatrix} 1 & -5 & -1 \\ 1 & 0 & \lambda \\ \frac{1}{2} & 0 & -1 \end{vmatrix} = 0$$

$$\Rightarrow 5(-1-\frac{\lambda}{2}) = 0 \Rightarrow \lambda = -2$$

$\Rightarrow\ |\lambda| = 2$

89. (0.50) Since A and B are two independent events

$\therefore\ P(A \cap B) = P(A)\,P(B)$

$\Rightarrow\ P(A)\,P(B) = \dfrac{1}{6}$ (given) ...(i) and P (neither of A nor B) = $P\,\overline{(A \cup B)} = \frac{1}{3}$

$\Rightarrow P(A \cup B) = 1 - P(\overline{A \cup B}) = 1 - \frac{1}{3} = \frac{2}{3}$

We know that

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad \frac{2}{3} = P(A) + P(B) - \frac{1}{6}$

$\Rightarrow \quad P(A) + P(B) = \frac{2}{3} + \frac{1}{6} = \frac{5}{6}$

$\Rightarrow \quad P(A) + \frac{1}{6P(A)} = \frac{5}{6}$ [from (i)]

$\Rightarrow \quad 6\,[P(A)]^2 - 5P(A) + 1 = 0$

$\Rightarrow \quad (2P(A) - 1)(3P(A) - 1) = 0$

$\Rightarrow \quad P(A) = \frac{1}{2}, \frac{1}{3}$

$\therefore$ Most $P(A) = \frac{1}{2}$

90. (0.05) $B\left(5, \frac{1}{2}\right) \Rightarrow n = 5, p = \frac{1}{2}, q = \frac{1}{2}$

$B\left(7, \frac{1}{2}\right) \Rightarrow n = 7, p = \frac{1}{2}, q = \frac{1}{2}$

Since X and Y are independent events

$\Rightarrow X + Y = 3$

$\Rightarrow X = 0, Y = 3; X = 1, Y = 2; X = 2, Y = 1; X = 0, Y = 3$

$\therefore P(X + Y = 3)$

$$= {}^5C_0\left(\frac{1}{2}\right)^5 \cdot {}^7C_3\left(\frac{1}{2}\right)^7 + {}^5C_1\left(\frac{1}{2}\right)^5\ {}^7C_2\left(\frac{1}{2}\right)^7 + {}^5C_2\left(\frac{1}{2}\right)^5\ {}^7C_1\left(\frac{1}{2}\right)^7 + {}^5C_3\left(\frac{1}{2}\right)^5\ {}^7C_0\left(\frac{1}{2}\right)^7$$

$= \frac{55}{1024} = 0.05$